19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas

CASE 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

CASE 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service

EVIDENTIARY HEARING

DATE: February 20, 2020

LOCATION: 19th Floor Hearing Room
Agency Building Three,
Empire State Plaza
Albany, New York

ALJ JAMES A. COSTELLO, DPS

ALJ MAUREEN F. LEARY, DPS

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19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas

(On record 9:01 a.m.)

A.L.J. COSTELLO: We're here on the 7th day of evidentiary hearings in cases 19-G-0309 and 19-G-0310. We'll begin with appearances and we'll start with the companies.

MR. MALONEY: Good morning, Your Honors, on behalf of the companies. My name is Kenneth Maloney of the law firm of Cullen and Dykman. Also appearing with me today are Phillip Decicco, Patrick O'Brien and Jeremy Euto for the companies.

A.L.J. COSTELLO: Department of Public Service Staff.

MR. FORST: For the Department of Public Service Staff, you have Nicholas Forst, Brandon Goodrich and Raquel Parks.

A.L.J. COSTELLO: Utility Intervention Unit.

MS. KASOW: For the Department of State Utility Intervention Unit, Jillian Kasow and Katie O'Hare.

A.L.J. COSTELLO: Good morning. Public Utility Law Project.

MS. WHEELOCK: For PULP we have Laurie Wheelock and Richard Berkley.

A.L.J. COSTELLO: Good mooring. City of New York.

MR. CONWAY: For the City of New York you have from the law firm of Couch White, LLP, Adam Conway, Russell King and this afternoon Justin Fung.

A.L.J. COSTELLO: Okay. Thank you. And is there anyone here who is making an appearance who I didn't call? No. Okay. Not hearing anything, I guess, we'll begin with the D.P.S. Staff Finance Panel. Okay.

And what I'm going to ask you to do is starting from the gentleman to my -- the furthest from me, just state your name and business address for the record.

MR. DUAH: My name is Kwaku Duah, K-W-A-K-U D-U-A-H., Principal Utility Financial Analyst. My address is 3 Empire State Plaza, Albany, New York 12223.

A.L.J. LEARY: Is your microphone on?

MR. DUAH: It's on.

A.L.J. LEARY: Okay. It needs to be closer to all the witnesses.

MR. QADIR: Abdul Qadir, 3 Empire State Plaza, Albany, New York.

MR. BULLOCK: Kenneth Bullock, Senior Utility Financial Analyst, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Okay. Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MR. DUAH: I do.

MR. QADIR: I do.

MR. BULLOCK: I do.

Kwaku Duah; Sworn

Abdul Qadir; Sworn

Kenneth Bullock; Sworn

A.L.J. COSTELLO: You have to say it out loud. Okay. Thank you. You may be seated. Counsel.

DIRECT EXAMINATION

BY MR. FORST:

Q. Panel members before you is a document entitled prepared corrected testimony of Staff Finance Panel consisting of a cover page and 134 pages of question and answers dated January 2020 and six exhibits submitted with your testimony labeled S.F.P.-1 through S.F.P.-6. Is that correct?

A. (Mr. Duah) Yes.

Q. These exhibits have been pre-marked as Exhibits 424 through 429. Panel members, do you also have before you one exhibit labeled S.F.P.-7 marked as Hearing Exhibit 430 and one exhibit labeled corrected S.F.P.-7 marked as Hearing Exhibit 530 dated January 2020. Is that correct?

A. Yes.

Q. Panel, did you also submit with your testimony 32 additional exhibits labeled S.F.P.-8 through S.F.P.-39?

A. Yes.

Q. These exhibits have been pre-marked as Exhibits 431 through 462. Was this set of testimony and exhibits prepared by you are under your direct supervision?

A. Yes.

Q. Do you have any changes or corrections to make to that testimony?

A. No.

Q. If you are asked the same questions today under oath, would you answer them the same way?

A. Yes.

Q. And do you affirm the information contained in your testimony and exhibits are true to the best of your knowledge, information and belief?

A. Yes.

MR. FORST: Your Honors, I would like to move the pre-filed corrected testimony of the Staff Finance Panel to be entered into the record as if given orally during the proceedings here today.

A.L.J. COSTELLO: The motion is granted and at this point the court reporter should insert the following file DPS Staff Finance Panel Corrected Direct testimony. #

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

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In the Matter of

Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

JANUARY 2020

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Prepared Corrected Testimony of:

Staff Finance Panel

Kwaku Duah
Principal Utility Financial Analyst

Abdul Qadir
Associate Utility Financial Analyst

Kenneth Bullock
Senior Utility Financial Analyst

Office of Accounting, Audits and Finance
State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Finance Panel (SFP), please state your names, current employer and business address.

A. Our names are Kwaku Duah, Abdul Qadir, and Kenneth Bullock. We are employed by the New York State Department of Public Service (Department). Our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Mr. Duah, what is your position at the Department?

A. I am a Principal Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I received my Master's degree in Business Administration with a concentration in Finance and Accounting from State University of New York Institute of Technology, now known as SUNY Polytechnic Institute, in 2005.

Q. Please briefly describe your current responsibilities with the Department.

A. As a Principal Utility Financial Analyst, my assignments involve analyzing the financial condition, financing mechanisms, risk, costs of

debt and equity, diversification, and the relative competitive position of utilities and their holding company parent(s). My assignments generally involve testifying in rate cases on financial issues, analyzing financing proposals and special projects that usually involve risk and return analyses.

Q. Have you previously testified in a regulatory proceeding before the New York State Public Service Commission (Commission)?

A. Yes, I have presented testimony before the Commission in rate cases concerning NYSEG/RG&E in Cases 09-E-0715, 09-G-0716, 09-E-0717, and 09-G-0718; Niagara Mohawk in Cases 08-G-0609,10-E-0500, 12-G-0202 and 12-E-0201; Long Island Water Corporation in Case 11-W-0020; United Water New Rochelle and United Water Westchester in Cases 13-W-0539, 13-W-0564 and 14-W-0006; Central Hudson Gas and Electric in Cases 14-E-0318 and 14-G-0319; Central Hudson Gas and Electric in Cases 17-E-0459 and 17-G-00460; Suez Water New York in Cases 16-W-0130, 19-W-0168, 19-W-0269; and New York American Water Company, Inc. in Case 16-W-0259.

Q. Mr. Qadir, what is your position at the Department?

A. I am an Associate Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I graduated from the University at Albany in 2004 and received a Bachelor of Science degree with honors in Marketing and Finance. In 2007, I earned a Master's degree in Economics with a concentration in Finance and International Trade from the University at Albany. Prior to joining the Department, I worked in the banking industry, primarily as a loan officer. I joined the Department in March 2012.

Q. Please briefly describe your responsibilities with the Department.

A. My responsibilities as an Associate Utility Financial Analyst include analyzing a company's financial condition, capital structures, financing mechanisms, risks, costs of debt and equity, diversification, and the competitive position of utilities operating in New York State. My responsibilities also include

analyzing financial petitions, performing rate of return analysis and other special projects.

Q. Have you previously testified before the Commission?

Yes. I have testified in the United Water New York Inc. rate case, Case 13-W-0295; NYSEG and RG&E rate cases, Cases 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-0286; Keyspan Gas East Corporation d/b/a National Grid (KEDLI) and The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) rate cases, Cases 16-G-0058 and 16-G-0059; Corning Natural Gas Corporation (Corning) rate case, Case 16-G-0369; and Central Hudson Gas and Electric in Cases 17-E-0459 and 17-G-00460.

Q. Mr. Bullock, what is your position at the Department?

A. I am employed as a Senior Utility Financial Analyst in the Office of Accounting, Audits and Finance.

Q. Please describe your educational background and professional experience.

A. I received a Master of Business Administration from the College of Saint Rose with an emphasis in entrepreneurship. Prior to joining the

Department, I worked in the telecommunications industry. My responsibilities in that industry included the construction and maintenance of plant, remote monitoring and diagnostics of various switching systems along with the outsourcing, design and costing of information technology networks for multi-national corporations. I joined the Department in July 2013.

Q. Please briefly describe your current responsibilities with the Department.

A. My responsibilities as a Senior Utility Financial Analyst include analyzing a company's financial condition, capital structures, financing mechanisms, risks, costs of debt and equity, diversification, and the competitive position of utilities operating in New York State.

Q. Have you previously provided testimony before the Commission?

A. Yes, I provided testimony before the Commission as a cost of capital witness in Cases 13-W-0539 and 13-W-0564, rate cases involving United Water New Rochelle and United Water Westchester; in Cases 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-

0286 rate cases involving New York State Electric & Gas Corporation and Rochester Gas and Electric Corporation; Case 16-W-0130 involving Suez Water New York; and in Case 18-G-0133 involving St. Lawrence Gas Company, Inc.

## Purpose of Testimony

Q. What is the purpose of your testimony in this proceeding?

A. The purpose of our testimony is to recommend the fair rate of return to be used by the Staff Accounting Panel to determine the revenue requirement for The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) and Keyspan Gas East Corporation d/b/a National Grid (KEDLI) (collectively the Companies). Our testimony will concentrate on the appropriate ratemaking capital structure and the cost of the various capital components in the capital structure for the rate year ending March 31, 2021. We will also respond to issues raised in the direct testimonies of the Companies' capital structure witness, Mr. Jonathan Cohen and cost of equity witness, Ms. Ann E. Bulkley.

Q. Does your testimony rely on interrogatory responses (IRs) provided by the Companies?

A. Yes. These IR responses are included as Exhibit___(SFP-1).

Q. Are you sponsoring any exhibits?

A. Yes. We are sponsoring 39 exhibits as labeled below.

| Exhibit Number | Description |
|---|---|
| Exhibit___(SFP-1) | Interrogatory Responses (IRs) of the Companies Supporting Staff Finance Panel Testimony; |
| Exhibit___(SFP-2) | KEDNY/KEDLI Capital Structure; |
| Exhibit___(SFP-3) | National Grid Corporate Structure; |
| Exhibit___(SFP-4) | New York Ring-fencing provisions; |
| Exhibit___(SFP-5) | Moody's Credit Opinion for KEDNY, April 8, 2019; |
| Exhibit___(SFP-6) | Moody's Credit Opinion for KEDLI, April 8, 2019; |
| Exhibit___(SFP-7) | Ms. Bulkley's Proxy Groups' Average Common Equity for 2014 to 2017 |
| Exhibit___(SFP-8) | Standard & Poor's (S&P) |

Credit Opinion on KEDNY, September 2018;
Exhibit___(SFP-9) Standard & Poor's (S&P)
Credit Opinion on KEDLI, September 2018;
Exhibit___(SFP-10) KEDNY/KEDLI Implied Credit
Metrics;
Exhibit___(SFP-11) S&P Corporate Rating
Methodology, Updated April 1, 2019;
Exhibit___(SFP-12) Moody's Rating Methodology
for Regulated Electric and Gas Utilities, June
23, 2017;
Exhibit___(SFP-13) Debt Cost Calculation;
Exhibit___(SFP-14) Summary of Staff's Cost of
Equity;
Exhibit___(SFP-15) Staff's Proxy Group
Exhibit___(SFP-16) Three-Month Stock Prices for
Staff Proxy Group;
Exhibit___(SFP-17) Discounted Cash Flow
Calculation for Staff Proxy Group;
Exhibit___(SFP-18) Blue Chip Economic
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Exhibit___(SFP-19) Merrill Lynch Quantitative
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Exhibit___(SFP-20) Staff Capital Asset Pricing
Model Results;

Exhibit___(SFP-21) Ambika Prasad Dash's Article: "Security Analysis and Portfolio";

Exhibit___(SFP-22) An Excerpt on Sample Size from Dr. Morin's book entitled, Regulatory Finance;

Exhibit___(SFP-23) Illustration of Different Accounting Policy Choices on Operating Income;

Exhibit___(SFP-24) Equity Analysts' Coverage on MGE Energy Inc., El Paso Electric and Otter Tail Corp.;

Exhibit___(SFP-25) Value Line Investment Survey, Quality Control Procedures;

Exhibit___(SFP-26) Energy Information Association (EIA): Annual Energy Outlook, January 2019;

Exhibit___(SFP-27) Generic Finance Case's DCF Results Recast;

Exhibit___(SFP-28) Trends in Utility Stock Valuation and Dividend Yields;

Exhibit___(SFP-29) Wall Street Journal Article entitled, "Can Low Rates Explain High Stock Prices?";

Exhibit___(SFP-30) "The Cost of Equity - A Practitioner's Guide (2010 Edition)" by David C.

Parcell;

Exhibit___(SFP-31) Wall Street Journal Article on Federal Reserve interest rates, dated July 19, 2019;

Exhibit___(SFP-32) Ms. Bulkley's Market Return Using the Federal Energy Regulatory Commission's (FERC) Approach;

Exhibit___(SFP-33) Article by Kent Baker et al. entitled "Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice,";

Exhibit___(SFP-34) "Whose Beta is Best?", Ms. Diana Harrington;

Exhibit___(SFP-35) Impact of Correcting the Flaws in Ms. Bulkley's ROE Approach;

Exhibit___(SFP-36) Regulatory Research Associates' (RRA) Report on State Regulatory Evaluations;

Exhibit___(SFP-37) RRA's, Report on New York State Regulatory Environment;

Exhibit___(SFP-38) Moody's Credit Opinion for National Grid USA, July 2017;

Exhibit___(SFP-39) Commission's Chair's Meeting with RRA dated October 2014;

## Rate of Return

### Summary

Q. Please summarize your overall rate of return recommendation for KEDNY and how it varies from the Company's request.

A. We are recommending an overall after-tax rate of return of 6.08% for KEDNY, as opposed to KEDNY's request of 6.76%. On a pre-tax basis, our recommended overall rate of return is 7.58% compared to KEDNY's request of 8.53%. The difference between our recommended rate of return and KEDNY's is mainly attributable to a reduction in the cost of equity from 9.65% to 8.20%.

Q. Please summarize your overall rate of return recommendation for KEDLI and how it varies from the Company's request.

A. We are recommending an overall after-tax rate of return of 6.04% for KEDLI, as opposed to the Company's request of 6.74%. On a pre-tax basis, our recommended overall rate of return is 7.54% compared to KEDLI's request of 8.51%. The difference between our recommended rate of return and KEDLI's is attributable to a reduction in the cost of equity from 9.65% to 8.20%.

Q. Please explain the revenue requirement impacts of your adjustments to the Companies' requested rates of return?

A. In terms of revenue requirement, the difference between Staff's and KEDNY's pre-tax rate of return (PTROR) and after-tax rate of return (ATROR) account for approximately $49.0 million and $34.0 million, respectively. Similarly, the difference between Staff's and KEDLI's PTROR and ATROR account for approximately $31.4 million and $22.8 million, respectively. For KEDNY, 10 basis point in ROE is worth nearly $3.3 million and 100 basis points in equity ratio is worth about $5.0 million. For KEDLI, 10 basis point in ROE is worth approximately $2.1 million and 100 basis points in equity ratio is worth about $3.1 million. Our recommended rates of return for both Companies are summarized in Exhibit___(SFP-2).

**Fair Rate of Return**

Q. Please elaborate as to what is meant by a fair rate of return.

A. A fair rate of return for a regulated utility is

the overall rate of return that enables the company to provide safe and adequate service to its customers, while at the same time assuring the utility continuing support in the capital markets for both its long-term debt and common equity securities at terms that are reasonable given the Company's risk. The overall rate of return is alternatively referred to as a weighted cost of capital.

Q. How is the rate of return typically calculated?

A. The rate of return for a utility company is calculated through a weighted average of the individual cost components of its expected capitalization during the rate year. Typically, there are three sources of capital. The two primary sources are long-term debt and common equity. Customer deposits, while a very small component, is generally reflected in a company's capitalization because it is a relatively permanent and stable source of capital employed by utilities. Investors in debt securities enter into contractual obligations with the utility and receive relatively fixed income streams. Common equity investment is non-contractual. Common

equity investors may share in, but are not guaranteed, a portion of the utility's residual earnings. A fair rate of return allows the utility company to recover its incurred cost of long-term debt while providing its common equity investors the opportunity to earn a return that is comparable to the return available with investments of similar risk.

Q. How are the cost rates of the components of the capital structure typically calculated?

A. The cost rates associated with the company's long-term debt and customer deposits are relatively simple to determine. The costs of existing long-term debt can be readily calculated by examining their contractual terms; i.e., the interest payment on the long-term debt and the amortization of issuance costs. The costs of any new long-term debt instruments, however, require estimation. The cost rate for customer deposits is simply a matter of applying the interest rate that is currently prescribed by the Commission. The cost of common equity, however, is neither contractual nor generically prescribed by the Commission. Its calculation is further

complicated by the fact that all variables affecting a company's cost of equity cannot be directly observed, and instead, some must be estimated.

Q. Is the cost of common equity typically more expensive than the cost of debt for a utility?

A. Yes. Even though both lenders and investors supply the utility with the funds it needs to build and operate its systems, the equity investors only earn a return after the payment of all other expenses. Because equity investors run the risk that a company's achieved returns will not equal their expectations, the return required by equity investors is usually higher than that of the utility's debt holders. Importantly, equity returns are profits and in a company's revenue requirement profits are taxed at the corporate level, which is an added cost to ratepayers that adds to the expense of the equity return.

Q. How can a utility's cost of common equity be measured?

A. The return requirements of a utility's common equity investors can be estimated through a cost of equity analysis. Historically, the Commission

has favored market-based methodologies such as the Discounted Cash Flow (DCF) and the Capital Asset Pricing Model (CAPM) to estimate the return required by equity investors.

**Capital Structure**

Q. Please discuss the corporate structure of KEDNY and KEDLI.

A. KEDNY and KEDLI are both direct subsidiaries of National Grid USA, which is an indirect subsidiary of the ultimate parent, National Grid Plc. (National Grid), a United Kingdom-based utility holding company. As illustrated in Exhibit___(SFP-3), there are two subsidiary tiers between KEDNY and KEDLI and National Grid.

Q. Do the Companies propose that their rates be set using their stand-alone capitalizations or the capitalizations of either National Grid Plc. or National Grid USA?

A. The Companies propose to use their stand-alone capital structures and propose capital structures that reflect a 48.0% common equity ratio.

Q. Did the Companies provide any rationale for proposing their stand-alone capital structure for

the rate year?

A. Yes. The Companies' capital structure witness, Mr. Jonathan Cohen and cost of equity witness, Ms. Ann E. Bulkley present two main reasons why it is appropriate to use the Companies' stand-alone capital structure for ratemaking purposes. First, the companies believe that they implemented strong ring-fencing measures that have been recognized by the major rating agencies. Secondly, the Companies believe that utility operating subsidiaries of their proxy group have for the last four years maintained average common equity ratios more than 8.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding.

Q. What is your response to the Companies first rationale that they have strong ring-fencing measures that have been recognized by the major rating agencies?

A. We agree with the Companies that they have implemented adequate ring-fencing measures that have been acknowledged by both Moody's and Standard and Poor's.

Q. Could you please elaborate on the ring-fencing measures implemented by the Companies?

A. Yes. As a condition of accepting the Commission's approval of the merger between National Grid and the Companies in 2006, both KEDNY and KEDLI agreed to a series of ring-fencing provisions. These provisions, as well as those agreed to by other New York State utilities, are contained in Exhibit___(SFP-4). We will discuss the specific provisions of the ring-fencing measures later in our testimony.

Q. Please discuss Moody's recent opinion about the strength of the Companies ring-fencing measures.

A. Contained in Exhibit___(SFP-5) is Moody's April 8, 2019, Credit Opinion for KEDNY. The last paragraph of the report states that "[w]hile there is significant debt at KEDNY's parent holding companies, the strong ring-fencing provisions applicable to New York utilities reduce the potential for debt to be pushed down into KEDNY. In particular, we view the explicit leverage restriction for KEDNY (to maintain a debt to capitalization ratio of less than 56%) as providing some credit support at the current

rating level. This provision compares favorably against other New York utilities within the National Grid group. Additional ring-fencing provisions imposed by the NYSPC for KEDNY, which we view as credit supportive, include a special preferred share provision that reduces the probability of bankruptcy in a distressed situation, and the requirement for KEDNY to hold an investment-grade rating." Moody's expressed similar view about KEDLI's ring-fencing measures by stating that "[while] there is considerable debt at KEDLI's parent holding companies, the strong ring-fencing provisions applicable to New York utilities reduce the potential for debt to be pushed down to KEDLI. In particular, we view the explicit leverage restriction for KEDLI (to maintain debt/capitalization at less than 58%) as providing the greatest credit support. Additional ring-fencing provisions imposed by the NYSPC on KEDLI that we view as credit supportive include: (1) the special preferred share provision that reduces the probability of bankruptcy in a distressed situation, and (2) the requirement for KEDLI to hold an investment-grade

rating." This Moody's report published on April 8, 2019, is contained in Exhibit__(SFP-6).

Q. Comparatively, to what extent has Standard and Poor's recognized that the financial protections adequately insulate the Companies from their parent and affiliates?

A. In its latest credit opinions about KEDNY and KEDLI issued separately on September 24, 2018, S&P stated that "…we assess [The Brooklyn Union Gas Co and KeySpan Gas East Corporation] as insulated, with existing insulation measures supporting a one-notch separation between the subsidiary and its parent. Key insulating measures include: a) Prohibition of dividend distributions if the company does not maintain investment-grade credit ratings with at least two of the three major rating agencies, b) Prohibition of dividend increases and payments if the company's debt-to-capital ratio exceeds a certain amount, 3) A golden share whose vote is required to file the utility into a voluntary bankruptcy, 4) Clear economic and strategic incentive from the parent to maintain [The Brooklyn Union Gas Company's] financial

strength."

Q. What is your conclusion regarding the financial independence of KEDNY and KEDLI?

A. As stated earlier, we believe that the rating agencies still view KEDNY and KEDLI as adequately ring-fenced and thus rate the Companies' credit quality substantially on a stand-alone basis, apart from their holding Companies.

Q. Please elaborate on the Companies' second rationale that utility operating subsidiaries with similar business risk characteristics to KEDNY and KEDLI have for the last four years maintained average common equity ratios more than 8.00 percentage points above the 48.0 percent equity ratio that the Commission approved for KEDNY and KEDLI in the Companies' last rate proceeding.

A. On page 98 of her direct testimony, Ms. Bulkley stated that the average common equity ratio (based on SNL Financial, a business unit of S&P Global, data source) of the utility subsidiaries for the years 2014-2017 was 56.65% for her Combined Utility Group and 57.94% for her Natural Gas Proxy Group.

Q. Do you have any comments about Ms. Bulkley's use of the 56.65% and 57.94% historical common equity ratios to support the reasonableness of the Companies' 48% common equity ratio proposal?

A. Yes. First of all, Ms. Bulkley's use of operating company equity ratios is inappropriate. Investors in her proxy group companies base their return requirements on the holding company's use of leverage, along with its other risk attributes, as they are unable to directly invest in the subsidiaries. Secondly, the capital structures for utility subsidiaries of holding companies may not reflect either rational capitalization policies, or actual common equity employed, and therefore may not be suitable for supporting an operating subsidiary's rate of return. The subsidiary's common equity balance may not, in fact, be financed by common equity at the holding company level. Rather, some of the utility common equity balance may instead be proceeds from debt issued at the holding company level and classified on the utility subsidiary's books as common equity at the time the proceeds were invested in the utility subsidiary.

Q. You stated that the relevant equity ratio to investors is that of the utility holding companies comprising Ms. Bulkley's proxy groups. Have you performed any analysis to ascertain the average common equity ratio of the Companies' witness' proxy group?

A. Yes. According to S&P Capital IQ Financial data, we found that the average common equity ratio of Ms. Bulkley's Combined Utility Proxy Group and Natural Gas Proxy Group for years 2014-2017 were 47.89% and 53.14%, respectively as shown in Exhibit___(SFP-7). This is informative because investors cannot invest in the equity of a wholly-owned subsidiary company. As such, equity investors look at, among other things, the overall capital structure of the consolidated parent company to determine their risk/reward outlook and make investment decisions.

Q. What are the proportions of common equity that the Companies project in the rate year and how were they developed?

A. Per the Companies witness Cohen's Exhibit___ (JC-1, Schedule 2), the rate year proportions of common equity for KEDNY and KEDLI were forecasted

by starting with their respective 2018 year-end capital structures and reflecting the impacts of their forecasted sources and uses of funds from the beginning of 2019 through the rate year ending March 31, 2021. In developing KEDNY's forecasted capital structure with a 51.3% common equity ratio for the rate year, KEDNY indicated that it plans to retain a portion of its earnings and use those earnings to address a portion of its funding needs, driven primarily by capital expenditures of approximately $866 million in the rate year. Further, there is also a $500 million planned equity infusion from the ultimate parent, National Grid during the bridge period. KEDNY plans to issue $1.0 billion of long-term debt during the bridge period and another $400 million long-term debt near the end of the rate year. For its customer deposits balance, KEDNY forecasts an amount of $21.055 million for the rate year compared to $23.413 million as of December 2018.

Q. How did KEDLI develop its forecasted 48.9% common equity ratio for the rate year.

A. In developing KEDLI's forecasted capital

structure with a 48.9% common equity ratio for the rate year, the Company indicated that it plans to maintain a majority of its projected earnings and use them to address its funding needs, which will be driven primarily by capital expenditures of approximately $586 million during the rate year. Unlike KEDNY, KEDLI did not forecast any equity infusion or long-term debt issuance during the bridge period and the rate year. For its customer deposits balance, KEDLI forecasts an amount of $20.581 million for the rate year compared to $14.962 million as of December 2018.

Q. Are the Companies requesting their forecasted common equity ratios be used for ratemaking purposes?

A. No, in order to mitigate their rate requests, the Companies propose to set rates based upon a capitalization with a 48.0% common equity ratio; the common equity ratio upon which their rates are currently set.

Q. What is the Commission’s policy on the appropriate capital structure to be used in ratemaking when a utility resides in a holding

company?

Q. The Commission has expressed a preference for using the capital structure of the utility's parent. The Commission elaborated its position in Case 05-E-1222, stating:

"The established regulatory practice in New York in fully litigated rate proceedings, like this one, is to use the consolidated capital structure of the holding parent company for ratemaking purposes. This practice has typically been applied to utility holding company structures in other regulated industries. Holding companies owning electric utilities have only recently appeared in New York and we must now consider whether the conventional approach should apply to the electric industry. We find no valid basis for excluding electric utility holding company structures from the time-honored and well-established regulatory practice. There is no rational basis for us to depart from the approach developed and used for consolidated telecommunication, natural gas and water companies when addressing the substantially similar issues pertaining to consolidated

electric company operations."

Q. Are there circumstances under which the Commission would allow the use of a stand-alone capitalization?

A. Yes. The Commission has indicated that it will consider using a stand-alone capitalization if a utility can demonstrate that it has adequately isolated itself from the risks of its affiliates.

Q. How can a utility demonstrate that it has isolated itself from the risk of its affiliates?

A. In its Abbreviated Order Authorizing Acquisition, issued September 9, 2008, and its Order Authorizing Acquisition Subject to Conditions, issued January 6, 2009 in Case 07-M-0906 where the Commission approved Iberdrola's acquisition of Energy East, the Commission stated it would consider using a stand-alone equity ratio upon a showing that the rating agencies consider the utility operating subsidiary in question adequately insulated from the risks of the parent company's other operations. A company can establish such insulation by demonstrating the presence of significant ring-fencing measures.

Q. Can you briefly explain the concept of ring-

fencing in this context?

A. Yes. Ring-fencing, with respect to a regulated public utility subsidiary, occurs when a subsidiary financially separates itself from its holding company/parent company to protect the subsidiary from financial instability or the bankruptcy of the parent. The intent of the ring-fencing measures is to insulate assets in the subsidiary from the risks of the holding company/parent and from the other subsidiaries in the holding company. In theory, if a subsidiary is ring-fenced, and the holding company or another subsidiary goes bankrupt, creditors would not be able to attach their claims to the assets of the ring-fenced subsidiary. In addition, a fully ring-fenced company should be insulated from its parent's influence regarding the financial decisions of the subsidiary. In theory, ring-fencing protects the assets of the subsidiary in the event of a bankruptcy of a parent or other affiliate. Moreover, ring-fencing may allow a subsidiary to be rated higher than the parent.

Q. Besides bankruptcy protection, what are the

advantages of ring-fencing a utility subsidiary?

A. If a utility subsidiary achieves a higher rating than its parent, there is a benefit of a lower cost of debt to the subsidiary and ultimately to ratepayers.

Q. Does the existence of adequate ring-fencing require the use of the subsidiaries' stand-alone capital structures in establishing the ratemaking capital structure?

A. No, it does not. While the presence of adequate ring-fencing provides the opportunity to use the utility's stand-alone capital structure, the financial policies of the utility must still be evaluated with an eye toward identifying whether a suitable financing mix, resulting from reasonable financing policies, is being deployed. The idea is to find the approximate ratios of debt and equity that lead to the lowest weighted cost of capital.

Q. Please describe what constitutes reasonable financial policies for a regulated utility.

A. A reasonable financial policy is one that provides an optimal mix of debt and equity financing. An optimal financing policy balances

the interests of both ratepayers and investors by having the lowest cost of capital for an acceptable level of risk. A regulated utility, through its financial policies, should obtain a balanced mix of capital sources that will result in the most economical financing of its assets over the long run. For a utility company, a balanced capital structure will contain a sufficient proportion of common equity that will enable the utility to raise capital most economically. While common equity is costlier than debt, it should still be employed in sufficient amounts that will provide an economically efficient credit rating.

Q. How is an economically efficient credit rating determined?

A. An economically efficient credit rating is one that provides the lowest long-run cost of capital. The Recommended Decision in Case 91-M-0509 regarding Generic Financing generally determined that the overall cost to ratepayers, as measured by the pre-tax rate of return, was minimized at either a "BBB" or "A" bond rating. When practical, the Commission has generally

supported rate levels that provide utilities the opportunity to achieve and maintain a bond rating within the "A" range.

Q. What credit ratings are assigned to KEDNY and KEDLI by Moody's and S&P?

A. As indicated in our previously referenced Exhibit___(SFP-5) and Exhibit___(SFP-6) as well as Exhibit___(SFP-8) and Exhibit___(SFP-9), Moody's assigns a credit rating of "A3" to KEDNY and KEDLI. S&P assigns a rating of "A-" to both Companies.

Q. Have their "A3/A-" credit ratings provided the Companies ready access to the capital markets at reasonable rates?

A. Yes. KEDNY's access to the capital market is evidenced by its February 2019 issuance of $550 million of 10-year fixed rate senior unsecured securities and $450 million of 30-year fixed rate senior unsecured securities at very favorable rates of 3.865% and 4.487% respectively. KEDLI has not issued any long-term debt securities since 2016. As previously discussed, KEDLI has the same "A3/A-" rating as KEDNY and is viewed as possessing very similar credit risk attributes,

and thus, should have similarly strong access to the credit markets.

Q. What capital structure do you propose for the Companies?

A. We accept the same 48% common equity ratio proposed by the Companies.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDNY?

A. We based the capital components on Staff's determination of Staff's adjusted earnings base for KEDNY. To Staff's adjusted earnings base of $4,834,497,000, we applied the 48% common equity ratio to determine a common equity component of $ 2,320,559,000 ($4,834,497,000*48%) in our recommended capital structure. We included in our recommended capital structure KEDNY's projected customer deposits balance of $21,055,000 for the rate year. After determining the common equity and customer deposits components, the residual of $2,501,430,000, or approximately 51.56% of Staff's adjusted earnings base, represents the projected long-term debt component for KEDNY.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDLI?

A. We based the capital components on Staff's determination of Staff's adjusted earnings base for KEDLI. To Staff's adjusted earnings base of $3,269,060,000 we applied the 48% common equity ratio to determine a common equity component of $1,569,149,000 ($3,269,060,000*48%) in our recommended capital structure. We included in our recommended capital structure KEDLI's projected customer deposits balance of $20,581,000 for the rate year. After determining the common equity and customer deposits components, the residual of $1,679,330,000, or approximately 51.37% of Staff's adjusted earnings base, represents the projected long-term debt component for KEDLI.

Q. Do you believe that your capital structure recommendation will be supportive of the Companies' financial integrity?

A. Yes, we believe that our capital structure recommendation will be supportive of the Companies' financial integrity.

Q. Have you performed any analyses to substantiate your assertion that your recommended capital structure with a 48% common equity ratio, together with your recommended return on common equity (ROE) will be supportive of the Companies' financial integrity?

A. Yes, our recommended common equity ratio and return on equity recommendation when combined with other quantitative metrics and qualitative factors produce credit metrics that are generally consistent with the performance required by Moody's and S&P to maintain the Companies' current ratings of "A3" and "A-" respectively, as shown in Exhibit___(SFP-10).

Q. Please discuss how you determined that the Companies' credit metrics are consistent with those ratings.

A. We based our analysis on the overall credit metrics used by Moody's and S&P in evaluating both the financial and business risks of regulated electric and gas utilities. We note that only S&P provides specific categories of financial and business risks.

Q. Please discuss the concept of financial risk.

A. Financial risk is the additional risk a shareholder bears when a company uses debt in addition to equity financing. Companies that employ greater leverage would have higher financial risk than companies financed mostly or entirely by equity. S&P financial risk profile categories include “Minimal”, “Modest”, “Intermediate”, “Significant”, “Aggressive”, and “Highly Leveraged” based on seven different financial metric ratios. Reflected in Exhibit___(SFP-11) is the S&P Corporate Rating Methodology report originally published on November 19, 2013, and most recently updated on April 1, 2019. As illustrated on pages 33 and 34 of the report, S&P calculates two core coverage ratios and six supplementary coverage ratios. Specifically, S&P examines the following financial risk ratios in assessing a firm’s credit risk: a. Funds from Operations(FFO)/debt (core metric); b. Debt/EBITDA (core metric, ); c. Cash Flow from Operations (CFO)/debt; d. Free Operating Cash Flow (FOCF)/debt; e. Discretionary Cash Flow (DCF)/debt; f. (FFO + interest)/cash interest (FFO cash interest cover); and g.

EBITDA/interest.

Moreover, based on the methodology guidelines for the above metrics, we used S&P's medial volatility financial benchmark table reflecting the Companies lower-risk, rate-regulated distribution operations.

Q. In addition to the financial risk metrics, did you consider other qualitative factors considered by S&P in its business risk analysis of the Companies?

A. Yes, we did. S&P's business risk profile analysis incorporates such factors as country risk, industry risk, regulatory environment, company competitive position, business and geographic diversification, and management strategy. S&P's business risk profile categories include "Excellent", "Strong", "Satisfactory," "Fair", "Weak", and "Vulnerable". In determining the "A-" rating, S&P has assigned a business risk profile of "Excellent" to the Companies. Therefore, we used an "Excellent" business risk profile in analyzing the credit metrics of the Companies.

Q. Please discuss S&P's rationale for assigning an

"Excellent" business risk profile to KEDNY?

A. As illustrated in previously referenced Exhibit___(SFP-8), S&P credits the low operating risk of KEDNY's regulated gas distribution operations to its predominantly residential and commercial customer base, economically robust service territory and generally constructive regulatory environment as the primary drivers of the "Excellent" business risk evaluation.

Q. What is S&P's rationale for assigning an "Excellent" business risk profile to KEDLI?

A. As shown in our Exhibit___(SFP-9) referenced earlier, S&P's "Excellent" business risk assessment for KEDLI is based upon the Company's low operating risk regulated distribution operations, its somewhat small service territory that lacks geographic and operating diversity, a predominately residential and commercial customer base, and a generally constructive regulatory framework.

Q. Please continue with your credit metrics analysis based on Moody's rating guidelines.

A. Our Exhibit____(SFP-12) contains the June 23, 2017, Moody's ratings methodology for regulated

electric and gas utilities. Page 4 of the Moody's report contains the following financial strength metrics examined by Moody's: a) Cash Flow from Operation (CFO), pre-Working Capital (WC) + Interest/Interest, with Moody's assigned weight of 7.5%; b) CFO pre-WC/Debt, with Moody's assigned weight of 15%; c) CFO pre-WC-Dividends/Debt, with Moody's assigned weight of 10%; and d)Debt/Total Capital, with Moody's assigned weight of 7.5%.

Q. Did you consider the Companies' business profile/qualitative factors (business risk profile) based on Moody's rating guidelines?

A. Yes, we did consider the business risk profile of the Companies based on Moody's rating guidelines for regulated electric and gas utilities as shown in our Exhibit___(SFP-12) referenced earlier. Moody's business risk profile analysis, which accounts for 60% of their overall indicative grid rating incorporates such factors as: 1) Legislative and Judicial Underpinnings of the Regulatory Framework; 2) Timeliness of Recovery of Operating and Capital Costs; 3) Consistency and Predictability of Regulation; 4) Sufficiency

of Rates and Returns; 5) Market Position; and 6) Generation and Fuel Diversity. Each of the six business risk factors is ranked as "Aaa", "Aa", "A", "Baa", "Ba", "B", and "Caa". Given that the Companies are rated by Moody's, we used Moody's qualitative scores for the Companies averaging close to "A3".

Q. How did you apply the 11 credit metrics consisting of Moody's four financial metrics and seven S&P financial benchmarks?

A. In our previously referenced Exhibit___(SFP-10), we applied the 11 credit metrics using Companies financial results based upon our capital structure, cost of debt, and cost of common equity recommendations. As the exhibit illustrates, the overall financial metrics when combined with the "Excellent" business risk profile of the Companies, produce credit metrics consistent with the rating guidelines of S&P rating of close to "A-" and Moody's rating of "A3".

Q. For comparison purposes, did Mr. Cohen provide a projection of the Companies' financial metrics based on the Companies' requested capital

structure and cost rates?

A. Yes, the Companies' response to IR DPS-590 provided the rate year credit metric projections. According to the Company's projections, KEDNY's forecasted core and supplementary credit metrics are generally similar to what KEDNY has historically maintained over the past three years. With the exception of CFO pre-WC-Dividends/Debt, KEDLI's other forecasted metrics will result in higher core and supplementary metrics than it has historically exhibited. As explained earlier, these credit metrics are based on the medial volatility/low-risk grid table of S&P/Moody's reflecting the low business risk profile of the Companies.

Q. Do you have any observations about the Companies response to IR DPS-590?

A. Yes. We noted that some of the inputs for the credit metrics contained in the IR DPS-590 are not consistent with the Companies filing. For instance, the common equity ratios in IR DPS-590 for KEDNY and KEDLI are approximately 63% and 62%, respectively compared to the 48% in their rate year request. Also, the Net Pension & Other

Postemployment benefits (OPEBs) accrual & deferral for KEDNY and KEDLI are negative $1.64 million and negative $4.76 million respectively, compared to positive $47.47 million and negative $36.39 million in the rate filing. Due to such inconsistencies, we recalculated the Companies credit metrics based on their filing and rating agencies adjustments provided by the Companies in their response to IR DPS-590 as shown in our previously referenced Exhibit___(SFP-10).

Q. How do your financial metrics and implied credit ratings compare with the recalculated financial metrics and ratings in the Companies' filing?

A. For KEDNY, our previously referenced Exhibit___(SFP-10) shows coverage ratios and implied "A2"/"close to A-" Moody's/Standard & Poor's credit ratings compared to "A2"/"A-" implied by the Company's filing. Similarly, for KEDLI, our implied Moody's/Standard & Poor's credit rating is "A3"/"close to A-" compared to "A3"/"close to A-" in the Company's filing. One core coverage ratio used by S&P to assess the relative financial risk of regulated electric and gas utility companies is leverage ratio (Debt to

EBITDA—Earnings Before Interest, Taxes, Depreciation and Amortization). The ratio indicates a company's ability to cover its fixed debt service (interest plus principal) obligations as well as the variability of its earnings net of interest payments. The lower the leverage ratio the better. As the exhibit illustrates, our fallout Debt to EBITDA coverage ratio is 5.13x and 4.99x for KEDNY and KEDLI, respectively compared with the implied 4.86x and 4.53x for KEDNY and KEDLI, respectively, as calculated from the Companies' filings. Per S&P's rating guidelines for regulated utilities, the appropriate benchmark for a regulated electric and gas utility in the single "A" range with an excellent business risk profile is between 2.5x-4.5x. Another core metric used by S&P to assess the financial health of regulated electric and gas utilities is Funds from Operation to Debt (FFO/debt). The higher the FFO/Debt ratio the better. Under this metric, our capital structure recommendation along with Staff's other adjustments produces FFO/Debt ratio of 19.4% and 14.1% for KEDNY and KEDLI,

respectively compared with 21.9% and 16.4% for KEDNY and KEDLI, respectively in the Companies' filings. The acceptable FFO/Debt benchmarks reasonably expected from a regulated electric and gas utility in the single "A" range with an excellent business risk profile is between 13-35% per S&P rating guidelines.

Q. Please continue with your core metric analysis relating to Moody's guidelines.

A. The core coverage ratio used by Moody's to assess the financial health of regulated electric and gas utility companies is Cash flow from Operation Pre-Working Capital to debt (CFO pre-WC/Debt). The ratio indicates a company's ability to cover its fixed debt service (interest plus principal) obligations as well as the variability of its earnings net of interest payments. The higher the coverage metric the better. As our previously referenced Exhibit___(SFP-10) illustrates, our fallout (CFO pre-WC/Debt) metric is 19.2% and 13.6% for KEDNY and KEDLI, respectively compared with the implied 21.7% and 15.9% for KEDNY and KEDLI, respectively, as calculated from the Companies' filings. Per

Moody's rating guidelines for regulated utilities, the appropriate benchmark for a regulated electric and gas utility in the single "A" range with a low business risk profile is between 19%-27%. While KEDLI's underperformance on this metric is of potential concern, we note that Moody's will often discount performance in the short run particularly if it is related to cash flow impacts that may not be durable. For instance, during fiscal years 2016 through 2018, KEDNY's CFO pre-WC/Debt was below the 13.7% figure forecast for KEDLI during the rate year. Due to higher than anticipated environmental remediation costs KEDNY only achieved successive CFO pre-WC/Debt ratios of 7.9%, 13.3% and 9.8% during this timeframe and was able to maintain an "A2" rating. In the unlikely event that Moody's downgrades KEDLI, there would be no immediate rate impact because the Company has no plans to issue new debt until after the end of the rate year.

Q. Did you review the most recent Moody's and S&P credit analyses to ascertain each rating agency's view with respect to the Companies' ratings

outlook and the factors each indicate could lead to a rating upgrade or downgrade?

A. Yes. As illustrated in our previously referenced Exhibit___(SFP-10) and Exhibit___(SFP-11), in September 2018, S&P's assessment indicated a "Stable" outlook for KEDNY's and KEDLI's "A-" rating. According to S&P, a rating outlook assesses the potential direction of a long-term credit rating over the intermediate-term (typically six months to two years). Therefore, it does not view KEDNY's and KEDLI's "A-" credit rating as likely to change over the intermediate-term.

Q. What is Moody's recent assessment of the Companies' outlook?

A. In its April 2019 credit opinion report, as shown in our previously referenced Exhibit___(SFP-5) and Exhibit___(SFP-6), Moody's assessment indicated a "Stable" outlook for the Companies' "A3" rating. Therefore, Moody's does not view KEDNY's and KEDLI's "A3" credit rating as likely to change over the intermediate-term. Moody's indicated that the Companies could be downgraded if their respective CFO pre-WC/debt is

consistently below the high teens.

Q. The Companies' capital structure witness, Mr. Cohen, indicated that there has been pressure on both the Companies credit metrics and bond ratings in recent years thereby putting KEDNY/KEDLI's objective of maintaining an "A3" bond/credit rating at risk. As a result, in February 2019, Moody's downgraded the Companies ratings by one notch to "A3" from "A2". Do you have any comment?

A. Yes. First, Moody's downgraded both Companies rating from an "A2" to "A3" in February 2019 largely because of the impact of the Federal Tax Cuts and Jobs Act (Tax Act) of 2017, as well as heightened capital investments, and in the case of KEDNY also because of higher spending on environmental remediation. The recent downgrade of the Companies by Moody's was not unique, the reduced cash flows resulting from the Tax Act led Moody's to downgrade many utilities over the past year and a half. As we have demonstrated, the Companies could maintain their current "A3" credit ratings with our implied credit metrics and access the capital markets at reasonable

rates. Moreover, both rating agencies (Moody's and S&P) have placed the Companies' current ratings on a "Stable" outlook. Per these rating agencies, a rating outlook assesses the potential direction of a long-term credit rating over the intermediate-term (typically six months to two years). Therefore, KEDNY's and KEDLI's "Stable" outlook suggests that the Companies' rating is not likely to change over the intermediate-term.

**Cost of Long-Term Debt**

Q. What costs of debt are the Companies requesting in this proceeding?

A. In its filing, KEDNY requested a rate year cost of debt of 4.11%. We came across an error in the calculation of KEDNY's long-term debt cost rate which was acknowledged by the Company in response to an IR number DPS-1002 dated August 21, 2019. Therefore, the Company's corrected debt cost is 4.14%. KEDNY's 4.14% cost rate reflects the weighted average cost of its outstanding long-term debt as well as a projected 3.83% interest rate for a planned $400 million issuance of 10-year senior unsecured debt obligations near the

end of the rate year. KEDLI is requesting a rate year cost of debt of 4.06% reflecting its embedded debt cost. KEDLI has no plans to issue any new long-term debt during the link period and the rate year.

Q. How did KEDNY determine the 3.83% interest rate for the $400 million planned debt issuances during the rate year?

A. Per Mr. Cohen's Exhibit___(JC-1), page 2 of Schedule 1, KEDNY determined the 3.83% interest rate for the new debt security by adding a 117 basis point spread associated with KEDNY's recent debt issuance in February 2019 to a 2.66% forward rate of US 10-year Treasury for March 1, 2021.

Q. Are you proposing any adjustments to the Companies' requested cost rates?

A. Yes. We are proposing one adjustment to KEDNY's request and no adjustments to KEDLI's calculation.

Q. Please discuss your adjustment to KEDNY's requested rate year cost of debt.

A. Rather than using the 10-year Treasury forward rate recommended by Mr. Cohen, we recommend using the most recent (June 2019) 10-year Treasury

yield of 2.07% in combination with the Company's 117 basis points spread to estimate an interest rate of 3.24% for KEDNY's proposed new debt issuance as shown in our Exhibit___(SFP-13). The interest rates should be updated based upon interest rates and yield spreads in existence just prior to the Commission's decision.

Q. Why did you use the most recent 10-year Treasury yield to estimate the interest rate for KEDNY's new debt issuance?

A. We used the most recent 10-year Treasury yield in this rate proceeding because the Commission considers current interest rates to be the best predictor of where interest rates will be in a given rate year going forward. Specifically, in Case 10-E-0050, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation for Electric Service, the Commission stated that, "[i]n as much as we are making a one-year rate determination, the current interest rates are reasonably representative for the immediate future."

Q. Did you make changes to KEDNY's proposed 4.14%

long-term cost rate for the rate year?

A. No. Although we used a 3.24% interest rate for KEDNY's projected new debt issuance, the fact that those funds will be raised so late in the rate year, there was only a di minimis impact on the overall debt cost rate of 4.14% proposed by KEDNY. As illustrated in Exhibit___(SFP-13) referenced earlier, we are accepting the 4.14% cost of debt for the rate year ending March 31, 2021.

**Cost of Customer Deposits**

Q. What customer deposits rate are the Companies proposing for the rate year ending March 31, 2021?

A. The Companies are proposing a 2.45% customer deposits rate currently established by the Commission for investor-owned utilities in Case 18-M-0611.

Q. Do you agree with the Companies' customer deposits rate proposal?

A. Yes, we agree, however, prior to the Commission decision in this case we also recommend that the customer deposits rate be updated to reflect any

new rate established by the Commission.

## Cost of Common Equity

### Summary

Q. Please summarize your analysis in determining your 8.20% cost of common equity.

A. Our common equity cost estimate is based on applying a DCF analysis and CAPM analysis to a proxy group of 28 electric utility holding companies. The DCF applied to the proxy group results in an average equity cost of 8.22%. We averaged two different CAPM analyses to produce an equity cost of 8.04%. Following the Commission's practice, we apply weightings of two-thirds to the DCF and one-third to the CAPM, which resulted in a ROE of 8.16% which we rounded up to 8.20%. A summary of these calculations can be seen in Exhibit___(SFP-14).

Q. Please discuss prior Commission precedent in estimating the cost of equity.

A. The Commission has consistently used the methodology of weighting the DCF result as two-thirds of the total equity cost and the CAPM result as one-third in estimating a utility's

cost of equity. For example, in Case 10-E-0362 involving Orange & Rockland Utilities, Inc. and in Case 10-E-0050 involving Niagara Mohawk electric rate case, the Commission authorized a cost of equity based on two-thirds DCF and one-third CAPM methodology. This methodology was also used in Case 16-G-0257, a rate proceeding for National Fuel Gas (NFG). Consistent with Staff's approach and prior Commission precedent, we applied the DCF and the CAPM approach to a proxy group of companies that were selected based on the criteria that we will discuss later.

**Staff Proxy Group**

Q. Why do you use a proxy group in your analyses to estimate the Companies' cost of equity?

A. The use of a proxy group to determine KEDNY/KEDLI's cost of equity is necessary because the Companies' common stock is not publicly traded, and thus, direct DCF and CAPM analyses of the Companies are not possible. Equally important is that the DCF analyses for individual companies are reliant on equity analysts' estimates of growth which are, by nature,

inaccurate and may also be biased. Similarly, beta determinations used in the CAPM methodology are based on historical observations, that due to circumstances such as corporate restructurings or industry transformations, may not represent the level of earnings volatility expected in the future. By employing a sufficiently large proxy group of similarly situated companies, the undesirable effects of biased (both upward and downward), inaccurate growth estimates or beta measures for any individual company is diminished. While the objective is to choose a group of companies whose risks closely match those of the company being examined, it is important that the group selected is also large enough to provide confidence in its results. The greater the number of suitable companies, the less sensitive the overall cost of equity estimate will be to the effect of any potential inaccuracies in the data for any particular company.

Q. Please describe how you developed your proxy group.

A. Even though the Companies are local gas

distribution companies, their ultimate parent is largely considered to be an electric utility by investors. Accordingly, we believe that the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be "electric utilities." The majority of these companies have operating subsidiaries that are combination electric and gas utilities and thus our analysis with this group will reflect the challenges and opportunities facing energy utilities in general. We believe that the two types of utilities are fundamentally very similar. Both are regulated public utilities that are subject to nearly identical ratemaking treatment. In addition, both invest primarily in capital-intensive physical networks that connect the customer to the source of supply; and sell their products and services at regulated rates to customers.

Q. How does your approach compare with the approach Staff uses to develop a proxy group for the Companies sister company Niagara Mohawk Power Corporation (Niagara Mohawk)?

A. Consistent with our approach in this proceeding,

and as Staff has done in all Niagara Mohawk rate proceedings over the past twenty years, Staff in Cases 17-E-0238 and 17-G-0239, employed a proxy group using companies screened from Value Line's electric utility companies.

Q. What criteria did you apply to the Value Line electric utility group?

A. To be selected for our proxy group, a company had to meet the following requirements: (1) an investment grade rating from both Moody's and S&P ("Baa3" and above by Moody's and "BBB-" and above by S&P); (2) regulated revenues that constituted at least 70% of that company's total revenue, as determined by each company's 2018 SEC annual report (10-K); (3) currently paying common stock dividends; and (4) not currently involved in any merger/acquisition (M&A) activity.

Q. Please describe how you selected your proxy group.

A. As shown in Exhibit___(SFP-15), our proxy group consists of 28 companies. Beginning with the 37 holding companies deemed by Value Line to be "electric utilities", we eliminated Avangrid Corporation because it has only been publicly-

traded since December 2015, which we believe is an insufficient period or track record to establish confidence in the financial forecasts needed for calculating a company's cost of equity. Specifically, Value Line usually calculates its value of beta over a five-year period, but Avangrid Corporation has only three years of stock price data from which to calculate beta. Similarly, we also eliminated Evergy, Inc. because it has only been publicly-traded for about a year, a time frame so short that Value Line hasn't attempted to calculate a beta. As for the remaining 35 companies, we eliminated five companies whose regulated revenues constitute less than 70% of total revenue. These companies are CenterPoint Energy Inc., DTE Energy Co., Exelon Corp., Otter Tail Corp and Public Service Enterprise Group Inc. Regarding the remaining 30 companies, we eliminated Fortis Inc. because it is a Canadian-based company, which, although traded on both the Toronto and New York stock exchanges, has its financial reports and Value Line projections based on Canadian currency which would introduce the possibility that

exchange rate variations could affect the DCF calculation. Finally, we eliminated El Paso Electric because of a pending merger/acquisition transaction. Therefore, we arrived at a proxy group comprised of 28 electric utility holding companies which met all four criteria stated previously.

**Discounted Cash Flow Methodology**

Q. Please describe the DCF methodology.

A. The DCF methodology assumes that the principles used to measure the cost of common equity are the same as those used in measuring the yield investors require on debt. However, while interest payments are known with relative certainty, future dividend payments are uncertain. The foundation of the DCF is that investors will price common stock to equal the present value of future dividend payments. Therefore, the valuation of a share of stock can be represented by the following dividend growth rate expression:

$$P_0 = \frac{D1}{(1+k)^1} + \frac{D2}{(1+k)^2} + \frac{D3}{(1+k)^3} \ldots \frac{Dn}{(1+k)^n}$$

Where:

$P_0$ = current stock price

$D_n$ = dividend in period "n"

k = cost of equity.

Q. Please describe your DCF methodology and its results.

A. As illustrated in Exhibit___(SFP-16), the DCF uses a three-month average stock price, calculated by averaging the high and low monthly prices for each company in the proxy group. Our analysis covers the period from April 2019 to June 2019. As shown in Exhibit___(SFP-17), our DCF methodology develops a stream of dividends for each proxy company. Each dividend stream is discounted at a rate which makes the discounted value of its dividend stream equal to its respective three-month average stock price. The discount rate required to discount the stream of expected dividend payments into the current stock price is the estimated cost of equity for that company. Our DCF methodology calculates an average return on equity of 8.22% for the proxy

group.

Q. How did you develop your dividend projections?

A. We employed a two-stage DCF methodology. For the first stage encompassing the 2019 through 2023 period, we used Value Line dividends per share estimates. Beyond 2023, we calculated a sustainable growth rate for each company in the proxy group based primarily upon its expected return on equity and projected retention ratio. Also included in the sustainable growth rate estimates is the external growth that can be realized when a company issues new shares at prices above book value.

Q. How did you calculate the short-term dividend growth rates?

A. We calculated the short-term dividend growth rate by solving for the compound annual growth rate between the base year value of our dividend stream of 2019 and the ending year value of our dividend stream in 2023. The median and average short-term growth rates for our proxy group are 5.51% and 5.06%, respectively.

Q. How did you derive the long-term sustainable dividend growth rates used in the DCF?

A. Sustainable dividend growth rates depend on a firm's expected internal and external growth rates. Long-term internal growth will be achieved based upon a company's earned return on common equity and on the portion of earnings retained within the business. That is, internal growth, also known as retention growth, is achieved based upon the growth in a company's retained earnings. In formulaic form, retention growth is measured by the following formula:

Retention growth = b*r

Where:

"b" = the fraction of earnings retained in the company

"r" = the expected rate of return on common equity.

For example, if a firm retains 40% of its earnings and earns 10% on equity, then its retention growth rate will be 4.0% (40%*10%). Thus, the growth rate is determined by the increase in the equity retained in the business and the expected return on that equity. Our analysis calculates median and average retention growth rates of 3.89% and 4.09%, respectively for

the proxy group.

Q. How is external growth measured in your DCF methodology?

A. As discussed earlier, external growth can be achieved if a company issues common equity above its book value. This is commonly referred to as the SV factor, where "S" is the increase in common shares outstanding that have been sold and "V" is the per share premium or discount to the book value on the shares sold. Typically, Staff evaluates the SV factor using the formula: (MBR-1) * growth in common shares, where: (MBR) is the market-to-book ratio. Thus, the formula for the sustainable growth rate is given by g = (b*r) + (SV), where SV is also given by (MBR-1)*growth in common stock. In our analysis, the median and average external growth rates for the proxy group are 0.57% and 0.77%, respectively.

Q. What is your sustainable dividend growth rate for the proxy group?

A. The aggregation of our retention growth and external growth produces median and average sustainable growth rates of 4.64% and 4.86%, respectively.

Q. How did you check the reasonableness of your sustainable growth rate?

A. We compared it with the projected overall growth rate of the economy. Rational utility investors expect growth to generally track the changes in output, or growth in the overall economy, as measured by growth in the nominal Gross Domestic Product (GDP). As reported in the March 10, 2019 edition of Blue Chip Economic Indicators shown in Exhibit___(SFP-18), the consensus long-range estimate of nominal GDP growth, going out to 2026-2030, is 4.0%, which is modestly lower than our median sustainable growth rate of 4.64%. Thus, if anything, our sustainable growth rate is more optimistic on utility growth prospects. It should be noted that the 4.0% nominal GDP growth rate estimate is comprised of two components: real GDP growth of 1.9% and an inflation rate of 2.1%. The long-run projections generally show annual real GDP modestly increasing from a rate of 1.7% in 2021 to a rate of 2.0% in 2025, while inflation is forecasted to hold between 2.1%-2.2% from 2021 and beyond into the long-run. This comparison is appropriate because the

nominal GDP rate reflects assumptions about future inflation, in addition to the real growth expected in the economy as a result of productivity gains. Therefore, it would not be unreasonable for investors in the market, to expect their future dividends to generally keep pace with overall inflation, as well as reflect productivity gains similar to those expected for the economy as a whole. Likewise, for investors in a mature sector of the economy, such as the utility industry with perhaps slower-than-average growth prospects, it is not unreasonable to expect future dividend growth to be roughly equivalent to that of the overall economy.

Q. What was the result of your DCF analysis?

A. Based on the inputs that we have just described, the DCF average cost of equity for our proxy group is 8.22%, while the median result is 8.11%.

**Capital Asset Pricing Model**

Q. Please explain the concept underlying the Capital Asset Pricing Model.

A. The basic concept underlying the CAPM is that risk-averse investors demand higher returns for

assuming additional risk. In other words, higher-risk securities are priced to yield higher returns than lower-risk securities.

Q. How does the CAPM measure risk?

A. The CAPM quantifies the additional return, or risk premium, required for bearing incremental risk above the risk-free rate of return. The model calculates a cost of equity based upon the premise that an investor who diversifies his security portfolio is only exposed to systematic risk (i.e. market risk). The systematic risk represents the risk of investment within a well-diversified portfolio. The remaining risks are not assumed by a rational investor since these risks can be eliminated through diversification. Hence, investors do not require compensation for diversifiable risks. In the CAPM, the beta of a stock measures the volatility of that stock in relation to the volatility of the stock market as a whole. It is produced by a regression analysis and represents the covariance of a stock with the market. A beta above 1 means the stock has greater risk than the market; a beta below 1 means that a stock has less risk than the market.

A beta of 1 means the risk of the stock is equivalent to the risk of the market.

Q. What inputs are used in your CAPM models and how did you derive them?

A. There are three inputs in the CAPM equation: the risk-free rate, the beta of a stock, and the market risk premium. We employed the monthly average of 10-year and 30-year Treasury bond yields over a three-month period from April 2019 to June 2019, as the risk-free rate. The resulting risk-free rate is 2.55%.

The beta that we employed is the average of the latest Value Line published betas for the 28 utilities in our proxy group. As shown on page 1 of the previously referenced Exhibit___(SFP-17), we have calculated the average Value Line beta for our proxy group to be 0.61.

Q. Why did you use the mean beta rather than the median beta of the proxy group?

A. As a practical matter, there is only a minor difference, as the median beta of the proxy group is 0.60. Employing the median beta rather than the mean beta in this case would not materially change our overall ROE calculation of 8.16%. Our

model produces an 8.16%, however, if we used the 0.60 median beta, the overall ROE falls by about two basis points to 8.14%. Nonetheless, we note that we have previously recommended the use of the median beta because it diminishes the undue influence of any outlying value of beta. In addition, Value Line reports its beta determinations in increments of 0.05, therefore, to be consistent, changes from month to month in the median value should only change in increments of 0.05 as well. Changes in the CAPM results based on the median beta are not as smooth as those based on the mean, because this attribute has the potential to introduce unintended volatility in the determination of the proxy group's beta, therefore, we have employed the mean beta as the appropriate input to the CAPM.

Q. How did you arrive at your 8.33% estimate of the market risk premium?

A. Our estimate of an 8.33% market premium was based on the market return estimates contained in Merrill Lynch's Quantitative Profiles reports less the 2.55% risk-free rate discussed earlier. Specifically, we used the May, June and July 2019

editions of Merrill Lynch's Quantitative Profiles and averaged the required and implied market returns of each of the three point-in-time estimates to arrive at an average required return for the market of 10.88%. Then we subtracted the 2.55% average risk-free rate to arrive at the 8.33% market premium. The Merrill Lynch Quantitative Profiles are attached as Exhibit___(SFP-19).

Q. Why did you use Merrill Lynch data for calculating market risk premiums?

A. Our market risk premium is calculated by subtracting the risk-free rate from Merrill Lynch's cost of equity for the market. The Merrill Lynch data provides two forward-looking returns on the market: a required return and an implied return. Merrill Lynch is a respected independent firm with many resources available to make the calculations. This market risk premium estimation methodology has been used by the Commission since 1996, in Case 95-G-1034, Central Hudson Gas & Electric Corporation - Rates, Order and Opinion No. 96-28 issued October 3, 1996, on page 14, where the Commission approved the use of

the Merrill Lynch estimate. In its Opinion, the Commission stated, "…the Judge's market return calculation based on Merrill Lynch data is a reasonable method of deriving a risk premium."

Q Why did you employ 10-year and 30-year Treasury bond yields as the risk-free rate in your analysis?

A. While there are not any true risk-free rate securities, Treasury securities backed by the full faith and credit of the United States government are very low-risk investments. As such, they serve as an excellent surrogate for the risk-free rate. Additionally, the blending of the yields on Treasury securities with a 10-year and 30-year maturity is reasonable because it approximates most investors' time horizon.

Q. Why did you use Value Line betas in your analysis?

A. Value Line is a well-known publication and its beta calculations are widely available to investors. In addition, the Commission historically uses the Value Line betas in estimating the CAPM cost estimates because Value Line calculates its betas over a five-year

period, thereby mitigating the inherent volatility of using beta estimates calculated over shorter time periods.

Q. Please describe the calculation of the traditional CAPM.

A. The traditional CAPM expression takes the form:

$K = Rf + \beta*(Rm-Rf)$

Where:

K = investor's required return or equity cost of capital

Rf = risk-free rate of return

β = beta, measure of both business and financial risk (systematic risk)

Rm = market rate of return

Rm–Rf = market risk premium

The traditional CAPM determines the return on common equity by adding a company's non-diversifiable risk premium to the risk-free rate. The non-diversifiable risk premium is measured by multiplying the proxy group's beta by the market risk premium.

Q. What is the result of your traditional CAPM analysis?

A. As calculated below, our traditional CAPM cost of

equity is 7.63%:

Required return = (10.88%-(2.55%)*0.61+2.55%

Q. Please describe the zero-beta CAPM.

A. The zero-beta CAPM is a two-factor version of the standard CAPM. The zero-beta CAPM determines the return on equity by adding two factors to the risk-free rate: first, a factor of 0.75%, multiplied by the average beta of the proxy group and the S&P 500 market risk premium; and second, a factor of 25% multiplied by the S&P 500 market risk premium. The zero-beta CAPM expression takes the form: K = Rf + 0.75*β*(Rm-Rf) + 0.25*(Rm-Rf), where all the variables are the same as used in the standard CAPM. The zero-beta CAPM was developed to address the alleged infirmity that the traditional CAPM understated the required return of stocks with a beta below 1.

Q. What is the result of your zero-beta CAPM methodology?

A. As the following calculation illustrates, our zero-beta CAPM produces an 8.44% cost of equity.

Cost of equity = 2.55% ((0.75 *.61 (10.88%-2.55%)) + ((0.25 x (10.88%-2.55%)).

Q. What CAPM result did you use in the calculation of the proxy group cost of equity?

A. Consistent with Commission's precedent in prior rate proceedings, including its most recent decisions and orders in fully litigated cases, such as the recent NFG Rate Case, Case 16-G-0257, we have averaged the results of the traditional and zero-beta CAPM analyses to arrive at a CAPM result of 8.04%. The CAPM calculation is shown in Exhibit___(SFP-20).

Q. What role should the CAPM play in the cost of equity analysis?

A. In prior rate proceedings, like those to which we have previously referred to, the Commission has recognized that the CAPM presents a conceptual framework for calculating a reasonable estimate of a firm's cost of equity, however, because of the relative weaknesses of the CAPM in comparison with the DCF methodology, it is given a lesser weight than the DCF. The Commission has adopted a 33% weighting for the CAPM, compared to a 67% weighting of the DCF in the overall cost of equity analysis.

Q. What are some of the weaknesses of the CAPM

methodology?

A. The first shortcoming of the CAPM is its reliance on historical betas as a surrogate for forward-looking betas of companies. Historical estimates only reflect the past riskiness of an equity security that may not be representative of the future riskiness that is relevant to equity investors. The second weakness is that recent history has shown that volatility in betas can be dramatic. This may lead to estimating errors in the cost of equity. Finally, the traditional model may not adequately explain the variation in stock returns. To underscore the fact that the CAPM does not appear to adequately explain the variations in stock returns, Ambika Prasad Dash explained in his book entitled Security Analysis and Portfolio Management that "…empirical studies show that low beta stocks may offer higher returns than the model would predict." This excerpt is attached as Exhibit___(SFP-21).

Q. What cost of common equity did you calculate for your proxy group?

A. Using a 2/3 DCF and 1/3 CAPM weighting, the estimated cost of common equity is 8.16% rounded

up to 8.20%.

Q. How does the Panel's 8.20% recommendation compare to the Companies' currently authorized ROEs?

A. The Companies' are currently authorized an ROE of 9.0% dating back to the multi-year rate plans approved by the Commission in Cases 16-G-0058 and 16-G-0059 in December 2016. It is important to note, however, that when the Commission evaluates the reasonableness of a multi-year rate plan's ROE, it recognizes the necessity of a premium above the ROE produced using Staff's methodology in recognition of the increased financial and business risks that are inherent when a company is operating within the parameters of a multi-year rate plan. By agreeing not to file for new rates for three years, the Companies reduced their ability to adjust rates in response to changing conditions. Accordingly, locking in rates for an extended period somewhat diminished the Companies' flexibility and thereby increased their level of risk. Additionally, the forecasts of expense items in years two and three of a multi-year rate plan are inherently more inaccurate and thus expose a company to

additional operating risk. Finally, it is also worth noting that the interest rate environment when the Commission approved the current rate plans' 9.0% ROEs was modestly higher compared with today.

Q. Is there any reason a rational investor would expect the Commission to authorize an ROE in these proceedings anywhere close to the Companies' 9.65% requested ROE?

A. No. Rational investors are well aware of the Commission's preference for a formulaic approach to the cost of common equity and are also aware of the Commission's preference for establishing rates over a multi-year period. Accordingly, they likely expect that rates in this proceeding will be set over a multi-year period, and that the ROE, after accounting for the additional financial and business risk, would likely be similar to the 8.8% to 9.0% range of recent Joint Proposals approved by the Commission.

Q. What are the Joint Proposals that the Panel is referring to?

A. We are referring to the three most recent Joint Proposals approved by the Commission. The Joint

Proposal for Orange and Rockland, approved in March 2019, adopted an ROE of 9.0%. Further, the Joint Proposal for Central Hudson, approved in June 2018, adopted an ROE of 8.8%. Finally, the Joint Proposal for Niagara Mohawk, approved in March 2018, adopted an ROE of 9.0%.

Q. In the event that a Joint Proposal is not reached, and the Commission will be setting rates solely for the rate year, are there additional factors that it may want to consider with respect to the level of the Companies ROE?

A. Yes. We believe that there are two recent developments that could modestly and negatively impact investors return requirements. Specifically, once it has had sufficient time to analyze the issue, we believe the Commission should consider the impact of the Companies' moratorium on processing new natural gas service applicants in their respective service territories until the Northeast Supply Enhancement (NESE) gas pipeline receives its necessary permits. This development implies that the Companies will be potentially foregoing growth opportunities in the future. However, the

Commission's ROE authorization should also incorporate the impact of the likelihood that the Companies Revenue Decoupling Mechanism (RDM) will also be changed to a revenue per class reconciliation from the current revenue per customer model. In light of the curtailing of natural gas expansion in the Companies' service territories, the switch to a revenue per class RDM would offer the Companies' greater protection from a decline in sales.

**Discussion of Companies' Witness Bulkley's ROE Approach**

Q. Please summarize the approach followed by the Companies' witness Ms. Bulkley?

A. Ms. Bulkley began her analysis by applying her cost of equity models to two separate proxy groups. The first consisting of both electric utility holding companies and natural gas distribution holding companies, which she referred to as her "Combined Utility Proxy Group," and the second proxy group was derived solely from natural gas distribution holding companies, to which she referred to the "Natural

Gas Proxy Group". Ms. Bulkley developed three multi-stage DCF models utilizing the low, mean, and high earnings growth rate estimates published by Yahoo Finance, Zack's, and Value Line as shown in her Exhibit___(AEB-1). These DCF calculations resulted in a low, mean, and high ROE estimate for each company in her proxy group. Ms. Bulkley utilized both the traditional form of the CAPM and the zero-beta CAPM applied to both Bloomberg and Value Line betas that resulted in two sets of 12 ROEs for both her Combined Utility Proxy Group and Natural Gas Proxy Group. Ms. Bulkley then concluded that an appropriate ROE for the Companies ranges between 9.65% and 10.55%. In this proceeding, the Companies are seeking a 9.65% ROE for the rate year.

Q. What are your principal points of contention with Ms. Bulkley's analyses?

A. Overall, we are concerned with three primary aspects of her analysis: 1) the composition of both of the Companies' proxy groups, 2) the use of excessive growth rates in the Companies' DCF analyses, and 3) the use of several flawed key inputs employed in her CAPM analyses;

particularly Ms. Bulkley's market return estimate, and to a lesser extent the risk-free rate. Finally, we strongly disagree with Ms. Bulkley's premise that current market conditions warrant that the CAPM be accorded equal weighting with the DCF.

**Criticism of The Proxy Groups Employed By Ms. Bulkley**

Q. Please describe Ms. Bulkley's proxy groups.

A. Ms. Bulkley developed two proxy groups: one which she referred to as a Combined Utility Proxy Group, and the other which she referred to as a Natural Gas Proxy Group. The Natural Gas Proxy Group consists solely of the seven gas distribution companies that are also included in her Combined Utility Proxy Group. The Combined Utility Proxy Group consists of those seven gas companies and 13 additional electric distribution companies.

Q. How did Ms. Bulkley develop her Natural Gas Proxy Group?

A. Beginning with 10 companies classified as natural gas distribution companies by Value Line, she applied the following screening criteria: 1)

eliminated companies that are not covered by at least two utility industry equity analysts; 2) eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's; 3) eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source; 4) eliminated companies with less than 70 percent of total operating income derived from regulated utility operations; 5) eliminated companies that derive less than 50 percent of total operating income from regulated natural gas distribution operations; 6) eliminated companies known to be party to a merger, acquisition, or other transformational transaction; and 7) eliminated companies that have owned generation comprising greater than 60 percent of the company's MWh.

Q. Do you believe that Ms. Bulkley's use of a Natural Gas Proxy Group is appropriate?

A. No, the primary reason Ms. Bulkley's use of a Natural Gas Proxy Group is inappropriate because National Grid is viewed principally as an

electric utility holding company by investors. Also, even in instances where it is appropriate to look at the gas utility industry such as with companies like NFG, Corning and St. Lawrence Gas, a Natural Gas Proxy Group, when constructed using Staff's long-running selection criteria, would only produce a five company proxy group, and a proxy group consisting of only five companies does not provide an adequate sample size to produce reliable results. An inadequate sample size introduces the possibility of a single analyst error or the presence of an outlier to influence the cost of equity results as evidence in Ms. Bulkley's use of Northwest Natural Gas Company's growth rate shown on page 6 of her Exhibit__(AEB-1).

Q. Can you provide any research indicating what a reasonable size of a proxy group should be?

A. Yes. In the book "New Regulatory Finance" by Dr. Roger Morin, he suggests that a proxy group size of between 15 and 30 is reasonable to support a reliable cost of equity estimate.

Q. Can you provide an excerpt from the book to support your assertion that a proxy group size of

between 15 and 30 is reasonable?

A. Yes. In discussing cluster analysis to select a proxy group from a universe of pure-play utility companies, Dr. Morin states on pages 225 and 226 of his book that, "With cluster analysis, comparable companies are selected on the basis of 'closeness' to the targeted entity in terms of such predetermined risk variables such as credit ratings, after-tax interest coverage, equity ratio, total capital, and variability of operating income… A cluster size of 15 to 30 firms is reasonable. Once a cluster of comparable companies is identified for each firm or division, the average DCF and CAPM/ECAPM (Empirical Capital Asset Pricing Model) cost of equity estimates for the publicly traded companies are used as estimates of equity costs for the divisions." This excerpt is shown in Exhibit___(SFP-22). Even though the above-mentioned quote by Dr. Roger A. Morin is relative to cluster analysis, the principle is equally applicable to any proxy group selection irrespective of the selection criteria and methodology employed because unreasonable sample

size produces unreliable results.

Q. What were the selection criteria for Ms. Bulkley's second proxy group, the Combined Utility Proxy Group?

A. Ms. Bulkley began by selecting companies classified as electric utilities by Value Line, and then applied the following screening criteria: 1) eliminated companies that are not covered by at least two utility industry equity analysts; 2) eliminated companies that do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's; 3) eliminated companies that have not paid regular dividends or do not have positive earnings growth projections from at least one source; 4) eliminated companies that have owned generation comprising greater than 60 percent of the Company's MWh sales to ultimate customers; 5) eliminated companies with less than 70 percent of total operating income derived from regulated utility operations and 6) eliminated companies known to be party to a merger, acquisition, or other transformational transaction.

Q. What is the impact of Ms. Bulkley's selection criteria?

A. In the Companies' response to IR DPS-249, Ms. Bulkley indicated that, based on her six selection criteria, she eliminated 29 companies from the universe of 49 Electric and Natural Gas companies in constructing her Combined Utility Proxy Group.

Q. Do you have any concern with any of Ms. Bulkley's selection criteria?

A. Yes. We have three concerns regarding her selection criteria: 1) relating to net operating income, 2) analyst coverage, and 3) generation MWH sales.

Q. Please explain.

A. First, we find Ms. Bulkley's requirement that a company must have 70% of its net operating income from 'regulated utility operations' to be too restrictive. Operating income is the amount of revenue left over after accounting for all the expenses necessary to keep the business running. Her methodology would result in the elimination of companies whose risk characteristics are closely comparable to KEDNY/KEDLI for example,

NextEra Energy, Inc. was eliminated from her proxy group using the 70% operating income criteria. Moreover, using operating earnings instead of a revenue criterion for establishing the proxy group may allow companies into the group that contain substantially greater risk than a typical regulated utility. Revenues more accurately reflect the exposure to riskier competitive operations, because it is not uncommon for competitive ventures to be unprofitable at times. Accordingly, it is quite possible that using the 70% of operating income criteria could introduce companies into the proxy group that have a higher inherent risk profile than is proper for a regulated utility proxy group.

Q. Is there any other reason why it is inappropriate to use operating income as a selection criterion?

A. As illustrated in Exhibit___(SFP-23), the difference in accounting policy choices such as a difference in inventory valuation policy options and a difference in depreciation choices across companies could have a significant impact on reported operating income than on reported

revenues. For these reasons, revenue remains a superior criterion for ensuring that a utility proxy group properly reflects the amount of regulated business that most closely approximates the risk of the regulated utility being analyzed, and we recommend that Ms. Bulkley's use of operating income as a criterion should be rejected, and instead our 70% regulated revenue criterion be adopted.

Q. Please explain your second concern regarding her selection criteria?

A. Our second concern relates to the elimination of companies that are not covered by at least two utility industry equity analysts. Based upon her application of this criterion, Ms. Bulkley eliminated three companies: Otter Tail, MGE Energy, Inc. and El Paso Electric. It appears to us, however, that either Ms. Bulkley is not appropriately applying her own selection criterion, or she is unaware that these three companies are followed by more than one equity analyst. Specifically, Otter Tail is covered by Value Line Investment, Williams Capital Group, L.P, and Maxim Group, while El Paso Electric is

covered by Merrill Lynch, Mizuho Securities and Williams Capital Group as shown in our Exhibit___(SFP-24). Moreover, MGE Energy is covered by more than one equity analyst including CFRA Equity Research, Ford Equity Research Jefferson Research & Management with their latest reports published on July 19, 2019, August 9,2019, and August 9, 2019, respectively. Therefore, there is no reason to exclude Otter Tail, El Paso Electric, and MGE Energy from her proxy group on the basis of number of equity analysts.

Q. Please explain your final concern regarding her selection criteria?

A. We find Ms. Bulkley's limiting her proxy group to primarily transmission and distribution utilities (T&D) by excluding those having owned generation comprising greater than 60% of MWH sales to ultimate customers to be unreasonably restrictive. This restriction resulted in the elimination of eleven companies, that are all comparable in risk and thus suitable surrogates for the Companies. By drastically reducing the size of her proxy group, Ms. Bulkley has

undermined the reliability of her results. Thee eleven eliminated companies, which are all in our proxy group are: Ameren Corp., American Electric Power Co., Duke Energy, Entergy Corp., IDACORP, MGE Energy, NextEra Energy, OGE Energy, Pinnacle West Capital Corp., PNM Resources Inc. and Xcel Energy.

**Criticism of Ms. Bulkley's DCF Methodology**

Q. Please describe Ms. Bulkley's DCF methodology.

A. Ms. Bulkley uses a multi-stage DCF analysis based on a three-month average stock price and a range of near-term growth rate assumptions that produces a mean ROE of 9.20% for her Combined Utility Proxy Group and 9.04% for her Natural Gas Proxy Group. Specifically, she uses a three-stage DCF analysis that relied on equity analysts' earnings growth forecasts published by Value Line, Zacks and Yahoo Finance.

Q. Do you have any concerns with Ms. Bulkley's DCF analysis?

A. We have several concerns. Foremost among them, we find her third stage growth rate to be excessive and believe that she applies an

inappropriate internal rate of return (IRR) tool for calculating her ROE estimate. As we will explain shortly, we also find her first-stage growth rates to be inferior to ours. Ms. Bulkley's use of an excessive third stage growth rate is the principal reason that her 9.20% DCF-based ROE recommendation is flawed. While the use of a three-stage DCF growth model is not necessarily problematic, our primary concern lies with her use of an excessive growth rate in her long run or third-stage growth rate, and to a lesser extent, some of the excessive growth rates she uses in the first stage of her analysis.

Q. Please elaborate why you believe that her first-stage growth rates are problematic.

A. Ms. Bulkley's model uses earnings growth rate estimates to determine short-term dividend growth rates, ignoring available dividend growth rates or forecasts. This is in direct contrast to the basic premise of the DCF which measures the present value of future dividends. In using earnings growth, Ms. Bulkley has simply assumed that dividend growth will match earnings growth but has presented no evidence why this

expectation is valid. Of greater concern is her use of a 25.5% earnings per share (EPS) growth rate for Northwest Natural Gas Company. We believe that a growth estimate of that magnitude is unreasonable and that she should have excluded it from her analysis. The impact of eliminating this 25.5% EPS growth rate would have reduced Ms. Bulkley's DCF results by nine basis points from 9.20% to 9.11%.

Q. Do you have any other concerns?

A. Yes. We also have a concern about her use of a historically derived real gross domestic product (GDP) rate, together with a forecasted inflation rate as the long-term growth rate of her proxy group. We will discuss this in detail later.

Q. Please describe how Ms. Bulkley derives her short-term dividend growth rates.

A. In the first stage of the three-stage DCF model, Ms. Bulkley takes the current annualized dividend for each company in her proxy group and escalates the dividend for a period of five years based on the average of the three-to-five-year earnings growth estimates reported by Yahoo Finance, Zacks and Value Line.

Q. Why is this approach problematic?
A. It is highly unlikely that investors would rely exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections. The 25-year old Harris study on page 59 of Ms. Bulkley's testimony purports that a "growing body of knowledge" shows that analyst earnings forecasts are indeed reflected in stock prices. While we agree that all relevant information is incorporated into a company's stock price, the linear relationship of earnings to dividend growth that she assumes is unlikely. That is, payout ratios will change based upon many factors including individual company cash flow requirements, future market conditions as well as other factors. Investors will not just look at expected earnings but also factor in all relevant information when estimating their required return.
Q. Does Ms. Bulkley have any concern with the use of Value Line short-term dividend projections?
A. Yes. Ms. Bulkley, on page 61 of her testimony, claims that since Value Line's projections of

short-term dividend growth do not explicitly include growth in retained earnings, their use is improper as inputs to the DCF analysis. What she has failed to recognize is that, while in the long-term, dividend growth is constrained to the combination of retention growth and issuing stock above/below book value, in the short run, dividend policy can change and result in dividends to grow above or below retention growth. Consequently, her short-term dividend projections are a direct product of the average earnings growth estimates of three different publications, without any consideration being given to the growth rates effect on future dividend payouts. Our use of Value Line dividend growth projections recognizes the impact of payout policy while her sole use of short-term earnings forecasts does not.

Q. Does Ms. Bulkley have other criticisms of using only Value Line dividend growth rates?

A. Yes. Ms. Bulkley claims Value Line's dividend projections are the growth expectations of a single analyst and attempts to discredit utilization of Value Line by claiming the

publication is based on a single analyst's viewpoint. While Value Line assigns a lead analyst in the electric utility industry who is then credited for each company's report, each report that Value Line develops goes through a continual evaluation and quality control process where 'multiple analysts' review the reports generated before they are ever posted. The Value Line document describing this process is included in Exhibit___(SFP-25).

Q. Is there another advantage to the continued use of Value Line projections in your analysis?

A. Yes. Staff and the Commission have reasonably relied on Value Line for many years. This methodology is generally understood by the investment community and lends a degree of predictability to the rate setting process in New York State. While this consistency does not help to identify the return that equity investors currently require, it is important in the sense that it provides predictability in the earnings level that investors in New York utilities can expect. This is particularly important to the major credit rating agencies, who view

unpredictability as an additional risk.

Q. You previously stated that Ms. Bulkley estimated long-term dividend growth based upon expected GDP growth. How did Ms. Bulkley calculate expected GDP growth?

A. Rather than using estimates of future GDP growth, Ms. Bulkley utilized a 3.22% historical average growth rate in real GDP for the period 1929 through 2018 based on data from the Bureau of Economic Analysis. Ms. Bulkley then calculated a 5.56% forecasted nominal GDP rate by taking this historical average growth rate, together with her expected average inflation rate of 2.27%. The 2.27% expected average inflation rate was calculated based upon an average of 2.2% expected inflation published by the Blue Chip Financial Forecasts, and 2.31% and 2.29% she derived using data from the Energy Information Administration (EIA). The EIA expected inflation was based on the compound annual Consumer Price Index (CPI) growth rate and the compound annual growth rate of the CPI for all urban consumers and the compound annual GDP Price Index. The derivation of her 5.56% long-term GDP growth rate is shown

on page 1 of her Exhibit__(AEB-2).

Q. Do you agree with Ms. Bulkley's 5.56% long-term growth rate estimate?

A. No. While we do not necessarily find it unreasonable to assume that long-run growth for utilities would approximate growth in the overall economy, what is unreasonable is the excessiveness of her estimate of long-term GDP growth. Specifically, we do not think the use of her historically-derived real GDP as the long-term growth rate is appropriate for reasons previously discussed, namely because historical growth rates are necessarily measured over a subjective time period, and in this instance, one that incorporates far different circumstances than exist today. Historical average growth rates by their very nature completely, and inappropriately ignore economic conditions today. In addition, the calculation presented by Ms. Bulkley does not accurately measure GDP growth. This inaccuracy occurs since the 2.27% expected inflation rate that she utilizes reflects expected price changes in the Consumer Price Index and in the CPI-Urban. The CPI measures

changes in the price level of a basket of consumer goods and services and, unlike the GDP deflator, does not measure inflation over the entire economy. Additionally, Ms. Bulkley's use of the real historical GDP growth rate from 1929 to 2018 is inappropriate because while historical averages provide insight into how past factors might have influenced past changes in GDP, they are poor indicators of future economic activity.

Q. Are there estimates of long-run growth in the economy that you believe provide a more reasonable estimate of long-run growth?

A. Yes, we are aware of two reasonable estimates of long-run growth. The first source is the Long-Range Consensus U.S. Economic Projections provided by Blue Chip Economic Indicators, which we previously discussed, and employed in ascertaining the reasonableness of our sustainable growth rate. Not only does this report project out into the future twice as far as any other reputable source of economic data, it also reflects the forward-looking consensus of approximately 50 of the financial community's most prominent economists. According to the

March 10, 2019 publication, which is the most recent and is also shown in our previously mentioned Exhibit___(SFP-18), the consensus long-run nominal GDP growth rate for the most distant period forecast, 2026-2030 is 4.0%, which includes both 2.1% real GDP growth and 1.9% expected inflation components. A reasonable investor would expect a similar long-term growth rate for an energy utility rather than the excessive 5.56% growth rate used by Ms. Bulkley.

Q. Please continue with your second source.

A. The second source as shown in our Exhibit___(SFP-SFP-26), is based on information and data from U.S. Energy Information Administration (EIA). On page 18 of its April 2019 Annual Energy Outlook, the EIA states that "The Reference, High Economic Growth, and Low Economic Growth cases illustrate three possible paths for U.S. economic growth. In the High Economic Growth case, average annual growth in real GDP is 2.4% from 2018 to 2050, compared with 1.9% in the Reference case. The Low Economic Growth case assumes a lower rate of annual growth in real GDP of 1.4%."

The EIA's optimistic long-run real GDP growth

rate of 2.4% suggests a high nominal GDP growth rate of about 4.5% (2.4% plus 2.1% inflation). Accordingly, in light of this information, Ms. Bulkley's use of an excessively high long-term growth rate of 5.56% totally undermines the reliability of her DCF results.

Q. Earlier, you indicated that you have a concern with Ms. Bulkley's use of the irregular internal rate of return in her DCF analysis. Please explain.

A. Our concern relates to Ms. Bulkley's inappropriate use of the extended version of the Internal Rate of Return (XIRR) in her DCF analyses versus the Commission preferred use of the regular IRR in our model. Her use of the XIRR overstates her mean DCF result of 9.20% for her Combined Utility Proxy group by approximately 15 bps and approximately 14 bps for her Natural Gas Proxy Group's mean DCF result of 9.04%.

Q. Please explain the XIRR and the IRR?

A. The XIRR is a Microsoft Excel function, or tool, used to calculate returns for a series of cash flows that are not regular. The IRR is suitable for regular cash flows that could be monthly,

quarterly, or annually. As an illustration, if an individual had six series of cash flows consisting of an initial investment of $1,000 made on June 30, 2014, and four dollar returns consisting of $200, $220, $400, and $300 on 2/28/2015, 1/28/2016, 2/15/2017 and 3/28/2018, respectively, this individual's investment return based on the irregular cash flow will be 5.0% compared to 4.3% using the regular IRR. Since dividend cash flow are typically paid on a regular basis, we do not believe that the XIRR is an appropriate tool for determining the DCF-based ROE.

Q. At various sections of her direct testimony, Ms. Bulkley, to some extent, cited the Recommended Decision (RD) issued in the Generic Finance Case (GFC) 91-M-0509 to support her assertion that the cost of equity methodologies that she employed are consistent with the goals of the Recommended Decision issued in the Generic Finance Proceeding. Did Ms. Bulkley provide any evidence indicating that the Generic Finance Case (both KEDNY and KEDLI were parties) either directly or indirectly relied on the use of the XIRR?

A. No. In response to IR DPS-895, Ms. Bulkley stated that she "…is unaware whether the Generic Finance Case RD relied on the IRR or XIRR formula."

Q. Panel, what is your opinion about whether the Generic Finance Case RD relied on the IRR or XIRR formula?

A. To understand whether the Generic Finance Case Recommended Decision relied on the IRR or XIRR, we reviewed and recast the DCF results for both the 34-member electric and the 11-member gas proxy groups in Appendix B of the GFC's "Return on Equity Consensus Document" prepared by the signatory members of the Electric and Gas Industry to which both KEDNY and KEDLI were signatory parties. For the 34-member electric proxy group, the GFC results shows a median and average DCF ROEs of 9.62% and 9.61%, respectively. Our recast using the regular IRR resulted in a 9.62% for both median and average ROE compared to 9.9% for both median and average ROE using the XIRR. For the 11-member gas proxy group, our recast of the GFC results shows a 10.70% and 10.40% average and median ROE,

respectively. Our recast using the regular IRR resulted in 10.71% and 10.40% average and median ROE, respectively compared to 11.0% and 10.67% average and median ROE using the XIRR. Our recast is shown in Exhibit___(SFP-27).

Q. Panel, what is your conclusion?

A. Our conclusion is that even though there was no specific discussion in the "Return on Equity Consensus Document" of the GFC concerning whether to use the regular IRR or XIRR, our analysis shows that the Generic Finance Case RD relied on the IRR rather than the XIRR for its DCF analysis at that time.

A. Has the Commission ever granted a cost of equity based upon a DCF analysis that discounted future dividends using XIRR?

A. No. Instead, both Staff and the Commission have relied on the regular IRR in DCF-based ROE calculations in both litigated and negotiated cases. Therefore, we recommend that the Commission reject Ms. Bulkley's use of the irregular internal rate of return and adopted Staff's position.

Q. Beginning on page 14 of her direct testimony, Ms.

Bulkley argues that due to the current market conditions, and in particular the Federal Reserve's monetary policies, the DCF model currently underestimates investors' required returns due to low dividend yields and high valuations on utility stocks, which she argues are not sustainable in light of her contention of investors' expectation of higher interest rates. How does the Panel respond to this assertion?

A. To gain better insight about trends in utility stock valuation and dividend yields, Staff compiled dividend yields, MBR and price-to-earning (P/E) ratio data from Merrill Lynch Quantitative Profiles from January 2005 to June 2019 as shown in Exhibit___(SFP-28). As the exhibit illustrates, the levels of utility valuation and dividend yields in the three years (2005-2007) preceding the 2008/09 credit and financial crisis are similar to or in some cases higher than the current levels. For instance, from 2005 to 2007, the average P/E ratio, MBR and dividend yields on annual basis were 16.49, 2.06 and 3.27%, respectively, compared to 16.23, 1.72, and 3.71%, respectively, for the post-financial

crisis period (2011 to June 2019). This trend suggests that low-interest rates as a result of the Federal Reserve monetary policies are not necessarily the only contributing factor to high valuation and low dividend yields in utility stocks. Therefore, Ms. Bulkley's belief that the current higher equity valuations for utility stocks have been driven by the Federal Reserve's monetary policies resulting in the DCF understating investors' required return is not grounded in any statistical analysis but appears to be based on a priori assumptions. What Ms. Bulkley is calling abnormal is really not all that different from the economic conditions that prevailed over several years before the financial crisis hit in 2008. Our interpretation is shared by one article that was recently published in the Wall Street Journal on July 8, 2019.

Q. Could you please cite the Wall Street Journal article.

A. Yes. In the article entitled, "Can Low Rates Explain High Stock Prices? Not So Fast", Mark Hulbert stated that "[t]oday's low interest rates justify above-average price/earnings ratios.

That's the theory, anyway. Unfortunately, that theory doesn't hold up when you look at it through the lens of history." This article is provided as Exhibit___(SFP-29).

Q. Please continue with your response.

A. Ms. Bulkley's assertion goes against the crux of the efficient market hypothesis (EMH) underlying the DCF model that posits that stock prices reflect collective judgment of all stock market participants and therefore, reflect all known factors influencing the price of a stock, including the Federal Reserve's monetary policies.

Q. Are you aware of any financial literature that has come out subsequent to the Financial Crisis that supports continuing adherence to the efficient market hypothesis?

A. Yes. In discussing the importance and interpretation of stock prices in his book entitled, "The Cost of Equity - A Practitioner's Guide (2010 Edition)", David C. Parcell, specifically stated on page 88 of his book that "[t]he market price of a firm's stock represents the collective judgement of all stock market

participants as to the value of a firm at a particular point in time. The stock price takes into consideration the participants' interpretation of all relevant factors, such as past present and future earnings, the risk of these earnings, dividend policy and other factors." This article is shown as Exhibit___(SFP-30).

Q. Is there any indication that the Federal Reserve is likely to increase interest rates in the near term as claimed by Ms. Bulkley?

A. We do not believe so because the market expectation is that the Federal Reserve is likely to cut or maintain the current level for the near term as shared by one article that was recently published in the Wall Street Journal on July 19, 2019, attached as Exhibit___(SFP-31). In discussing recent signals from Federal Reserve Chairman and other officials, the author, Nick Timiraos stated that "Fed Chairman Jerome Powell set the stage last week for the first interest rate cut in over a decade during congressional testimony, when he signaled concern about global growth and the risk of a more prolonged shortfall

in inflation from the Fed's 2% target. Those developments strengthened the case for a somewhat easier policy stance." The author continued by saying, "Federal Reserve officials signaled they are ready to lower interest rates by a quarter-percentage point later this month, while indicating the potential for additional reductions, despite the recent surge in market expectations of a half-point cut. Officials aren't prepared for bolder action now, according to the officials' recent public statements and interviews, as they weigh concerns about a slowdown in global growth, an increase in trade-policy uncertainty and a pullback in inflation." Accordingly, the Federal Reserve cut the interest rate on July 31, 2019, by 25 basis points, for the first time since December 2008.

Q. Panel, what is your overall conclusion?

A. Our overall conclusion is that Ms. Bulkley's use of historically-derived long-term growth rate, 25.5% earnings per share (EPS) growth rate for Northwest Natural Gas Company, irregular IRR and expectation of higher interest rate are inappropriate and should therefore, be rejected

by the Commission.

**Criticism of Ms. Bulkley's CAPM Methodology**

Q. Would you please summarize Ms. Bulkley's CAPM approaches?

A. Ms. Bulkley provided a total of 24 CAPM-derived ROE estimates for her Combined Utility and Natural Gas proxy groups using the same CAPM methodologies that we employ. Ms. Bulkley calculated 12 ROEs using the traditional CAPM methodology and 12 ROEs using the zero-beta CAPM methodology, using two different beta sources in combination with three different risk-free rate estimates for each version of the CAPM.

Q. Please explain how Ms. Bulkley derived each of the three major inputs necessary for her CAPM methodologies: the risk-free rate, beta and the market risk premium.

A. As shown in her Exhibit__(AEB-4), Ms. Bulkley's traditional and zero-beta CAPM methodologies use a risk-free rate based on the current three-month average yield (December 2018 to February 2019 yields) on 30-year Treasury bonds (3.05%), the near-term projected 30-year Treasury yield for

the five forward-looking quarters (2Q-2019 through Q2-2020) (3.28%), and the long-term projected 30-year Treasury yield for the period 2020-2024 (3.90%). Similarly, Ms. Bulkley developed a market risk premium based on an estimated S&P 500 required market return of 13.77% and the subtraction of each of the three risk-free rates we discussed above. Ms. Bulkley utilized two different beta determinations within each of her CAPM methodologies. For her Combined Utility Proxy Group, Ms. Bulkley applied its 0.691 Bloomberg average beta and its 0.645 Value Line average beta. For her Natural Gas Proxy Group, she applied its 0.70 Bloomberg average beta and its 0.686 Value Line average beta. Given these respective inputs, Ms. Bulkley then developed six traditional CAPM estimates of the cost of common equity for KEDLI and KEDNY for both proxy groups, ranging from 9.96% to 10.81%, and six zero-beta estimates of the ROE for both proxy groups ranging from 10.92% to 11.55%. By essentially averaging all of these results, Ms. Bulkley's CAPM methodology produced a mean ROE estimate of 10.95% for her Natural Gas Proxy

Group and 10.72% for her Combined Utility Proxy Group.

Q. What concerns do you have with Ms. Bulkley's CAPM analyses?

A. We have three principal concerns regarding Ms. Bulkley's CAPM analysis: 1) relating to her derivation of a 13.77% market return, 2) her beta determinations, and 3) her risk-free rate listed in order of magnitude of their impact. Collectively, the impact of these infirmities is an overstatement in her cost of equity estimate of a little over 200 bps.

Q. Please explain your concern with the approach she used to determine her 13.77% required market return.

A. The primary flaw with her CAPM approach is that it relies entirely upon a constant growth DCF analysis of the S&P 500. Her use of an excessively high 13.77% market return versus our 10.88% market return is the main reason for the large differences in her CAPM results and ours. Quite simply, the basic assumption of this approach, that the Bloomberg earnings growth rate estimates formulated for the next three-to-five

years will last into perpetuity, is not reasonable. This is precisely why Staff relies on the ex-ante estimate of the required return of the S&P 500 provided by Merrill Lynch, in our previously mentioned Exhibit___(SFP-19). Merrill Lynch's multi-stage DCF-derived required return does not make this unrealistic assumption, but rather uses three specific stages with varying durations depending on the facts and circumstances of the company being examined. The unreasonableness of using a constant growth DCF calculation to estimate the required market return is perhaps best illustrated by considering the fact that 39 of the S&P 500 companies in Ms. Bulkley's market return calculation have near-term earnings growth estimates in excess of 20.0% and range as high as 96.36% (Exhibit__(AEB-1)). It is plainly unreasonable that investors would assume that those companies would be able to maintain those extraordinary growth rates forever.

Q. At various sections of her direct testimony, Ms. Bulkley endorses recent conclusions the FERC has made with respect to ROE to support the

reasonableness of her approach to ROE. For example, on page 69, line 6 to 8, she stated that "…FERC now relies on analysts' estimates of earnings growth in the short-term and a long-term GDP growth rate as the measure of growth in the second stage." Could Ms. Bulkley have also equally considered the FERC's proportionate weighting of short-term and long-term growth rate to check the reasonableness of her 13.77% derived market return?

A. Yes. Ms. Bulkley could have considered the FERC's proportionate weighting of short-term and long-term growth rate at two-thirds and one-third, respectively to check the reasonableness of her 13.77% derived market return.

Q. What would have been Ms. Bulkley's market return if she had used the FERC's weighted growth rate approach?

A. As shown in our Exhibit__(SFP-32), Ms. Bulkley's market return would have been 11.37%, which is 240 basis points lower than her 13.77% market return.

Q. Does Ms. Bulkley's market return calculation have any other flaws?

A. Yes. Many of the companies in the calculation of her excessive 13.77% market return are not paying any dividends. The DCF calculation is premised upon the present value of future cash flows, in this case dividends. If companies are not paying dividends, then the whole DCF calculation is flawed because a DCF cannot be performed and therefore her 13.77% market return should be rejected because of flaws in her calculation.

Q. What is the impact of her use of the excessive 13.77% market return?

A. For her Combined Utility Proxy Group, the impact is an increase of about 166 basis points.

Q. Please explain your second concern regarding the derivation of Ms. Bulkley's beta estimates.

A. The Commission has consistently utilized Value Line betas, and one of the principal reasons for doing so is Value Line's superiority over other sources like Bloomberg.

Q. Are there any independent sources that indicate that Value Line Beta is superior?

A. Yes. In their recent book entitled Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice, H. Kent Baker and

Gerald S. Martin also wrote about the superiority of Value Line betas over Bloomberg betas. They wrote that "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg, which uses the following formula:
adjusted beta = historic beta*(0.67)+0.33."
After plotting historic betas, Value Line adjusted betas, and Bloomberg adjusted betas, the authors concluded that "the [Exhibits] convey a key message for analysts: be wary of blindly using historic betas from individual company data. The tails of the distribution for historic betas contain significant measurement error. A clear advantage of Value Line betas (or those from other providers who do more than just historic regression) is that they reflect professional judgment and attention to detail."
This report is attached as Exhibit__(SFP-33).

Q. Has the Value Line beta accuracy been tested?

A. Yes. In her empirical research work entitled "Whose Beta is Best?", Ms. Diana Harrington investigated the accuracy of Value Line betas and those betas reported by Merrill Lynch, Wilshire

Associates, Rosenberg and Associates (now BARRA) and others. For the period studied, Value Line consistently outperformed all other services over the forecast horizon of one to four years for public utilities. The study also indicated that increased data points increase the beta accuracy. The report is attached as Exhibit__(SFP-34). We have not come across any empirical research work that shows that Bloomberg beta estimates are superior to Value Line betas.

Q. What is the impact of using the blended Value Line and Bloomberg betas?

A. For her Combined Utility Proxy Group, the impact is an overstatement of about 21 basis points.

Q. Please explain your third concern relating to Ms. Bulkley's use of the 30-year Treasury and future estimates of its rate as the appropriate risk-free rate.

A. Ms. Bulkley argues that the sole use of the yield on the 30-year Treasury is appropriate because, in her view, utility companies represent long-duration investments. However, it has long been Commission policy to rely on the average of the 10-year and 30-year Treasury yields to arrive at

the risk-free rate as investors have a range of investment horizons. Furthermore, as the Commission has maintained for many years, current rates are the best indicator of future rates as they are based upon the latest available information to investors. The rationale for this approach is well-established and reflects the reality that there are investors with intermediate-term as well as long-term investment horizons.

Q. Ms. Bulkley argues that the Commission's preferred approach is flawed because it does not address the expected lives of the Companies' assets, the equity duration of the utility industry, or what Morningstar suggests is "the time horizon of the chosen Treasury security is that it should match the time horizon of whatever is being valued," as stated on page 72. Do you agree with her arguments?

A. No, we do not. While Ms. Bulkley is correct that utility plant assets have very long lives, and we would agree that sound financing practices generally dictate these long-lived assets be financed with similarly long-lived securities,

her conclusion that all utility equity investors have an investment horizon of 30 years is unsubstantiated and erroneous. One needs to look no further than the long-term debt obligations supporting the Companies own rate bases to understand that investors' have different time horizons.

Q. Is Ms. Bulkley's use of Treasury yield projections appropriate for use in the CAPM?

A. No. Her use of interest rate projections should be rejected. As Staff and the Commission have maintained for many years, and as mentioned earlier, current rates are the best indicator of future rates as they are based on the latest available information to investors.

Q. What is the combined effect on Ms. Bulkley's CAPM results if she had used a blended yield of 10-year and 30-year Treasury yields similar to the Commission's preferred approach instead of her use of projected 30-year Treasury yields?

A. The impact is an increase of about 13 basis points.

Q. Panel, what is your overall conclusion relating to your concerns of Ms. Bulkley's excessive

13.77% market return, her beta determinations, and risk-free rate derivation?

A. Our overall conclusion is the impact of these infirmities is an overstatement in her cost of equity estimate of a little over 200 bps. Therefore, we recommend that the Commission reject Ms. Bulkley's 13.77% market return, her beta determinations, and risk-free rate derivation and instead accept Staff's appropriate CAPM inputs for the risk-free rate, market risk premium and beta determination.

**Equal Weighting of DCF and CAPM Results**

Q. On page 79 of her direct testimony, Ms. Bulkley argues that "…it is reasonable to apply equal weighting to the DCF and CAPM methods when determining the ROE for the Companies." Why does Ms. Bulkley argue that the CAPM methodology should be accorded an equal weighting with the DCF methodology?

A. Ms. Bulkley's explanation to justify her equal weighting is based on three main points: 1) the effect of the Federal Reserve monetary policies on market conditions and its corresponding impact

on utility stock valuations, 2) other utility regulatory bodies' recent response to the alleged impact of current capital market conditions on the DCF model, and 3) market expectations of higher interest rates. She then concluded that the DCF method is producing ROE results that are significantly lower than the results derived from her CAPM calculations, and therefore, it is reasonable to place less reliance on the DCF results (page 79 lines 8-10).

Q. Please what is your response?

A. First, as we pointed out earlier, the MBRs and P/E ratios over the past nine years or so, have actually generally been lower than they were in the three-year period preceding the financial crisis. The significance of that timeframe is that it was a time period well before the Federal Reserve had begun its Quantitative Easing, and yet we had MBRs even higher than they are today. Therefore, Ms. Bulkley's whole contention that utility stock prices are extraordinarily high specifically because of the Federal Reserve's monetary policies is misguided. Moreover, Ms. Bulkley would have us believe that investors

believe these "anomalous" conditions will significantly change in the not too distant future. Conditions that have persisted for nearly a decade should not be considered anomalous instead they should be viewed as the new normal. If the valuations for dividend-paying stocks have increased, it simply means that investors view dividend payments favorably and perceive less risk in an investment in companies with higher dividend payouts as compared to companies paying lower or no dividends. As such, lower risk should translate into lower return requirements and should be reflected in required ROE estimations. This would hold true in a DCF or any analysis quantifying required returns.

Q. Please elaborate on her second argument that other utility regulatory bodies have responded to the impact of the current market conditions on the DCF results.

A. Ms. Bulkley stated that other utility regulatory bodies like the Illinois Commerce Commission, the Pennsylvania Public Utility Commission, and the Missouri Public Service Commission have all "considered" the current market conditions impact

on DCF results. She also stated that the FERC, in particular, has determined that current capital market conditions have caused the DCF model to understate equity costs for regulated utilities at this time. She continued by saying that the FERC recently proposed a methodology that reflects the commission's current view that investors rely on multiple ROE estimation models and as a result the FERC proposed ROE methodology that includes equal weighting of the DCF, CAPM, Risk Premium and Expected Earnings.

Q. What is your response to her second argument?

A. Ms. Bulkley's characterization that other utility regulatory bodies have recognized the understatement of the DCF is misleading and shows selection bias. In any case, the opinion of another commission should have no bearing on a New York State proceeding. Furthermore, Staff is not compelled to abandon its approach, nor would we recommend to the Commission to deviate from its approved approach.

Q. Please respond to Ms. Bulkley's contention that because her CAPM results are significantly higher than the results of her DCF analyses, her DCF

results must be understated.

A. To begin with, we have already shown that her DCF results are not understated. However, because she has chosen to employ a methodology that yields such an egregiously high market return in her CAPM analysis, it is not surprising that her DCF analysis yields a lower ROE determination. However, such a finding would not be the case if her CAPM analysis employed reasonable assumptions as does ours. Under Staff's cost of equity model, the 8.10% CAPM result is currently lower than the 8.22% DCF result.

Q. Why should the CAPM be given less weight than the DCF in the weighting of the ROE methodologies?

A. As recognized by the Commission in prior rate proceedings, the CAPM presents a conceptual framework that provides a reasonable estimate of a firm's cost of equity, however, given some of the perpetual weaknesses of the CAPM discussed previously in our testimony, the Commission should continue its weighting of 33% compared to the DCF weighting of 67%.

Q. Has the Commission ever addressed the issue of equal weighting of the DCF and CAPM?

A. Yes. The Commission has consistently used the DCF as its primary methodology for determining ROE by applying a 2/3 weighting in the determination of the cost of equity. About a decade ago, some company witnesses in Cases 13-W-0539 & 13-W-0564 involving Suez Water Westchester, formerly known as United Water Westchester and New Rochelle, raised concerns about the weighting accorded the DCF and argued for equal weighting of the DCF and CAPM when the market-to-book ratio exceeds one or when the DCF was producing lower results than other methodologies the witnesses were advocating. In all the cases that we reviewed, the Commission repeatedly affirmed its preference for the two-thirds and one-third weighting of the DCF and CAPM, respectively. For example, we noted that the 2/3 DCF and 1/3 CAPM weighting was affirmed in 2008 by the Commission's Order in Case 07-E-0523 Consolidated Edison Company of New York, Inc.- Electric Rates, and in 2007 by the Commission's Order in Case 07-G-0141, National Fuel Gas Supply Co. – Gas Rates. There, the Commission stated, "[w]e also agree with Staff,

CPB, and Multiple Interveners that the Company (NFG) has not provided any compelling reasons to provide equal weight to the DCF and the CAPM methods... Accordingly, we will continue to use the two-thirds DCF Method and one-third CAPM method weighting in this case." Similarly, on pages 14 and 15 of the Commission's Order in Cases 06-E-1433 and 06-E-1547, Orange & Rockland Utilities, Inc. - Electric Rates, the Commission stated, "We will continue to accord two-thirds weight to the DCF result and one-third to the CAPM result as we have in past decisions." It is worthy of note that the Commission affirmed the proportionate weighting of the DCF and the CAPM when the utility stock valuations and dividend yields were similar to the current levels as discussed earlier. Therefore, we believe that Ms. Bulkley has not presented any new evidence that would compel the Commission to reject its long-standing policy of a two-third and one-third weighting of the DCF and CAPM and as such Ms. Bulkley's approach should be again rejected.

**Impact of Correcting the Flaws in Ms. Bulkley's ROE Approach**

Q. What is the combined effect on Ms. Bulkley's results if she had corrected the principal flaws in her DCF approach including 1) the use of the extended version of internal rate of return(XIRR) instead of the Commission preferred IRR version, 2) eliminated the excessive growth rate of Northwest Natural Gas Company, 3) properly relied on only Value Line betas, 4) used Merrill Lynch market returns, and 5) weighted her DCF results at two-thirds and CAPM results at one-third?

A. The combined effect of correcting the flaws in her DCF and CAPM approach including the inappropriate use of the extended version of IRR, rely on only the Value Line betas, and weight her DCF and CAPM results at two-thirds and one third, respectively is a reduction of 131 basis points from 9.71% to approximately 8.40%. The summary recalculation is shown in Exhibit___(SFP-35).

**Regulatory Framework in New York**

Q. On page 90 of her direct testimony, Ms. Bulkley stated that New York's regulatory framework has a

somewhat greater risk than the jurisdictions in which her proxy group companies provide service. Did Ms. Bulkley perform any qualitative analyses of state regulatory jurisdictions to support this assertion?

A. Ms. Bulkley performed three analyses as contained in her Exhibit___(AEB-9) and Exhibit___(AEB-10) to support her argument that the New York regulatory environment is relatively riskier. While Exhibit___(AEB-9) relates to KEDNY/KEDLI and Proxy Group Companies' Regulatory Research Associates (RRA) Rankings, Exhibit___(AEB-10) relates to S&P's assessment of regulatory credit supportiveness of the Companies and her proxy group.

Q. What is your opinion of Ms. Bulkley's statement that New York's regulatory framework has a somewhat greater risk than the jurisdictions in which her proxy group companies provide service?

A. Recent industry reports regarding the relative business and regulatory risks of New York's regulatory regime appear to find otherwise. For example, in May 2017, Regulatory Research Associates, a business unit of S&P, published the

results of its comprehensive audit of regulatory rankings in a report entitled, "State Regulatory Evaluations: Regulatory Climate for Energy Utilities Including an Overview of RRA's ranking process". Contrary to Ms. Bulkley's contention of higher business risk in New York, RRA viewed regulatory risk in the State as decreasing, as evidenced by its upgrading of New York's ranking from "Average/2" to "Average/1". This report is attached as Exhibit___(SFP-36).

The table on page 3 of that report only ranks nine, out of 53, national regulatory jurisdictions with higher evaluations relative to New York, which implies that RRA finds New York to have a more-favorable-than average regulatory environment.

Q. What aspects of New York's regulatory environment has RRA highlighted in its assessment?

A. In regards to New York's authorized electric and gas ROEs, RRA states that the Commission, "…in rate cases decided in recent years, has authorized electric and gas ROEs that are lower than the nationwide industry averages, for the most part, these decisions were based on multi-

year settlements that incorporated increasing rate bases over the term of the plans, revenue decoupling mechanisms and deferral accounting for increases in such items as net plant, pension expense, and labor costs. Additionally, other factors in the rate-setting process, including the incorporation of fully forecasted test periods improve the utilities' opportunity to earn the authorized ROE". This is shown as Exhibit___(SFP-37).

Q. Have any of the major credit rating agencies commented on New York's regulatory environment in recent credit reports?

A. Yes. Moody's issued a credit opinion for National Grid USA on July 20, 2017, which is provided in Exhibit___(SFP-38). On page 3 of that report, Moody's stated that "[w]hile our view of regulation in the US has improved generally, we regard New York as one of the most creditor-friendly jurisdictions. Recent rate case settlements have allowed rates to increase and included de-risking provisions such as timely cost recovery, forward-looking test years for operating expenditure and revenue decoupling

(designed to provide stable fixed cost recovery). Collectively, these provisions have provided more transparency and will likely result in lower cash flow volatility going forwards. In addition, a number of utilities regulated by the [Commission] are required to maintain a certain financial profile either through explicit leverage restrictions (KeySpan Gas East Corporation, known as KEDLI, and Brooklyn Union Gas Company, known as KEDNY) or are at least required to maintain an investment-grade rating (KEDLI, KEDNY and Niagara Mohawk Power Corporation, NiMo, A2 stable). KEDLI, KEDNY and NiMo are also subject to a 'golden share' provision which limits the potential for financial distress at the parent negatively affecting individual subsidiaries."

Q. Does New York regulation incorporate other risk-reducing elements in addition to those cited by Moody's?

A. Yes, New York regulation incorporates many important elements that have a direct impact upon a utility's ability to achieve its authorized return on equity. For instance, New York uses a fully forecasted test year to determine revenue

requirement, which recognizes changes in revenues and expenses expected through the end of the rate year. The Commission also generally allows for a high level of expense reconciliation for such material cost items such as environmental remediation costs and pension and OPEB expenses. New York also uses purchase power adjustment clauses that allow full and timely recovery of this large and volatile cost element. Many other jurisdictions do not incorporate these mechanisms into their rate plans. For overall blanket-type protection, New York State, unlike most other states, allows utilities to request deferral accounting for material items actually incurred but not provided for in the base forecasts.

Q. Please discuss the impact of a fully forecasted rate year upon a utility's risk profile.

A. During periods of rising operating and maintenance costs, combined with the need for significant capital additions, we believe that New York State's methodology of a fully-forecasted rate year provides utilities with a much greater opportunity to earn their authorized return than jurisdictions that establish rates

based on an adjusted or unadjusted historic test year. In addition, New York State regulation allows utilities to revise their rate filings during the suspension period for known changes in cost rates, changes in estimates, and for wholesale revisions because of changed circumstances beyond the company's control. This significantly reduces risk and improves earnings.

Q. Do you believe that investors are aware of such risk reducing elements of NYS utility regulation?

A. Yes. The NYS regulatory environment was presented by Commission Chair in a message to RRA clients, dated October 29, 2014. We note two main points in the report. First, the Commission's reliance on fully forecasted test years provides NYS utilities with mechanisms whereby expenses, capital additions, and returns are forecast through the end of the year for which rates are being set. Therefore, earnings are not adversely affected by inflation, new programs, and plant additions. Second, New York utilities have a strong history of being able to achieve healthy returns largely because of New York's risk-reducing mechanisms such as fully

forecasted test years, revenue decoupling mechanisms, deferrals and true-up mechanisms for pension, other post-employment benefits, and property taxes. This report is shown in Exhibit___(SFP-39).

Q. How many of their existing true-up and deferral mechanisms have the Companies requested be continued in this proceeding?

A. The Companies have requested for continuation of all of their existing true-up and deferral mechanisms.

Q. Please comment on the Companies' requested deferral mechanisms in this proceeding.

A. Even though not all the requested true-ups have been blessed by Staff, we believe that some of KEDLI/KEDNY's revenues and costs would be reconciled and therefore there will be no substantial risk for investors. Here are a few examples. First, Staff Efficiency and Sustainability Panel recommends an energy efficiency (ETIP) reconciliation that aligns with the Commission's direction in the March 2018 ETIP. Second, the Staff Consumer Services Panel agrees with the companies' proposal to fully

reconcile credit card and debit card costs but with some modifications. Third, Staff witness for the New York Facilities System Surcharge (NYFS) agrees with the Companies' proposal to change the timing of their NYFS reconciliations. Fourth, the Staff Gas Supply Panel agrees with the Companies' proposal to use 100 percent of options as a financial hedge against volatility in gas cost because such change will provide an added level of diversification for mitigating gas cost volatility. If granted by the Commission, the resulting risk reduction would reduce the need for thicker common equity ratio and the equity return requirements.

Q. Does Ms. Bulkley take these deferral mechanisms into account in her ROE recommendation?

A. No. Ms. Bulkley did not take into account the various deferral mechanisms in her ROE recommendation.

Q. Does Ms. Bulkley's return on equity request accurately reflect the Companies' deferral requests?

A. No, Ms. Bulkley has not made an attempt to compare the extensive suite of true-up and

deferral mechanisms being requested by the Companies with such mechanisms in place for her surrogate group. Absent such an analysis, it is very likely that Ms. Bulkley has overestimated the cost of equity for the Companies.

Q. Have the Companies proposed any Earnings Adjustment Mechanisms (EAMs) as part of the REV initiative?

A. Yes. The Companies propose three categories of EAMs, totaling six separate metrics. The three categories consist of: System Efficiency, Energy Efficiency, and Carbon Reduction. KEDNY could earn up to 78 basis points at maximum attainment levels for calendar year (CY) 2020 and 63 basis points for CYs 2021, 2022, and 2023. KEDLI could earn up to 77 basis points for CY 2020, 62 basis points in 2021, 63 basis points in 2022 and 64 basis points in 2023. Staff is recommending EAMs relating to Peak Reduction and Incremental Energy Efficiency worth about 25 basis points for the rate year.

Q. If approved by the Commission, how might these EAMs impact the Companies?

A. Successful implementation of these new

initiatives would likely enable the Companies to enhance their earnings above and beyond their authorized ROE.

Q. Does Ms. Bulkley take into account the opportunities afforded by the EAMs in her ROE recommendation?

A. No. Ms. Bulkley did not take into account the additional potential new sources of earnings resulting from the EAMs in her ROE recommendation. Rather, she discussed only the number of customer service quality and gas safety performance metrics that could negatively impact the revenue adjustments of the Companies. Ms. Bulkley's failure to account for the EAMs incremental earnings is inconsistent with the Companies' presentation for a reasonable opportunity to earn extra basis points above and beyond the Companies requested 9.65% ROE. To be conservative, however, our recommended cost of equity and credit metrics analysis do not reflect any of the potential incremental earnings associated with the EAMs.

**Conclusion**

Q. What is your recommended cost of capital for the Companies?

A. As summarized in Exhibit___(SFP-2) referenced earlier, on a pre-tax basis, our recommended overall pre-tax rate of return for KEDNY is 7.58% and 7.54% for KEDLI based on a capital structure with a 48% common equity ratio and an 8.20% cost of equity. As we stated earlier, we dismiss Ms. Bulkley's recommended cost of equity range of 9.65% to 10.75% as too high. Therefore, we recommend that the Commission reject both Ms. Bulkley's recommended cost of equity range and the Companies' requested 9.65% cost of equity and instead adopt our recommended 8.20% cost of equity.

Q. Does this conclude your testimony at this time?

A. Yes, it does.

MR. FORST: Your Honors, I now proffer the Staff Finance Panel for cross examination.

A.L.J. COSTELLO: Mr. Maloney.

MR. MALONEY: Yes.

CROSS EXAMINATION

BY MR. MALONEY:

Q. Good morning, panel. My name is Kenneth Maloney. I'll be asking you some questions on behalf of the companies.

MR. MALONEY: Before I start, Your Honors, in the -- in the interest of trying to do this efficiently, I've handed to you and to the panel and passed out to counsel around the room 15 documents that I'm going to refer to during my cross examination. And if I could have each of those marked for identification, I think it would speed things along.

The first one is N.G.A. -- the response to N.G.A.-58.

A.L.J. COSTELLO: Okay. That'll be 645.

MR. MALONEY: The second one is the response to N.G.-60.

A.L.J. COSTELLO: That'll be 646.

MR. MALONEY: The third one is the response to N.G.-62.

A.L.J. COSTELLO: That's Exhibit 647.

MR. MALONEY: The next document is the response to N.G.-63.

A.L.J. COSTELLO: That will be 648.

MR. MALONEY: The next one is the response to N.G.-64.

A.L.J. COSTELLO: 649.

MR. MALONEY: The next one is the response to N.G.-65.

A.L.J. COSTELLO: 650.

MR. MALONEY: The next one is the response to N.G.-67.

A.L.J. COSTELLO: 651.

MR. MALONEY: The next one is a document dated June 7, 1993. It purports to be a letter to the -- the commission enclosing two copies of the Nine Energy Utilities comments in support of the return on equity consensus document.

I -- I will say that this document was taken from the commission's website and what's in here is the return on equity consensus document, but I'd ask that that be marked the next number.

A.L.J. COSTELLO: It'll be 652.

MR. MALONEY: 52. The next one is a document dated January 31, 2020, entitled regulatory research, R.R.A. regulatory focus, major rate case decisions January to December 2019.

A.L.J. COSTELLO: That will be 653.

MR. FORST: Your Honors, is counsel preparing to prepare a foundation for these documents?

A.L.J. COSTELLO: Yeah, we're just marking them at this point and then I believe he's going to use them during cross, so any -- we're not admitting them into evidence at this point. So any objections you want to make, you'll be able to make as we proceed.

MR. MALONEY: The next document is a document date -- under a cover sheet that says, value line information for staff proxy group April 2019 to June 2019.

A.L.J. COSTELLO: That is marked for identification as 654.

MR. MALONEY: The next document is certain excerpts from a -- a book by Dr. Roger Morin, PhD, new regulatory finance.

A.L.J. COSTELLO: 655.

MR. MALONEY: The next document is a document from the commission's files that is exhibit -- labeled Exhibit 5 witness R.A. Morin in the Niagara Mohawk Power Corporation Gas Rate Case 08-G-0609.

A.L.J. COSTELLO: That will be 656.

MR. MALONEY: The next document is a Moody's Investor Service Report dated November 18, 2019 that says, threat to revoke National Grid's Operating License is credit negative for utilities.

A.L.J. COSTELLO: That's 657.

MR. MALONEY: The next one is a credit opinion from Moody's dated December 30th, 2019, the Brooklyn Union Gas Company update following rating confirmation with negative outlook.

A.L.J. COSTELLO: That's 658.

MR. MALONEY: And finally, the last document is labeled KeySpan Gas East Corporation, it's a Moody's document dated December 30th, 2019 and saying update, following rating confirmation with negative outlook.

A.L.J. COSTELLO: And that will be marked as 659.

MR. MALONEY: 659. Thank you, Your honor.

BY MR. MALONEY: (Cont'g.)

Q. Panel, so that I -- I don't forget. Good morning. You've been handed copies of responses that have been marked between Exhibits 645 through 651. Do these documents represent correct and accurate copies of responses that were provided by the Finance Panel to National Grid?

A. (Mr. Duah) Yes.

Q. And -- and to -- to the best of your knowledge, is the information contained in them true and correct?

A. Yes.

Q. Okay. Thank you. So panel, it's correct, isn't it that the primary purpose of your testimony is to recommend the fair rate of return to be used to determine the revenue requirements in this case.

A. As we stated on page 6 of our testimony, the purpose of our testimony is to recommend a fair rate of return to be used by the staff accountant panel to determine the revenue requirements. So the simple answer is yes.

Q. Okay. And one component of that rate of return is the rate of return on common equity. Is that correct?

A. Yes.

Q. And the other major component of the rate of return is the return on long-term debt. Is that correct?

A. Yes.

Q. So in order to compute the overall return, you need the proportions of debt and equity and the cost rates applied to each of those components. Is that correct?

A. Yes.

Q. Okay. And is -- is it also correct that the cost of equity is greater than the cost of debt?

A. Yes, I agree with you.

Q. And is it also correct that both the companies and the panel are proposing to set rates in this proceeding using a capital structure consisting of 48% common equity?

A. That's correct.

Q. Nonetheless, is it correct that if we wanted to increase the return dollars included

in the company's revenue requirement, we could do that by either increasing the return on equity or increasing the amount of equity in the capital structure?

A. Can you repeat the question?

Q. Sure. Is it correct that if we wanted to increase the return dollars included in the company's revenue requirements, we could do that by either increasing the return on equity or increasing the amount of equity in the capital structure?

A. The panel believes that you can increase the equity -- equity ratio, or you can increase the debt. But the overall cost of capital should say that it is at the lowest level.

In terms of revenue requirement, it should be such that it should not lead to an excessive revenue requirement. So that is my simple answer. So you should have protocols that they would impact.

Q. I'm sorry, I had trouble hearing that.

A. You can either increase the equity or you can increase the equity ratio, increase the return on equity. But it must be in such a way

the overall rate of return should be at the lowest level and affordable to ratepayers.

Q. Right. But -- but overall, all I'm asking you is, if -- if you want to increase the dollars you can do it by increasing the equity ratio or you can do it by increasing the return applied to that equity ratio. That's correct?

A. Correct.

Q. Okay. Yes. Okay. Now, from the standpoint of what we mean by a fair rate of return is such a return one that will enable the companies to provide safe and adequate service to their customers?

A. Can you repeat the question?

Q. Yeah. When we speak of a -- a fair rate of return, will such a return enable the companies to provide safe and adequate service to their customers?

A. We respond to that on -- on page 12, line 24, is a fair rate of return for a regulator utility is overall rate of return enables the company to provide safe and adequate service to each customer while at the same time assuring that the utility

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A.L.J. LEARY: So I need to ask the witness to speak really slowly so the record can pick up more precisely what your testimony is and the same for you Mr. Maloney, just be a little aware of how quickly you're speaking for purposes of making sure the record is clear. Thank you.

BY MR. MALONEY: (Cont'g.)

Q. And you were referring to your testimony -- I'm sorry, on 12 and 13 of the corrected testimony. Is that correct?

A. I'm referring to page 12, line 24 to page 13, line 1 up to 5.

Q. Yes. Thank you. Is it also correct that a fair return on equity amounts should be consistent with returns on investments having corresponding risks?

A. We respond to that on page 14, if you go to line 3, a fair rate of return allowed the utility to recover its incurred costs of long-term debt while providing its common equity with a great opportunity to earn a return that is comparable to

the return available for -- with investment of similar risk.

Q. Okay. And when we use the term risk, is it fair to say that another word for the term risk is uncertainty such that the greater uncertainty about the ability of shareholders to realize their returns, the greater the risk?

A. Can you repeat the question?

Q. What -- when we use to -- the term risk, is it fair to say that another term for -- another word for the term risk is uncertainty such that the greater uncertainty faced by shareholders in attempting to earn the rate of return, the greater the risk?

A. I would say the panel believes the uncertainty is one of risk, but that isn't the total risk because if you want to look at total risk you look at the financial risk and also look at business risk of the company. So it's one of them.

Q. I -- I understand that there are business risks and there are financial risks. What I'm asking you to -- whether you agree or not is do you agree that that means essentially uncertainty? There's uncertainty surrounding the company's future

business prospects. There's uncertainty surrounding its finances.

A. Yeah, the panel would believe that that is the case.

Q. Thank you. Now, you're proposing a return on equity for the companies of 8.2%, is that correct?

A. (Mr. Bullock) Correct.

Q. I think you need to turn your mic on.

A. Correct.

Q. Okay. And would you agree that if the commission adopts your recommendation the 8.2% would represent the lowest return on equity adopted by the commission in the last 30 years?

A. (Mr. Duah) Can you repeat the question?

Q. Yeah. Would you agree that if the commission adopts your recommended 8.2% return on equity that would represent the lowest return on equity adopted by the commission in the last 30 years?

A. I would say yes, but we need to also to understand that over the last 30 years, we

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas had the lowest interest rate environment. Well, if you look at the use on corporate bonds, property equity bonds, we are the lowest at this point.

In fact, the triple B is a yield in 3.5% was the lowest so far we have ever seen.

Q. Is it correct that you used a proxy group of 28 holding companies that own various combinations of either electric and gas utilities or electric only and gas only utilities to establish the rate of return in this -- in this case?

A. (Mr. Bullock) Correct.

Q. Are you aware of whether any of the regulated electric or gas utilities that are owned by the members of the Staff Proxy group have authorized returns on equity of less than 8.9%?

A. (Mr. Duah) Can you repeat the question?

Q. Yeah. Are you aware of whether any of the regulated electric combination, electric and gas or gas utilities owned by the members of the Staff proxy group have authorized returns on equity from their public service commissions or regulators that are less than 8.9%?

A. Can you repeat the question?

Q. Are you aware of whether any of the regulated electric or combination electric and gas or gas only utilities owned by the members of the Staff proxy group have augh -- authorized return on equity of less than 8.9%?

A. We are not aware. What we are aware is that each jurisdiction has their own approach in establishing return on common equities.

Q. So would it be correct to say that in setting your return on equity, you did not inquire or consider information as to what the authorized returns on equity were for the regulated companies that were owned by your proxy group? Please don't ask me to repeat that question.

A. I'm sorry. Can you rephrase the question?

Q. I -- I could try. Is it correct that you were not aware and did not make inquiry about what the authorized returns on equity were for the regulated electric, combination gas and electric or gas only distribution companies that were owned by the members of your proxy group?

A. I will say we -- we are aware that the allowed return equity for the companies in

our proxy were generally higher than what we're recommending. But as I said earlier, the methodology of setting those allow return on a common equity, that methodology is a different from each jurisdiction.

And as we said in our testimony at in page 64 investors are aware of what the commission's approach is when it comes to a return on common equity determination.

Q. So -- so your answer then is generally you are aware that the returns for the regulated companies are higher than what you're recommending here?

A. Correct.

Q. Okay. Now, the equity returns that we're seeking to determine in these proceedings are for two local gas distribution companies. Is that correct?

A. Yes.

Q. It's correct that we're not seeking to establish the return on equity for National Grid P.L.C., the companies ultimate parent company or any other affiliate company in the National Grid corporate structure. Is that correct?

A. Yes.

Q. Could you refer to your testimony on -- beginning on page 53 at line 22. You have that reference?

A. Yes.

Q. Okay. And is it correct that you say, even though the companies are local distribution companies, their ultimate parent is largely considered to be an electric utility by investors?

Accordingly, we believe that the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be electric utilities. Is that a correct reading of your testimony?

A. Yes.

Q. Now, when you use the word believe to describe your conclusion that electric utilities are the most suitable proxy group, was the panel's decision to use a group of utilities deemed by Value Line to be electric utilities, the product of some preconceived notion you had or that that was the most suitable option?

Or did you do some analysis of various possibilities and then decide to use the group of electric utilities?

A. Can you repeat the question?

Q. In your testimony, you used the word believe and I don't want to mischaracterize your testimony. We believe the most suitable proxy group would be divided from a universe of 37 holding companies deemed by Value Line to be electric utilities.

And what I'm asking you is when you use the word believe, did you simply have a preconceived notion that this was appropriate, or did you do some analysis of various possibilities and then decide that electric utilities were the most suitable proxy group?

A. We believe the electric proxy group is the most suitable for the proxy group because we look at the parent company, National Grid P.L.C., and then determine that investors, first of all, do not invest in KEDLY and KEDNI, they do invest in the parent -- ultimate parent company.

And the parent company is viewed as electric by investors and also if the parent company

were to be covered by Value Line, Value Line will have classified as electric.

Q. Now, I thought you told me before that we were setting the return on equity for gas distribution companies, not for National Grid P.L.C., are you -- or an electric utility. Are you changing your testimony now?

A. No.

Q. Is it your testimony that if -- strike that. Are you aware that KEDNI, the Brooklyn Union Gas Company, was once a standalone gas distribution company?

A. Yes.

Q. Okay. Is it your testimony that as a standalone distribution company, its cost of equity would be different -- if that were the case today, its cost of equity would be different than it would be because it's owned by National Grid?

MR. FORST: Objection. I think that calls for speculation. They're not currently independent or they're not their own gas distribution utility. They are owned by National Grid P.L.L.C.

BY MR. MALONEY: (Cont'g.)

Q. And I'm asking you the panel as experts as to whether as a standalone gas distribution company they would have a different cost of equity, they would be if they weren't owned by National Grid. The panel is experts in finance. This is -- seems to be a pretty basic question.

A.L.J. LEARY: As long as the record is clear that this is a question posed to the expert panel that is not sourced in factual information in this case, the panel can answer. But I want to make sure that we don't cloud the record with a lot of examples or speculation about what KEDLY or KEDNI are not.

So I'm just going to caution, but I'm going to let the panel answer.

THE WITNESS: (Mr. Duah) The panel believes that the circumstance – the circumstances would be different because in the first instance that you mentioned, equity investor will have -- will invest directly if KEDLI were to be on a standalone entity in the stock market.

BY MR. MALONEY: (Cont'g.)

Q. So in effect, what you're saying is you're concerned about the return that an equity

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investor and National Grid P.L.C. will earn what they're expecting, not what the cost of equity is for the gas distribution properties within National Grid?

A. Can you rephrase the question?

Q. Okay. I -- I believe you just said that investors could only invest in National Grid and, therefore, you're concerned about what those investors could earn in National Grid.

Correct of what we're trying to measure is the cost of equity of the gas distribution assets and properties owned by National Grid in downstate New York. Not National Grid as a whole which would be a different answer I would think. Yes or no?

A. (Mr. Duah) The -- the panel believes that in order to determine or recommend the return on common equity for the two companies KEDLI and KEDNY, the appropriate proxy group should base on the electric proxy group because investors do not invest directly into the company KEDLI and KEDNY, they're under a parent company.

And we also say on page 54 of our testimony that this approach is also consistent of

what staff did in the Niagara Mohawk case which is a sister company.

Q. Okay. Its correct isn't it that Value Line is a source of a significant valid data in your -- in -- in your forecast analysis?

MR. GOODRICH: Objection, please define significant.

BY MR. MALONEY: (Cont'g.)

Q. Does the panel use Value Line information in -- did the panel use Value Line information in determining the cost of equity for the companies of this proceeding?

A. Yes, we did use.

Q. Are you aware the Value Line considers natural gas utilities and electric utilities to be separate types of entities for investment purposes?

A. I'm not sure if you are referring to National Grid -- National Grid P.L.C. or you're referring to which one -- are you refer -- the ultimate parent?

Q. I'm referring to the categories natural gas utilities and electric utilities. Are those separate categories within Value Line?

A. Value Line categorization for utilities that electric consist of companies that are -- combination electric and gas and also companies that are also electric. And then for the gas is a separate categorization.

Q. Could you turn to the document that's been marked for identification as -- I'd great if I can read my on handwriting, 652?

A. Okay. Got it.

Q. Is it correct that this --

A.L.J. LEARY: Excuse me, you need to lay a foundation about whether this panel has ever seen this document.

BY MR. MALONEY: (Cont'g.)

Q. Has the panel ever seen this document?

A. I'm generally familiar with it -- familiar.

Q. You are familiar with it?

A. Yeah.

Q. Yes, thank you. And -- and in fact, in this case you performed a back cast of certain data that was contained in this document, isn't it -- in examining the relationship between

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X.I.R.R. and I.R.R functions in excel spread sheet. Is that correct?

MR. FORST: Objection, Your Honor. I don't think that the counsel has laid a proper foundation for this document, while they've considered or ask the panel whether they're familiar with this document. I don't believe that they've certified that this is a true and accurate copy of the documents that's been provided.

A.L.J. LEARY: I'm also not hearing a reference to anything in their testimony indicating that they actually use this, but do you have a reference that says that they used it for -- I think you've said X.I.R.R. and --

MR. MALONEY: Well, that's what I just asked was did -- did you, in fact, perform an analysis?

A.L.J. LEARY: No, you did not ask that. What you asked was slightly different and you can ask that question, but you asked them -- you jump over that question so you -- you assumed in your question that they had used this.

So what I'm asking consistent with -- I think the objection here is for you to lay the

foundation that they actually used it. Generally familiar is one thing, use of this document in the preparation of their testimony is another. So let's go there first before you get into this document.

BY MR. MALONEY: (Cont'g.)

Q. Did you -- did you use information that was part of the equity consensus document in your testimony -- in preparing your testimony?

A. Yes, if you go to page 99 of our testimony --

A.L.J. LEARY: Would you repeat the page again?

THE WITNESS: Page 99 nine, nine.

A.L.J. LEARY: Thanks.

THE WITNESS: The question was, panel what is your opinion about whether the general finance case either they relied on the I --- I.R.R. which is internal rate of return or X.I.R.R. formula.

BY MR. MALONEY: (Cont'g.)

Q. And -- and you refer there to the fact that you reviewed and recast the D.C.F. results for both the 34 member electric and 11 member gas proxy groups. Is that correct?

A. Yes, that is line -- line 11 on page 99 of our testimony.

Q. Now, if I could refer you to the document on page 5. There's a discussion of those proxy groups. Is that correct?

MR. GOODRICH: Still -- still objection though because counsel still has not adduced whether or not this is a true and accurate copy and -- and given of -- of the document it purports to be and given that he has provided this to -- to the panel today and the panel doesn't know where the document came from, I don't see how the -- the panel could adduce without reading every word and comparing it to their own copy somewhere that it is a true and accurate copy.

MR. MALONEY: This -- this is a public document from the commission website that we obtain.

A.L.J. LEARY: I -- I agree. I mean this is in case 91-M-0509. I think the testimony, Mr. Goodrich, is that they're generally familiar when ask whether they used it, they referred to the discussion on page 99 of their testimony.

So is your point that they -- in -- in response to this question I'm not exactly sure what your point is.

MR. GOODRICH: Counsel is seeking to have a particular copy of this document entered as an exhibit and counsel hasn't demonstrated that this particular copy is true and accurate.

A.L.J. LEARY: It's a publicly available document on the department's website. Now I would say we -- we said at the beginning of the hearing that we would make things like this subject to check.

I -- I don't think that the company has offered a document that would be in any way incomplete. Is that right, Mr. Maloney?

MR. MALONEY: That's correct. This is the document we were able to obtain with -- the company would be willing to stipulate that whatever the correct -- if this is not the correct version of this document, the commission's records, whatever that document is we would -- we would stipulate to allow you at the record.

A.L.J. LEARY: Well, that doesn't help me. What will help me is where you got this document and when you got it?

MR. MALONEY: We -- we -- we took it from the commission's website as I understand it.

A.L.J. LEARY: As you understand it?

MR. MALONEY: Yeah.

A.L.J. LEARY: All right. So let's -- we're not going to -- I think you're objection is about admission of this document if that is the proper interpretation of your objection and what I would say is that we can save those objections when we deal with all of the hearing exhibits.

Generally familiar and this panel's testimony so far appears that they can at least testify about this document whether it's going to be actually admitted in the record is a separate issue. So why don't we punt on that today and revisit it, and let's allow Mr. Maloney and the panel to move forward with the testimony.

BY MR. MALONEY: (Cont'g.)

Q. Panel, I'm referring you to page 5 through 7 of this document, and if you'd take a

moment to review that. And let me know when you have completed that review.

A.L.J. LEARY: I just want to identify for the record this has been pre-marked as hearing Exhibit 652. It's subject to later admission and it's dated June 7th, 1993 in case 91-M-0509. It is captioned a return on equity consensus document in a proceeding on motion of the commission to consider financial regulatory policies for New York State Utilities.

THE WITNESS: Okay.

BY MR. MALONEY: (Cont'g.)

Q. As -- as a general matter, you're familiar with what's known as the generic financing case?

A. Generally, we are familiar, but not into detail because we are not part of the proceedings.

Q. But you believe that the methodology that you followed in this case follows what's known as the generic financing method. Is that --

A. Yeah, it has -- it's rooted in the -- the general finance case, yes.

Q. Would you agree that the section that I just ask you to read reflects agreement among the -- the parties who signed this document that natural gas company -- that the proxy group used to determine the cost of equity for natural gas companies would be to -- determined using a -- a -- a group of comparable gas companies?

A. Can you point me to the specific paragraph you're reading through?

Q. Sure. I'm looking at the first paragraph beginning on -- on 5 and carrying over to 6. The D.C.F. and CAP-F -- CAP-M calculations will be performed on proxy groups of electric and gas companies.

The electric company proxy group will be used to determine the cost of equity for the 7 electric and gas combination utilities. The gas proxy group will be used for the two gas only utilities.

A. And the question -- the question again?

Q. Would you agree that this reflects agreement that a gas proxy group would be

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used to determine the cost of equity for gas distribution companies?

A. This is what it says.

Q. Okay. And if -- if -- is it also correct that there's no suggestion in the pages that I asked you to read that a group of electric or combination gas and electric utilities should be used to establish the R.O.E. for the two gas distribution companies in New York at that time?

A. Can you point me to the exact term or the specific paragraph you're reading from?

Q. I think you'd have to read the whole thing but I guess I -- I would refer to page 7 and it says, if at any time -- if at the time of any calculation of return on equity there will be applied to a gas utility, fewer than 10 companies meet all 3 criteria, i.e., percentage of revenues derived from gas operations will be relaxed to pick up one or more companies that have 96% or less of their total annual revenues derived from gas utility operations.

So as to bring the number of proxy group companies up to 10. If it's a result of doing this the simple average percentage of revenues from gas utility operation of the entire proxy group drops

below 95, the parties will meet to discuss whether it is appropriate to make further adjustment to the proxy group selection criteria or if it is determined that an acceptable proxy group have at least 10 pure play gas utilities cannot be constituted whether the cost of equity result should be adjusted to account for the impact on the cost of equity of investment to the unregulated activities, and if so how much -- how such an adjustment would be made.

There is no suggestion in there that gas and electric company -- electric companies would be substituted for the gas companies, is there?

A. We -- as stated in the testimony on page 80, we will have included the gas companies, but in our proxy group, but we had really few companies that meet the selection criteria. We will have used their gas companies, but only five of them that meet staff criteria and then we believe it's an insufficient size to produce any reasonable results.

Q. But going back to my question, it’s fair to say that at least at the time this document was executed, it appears that gas companies were intended to be used to establish the -- a proxy group of gas companies was intended to be used to

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establish the cost of equity for gas distribution companies. Is that correct?

A. Yeah, that is correct. But in our case you don't have enough gas companies to run the analysis to produce a reasonable result.

Q. Now, I think we established before that you used a -- a group of 28 electric or combination gas and electric utilities to -- as a proxy group to derive your recommendation of the appropriate return on equity for companies. Is that correct?

A. Correct.

Q. And if you could look at the response to N.G.-67 which has been marked for identification as Exhibit 651. N.G.-67, 651 is the exhibit number.

A. Ready.

Q. Is it correct that 12 of the 28 proxy group companies have no gas operations?

A. Yes, 12 companies we agree with you, but our proxy group has companies that have more vertical integrate companies and those companies have more risk or they're riskier than the gas companies.

So our proxy group you always see on that basis is riskier than the pure gas companies.

Q. Is it also correct that none of the companies in your proxy group derive a majority of their revenue from gas operations?

A. There is -- is about 50 -- 58% of its revenue comes in from gas operation.

Q. So there is one -- one --

A. Yes, yes one over here and again, our proxy group of 28 companies, there are more vertically integrated companies in that proxy group and those vertical integrate company have more risk than gas operation. So our proxy group is more riskier than pure -- pure gas companies.

Q. But just so I'm reading this document correct. The -- the average of gas revenue, the 13% which reflected on page 2 of this document, that represents the average of all 28 companies?

A. Yes.

Q. Thank you. Let me ask you about the -- of your opinion about the future prospects for gas and electric companies. I -- I think we have heard testimony, we can all agree that the recent governmental responses to climate change is something

that's relatively new and that there is uncertainty surrounding the ultimate steps that might be taken to address climate change. Is that -- is that correct?

MR. FORST: Objection. Could counsel point to the specific testimony that we've heard that he is referring to?

A.L.J. LEARY: Is there such a testimony Mr. Maloney, or are you asking more of a general question?

MR. MALONEY: I'm asking more of a general question.

A.L.J. LEARY: And I think that's acceptable to the extent that the finance panel can respond about climate change, which is not a subject of their testimony as I read it.

Let's -- let's give Mr. Maloney some latitude on this. The panel can answer.

THE WITNESS: Can you repeat the question?

BY MR. MALONEY: (Cont'g.)

Q. Well, let -- let me -- I'll ask it this way. Would the panel agree that the potential for the electrification of transportation

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A. We respond to that in I.R. response N.G.-59, so I'll point you to that.

A.L.J. LEARY: This is not among your exhibits, I don't think.

MR. MALONEY: No, it's not.

THE WITNESS: Is a -- is a part --

MR. MALONEY: I mean, I'm looking to hear the answer, but no, it's not --

THE WITNESS: Yeah.

MR. MALONEY: -- among the exhibits.

THE WITNESS: For the question in the I.R. was that the Staff Finance Panel have an opinion as to whether climate change such as that passed recently in New York is likely to require gas utility such as KEDLY and KEDNY to undergo business transformation in foreseeable future.

If not, please explain why not. If so, explain whether the Staff Finance Panel believe such information will create additional risk for gas utilities that is not captured in proxy group that are used by the panel to determine the recommended R.O.E. in these proceedings.

And our response was, we believe that the New York State Climate Leadership and Community Protection Act and thereby the Climate Action Council will discuss various topics and engage in various opportunities and challenges, which may or may not impact electric and gas utilities. And which may or may not require certain business transformation.

So we believe that is, is pretty much here to suggest that it may have an impact.

BY MR. MALONEY: (Cont'g.)

Q. I think that that is the answer that you gave to that discovery response. But the -- the question I asked was, whether the electrification of the transportation of -- the electrification of transportation holds the potential for significant growth for the electric industry. Do you believe that's true or not?

A. I think it's -- it's too early for me to or for the panel to speculate on that because there are many things as we just mentioned, of may or may not.

Q. And would your answer be the same as -- as it applies to the electrification of heat?

A. Yes.

Q. So it's fair to say that -- well, let me ask you. In -- in preparing your -- in your discounted cash flow analysis that's used to, in part to establish your recommended cost of equity, it's correct that you consider dividend growth over a period of perpetuity or at least the next 100 years and feel free to tell me what the data is that you actually use.

A. Actually the projection for the dividend group, it go up to 200 years.

Q. 200 years.

A. Yes.

Q. So it's fair to say that in -- in making that projection you -- you haven't given any thought to whether the electric industry might be subject to significant growth during that period as a result of -- of responses to climate change such as increased electrification of heat or increased electrification of transportation?

MR. FORST: Objection. I think that the panel already stated that they've considered it, but they find that the results to be too speculative. I think this has been asked and answered.

A.L.J. LEARY: Slightly different question, but let's hear from the panel.

THE WITNESS: Can you repeat the question?

BY MR. MALONEY: (Cont'g.)

Q. I don't know if I can.

A.L.J. LEARY: It was, did the panel consider growth over -- growth of the gas industry over 200 years?

MR. MALONEY: The -- the growth of the electric industry --

A.L.J. LEARY: Electric industry.

MR. MALONEY: -- as a result of the electrification of heat.

BY MR. MALONEY: (Cont'g.)

Q. Is it -- is it fair to say that you didn't take that into account, the potential for growth of electrification of heat and of transportation in contemplating the growth of the industry over the next 200 years?

A. And I'll point you -- the panel would point you to page 103, line 8 to line 11 of our testimony. There is -- we said that the D.C.F. model processes as its top prices reflect collective

judgment of all stock market participants and, therefore, all known factors influence in the stock. So we believe that is what investors may consider that.

Q. So basically whatever is built into the stock price today that reflects the -- I mean, the judgment of investor is reflected in the stock price today. Is that -- that's your testimony?

A. Yes, that's what we're saying on page 108 of our testimony.

Q. And -- but let me ask you just in terms of the natural gas business does the panel have a opinion as to whether the prospects for growth of the natural gas business are as great as the prospects for growth of the electric industry?

MR. GOODRICH: I would say, objection, vague. I mean, the prospects for growth over what time period, 100 years --

MR. MALONEY: I think we've been establishing 200 years is the -- the basis of what we've used.

A.L.J. LEARY: Panel can answer.

THE WITNESS: Okay. As we said earlier, investors do consider all available

information and price on a stock. So as long as these things out there we believe stock prices reflect such issue that you're talking about.

BY MR. MALONEY: (Cont'g)

Q. I'd like you to refer now to your response to N.G.-62, which has been marked for identification as Exhibit 647.

A.L.J. COSTELLO: Did you say 6 --

MR. MALONEY: 47 I believe, yes.

BY MR. MALONEY: (Cont'g.)

Q. Do you have that before you?

A. (Duah) Yes, I'm ready.

Q. Now, in attachment 2 to that, is it correct that you utilized R.R.A. Regulatory Focus major rate case decisions January to December 2018, to support a conclusion that in the last 2 years of the average returns on equity for gas and electric utilities are virtually identical, indicating an overall convergence of risk?

MR. FORST: Can counsel point to a specific place in the attachment they're referring to?

MR. MALONEY: I'm -- I'm pointing to the response to the discovery request N.G.62 in the

second paragraph. I believe, it's maybe 4 sentences down and then I'm also referring to attachment 2, which is the paper that we're discussing.

MR. FORST: Thank you.

THE WITNESS: Yes, I believe there's a convergence.

BY MR. MALONEY: (Cont'g.)

Q. Okay. And was that testimony or was that response, that portion of your response based on consideration of the -- and I'm referring to Attachment 2 now and on the right-hand side of that there is a list of various returns under various circumstances.

And I think it's fair to say if you look at that, you'll find that for electric it says fully litigated cases 2018, 9.61 for gas fully litigated cases, 9.59, is that correct?

A. It is what the document says.

Q. That's what you were -- that's what you were relying on? And you -- and you consider this to be an authoritative source for this information?

MR. FORST: Objection. There's two questions there and I'd ask that the panel be allowed to answer them individually.

MR. MALONEY: I apologize. That's correct.

BY MR. MALONEY: (Cont'g.)

Q. So I think my first question was that is what you relied upon, is that correct?

A. Yes, we relied on page 7 of 16 of Attachment 2. What is the last line where it says 9.59 for electric or the average R.O.E., and then on the other 9.59 which is for gas utilities, the average.

Q. I'm sorry, were you looking at page 7?

A. Yeah, Page 7 of --

Q. Okay.

A. -- Attachment 2.

Q. Okay. Okay. Now, if I could refer you to page 3 of this document. There's a 3 paragraph summary that begins, capital structure trends, is that correct?

A. Yes.

Q. And it suggests that for 2018, the ac -- the average authorized equity ratio -- I'm reading from the second paragraph, for electric utilities was 50.53 during 2000 -- for cases decided during 2018. And for gas utilities, it was 51.47, is that correct?

A. Yes, that's what it says over here.

Q. So would the combination of an equivalent return on equity, and a higher equity ratio for gas utilities, read you -- lead you to believe that the decisions indicate that gas utilities are somewhat riskier than electric utilities?

A. Can you repeat the question?

Q. Would you agree that the combination of equivalent rates of return on equity, but is somewhat greater equity ratio for the natural gas utilities indicates that the natural gas utilities were viewed as somewhat riskier during 2018 in terms of the authorized returns that were set?

A. I believe so and you also said in --.

Q. Sir, I couldn't understand the first thing you said.

A. I believe so.

Q. Thank you.

A. And you also said in response to N.G.62 where we measure the electric and gas utilities risk issues. In the last -- on page 2, we say in the past couple of years, staff has observed investors sharpen their focus on the overall impact of decarbonization as to wind industry. So it would be surprising if this convergence in perceived risk was attributable to the diminishment of a key advantage that a gas utilities has had for some time relative to its electric counterpart. So in the past, the gas was less riskier, than gas use on safety going on.

Q. Now, you attach to this response the R.R.A. Regulatory Focus, do you have access to that database or their publications?

A. (Qadir) Yes.

Q. And so you review them from time to time?

A. Yes.

Q. And you consider the data that they provide reliable?

A. (Duah) The panel has no reason to doubt the accuracy or the reliability of that -- of the data from R.R.A.

Q. I'd like -- I'd like to refer now to Exhibit 653 which is an updated version of the same information that you provided in Exhibit N.G.62.

A.L.J. LEARY: Let's -- let the panel characterize the document, recognize it first, rather than you Mr. Maloney just for the record, so Hearing Exhibit 653.

MR. MALONEY: 653. Yes, Your Honor. Thank you.

THE WITNESS: Ready.

MR. MALONEY: Your Honor had asked the panel, they say they're ready.

A.L.J. LEARY: Well, I -- I want you to lay the foundation of asking the panel members if they've seen this document and so forth.

BY MR. MALONEY: (Cont'g.)

Q. Well, have you seen this document before?

A.L.J. LEARY: Or are they familiar with it and so forth.

THE WITNESS: Yes, the panel is familiar of it.

BY MR. MALONEY: (Cont'g.)

Q. Okay. Is it -- is it correct to say that for 2019 and I'm looking at page 1 of the document. That for fully litigated cases, the gas average return on equity was 9.74 percent, whereas for fully litigated cases on the electric side, the return on equity was 9.58 percent.

A. Yeah, this is what it says and the panel --.

Q. And -- and -- and would you agree that in terms of comparing authorized returns fully litigated cases is the one to look at as opposed to settled cases, which can reflect various agreements among parties?

A. Yes, we look at the litigated case, but as we said earlier, each state regulatory jurisdiction, each of them has their own approach in setting the return on common equity and investors are aware of the commission's approach in setting the return on common equity. In fact, Bank of America

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equity analysts and Moody's says that they are even expecting 8.8 in equity in this case which isn't a surprise to them, something the -- they were expecting below 9%, and now this latest report also suggests the others were aware of the commission's well-established formulaic approach and the split are already below 9%.

Q. And so you're saying that investors are expecting 8.8 percent and you're recommending 8.2 percent, is that correct?

A. No, what I'm -- in that case, it was a settled case, negotiated case. So what we are recommending now is a one-year case, so it's different.

Q. Just one more thing on the document that's been marked Exhibit 653. If I go to page 3, again, there's a -- a section that begins capital structure trends.

A. Yes.

Q. And it says that the second sentence of that or the second to last sentence, I'm sorry, that the average authorized equity ratio for electric utility cases nationwide was 4.94% in 2019. Whereas in the next sentence it says the average

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas allowed equity ratio for gas utilities nationwide was 5 -- 51.75%, is that correct?

A.L.J. LEARY: Let's finish the sentence, please, that you're quoting. In 2019 I think it says, is that right?

MR. MALONEY: In 2019.

A.L.J. LEARY: And then 50.09.

MR. MALONEY: I'm sorry, I may have read that wrong, Your Honor. I -- I did read it wrong. Let me try it again.

A.L.J. LEARY: Okay.

BY MR. MALONEY: (Cont'g.)

Q. The average authorized equity ratio for electric utility cases nationwide was 49.94% in 2019, 49.02% in 2018 And 48.9% in 2017. The average allowed equity ratio for gas utilities nationwide was 51.75% in 2019, 50.09% in 2018, and 49.88% in 2017. Is that correct?

A. Yes, that's what it says over here.

Q. So would you agree that to the extent that the average gas case reflected an R.O.E. of 9.74% and an average equity ratio of 52.76%, that and -- and the average electric litigated case

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reflected an R.O.E. of 9.58% and an average equity ratio of 51.55%.

The conclusion on average was that gas utilities were riskier and required higher returns than electric utilities?

A. I think we did answer that one earlier.

Q. And this is just an update of that for this further information.

A. Can you repeat your question, please?

Q. Okay. I need to re-find the exhibit to do that. Would you agree that the fact that the -- in 2019 the average fully litigated rate case resulted in an R.O.E. of 9.74% and an average equity ratio of 51.75% for gas companies, while at the same time for electric companies, the average R.O.E. was 9.58% and the average authorized equity ratio was 49.94%.

Would you believe that those 2 -- the comparison of those 2 data points indicates that gas utilities were judged to be riskier than electric utilities by regulators on average in 2019?

A. As we said earlier, on that basis we see that the -- if we're putting the I.R. response 62 which is page 2, then now we see that the gas utilities looks like, become riskier compared to the electric is what we said on page 2. And if you look at a better trend you see the electric is around a little bit -- a little bit lower than the gas companies.

Q. Okay. Now, could I refer you to your response to N.G.64, which has been marked for identification as Exhibit 649?

A. And your question, please?

Q. So you're --- okay. You have the document in front of you. Is it correct that what you provided here was that you added 5 natural gas companies to your proxy group and recalculated your discounted cash flow and capital asset pricing model results using those 5 additional companies? 5 additional gas companies, I should say.

A. Can you repeat it?

Q. Is it correct that what you did here was to recalculate your discounted cash flow results and capital asset price -- pricing mechanism results by adding 5 additional gas companies to your

proxy group and those companies were Atmos Energy, NiSource Inc., Northwest Natural Holding, ONE Gas, Inc. and Spire Inc. Is that correct?

A. Yes.

Q. Okay. And it's your conclusion that the addition of those five companies does not materially change your cost of equity recommendation, is that correct?

A. At the time of our testimony, we related that there is no difference -- material difference between the pure play -- between the 28 electric companies which we rely on their testimony. This is adding extra augmenting a proxy group of 5 more pure play gas companies, you didn't see any material difference. When you round them up, there came to 8. -- 8.2, but since then we have seen some variance between the 28 companies and it's 33 companies, including the 5 pure play gas companies. It has been around -- the variance is about around 15 to 20 basis points, our latest analysis is what it shows.

Q. Yes. So just to confirm. If I -- if I were to look at page 4 of the attachment, and I were to take the --

A.L.J. LEARY: Attachment 1 or Attachment 2, or am I not --

MR. MALONEY: I believe --

A.L.J. LEARY: -- on it --

MR. MALONEY: -- there's only one attachment that we're in -- we're in N.G.64, Your Honor.

A.L.J. LEARY: And that's been marked --

MR. MALONEY: 649, I'm sorry.

A.L.J. LEARY: -- as Exhibit 649.

MR. MALONEY: Right.

A.L.J. LEARY: Okay, just make sure the record is clear on what you're looking at and asking the panel about. Thanks.

BY MR. MALONEY: (Cont'g.)

Q. So now, if -- I'm looking at the last column on page 4, which says long form R.O.E. Would you accept, subject to just checking the math, that for those 5 companies, the average D.C.F. result is 9.074% and the medium D.C.F. result is 8.46%?

A. I haven't checked that math. I have not checked it.

Q. How could you check it?

MR. GOODRICH: He said I've not checked.

BY MR. MALONEY: (Cont'g.)

Q. You have not checked?

A. No, I have not.

Q. But would you accept subject to check, that if I take those 5 numbers and divide -- and add them up and divide them by 5, I'm going to get 9.074% with the median and the median is the one in the middle, which is 8.46%.

MR. GOODRICH: May I ask? I -- I -- subject to check becomes a bit of an issue.

A.L.J. COSTELLO: It does become, and we asked that that not be done. So it looks like he's calculating it now.

MR. MALONEY: It's -- it's a 5 number calculation, so it's -- I understand subject to check is problematic.

A.L.J. COSTELLO: We'll –we'll can I ask you. Yeah, can I ask you a question too?

MR. MALONEY: Yes.

A.L.J. COSTELLO: Do you have -- do you have a sense of how much more you have with your cross because after you get a response to your

question, we may take a break. If this is a good part -- point to take a break.

MR. MALONEY: Yeah.

A.L.J. LEARY: If the panel needs more time to do the math, we can do this now. Because is this something that you can do quickly, panel?

A.L.J. COSTELLO: If we take --

MR. MALONEY: I probably have another hour of cross --

A.L.J. LEARY: Yeah.

MR. MALONEY: -- taking a break right now would be fine.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: So do you want them, you know, --

BY MR. MALONEY: (Cont'g.)

Q. We can -- I -- I'd ask you to check on, one is that the median data of your calculation is .65 and then the other numbers that I outlined for you which is the average is 9.07, 9.074 and the median is 8.46%.

A. Would it be then the long form you're looking for those numbers is what you --

Q. That's for the 5 companies, the 5 gas companies.

A.L.J. LEARY: And which five gas companies --

MR. MALONEY: The 5 that are --

A.L.J. LEARY: -- you have to say.

MR. MALONEY: If you look at Exhibit 4, if you look -- I'm sorry, Attachment 1.

A.L.J. LEARY: Oh, I see the last highlighted.

MR. MALONEY: Yes, yes. They are highlighted.

A.L.J. LEARY: Blue.

MR. MALONEY: Yes.

A.L.J. LEARY: Okay. Again, this is really important. Just so the record is clear when you're talking about this exhibit, there's is a lot of companies listed here. I do see those 5 companies highlighted or grayed, if you will.

MR. MALONEY: Yeah.

A.L.J. LEARY: Just make sure that you're clearly identifying for the record what you're referring to.

MR. MALONEY: I think before I did identify each company. Yes, Your Honor.

A.L.J. LEARY: Oh, you did, okay.

MR. MALONEY: Yes.

A.L.J. LEARY: I missed that. About 10 minutes -- 10, you want 15?

A.L.J. COSTELLO: If they're going to be doing math and they want to get a calculator --

MR. FORST: We'll take the 15 minutes, Your Honor.

A.L.J. LEARY: 15 it is, okay. See you in 15.

(Off the record; 10:23 a.m.)

(On the record 10:47 a.m.)

A.L.J. COSTELLO: You may proceed.

BY MR. MALONEY: (Cont'g.)

Q. I believe when we went off the record, I had asked the panel some questions for some math calculations. Does the panel have that information?

A. (Qadir) Yes.

Q. Could you explain what the average D.C.F. result is and -- and the medium, beta and medium D.C.F. result?

A. (Duah) So the average for the 5 companies is 9.07%. And median is 8.46%. And the beta for the 5 companies -- gas companies is the average is .62 and median is .65.

Q. Thank you. Now, I take it that in arriving at your -- your return on equity recommendation, you conducted an analysis that relied primarily on a discounted cash flow methodology. Is that correct?

A. Yes.

Q. And it's correct that you employed at what they referred to as a 2 stage D.C.F. or 2 stage discounted cash flow, I'm going to try to avoid acronyms entirely?

A. Yes.

Q. And the second stage -- I'm sorry. And is it correct that in the first stage of your calculation you rely on Value Line dividend growth projections?

A. Yes.

Q. And when I say that, I mean, you -- you take the data that's provided by Value Line and simply use it in your calculations, you don't

make an independent evaluation or modify it in any way?

A. We rely on the Value Line 3 to 5 year short-term growth rate for our short-term growth rate. And then we use sustainable growth rate to project the dividends through for the fifth to the 200 years.

Q. Okay. And -- and which you just referred to the -- the second stage of the discounted cash flow analysis has a greater impact on the overall result than the first stage, is that correct? And by that I mean, it's a larger part of the growth number that falls out of it?

A. We did not run analysis to determine the impact of the sustainable growth rate on overall, but we believe it is only combined this what we called the first stage growth rate, and this is normal growth rate, it produces reasonable results, which is the commission's approach for the past, about 25 years or something like that.

Q. So -- so you don't know which was larger, but it is essentially the sum of one -- the one and the second one, the -- the -- the dividend

growth rate for 5 years and then the sustainable growth calculation for the next 195 years, I guess.

A. Yeah. The dividend, the sustainable growth rate went from the 5th year to 200 years.

Q. Thank you. And is it correct that in order to calculate sustainable growth in the second stage, you need an estimate of the long-term return on equity that will be earned by each member of the proxy group?

A. Can you repeat the question?

Q. Is it correct that in order to calculate the sustainable growth rate, you need an estimate of the long-term return on equity that will be earned by each member of the proxy group?

A. As we saw in S.F.P.17 page 2 the second column or third column content from the left. We see the return of commodity 2023 is used as one of the input for the internal growth rate, which is captured in the fourth column from the left and is also the internal growth in one combined with the external growth rate, which is in the heading as M.B.R.-1 that we get the sustainable growth as a

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combination of the internal growth rate and the external growth rate.

Q. Okay. The return on equity in column 2 that you referred to, is it correct that the source of that data is Value Line?

A. That is from -- yes, from Value Line.

Q. Yes. And -- and so each estimate in that column represents an individual determination of the R.O.E., the return on equity -- the expected return on equity for each company in the proxy group, right, into the individual estimate for each company?

A.L.J. LEARY: I'm sorry, are you referring to S.F.P.17?

MR. MALONEY: Yes, I'm sorry, Your Honor. S.F.P.17 has been marked, I believe, as Exhibit 440. And I'm referring to column 2.

A.L.J. LEARY: Thank you.

A. (Duah) Please, can you rephrase your question?

BY MR. MALONEY: (Cont'g.)

Q. I -- yeah, I guess. If I refer to Column 2, what that contains is an individual return on equity estimate for each company in the

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A. Yes.

Q. Could you refer to the exhibit that's been marked for identification as Exhibit 654?

A. Yeah.

Q. Now, you are – you -- I don't know if you utilize this specific value. Now, this -- this Value Line information is from April to June of 2019. And I believe you stated in your direct testimony that your analysis in this case was based on information developed between April and June of 2019. Is that correct?

MR. FORST: Objection, Your Honors. I don't believe Counsel's laid a foundation for this document.

MR. MALONEY: I'm not asking about the document --

MR. FORST: Okay.

MR. MALONEY: -- I'm asking about the Value Line information that they used.

A.L.J. LEARY: I think -- I think that's right.

THE WITNESS: Can you rephrase your question, please?

BY MR. MALONEY: (Cont'g.)

Q. Okay. Is it -- is it correct -- this -- this document purports to contain Value Line information from April to June of 2019. And it's my understanding from your testimony that the Value Line information that you used in developing your recommendation was derived from Value Line during the period, April to June of 2019. Is that correct? I could probably find a reference to that, but it might take a while.

A. Yeah, I'm checking my testimony to see that -- let me check. Page 57.

MR. FORST: If I may, Your Honors, just to speed time here. It's Page 65 of the panel's testimony.

MR. MALONEY: Thank you, Mr. Forst.

THE WITNESS: Yes, it's from April to June that is from page 65 of our testimony.

BY MR. MALONEY: (Cont'g.)

Q. Now, do you have access to electronic Value Line database, or do you rely on these individual documents to obtain your data?

A. Both.

Q. Okay. So would you confirm for me -- and I apologize that these are not in perfect alphabetical order, that this document contains the Value Line information for each of the 28 members of your proxy group. Does that appear to be what it is?

A. I tried to count, and Con Ed --

Q. There are definitely 28 pages and I believe they're all there although I know Con Ed is out of order and so is Ameren, but I don't know how that happened. It's because I'm incompetent at most tasks like this.

A.L.J. COSTELLO: How many pages did you say?

MR. MALONEY: 28. There are 28 pages under the cover sheet.

A.L.J. LEARY: Thanks.

THE WITNESS: (Qadir) I think you said that these are 3 months, so it should be like 28 times 3 then?

BY MR. MALONEY: (Cont'g.)

Q. No, it's just -- it's whatever month that -- that was available for each one, but it's either from April --

A. Okay.

Q. -- May or June, depending on the company.

A. The count.

Q. Sure.

A.L.J. LEARY: I just -- I just want to clarify the cover page. So it says Value Line information for staff proxy group April 2019 to June 2019. Did this document -- was this document created by the D.P.S. Staff Finance Panel?

MR. MALONEY: No, my -- my understand -- this is just simply data from Value Line and the -- the dates of each individual report are listed in the bottom right-hand corner of each report. And so some are from April, some are from May and some are from June.

Now, my understanding of that is Value Line doesn't do every company every month. And so over the 3-month period that creates the data.

BY MR. MALONEY: (Cont'g.)

Q. Is that -- is that correct, panel?

A. (Qadir) Yes, so I counted. It's 28.

Q. Yes. And I just want to understand from these -- looking at these documents, the data that you use to derive the return on equity, and some other inputs into what has been marked as Exhibit 440.

So with respect to the return on equity for 2023 if I were to look at the first page and it's Ameren Company. Could you tell me where you get the number that then shows up as 10.69 -- 10.69 for Ameren?

A.L.J. LEARY: You're going to have to help me out here, Mr. Maloney. I'm looking at page one of S.F.P. 17 marked as Exhibit 440.

MR. MALONEY: And if you're looking at --

A.L.J. LEARY: I'm looking opposite Ameren and you just said 10 point something.

MR. MALONEY: Yeah, if you look at page 2 of 2.

A.L.J. LEARY: Okay. There you go, so.

MR. MALONEY: And there in the second column is 10.69.

A.L.J. LEARY: Okay. Just --

MR. MALONEY: And if -- if I done this competently we would have used Alead, but we didn't, so.

A.L.J. LEARY: Yeah, they're not appearing in this document as far as I can tell.

MR. MALONEY: They are -- Alead is in there, but unfortunately --

A.L.J. LEARY: It's somewhere else.

MR. MALONEY: -- it ended up third.

A.L.J. LEARY: Okay.

MR. MALONEY: Because like I said the brain doesn't work so well late at night.

THE WITNESS: (Duah) Can you rephrase your question?

BY MR. MALONEY: (Cont'g.)

Q. Yes. So from looking at column 2, and I'm looking at Ameren and I see a return on equity number of 10.69. And I'm looking at the Value Line sheet for Ameren. And I could see off to the -- I will say off to the right-hand side of the page, there is a Value Line set of data that appears to be for 2022 to 2024.

And -- and there's a number below that, that says all the way down at the bottom of the

column that says return on shareholder -- or return on S.H.R. equity return on C.O.M. equity. Are -- are those -- simply stated, are those the data that you used to derive the 10.69 for Ameren in your column 2 of what's been marked Exhibit 440?

A. That one is based upon the earnings per share which is in column page -- page 1 of 2 of S.F.P. 17, the earnings per share column which is 4 from the left divided by the average book -- book value per share which is also on page 1 of S.F.P. 17.

Q. Okay. So you used the earnings per share and the book value per share to derive your individual R.O.E. for 23. And that number I think you would agree is close to what Value Line reports but not completely the same because you do it somewhat differently, is that?

A. Yeah, this is how we came out. We came out with that formula, yeah, earnings per share and average book value per share, this is how we came. And yeah, coming with Ameren. Yeah, come in with 10.69 and they are also come in with 10.5.

Q. But -- but all this information comes from that far right-hand column on the Value

Line sheets that we're looking at. So for Ameren, on those -- those sheets right there. The far -- I'm sorry, the far right boxes, if you will?

A.L.J. LEARY: So just identify it for the record those far right boxes and what they are.

BY MR. MALONEY: (Cont'g.)

Q. Yeah. It -- it says under the target price range, if you keep looking down it says Value Line P.U.B. L.L.C. 22 to 24. And then there are a series of numbers there including the book value per share, the revenues per share, cash flow per share, earnings per share. And if I think what you told me you used the earnings per share and the book value per share, to derive the return on equity, is that correct?

A. Average but we didn't know how Value Line also developed the returning of common equity here. What is the year end -- what you call it, what is the year-end book value per share? What is the year-end book value per share, what do you use the average book value per share, and we use average book value per share.

Q. Right. So -- so you're relying on Value Line inputs, you're just not relying on the return on equity input per se, right?

A. That's what we rely on, yeah, five -- due to 5-year forecast.

Q. Right. And so when I look at this then, if I were to look at all of these column 2 numbers, if I were to look at each of these 28 sheets, I could re-calculate that. And I would end up at the bottom of page 2 on what's been marked as S.P. 440 with an average return on equity of 11.14% and a median of 10.64% for the proxy group, right?

A. That is what staff methodology produces.

Q. Yes. Now, how does the staff then calculate the -- what inputs the staff use, I'm sorry, to calculate the retention amounts that -- which I assume represents the portion of earnings that are retained as opposed to paid out as dividends, is that correct?

A. That's right. That is also how much earnings per share the company generates and how much is paid out as dividends. So if if -- if you go to page 1 of S.F.P. 7 we have earnings per share

which is third -- fourth column from left and then let's, I'm taking Alead, Alead has 4.25 earnings per share. And then it has a dividend forecast of 2.85 -- 85 --

Q. Correct.

A. -- which is between 2022 to 2024. So the earnings of 4.25 and the paid dividend of 2.85. So what is retained is 4.25 minus 2.85. So that's how we run the retention rate in column 2 from the left on page 2.

Q. And so then you assume that that percentage is essentially what's then retained for the next 195 years. That's the way the model runs out?

A. Yeah. I would say yeah. So here the retention rate will be used in Alead that is 32.97. And then you look at the total ratio that will be one line above that, I think it should come somewhere around that.

Q. And -- and finally, where do you derive the estimates of the issuance of stock above book value, which I guess is the S.V. term of your sustainable growth method, is that correct? I realized there was a question in a statement in

there. Let me -- let me ask you. This is -- the -- the S.V. term represents the accruals to dividends that will occur as a result of issuance of stock at prices above book value?

A. Correct.

Q. And where do you derive that information from on these Value Line pages?

A. That is coming from -- if you look at the second and the third column content from the right on page 1 of S.F. – S.F.P. 17. It is coming from the number of shares. So for instance, I'm using Xcel, which is last Xcel Energy. For 2019 Xcel Energy is forecasting 515.5 shares and then it's forecasting issuance of 525, which is an accretion of value, so this is how we derived the S.V. factor.

Q. And -- and so where I would get that? If I'm looking at the Xcel Energy page from the Value Line pages, I've handed out is Exhibit 654 it would be -- there's a -- there's a column in the box about 7 lines down to the far right that says, common shares outstanding. And is it the difference between the 517 and the 521.50?

A. Yes, it's coming from 2019. There's a box there which -- the Value Line described

as common shares as -- each as outstanding and then it has superscript D. So it is a common share. So we took the 2019 and then we also do the 2022 to 2024, 3- to 5-year forecast.

Q. Okay. Thank you. Does the panel believe that given the uncertainties surrounding the future of the electric business, that the commission needs to exercise caution in accepting a growth estimate for 195 years, that's based on data that really is derived from a Value Line forecast for 20 -- - calendar year, 22 to 23?

MR. FORST: Objection. That's a very vague question. What uncertainties is counsel referring to?

MR. MALONEY: Well, uncertainty is about growth over the next 195 years.

A.L.J. LEARY: Yeah, I'm wondering if you could break that question down into two pieces. Is that something that --

MR. MALONEY: Well, I can try.

A.L.J. LEARY: Yeah, try -- try to do that. Mr. Forst, does that -- can we just wait and hold your objection to see whether counsel is rephrasing this question in two pieces works for you?

MR. FORST: Yeah, I think we can afford them that opportunity. Thank you.

A.L.J. LEARY: Okay.

MR. MALONEY: Thank you, Mr. Forst.

BY MR. MALONEY: (Cont'g.)

Q. The -- the -- does the panel agree -- well, does the panel agrees, doesn't it, that you're basically using 1 year of data or change in data over a 1-year period as forecast by Value Line to forecast sustainable growth for the next 195 years. Is that correct?

A. As we -- as the panel pointed out to page 62 of our testimony, we did compare the -- or check the -- the reasonableness of sustainable growth rate when comparing it with the G.D.P. which is Gross --

A.L.J. LEARY: Domestic product.

THE WITNESS: Domestic growth rate, so we did compare that and that has been the commission's preferred approach by comparison of the growth rate to the G.D.P. So this is what we did, and we came up with 4.64 for sustainable growth rate. And the G.D.P. forecast also is 4.0%. So we believe ours was a little bit optimistic.

Q. That's pretty good answer. Referring -- referring to page 53 of your testimony -- your corrected testimony. Is it correct that the reason that you employ a 28 company proxy group is that you hope to eliminate biased or inaccurate estimates of growth? And I refer you to 53, lines 7 to 12, is it?

A. Can you rephrase your question?

Q. Yeah. Is it -- is it correct that the reason that you employ a proxy group of 28 companies, a relatively large proxy group, is that you hope to eliminate biased or inaccurate estimates of growth?

A. We believe that the 28 companies is an adequate sample size, and when you don't have adequate sample size, it introduces unreasonable results. So that is why we -- one reason why we rely on the electric proxy group.

Q. And --

A. And --

Q. I'm sorry, I didn't mean to interrupt you.

A. And we didn't have sufficient gas companies as we discussed earlier. The gas companies

are only 5 companies. So that was not enough to do that. So I think having sufficient sample size would produce reasonable results.

Q. When you say produces reasonable results, the hope is that it eliminates biases one way or the other, could be upward or downward or inaccurate results, inaccurate forecasts, is that correct?

A. Yes.

Q. Yes. At the same time, it's correct in selecting a number of inputs in your return on equity model, you rely on a single source of data, is that correct? And -- and specifically, I'll refer to the forecast of dividend growth, the source of data for sustainable growth, the source of data for beta in the capital asset pricing model, and the source of data for the market risk premium in the capital asset pricing model.

A. We rely on the Value Line data because we believe, and consistent with the commission's practice, the Value Line produces a very superior to any other source that we know of. Bloomberg has been issuing 2 years of their data and

the company used 10 years and 10 years has a problem to us, to the finance panel.

It can generate more data, but our analysis suggests that it can bias the D.C.F. protocol, the beta results, so we don't believe the 10 year is a good estimation period to amended data analysis. On the Merrill Lynch, also consistent with the commission's approach. In estimating the market risk premium Merrill Lynch uses dividend growth model, 3-stage D.C.F. to generate the market returns. What the company did was using the single stage D.C.F. Single stage D.C.F. and what do you call it, 3-stage D.C.F.. The 3-stage D.C.F. is more superior because it has 3 different growth rates. So that is more superior to estimate the market return which Merrill Lynch uses.

So we believe the market risk premium that we rely on based upon Merrill Lynch is superior because it’s based upon 3-stage D.C.F. The company also used single stage D.C.F. to estimate -- to estimate the market risk premium which, by the way, the company used any growth rate to estimate the -- what we called it, D.C.F. resource and the market

return. So we don't believe that is a reasonable approach.

Q. Well, I understand that you have reasons for selecting each of the publications that you -- or data inputs that you use. My question, though, is a little different. Why -- when -- is -- when the company -- clearly the panel is concerned, correct, that in selecting a proxy group, you need a significant number of inputs to eliminate the potential for bias or inaccurate forecast, is that correct?

And my question simply goes to -- that go to sort of the ah-ha question. Why isn't the same thing true when you're selecting the inputs for the beta calculation, the market risk premium, the sustainable growth methodology, the short-term growth methodology to be used in the discounted cash flow?

A. This is what the panel discussed earlier. That you have to look at which one produces which of these sources produces reliable results. We talk about the beta source. I believe the Merrill Lynch is more superior to using a single stage D.C.F. just what a company is using, the company is using single stage D.C.F. to calculate the market return.

And then we look at the rebuttal testimony. The company used sources from Yahoo Finance, and I think 3 different sources, all of those sources the company rely on dividend yield and growth rate and growth rate was based upon earnings per share. I don't believe that earnings per share is a proxy for dividend per share and growth rate.

Q. But going back to your selections, is it the panel's testimony that there's no potential for bias or inaccurate results in the various inputs that you used that I identified before?

A. The panel does not believe so.

Q. Let me ask you a few questions about your capital asset pricing analysis. Capital asset pricing mechanism analysis. Now, you stated before that you rely on Value Line betas, is that correct?

A. Yes.

Q. And is it correct that the beta of a stock measures the volatility of the stock in relation to the stock market as a whole? And I could refer you to page 64 of your corrected testimony.

THE WITNESS: On page 64, line 17 coming down, we said a beta or the comparing of the data of a stock measures the volatility of that stock in relation to the volatility of the stock market as a whole, so the answer is yes.

BY MR. MALONEY: (Cont'g.)

Q. Thank you. And is it correct that the Value Line calculates beta using a regression analysis and represents the co-variance of a particular stock to the market as a whole?

A. Yeah, Value Line uses a -- Value Line co-variances and then makes some adjustment.

Q. And is it correct that Value Line betas are generally calculated using the most recent 5 years of data?

A. Correct.

Q. And is it also correct that a beta of less than one indicates that a particular stock is less volatile in the market as a whole?

A. Yeah. Yeah, that's what it said on page 64 line 22, beta therefore not only is the stock has bigger risk than the market, so the answer is yes.

Q. Would you agree that utility betas have declined significantly since the company's previous rate case in 2016 and specifically what -- were you a witness in that case? Let me -- let me ask it differently, were you a witness in that case?

MR. FORST: Objection, can the Counsel define what he means by significantly?

MR. MALONEY: Sure. I believe the betas have declined from 0.75 in your testimony in that case to 0.60 and you could check that.

MR. FORST: I would also just note there's two questions there, so there is one about the significant decline of betas and then there is also whether the with -- whether one of the panel members was a witness in another case.

A.L.J. LEARY: All right. Does the panel want to answer the witness question first?

THE WITNESS: (Qadir) From the panel, I was the wit -- witness on the last case.

A.L.J. LEARY: You were?

THE WITNESS: Yeah.

A.L.J. LEARY: Anyone else?

THE WITNESS: (Duah) I was not in that case.

THE WITNESS: (Bullock) I was not.

A.L.J. LEARY: Okay. Good. And I think Mr. – you were referring to the 2015 rate case?

MR. MALONEY: 2016, I believe, yeah.

A.L.J. LEARY: 2016 rate case?

MR. MALONEY: Yeah.

A.L.J. LEARY: That result -- I think that's --

MR. MALONEY: Resulted in a joint proposal.

A.L.J. LEARY: What's the case number 0058 and -59, but what's the year.

MR. MALONEY: 16-G.

THE WITNESS: (Duah) G.

A.L.J. LEARY: 16 E and G.

MR. FORST: Just G.

MR. MALONEY: G.

A.L.J. LEARY: Just G.

MR. FORST: Yes.

A.L.J. LEARY: Sorry.

MR. MALONEY: 0058 and 0059.

A.L.J. LEARY: Okay. Thanks.

THE WITNESS: (Duah) Okay.

MR. MALONEY: I'm sorry.

THE WITNESS: What is your question?

BY MR. MALONEY: (Cont'g.)

Q. Would you agree that utility betas declined from 0.75 at the time of the last case to 0.60 today?

A. Based upon a proxy group or based upon the --

Q. Based upon the proxy group.

A. Yeah. 2016, when I checked the proxy group 0.75 and 2019 our proxy group is 0.61.

Q. Okay. Could you refer to your testimony at page 46 beginning at line 17?

A. Okay.

Q. And it states there, doesn't it, that the recent downgrade of the companies by Moody's was not unique. The reduced cash flows resulting from the tax act led Moody's to downgrade many utilities over the past year-and-a-half. Is that correct?

A. Yes.

Q. Is it correct that the Tax Cut and Jobs Act which is the tax act that you're referring to there, correct?

A. Yes.

Q. Increased the cash flows for most companies?

A. I'm not sure what you mean by most companies.

Q. Well, other than -- it's correct that most utilities not only had to reduce rates, but also to refund previously collected deferred taxes as a result of the passage of the tax act?

A. So you are referring to the electric industry. Is that what you're referring --

Q. Electric utilities, gas utilities, utilities generally experienced a decrease in cash flow. Correct?

A. Yeah, I agree with you.

Q. Yes. But other companies, in all other four -- all other businesses evaluated by Value Line, generally experienced an increase in cash, in -- in their ability to derive earnings and returns as a result of the passage of the Tax Cuts and Jobs Act. Isn’t that a fair conclusion?

A. Can you repeat the question?

Q. With respect to the rest of the S. and P. 500 if you will, or with the exception of utility companies, didn't other companies as a result

of the passage of the tax act likely experience an increase in cash flow and an increase in earnings?

A. The -- the -- because they are competitive businesses, the impact on competitive business was different from the impact on regulated electric utilities, the -- the tax act had an impact on the utilities' cash flows.

Q. Right. But -- whereas for other companies, they had paid taxes at 35% and now they only had to pay them at 21%, so everything else being equal, they got to keep the other 14%, right?

A. Yeah, I understand that also the pass back of the -- what do you call? This -- the impact of -- what do you call? Pass back on the different income taxes as a result of the T.C.J.A., it varies from companies to company.

Q. Right. And -- and -- so utility companies were impacted by that, but generally speaking, companies in other sectors that are not regulated, they don't have to pass back different taxes, correct?

A. They don't, but the impact varies somewhat they -- they -- some of the pass back will shorten -- some of them have a longer amortization

period, some have a shorter amortization period. So it vary from company to company on their cash flows.

Q. But in the panels of -- opinion, what affect would a reduction in cash flow for a utility company compared to an increase in cash flow for the rest of the market have on the regression analysis that determines beta?

A. Beta determination is based upon three factors. One is the stock prices, one, which I say earlier, the stock prices reflects a collective judgement of all market participants and it's also based upon estimation period. It depends -- it depends also upon the index that is selected.

So it depends upon these three variables and it may be different -- because there are different betas out there, if you go to Yahoo Finance, their beta is different, if you go to Capital IQ their beta is different.

A.L.J. LEARY: Excuse me. I need you to speak much more slowly.

THE WITNESS: Thank you. There are different sources of beta. Yahoo Finance provides beta, Capital IQ provides beta, Bloomberg also provides beta. So and each of these sources have a

different way of calculating it and also making an adjustment.

Bloomberg adjustment is different from Value Line.

BY MR. MALONEY: (Cont'g.)

Q. But I -- I guess my question is the changing cash flows resulting from the passage of the Tax Cuts and Jobs Act, as it affected utilities as compared to the way that it affected non-utilities, couldn't that have affected the correlation of returns realized by utility companies and resulted in the reduction in the beta.

A. It -- it will reflect the stock prices because investors will perceive whatever, no pricing. All these impacts on the company's cash flows, the pricing and the stock -- you put in a price -- pricing of the company's stock prices. So those risks and the cash flow impact will be reflected in a company's stock prices, which will be used by these beta sources to determine the beta. So it's -- it kind of captured in the beta calculation.

Q. I'm sorry. I didn't understand the last thing --

A. The ca -- the cash flow input can be captured in the beta calculation post stock price.

Q. Okay. So the drop in betas could be attributable to the impact -- the different impacts of the Tax Cuts and Jobs Act on the companies, at least in part.

A. I'm not sure which drop in beta. Are you referring -- which are you referring to, I'm not sure of that.

Q. Referring to what we said before was a decrease from 0.75 at the time of the last rate case to 0.60 today.

A. Yes. But it could be part of it that many thing that goes into the beta calculation.

Q. It's correct isn't it, that Moody's and S -- Standard and Poor’s, the company's credit rating agencies generally regarded the impact of the Tax Cuts and Jobs Act on utilities as a negative – a credit negative that increase risk?

A. Do you have any document you can show me or something like that?

Q. I think -- I -- I think if you could look at page 46 of your direct testimony.

A. 46.

Q. If you look at lines 11 through 14.

A. And what is your question?

Q. That the ratings agencies in general, this only refers to Moody's, but for the – saw the impact of the Tax Cuts and Jobs Act on utilities as generally a credit negative.

A.L.J. LEARY: I just want to make sure that you are not mischaracterizing the testimony because this testimony refers to the companies, not the utility sector as a whole. So just make sure that your question if -- I mean you can ask that question.

MR. MALONEY: You can go to line 17, Your Honor. Thank you, I -- I appreciate your -- your concern. The recent downgrade of the companies by Moody's was not unique. The reduced cash flows resulting from the tax act led Moody's to downgrade many utilities over the past year-and-a-half.

A.L.J. LEARY: Thanks.

THE WITNESS: And your question is what?

BY MR. MALONEY: (Cont'g.)

Q. So -- so the -- the tax act was viewed as a negative by the credit ratings agencies, that was the question I asked. Something that increased risk for utility companies.

A. I would say it’s going to be a permanent situation because it could change somewhere along the line and now this is on report that shows that the company's cash flows -- the utilities cash flows will -- will get better after sometime.

So I would say it's going to be -- it's not going to improve, it is going -- it is likely to improve according to one of the reports I came across. So I would say it's going be a permanent situation.

Q. But it was reasonable to -- is it reasonable to conclude that it was negative for the past four to five years or since 2017?

A. As is said in page 46 of our testimony, it leads to think about 25 companies -- 25 utilities I think with -- has a return action. It looks like they will put a negative -- I'm trying to recall -- they will put a negative outlook or something of that sort.

Q. Okay.

A. But not all of them were downgraded.

Q. Now, I'd like to refer to page 80 of your testimony.

A. I'm ready.

Q. Is it correct that at lines 19 to 22, you cite Dr. Roger Morin's book, the New Regulatory Finance in support of proxy group size that -- that a proxy group size of between 15 and 30 is reasonable to support a reliable cost of equity estimate?

A. Yeah.

Q. And do you regard the text that you cited by Dr. Morin as authoritative on the subject of determination of rate of return?

A. I would say Dr. Morin, I think -- I wouldn't say everything I agree with him in the book because in the Ni -- Niagara Mohawk case he quoted some aspect of his -- of his book saying that staff approaches of determining the return of common equity secular and I don't believe in that. But --

Q. But overall, do you agree his book is considered an authoritative exposition on the

-- on rate of return? You cited it in your testimony, in support of your testimony.

MR. GOODRICH: So objection, I believe that when asked the question of -- when asked that question, the panel answered that they do disagree with certain aspects of Morin's analysis.

A.L.J. LEARY: Why don't you ask about the areas that they do find authoritative? This -- this fin -- Staff Finance Panel does find authoritative or you could ask --

BY MR. MALONEY: (Cont'g.)

Q. Well, let me ask the question in a different way. If we could look at exhibit what's been mark as Exhibit 655.

A. Exhibit 655. The panel is ready.

Q. Let -- let me start off by just referring to your testimony at -- at page 113. At -- at that point, is it correct that you criticize Ms. Berkley's analysis for using sole -- for solely using a 30-year treasury yield in -- in determining the risk free rate for the capital asset pricing methodology?

A. Yes.

Q. Now, Dr. Morin, if I could refer you to the second page of what's been marked as Exhibit 655. Under the paragraph that says CAP-M application risk free rate, it states there doesn't it, that at the conceptual level, because common stock is a long-term investment and because the cash flows to investors in the form of dividends last indefinitely, the yield on very long-term government bonds, namely the yield on 30-year treasury bonds is the best measure of the risk free rate for use in the capital asset pricing methodology. It says CAP-M and just premium methods.

MR. FORST: Objection, Your Honors. I don't believe counsel has laid a foundation for the panel's familiarity with this document nor whether the sections of this book that they purport to be the exhibit are true and accurate.

A.L.J. LEARY: Okay, okay.

MR. MALONEY: Your Honor.

A.L.J. LEARY: It's almost lunch time.

MR. MALONEY: You Honor, the panel attached sections of Dr. Morin's book as exhibits and --

A.L.J. LEARY: And reference Dr. Morin's book --

MR. MALONEY: -- reference Dr. Morin --

A.L.J. LEARY: -- in their testimony so --

MR. MALONEY: And I'm perfectly happy asking them how -- that they disagree with Dr. Morin, but I believe we've established that Dr. Morin's testimony is relevant to this – or the book is relevant.

A.L.J. LEARY: All right. Why don't you -- can you ask the panel if they recognize this document et cetera, because one of the things I don't see is the -- where were we; 80?

MR. MALONEY: Yes.

A.L.J. LEARY: I don't see the version. So this is the two thou -- what's been marked as Hearing Exhibit 655 is the 2006 version of Dr. Morin's book.

MR. MALONEY: And -- and I believe if we go to exhibit -- excuse me.

A.L.J. LEARY: Well, you don't have to do that. Is it your understanding that this is exactly the version --

MR. MALONEY: My understanding is this is the -- and I could ask the panel, is this the version that you were using in -- quoting in page 80?

THE WITNESS: No, I use the 2009 version -- edition.

MR. MALONEY: The 2009 edition.

A.L.J. LEARY: We don't know as we sit here how those two editions may have changed. I'm going to give the company the opportunity to -- let me just confer with Judge Costello a sec.

All right, we have an idea I would like to throw out to the company and counsel for D.P.S. staff. I would like to give the company the opportunity to get the right version and ask the questions that are needed to be asked in this area.

MR. MALONEY: Could I just -- just if I -- sorry to interrupt, but if I can refer to S.F.P.-22.

A.L.J. LEARY: Would the panel just turn to S.F.P.-22 as well as counsel so we can -- go ahead, I'm sorry.

MR. MALONEY: If you look at S.F.P.-22, it says that the pages that were sponsored by the Staff Finance Panel were the 2006 public utilities report version of Dr. Morin's New Regulatory Finance, which is the precise same document that I believe I'm using here. And I didn't include the copyright page, but it's says Public Utilities Reports 2006.

A.L.J. LEARY: Bear with me. Oh, I see. So I didn't realize this S.F.P.-22 is in fact the 2006 edition even though I think the testimony is clear that they also relied on the 2009 version, but I think you can clear this up.

On the record, is that acceptable to counsel for D.P.S. Staff Finance Panel? It sounds like they actually have relied on both versions of this document.

THE WITNESS: And what is the question again?

MR. MALONEY: Well, I did --

A.L.J. COSTELLO: So hold on one second.

A.L.J. LEARY: Hold on, hold on.

MR. MALONEY: I -- I don't know that there is a two -- can we go off the record?

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: Sure.

(Off the record 11:48 a.m.)

(On the record 11:52 a.m.)

A.L.J. LEARY: So there was a discussion off the record about some confusion that we all agree has been created in the record and what I'd like to do is impose on Staff Finance Panel to take the lunch hour which we're not taking yet to -- in consultation with their counsel, clear up the apparent lack of clarity about Dr. Morin's publication.

Counsel for the company, Mr. Maloney, has agreed to defer questions on that until after lunch. He is going to continue with a line of questioning and -- and decide when we're taking a lunch break. And if he doesn't decide in the right time, I -- I will decide with -- in consultation with Judge Costello. So let that be a warning to you. I'm just --

MR. MALONEY: Thank you, no pressure.

A.L.J. LEARY: -- being -- I'm -- I'm just for the -- just for the record, I'm just having a little levity here, so you want to proceed?

MR. MALONEY: Thank you, Your Honor.

BY MR. MALONEY: (Cont'g.)

Q. Panel, turn -- turning to another area. I believe you testified earlier today that one of the goals of a fair rate of return is that would -- it would assure the utility continuing support in the capital markets for both its long-term debt and common equity security at terms that are reasonable given the company's risk. Is that correct?

A. Yes.

Q. And it's correct isn't it that the company's cost of debt is determined in a significant part by the credit ratings that are assigned to it by the major ratings agencies?

A. I would say it depends upon the credibility list on the company, so to that I will say yes.

Q. These companies are publicly rated by Moody's and --

A. Yeah.

Q. -- Standard and Poor’s. Is that correct?

A. Yes.

Q. Okay. And is it your understanding that the company's current long-term debt issuances are assigned an A. minus rating by Standard and Poor’s and an A.3 rating by Moody's?

A. Yes.

Q. Do you believe that the commission should continue to offer rate support for low A. credit quality for the companies and by that I mean the current A. minus rating by Standard and Poor’s and A.3 rating by Moody's?

A. If you turn to page 30 of our testimony --

Q. I'm sorry, what pages?

A. Thirty, 30.

Q. 30, yes.

A. Line 24. So when practical, the commission has generally supported a variable that provides utilities the opportunity to achieve and maintain above rating within the A. range.

Q. And so that -- that's what the commission has done. My question was does the panel recommend that continue -- that that policy continue to be implemented.

A. If you turn to page 34, lines 7 to 14. It say yes, our common -- common equity ratio when combined with the return on common equity recommendation or on combined with either quantitative metrics or qualitative factors produce credit metrics that are generally consistent with the performance required by Moody's and S. and P. to maintain the company's current ratings of A.3 and A. minus as you've shown in -- S.F.P.-10.

Q. Could you refer to your testimony at page 45?

A. Ready.

Q. Is it correct there that you refer to April 2019 credit reports from Moody's and you state that Moody's identified a stable outlook and therefore it's your opinion that Moody's does not view KEDNY and KEDLI's creating -- credit rating is likely to change over the intermediate term?

A. Yes, it's what it states on page 45, line 20 to 24 that Moody does not view KEDLI and KEDNY a rate incre -- credit rating as likely to change over the intermediate term. Also for April 2019, our submit -- our understanding of Moody's opinion.

Q. Now, I -- I've passed out three documents that have been identified as -- marked for identification as Exhibit 657, 658 and 659. Do you have those documents?

THE WITNESS: Can you repeat that?

MR. MALONEY: 657, 658 and 659, what is -- there are two -- three Moody's credit reports. One has to do with the threat to revoke National Grid's operating license and the other two specifically have to do with Brooklyn Union and -- and KeySpan Gas East.

A.L.J. LEARY: Can we just get a little bit more specific? So the first Exhibit 657 is a sector comment dated November 18th, 2019, which as counsel stated is captioned quote, Threat to revoke National Grid's operating license is credit negative for utilities.

The other two are credit opinions both dated December 30, 2019. Hearing Exhibit 658 is for KEDNY Brooklyn Union Gas Company and the Hearing Exhibit 659 is for KEDLI KeySpan Gas East.

BY MR. MALONEY: (Cont'g.)

Q. Are you aware of whether -- you testified earlier that the outlook for the companies

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according to Moody's was stable. Is it correct that this is no longer the case?

A. Yes.

Q. And -- and these type -- these are reports by Moody's. Are these the type that you reviewed and -- and relied upon in -- in producing your testimony? Have you seen these -- I'm sorry, I'll let you answer this.

A. What's the question again?

Q. These reports from Moody's, are these the type that you relied on and in some cases included in preparing your testimony in this case?

A.L.J. LEARY: So, let's get specific. Are these the type or are these the documents?

MR. MALONEY: Are these the type?

A.L.J. LEARY: Are these the type of documents that you relied on?

MR. MALONEY: Right. They're Moody's reports.

THE WITNESS: First of all, these reports came in after the filing of our testimony.

BY THE WITNESS: (Cont'g.)

Q. Right. Yes, and I'm casting no aspersions on your testimony as not being true at the

time that you produced it. But these reports have since come in, are you familiar with these reports?

A. Yes.

Q. Yes, okay. And -- and is it fair to say that the recent actions that occurred between National Grid and the governor were seen as actions that -- by Moody's that created a credit negative for all utilities?

MR. FORST: Objection. I believe that question is very vague. Could counsel point to specific interactions between the companies and the governor that they're referring to?

A.L.J. LEARY: And I'd also like to have some reference or nexus with these particular documents created so --

MR. MALONEY: Okay.

A.L.J. LEARY: -- let's take this apart.

MR. MALONEY: Let's -- can we look at that -- the document that been marked 657, sector comment, November 18th, 2019.

THE WITNESS: And what's the question, please?

BY MR. MALONEY: (Cont'g.)

Q. Would you agree that what this document says is that the -- the letter that was sent by Governor Andrew Cuomo to National Grid in which it was -- he intends to direct the New York Public Service Commission to revoke National Grid's certificate to operate its downstate gas franchise within 14 days unless the company presents meaningful and immediate remedial actions to end its moratorium on providing new natural gas hook ups to customers.

It was viewed as a credit negative for not only National Grid, but the state's other electric and gas utilities.

A. Yeah, it says the -- it says that there is increasing risk and I said too, but I also want to point out to another report. You did not record this report over here, you look at S. and P.'s also report which came, you know, February 18th, but it was two days ago.

A.L.J. LEARY: I'm sorry, February 18th, 2020?

THE WITNESS: Yes, please.

A.L.J. LEARY: Thanks.

THE WITNESS: I would direct you to, excuse me, I don't have a copy, but we enter in the record.

MR. MALONEY: Your Honor, I -- I don't have a copy of what --

A.L.J. LEARY: Well, let --

MR. MALONEY: I -- I -- I'm not going to -- think we're going to have an objection --

A.L.J. LEARY: Hold on, hold on.

MR. MALONEY: -- but I do -- we would like to see the document.

A.L.J. LEARY: Okay. Hold on. You asked this question an open this door and what the panel is about to do is tell you something that I agree you should have access to, but I want the panel to be able to answer the question that you posed about Hearing Exhibit 657, the sector comment and the negative for all utilities in New York.

Let the panel respond to this question. Again, I'm going to -- I agree with you, you should have a copy of this. We'll get you a copy of this after lunch, let the panel testify about this.

THE WITNESS: As I say -- it just says that this is credit negative to all New York investor owned utilities.

MR. MALONEY: I'm sorry, can I have that repeated? Can you repeat that I can --

A.L.J. LEARY: Could you read that -- could you read that again?

MR. FORST: Your Honors, if I may interject. I think there's some confusion based on the colloquy between counsels and Your Honors and now I believe that the panel is not aware of where we are and what documents you're referring to.

A.L.J. LEARY: I see. Okay. I -- you started to respond to Mr. Maloney's question, and I want to give you the ability to provide that response. Apologizing for the interruption, you are referring to a February 18th, 2020 document and I -- we would like to hear what you were about to say.

THE WITNESS: The panel also reviewed credit opinion that was issued on February 18th, 2020 which is about three days ago, by Standard and Poor’s and this is what it said, However, we believe -- that's Page 5, we believe the company is somewhat

less effective at managing the political red letter risk in the New York compared with its peers.

As largely informed by the recent political focus on National Grid and continues, while the company reached an agreement for the state that relieves it's -- that relieves it from claims that its franchise or licenses be revoked in connection with the moratorium, recent events have led us to believe that National Grid will not re-establish that trust and relationship or in this case stakeholders in the state.

In order to effectively advance a strategy for all office New York subsidiary consistent of how it has done so in the past, so we reviewed that credit opinion for someone post. So I would say intelligence believes the political or the consequence of the moratorium is the company is likely to be blamed, that's my understanding.

MR. MALONEY: If I could refer to pan -- I'm sorry, if I could report -- refer the panel to Exhibits 658 and 659 and it's correct isn’t it that these opinions discuss the specific circumstances of Brooklyn Union in the case of 659 KEDNY and in the case of 659 KEDLI. Is that correct?

A.L.J. COSTELLO: Just I think you said for 659 both times, so just to clarify 658 for KEDNY and 659 for KEDLI.

MR. MALONEY: Thank you, Your Honor. It's time for lunch.

A.L.J. LEARY: Sure you're not ready for lunch?

MR. MALONEY: If I could just get through this.

BY MR. MALONEY: (Cont'g.)

Q. Could -- could I refer the panel to page 2 of each of these documents?

A. Okay.

Q. With respect to the document marked 658, is it correct that one of the credit challenges that Moody's believes KEDNY faces is persistently weak cash flow, base metrics with pressures accentuated until a new rate plan becomes affected by U.S. tax reform and material and growing spending on environmental remediation that may not be immediately recovered through revenue and that's under credit challenges.

A. Yes, that's what it says.

Q. And is it also true that as you go down the page, it says factors that could lead to a downgrade. The ratings could be downgraded if there was -- if one -- there was a deterioration in the regulatory or political environment or two, it appeared unlikely that KEDNY would be able to achieve, and I'm going to spell this out, cash flow from operations pre-working capital over debt of high teams in percentage terms on an underlying basis from fiscal '21 onwards. Is that correct?

A. Yes.

Q. And similarly for KEDLI, Moody's identifies a credit challenge as relatively weak cash flow based metrics with pressure accentuated by U.S. tax reform. Is that correct?

A. Yes, that's what it says.

Q. And once again, it says the factors that could lead to a downgrade would be the ratings could be downgraded if there was a deterioration in the regulatory or political environment and it appeared unlikely that KEDLI would be able to achieve cash flow pre-working capital over debt of high teens in percentage terms on an underlying basis from fiscal 2021. Is that correct?

A. Yes, that's what it says.

Q. Okay. And when we're speaking of fiscal 2021, we are talking about the rate year of this proceeding. Is that your understanding?

A. Yes, that's my understanding.

Q. Finally, in both the KEDNY and KEDLI reports it's correct isn't it, that a material strengthening of the metric, the cash flow from operations pre-working capital of debt is identified by Moody's and I'm reading from page 1 as, dependent on a timely and favorable settlement of the rate case currently ongoing, incorporating increased, operating cost allowances, and timely recovery of growing environmental remediation spend and its continued large CAP-X program. Is that correct?

A. Can you point just to where you are reading from?

Q. I'm sorry, I'm on page 1 of the document, the third paragraph down under summary.

A. There are two document -- there are two documents, which --

Q. Each document is -- it's in the same place, both 658 and 659.

A. Uh-huh. Yeah, it is what it says.

Q. Okay. Now in the end, it matters what -- in terms of whether the company is downgraded or upgraded, it -- it matters what Moody's thinks and what S. and P. thinks. It's not necessarily depending on what any -- anybody else thinks. Is that correct?

A. The -- the question again?

Q. It -- the determination of whether KEDNY or KEDLI's debt is ultimately downgraded, that's a rating -- that's a decision that's made solely by the ratings agencies, Moody's and S. and P. They don't rely on any other -- anybody else for that determination. Is that correct?

MR. FORST: Objection, Your Honors. I think that calls for a degree of speculation. I mean I don't think the panel can testify to the extent that Moody's may consider a variety of factors beyond what is evidenced by what they include in their credit ratings when they issue them.

MR. MALONEY: I -- I didn't ask them what Moody's considered, I asked them whether the

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company's debt ratings were in the end determined by Moody's and S. and P., not by any third parties. In other words, it doesn't matter what I think or anybody else thinks, that is what they think.

A.L.J. LEARY: I heard more -- a question that was more along the lines of what Mr. Forst is talking about which asks the panel about what Moody's and Standard and Poor's rely on in setting their rating. So, you want to just --

MR. MALONEY: I withdraw the question and then re -- try to -- try to restate it.

BY MR. MALONEY: (Cont'g.)

Q. Is it correct in the end that it's Moody and S. and P. who determine the credit ratings for the company's debt, not any other third parties?

A. Yeah, we believe it's Moody's and S. and P. who are the rating agencies.

Q. Yes.

A. They would do that.

Q. This is a convenient point to stop for lunch if we're going to. Thank you, panel.

A.L.J. LEARY: We're going to go off the record, but I would like to --

(Off the record 12:14 p.m.)

(On the record 1:06 p.m.)

A.L.J. COSTELLO: Just during the break, I don't know if staff had the opportunity to clarify the version of the -- that book that you're referring to --

MR. FORST: Your Honors, we've conferred with the staff panel and we've conferred with counsel for the companies and the correct year is 2006 as indicated in, I believe S.F.P.-17 and that should clear up the record that is the correct date.

There is one version from 2006, not 2009 and we're all in agreement on that.

A.L.J. COSTELLO: Okay. Thank you.

MR. MALONEY: And -- and I will also say, Your Honors, that we had a discussion with staff counsel, and I think in return for their agreement that they will withdraw any objections to Exhibit 655 and 656. I will have no further question on these exhibits and we'll just allow them to come into the records as long as no one else objects so --

MR. FORST: That is correct, Your Honors. We can agree to that.

MR. MALONEY: So we're moving along.

A.L.J. LEARY: Sounds good.

BY MR. MALONEY: (Cont'g.)

Q. Good afternoon, panel.

A. Afternoon.

Q. Could you turn to your corrected testimony at page 43?

A. Okay. There.

Q. Is it correct at lines 10 to 13, you state that the core coverage ratio used by Moody's to assess the financial health of regulated electric and gas utility companies is cash flow from operation pre-working capital to debt, what we were talking about earlier C.F.L. pre W.C. to debt. Is that correct?

A. Yeah, that's the core metric.

Q. Yes. And now, I'd like you to refer to the document that's been marked for identification as Exhibit 645, which is the staff response to N.G.-58. Do you have that in front of you?

A. Ready. Ready.

Q. Okay. Now, it's correct there that you were asked to confirm whether the -- some information presented by the companies reflected an

appropriate calculation of the cash flow from operations pre-working capital over debt metric based on staff's revenue requirement in this case?

A. Yes.

Q. And your response was that the information presented in the request was not correct and you presented an alternative version -- alternative versions of the calculations. Is that correct?

A. Yes.

Q. And those are set forth on -- on page 3 of the I.R. response or the exhibit. Is that correct?

A. Yes.

Q. Now, the difference between the calculation at the top of the page and the one at the bottom of the page, is that the presentation excludes cash flow impacts associated with net pension and OPEB accruals and certain cash flow impacts associated with uncollectable amounts. Is that correct?

Let me just try to restate that. The one at the bottom excludes it and -- I'm sorry, the

one at the top excludes it, the one at the bottom includes it. Is that correct?

A. Yes.

Q. Okay. Now, is -- in the staff revenue requirement, is there an amount associated with pension and OPEB expense?

A. Staff does not have in front of us now what we included in the cash flow analysis is what the company provided in one of its exhib -- exhibits, which we refer to on bullet number D on page 2.

So we made references to R.R.P.7 C.U. corrections and update and then you also mentioned uncollectable accounts expense also R.R.P. with the companies' 11 C.U. So this is where we pull information from.

Q. Right. And -- and that's reflected in the analysis of the metric that's presented at the bottom of page 3, correct? The data that was provided in that response?

A. That's right. The company's exhibit is what is reflected in the -- in page 3.

Q. Right. Now, in this question we're asking you to calculate the metric based on the

staff revenue requirement and I know that you said you use data that the companies provided, but with respect to the amount reflected in the staff revenue requirement, would you agree that there is an amount for pension and OPEB expense?

A. We -- the Staff Finance Panel run these numbers with the accounting panel, and they confirm these are the numbers that they -- they confirm these numbers.

Q. So with respect to pension and OPEB expense, is it the panel's understanding that the amount that's reflected in the revenue requirement within the staff revenue requirement is recovered by the company and in accordance with the commission policy statement on pensions and OPEBs is then used to fund the pension and OPEB trusts that the company manages?

Or that the company has created for the purpose of administrating its pension and OPEB funds?

A. The staff finance panel believes that to the extent the company's cash contributions and then expands the difference that should be reflected in the cash flow metrics and that is

consistent on the company's financial statement 2009 to 2018.

You also look at the company's P.S.N.R. report in which they do reflect those, what do you call net pension and OPEB accruals or deferrals.

Q. But wouldn't the staff expect that the company would comply with the policy statement by taking in the amount of expense and rates and using it to fund the trust which is -- now there are possibilities where that doesn't happen, but is that in the forecast?

A. The staff panel believes that whatever is the contribution and whatever the expense which is -- which is -- the pension expense which is this important as S.F.E.S. -- I've got the acronym, what's its name.

The FASB which is the accountant methodology of calculating pension expense. If there is any variance between what a company contributes, that variance is always captured in the cash flow and the company also in response to I.R.-590 also provided projection of the company's pension and -- net pension and OPEBs deferrals.

So you look at that I.R. response and you also look at the company's history and also look at the P.S.N.R report and we believe that it shouldn't be zero. So what we did was that, okay, assuming, no one is excluded, those two items are excluded, what would be the cash flow metric or the core metric. And if you include it what would be the cash flow metrics. So we look at the two scenarios.

Q. So -- so it's fair to say you looked at it both ways --

A. That's right.

Q. -- based on an understanding that the company might not contribute dollar for dollar what it collects in rates.

A. So, look at a two extremes, whatever the case may be.

Q. And -- and now with respect on collectables, is it fair to say that you did something of the same thing. It's correct isn't it that the staff revenue requirement would have an amount in rates that would be equal to the amount of uncollectable expense that the staff would expect the company to incur during the rate year?

A. It -- again, the same idea that we talk about pension and OPEBs, so we look at a two way -- it's two scenarios with and without and the company's cash flow statement based upon the P.S.N.R. report and based upon their own financial statement and based also upon the 590 response, where they capture this kind of -- what do we call it numbers for the two items, which is the net pension and OPEBs deferral and accountant -- what you call, uncollectables.

Q. Could -- I want to ask you about one more line in this which is the differed tax amount. Now, in the upper amount you show -- in the upper portion of page 3 of N.G., you show a $111,376,000 of deferred taxes for KEDNY and $34,280,000 for KEDLI. Is that correct?

A. Yes.

Q. Is it your understanding that KEDNY and KEDLI are in a net operating loss position with respect to income taxes?

A. This one, we also run the one of the accountant folks and they also confirmed these numbers.

Q. The accounting, I'm sorry, who?

A. Staff accounting panel.

Q. Well --

A. So these numbers is coming from the company's -- the deferred tax over here is coming from the company's file and which is -- I -- which the company responded in D.P.S., I think 35. We said in bullet number 2, plan number A -- number A, A as in Apple, page 2 of 6 -- 645.

We said that the company responds they project deferred income taxes of about 111 and 34 million for KEDLI, this in response to D.P.S. 35.

Q. But those were amounts based on the company's filings in this case. Isn't that correct?

A. Yeah, this was what the company provide in response to this I.R., what is the current, not accumulated deferred income taxes.

Q. With respect to the staff revenue requirement, wouldn't it be the case that the amount of cash that the company would take in with respect to taxes would be equal to the amount of income tax expense built into the staff revenue requirement?

A. As we said earlier, we run this one with the accountant panel because the accountant

-- the staff accounting panel, they checked these numbers for us.

We produce a special -- we pulled this number from our accounting staff and they confirmed this is the right number.

Q. Would you agree that the only cash flow from operations that the company can take in that it -- that affects the F.F.O metric, if you will, is the actual cash the company takes in or recovers?

A. Yeah, we're looking at the cash flows and this is -- we are looking at actual cash and this is what the company is projecting for the current deferred but not for the accommodated deferred income tax. What is the cash flow to the company?

Q. But if -- and I'm looking at the staff revenue requirement for the Brooklyn Union Gas Company and I can pass this out. This is Exhibit 515, which is the corrected Exhibit S.R.R.P.-6 that was posted in D.M.M. on February 13th, 2020.

There is no header on it, but my understanding is this is Exhibit 515. And that's

because I printed it off. That's why -- I think that was it.

A. Yeah, ready.

Q. Yeah, so if you look at that exhibit, it shows total income taxes in the revenue requirement of $54,313,000. Is that correct? And that's near bottom of the page.

A. Yes, we see that.

Q. Is there any other amount of cash in the revenue requirement which is listed and would be everything below operating revenues that -- that is associated with income taxes or deferred income taxes?

A. Can you rephrase your question?

Q. Yes. Other than the $54,000,000 that I identified, is there any other recovery amount reflected in that revenue requirement that's associated with income taxes such that it would become deferred taxes?

A. We see here two total income taxes of $54 million for the rate year and then 2021. That's what we see over here, $54,000,313. And of course --

Q. Other than that there is no other amount for taxes that you're aware of in this revenue requirement, is --

A. I see --

Q. -- or income taxes I should say.

A. Yeah, because the taxes would have done through revenue as well.

Q. There were other taxes --

A. That's right.

Q. -- I see, but the answer on income taxes is no or deferred income taxes is no.

A. Yeah, I don’t see anything described as income -- deferred incomes taxes.

Q. Okay. Thank you. One final question on -- on credit metrics and it's -- if the commission decides that the -- the staff revenue requirement creates a likelihood that KEDNY and KEDLI would be downgraded, do you recommend that the commission provide any change -- do you recommend any changes to your return on equity or capital structure to offer additional rate support for KEDNY and KEDLI's current credit ratings?

A. First of all, we believe if the companies are downgraded, it won't be two notches at

least, it may be one notch downgrade and if it's one notch downgrade, our proxy group is also triple B plus.

What that means -- let me try rephrase it again. The company is now rated A.3 A. minus and if it's downgraded, we believe it will be triple B. plus. If it's triple -- triple B plus, our proxy group average credit rating is B plus, so we don't believe that it -- it's -- it's necessary to recommend to the commission to increase the company's return on common equity as a result of a downgrade because our proxy group at this point is triple B plus and the company's A minus.

So on that rating basis the proxy group is relatively riskier than the company.

Q. Okay. Just one follow-up, I mean if the company is downgraded it's not a given is it, that would be one notch downgrade?

A. Please repeat your question.

Q. If the company is downgraded, would you agree it's not a given that that would be a one notch downgrade?

A. (Duah) -- the company's rating history. I didn't see the 2 notch downgrade just at

once. What I see 1 notch downgrade or upgrade, I didn't see 2 notch downgrade or upgrade at a point in time. So I don't expect that Moody's or S. and P. downgraded companies two notches right away.

Q. Okay. Could you -- turn to page 97 of your testimony. Now, this is a great subject for the end of the cross examination which is, it's correct that at line 11 you expressed a concern about Ms. Bulkley, what you term her inappropriate use of the extended version of the internal rate of return X.I.R.R. in her D.C.F. analyses, is that correct?

A. Yes.

Q. And for the record, X.I.R.R. and I.R.R. are two excel functions, as I understand it, that allow you to reflect differing cash flows in a model?

A. Yes.

Q. And it's correct, isn't it, that all the companies in your proxy group paid dividends quarterly, is that correct?

A. They all paid dividend of a quarter, so that is yes.

Q. Yes. And would you agree that an investor would value for example, payments of 25

cents that were received on March 31, June 30, September 30 and December 31 differently than an investor would value a payment of a dollar on December 31st?

A. If an investor -- if an investor is -- had the option of receiving $25 or 25 cents per quarter which amounts to $100, versus, another investor given the option to receive $100 at the year-end, just one-time payment, I will say that one has a quarterly dividend, -- will get the $100.

But then, his required return on common equity would be lower because the company that pays $25 in first quarter has no retained earnings, reduce the company's retained earnings. And when the company's return earnings reduced -- it's re -- is reduced, it reduces the company's sustainable growth rate. And when the company's sustainable growth rate is reduced, the cost of equity or required rate of return is also reduced.

So the company that has the 2 options would be, I would say would be indifferent. And this argument was also brought up sometime back in case 08-00539 when -- in a Con Edison's case. 1.2 -- 125 of that order, the commission mentioned the company

was -- given that the use of staff approach was the -- which is this annual dividend growth -- annual dividend D.C.F. ignores the entire value of money. And the commission didn't really buy into that argument.

In fact, on page 126, the commission said, we are not adopting a judge's recommendation, because the judge recommended that should be the case. But the commission said that the judge's recommendation that our D.C.F. is toward a further payment of dividends quarterly rather than early on the end of the -- on the end of each year.

In the excel tend to be achieved on account of quarterly dividend, re-investment would be achieved --

A.L.J. LEARY: Excuse me -- excuse me. You need to speak much more slowly.

THE WITNESS: Thank you.

A.L.J. LEARY: So I think you're quoting some -- something that you're reading, is that right?

THE WITNESS: Yes.

A.L.J. LEARY: Would you just start back at the beginning of that quote if it's not too long ago and just speak more slowly?

THE WITNESS: Thank you. On page 125 of the order, which is case 08-E as in echo -0539. The commission said, and I quote, The company replied that the insistence on the use of an annual dividend were half as improperly ignored the time value of money.

And on page 126, the commission continued, We are not adopting the judge's recommendation that our D.C.F. estimates reflect the payment of dividends quarterly rather than annually at the end of each year. And the tend to be achieved on account of quarterly dividend, re-investment would be achieved by those who actually invest their the dividends in the company stock. And any additional allowance will be duplicative for those who actually reinvest dividends un -- unnecessarily and generous to those who do not. Accordingly, the company proposed upward adjustment is rejected, unquote.

BY MR. MALONEY: (Cont'g.)

Q. So that was an interesting reading of the commission order and I wouldn't

dispute that it's -- I wouldn't say it's inaccurate, but having said that, the question I asked you was if you were an investor and I gave you the choice of receiving $25 a quarter or a dollar at the end of the year, which would derive the better return?

A. As I said --

Q. A hundred dollars, I should say.

MR. FORST: Objection, Your Honor. I think it's ultimately speculative. It's asking the panel to take the place of an investor and speculate as to which the investor would favor and that is solely within the mind of each individual investor.

A.L.J. LEARY: So I think the -- the question that you're asking deals with what the panel's opinion is about a hypothetical investor?

MR. MALONEY: Well, the panel's entire testimony is devoted toward estimating the cost of capital which is what I think we believe would be what a rational investor would want to be compensated for for an investment with the company. So I'm asking of these 2 cash flows, which one does the panel opine, would -- would be -- which one do they believe a rational investor would prefer?

A.L.J. LEARY: So rational investor --

MR. MALONEY: Yes.

A.L.J. LEARY: --in your expertise? It -- I agree it's speculative but I think the com -- the panel's position on this would be in -- would be informative for the record.

MR. FORST: So I would just distinguish in terms of counsel's restatement of the question. I think the original question spoke about what would investor value more in terms of quarterly dividends versus an annual dividend. And now the restatement of the question or at least as it's being discussed currently relates to what is the preferred cash flow. And I'm not sure there's really a --

A.L.J. LEARY: I didn't hear cash flow, I heard rational investor. This panel's opinion about which of the 2 scenarios Mr. Maloney laid out would be, in their expertise and their opinion, preferable to a rational investor? Which of those 2 scenarios which is 25 cents -- or $25 a quarter or a $100 at the close of the year? So the panel can answer.

THE WITNESS: Can you restate your question?

BY MR. MALONEY: (Cont'g.)

Q. From the standpoint -- does the panel have an opinion on whether from the standpoint of a hypothetical rational investor, would the investor prefer to get 4 payments of $25 equally spaced through the year or 1 payment of $100 at the end of the year?

A. The panel believes the rational investor would be indifferent.

Q. Okay. It's correct, isn't it, that with respect to the I.R.R. and X.I.R.R. model, that those models can be operated in a way that achieves the precise same results?

A. The panel believes it can be managed to achieve the same results provided there is no mismatch in the cash flows and the timing of the cash flows.

Q. Is it your testimony that Ms. Bulkley's -- the timing of the cash flows that she reflected in her model was inconsistent with the timing of the cash flows of her proxy group?

A. Do you have any specific reference for us, in our testimony?

Q. No. I -- I was asking you if you believed -- you -- you analyzed Ms. Bulkley's use of the X.I.R.R. function didn't you?

A. Yes, I did review it.

Q. Okay. And is it your testimony that she entered the cash flow payments into the excel spreadsheet that provides this function in a manner that's inconsistent with the cash flows of the companies that she was evaluating in her proxy group? Is it -- is that your understanding?

A. Yeah. I think there's a mismatch there. I see some mismatch because the company stock price was -- was taken at the end of February or specifically February 28th, 2019. The annualized dividend growth and the annualized dividend -- put it in this way, they were annualized, they were not semi-annual. And then when she was forecasting the first dividend payment was based upon 6 months period. Meanwhile, the dividend was annualized, so I see a mismatch there..

Q. Thank you, panel. I have nothing further.

A.L.J. COSTELLO: Do you want to take a few minutes?

MR. FORST: If we could, Your Honors, that would be great.

A.L.J. COSTELLO: 10 minutes?

MR. FORST: I think that'd be appropriate. Thank you.

A.L.J. COSTELLO: Okay. Go off the record.

(Off the record, 1:40 p.m.)

(On the record 1:56 p.m.)

THE REPORTER: On the record.

A.L.J. COSTELLO: Do you have any re-direct?

MR. FORST: Are we back on the record, Your Honors?

A.L.J. COSTELLO: We are back on the record.

MR. FORST: Yes. We do have a few questions for re-direct.

A.L.J. COSTELLO: Okay. Thank you. Proceed.

RE-DIRECT EXAMINATION

BY MR. FORST:

Q. Panel, there's a document being passed around titled response to N.G.59. It's a 1-

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas page response document. It was issued on November 12th, 2019 and it was responded to on November 27th, 2019. Do you have that document before you?

A. (Duah) Yes.

Q. And are you familiar with this document?

A. Yes.

Q. And would you say that the content of the document is true and correct to the best of your knowledge?

A. Yes.

MR. FORST: So I would ask that this document be given a pre-marked exhibit number.

A.L.J. COSTELLO: It'll be marked as Exhibit 660.

BY MR. FORST: (Cont'g.)

Q. Earlier panel, you were asked a number of questions regarding the C.L.C.P.A., known as the Climate Leadership and Community Protection Act, is that correct?

A. Yes.

Q. And in response to those questions you referred to the response to N.G.59, is that correct?

A. Yes.

Q. And based on the document you have before you, were your answers to those questions consistent with the response provided in this document?

A. Yes.

Q. Excellent, thank you. We're passing around another document. Panel, the document being passed around is a 5-page document containing a number of figures and tables. Could you -- do you have that document before you?

A. Yes.

Q. And could you please explain what is included in this document?

A. Yes.

Q. If I could just direct the panel's attention to the document that my co-counsel Raquel Parks just passed out. If you could just refer to that document and the question asked was, could you explain what's contained in that document?

A. Staff panel D.C.F. cost of equity calculations, electric and gas, 33 companies.

Q. And is this document similar to -- or an update to the document that was included with

-- and I'm going to find the hearing exhibit number here right away. Pardon me, Your Honors, we're just looking for the hearing exhibit number. That would be Hearing Exhibit 649. So is the document before you panel, an update to the attachment to N.G.64 otherwise known as Hearing Exhibit 649?

A. Yes.

A.L.J. LEARY: This is a different document than 649, is that correct?

MR. FORST: That's correct.

A.L.J. LEARY: So we need to give this a separate hearing exhibit number.

MR. FORST: Correct. You've predicted my exact next question.

A.L.J. LEARY: That's where you were going, I just --

MR. FORST: Correct.

A.L.J. LEARY: -- wanted to write this down correctly so --

MR. MALONEY: Your Honor, I'm going to object to the admission of this document or any further direct on it. This is a whole new analysis being propounded on -- being dropped on the company in re-direct. This -- this includes now a 33 member

proxy group that the staff never supported before. They -- they did give us some information on that, and we did ask some questions about it. But that it -- that wasn't their proposal. That was something they gave to us at discovery. Now, we're seeing a new proposal?

MR. FORST: And the company has opened the door for this type --

MR. MALONEY: We didn't open the door for this.

MR. FORST: -- of exhibit when you asked about the 5 additional gas proxy groups which this contains. In addition, this includes additional time, you know, time varying information which is extending off of the exact document that the company's introduced earlier today.

A.L.J. LEARY: Okay. Both points well-taken. And I think what we want to do is to provide the company -- we're not going to keep this out, as it is an update to a discovery response. And those updates can come in at any time. That's the way I view this document. I'd like to know when it was prepared, what's the date of prep -- when it was prepared?

MR. FORST: I mean, I -- I don't know exactly when it was prepared. It was prepared for this hearing though.

A.L.J. LEARY: Okay. The company needs to have an opportunity to digest this, consult with their own experts on it. And unfortunately, that could involve having the Finance Panel return. Judge Costello and I don't want to force the company to have to look at this, I believe, 5-page document with lots and lots of companies and numbers on it and just be reactive at this point.

Again, your -- your objection to this is noted. Again, it goes to whether this gets admitted. We're going to allow the testimony to be elicited based on this document that has nothing to do with whether this document ultimately comes into the record, that's for another day. But we are giving you the opportunity, to take a look at it and if you think you have to recall the Finance Panel, I'm just telling D.P.S. staff, the Finance Panel's coming back.

MR. FORST: We understand that, Your Honors.

A.L.J. LEARY: Okay?

MR. MALONEY: Your Honor, we're going to need the opportunity to do some discovery on it because there's a whole set of new assumptions in here. There's -- there's --

A.L.J. LEARY: We have not closed discovery.

MR. MALONEY: Okay.

A.L.J. LEARY: If you think you need discovery, you can serve discovery.

MR. MALONEY: And I just went through a 4-hour cross examination on a set of numbers, that now the numbers have changed entirely. And -- and that includes the metrics, that includes things that -- that I think the commission will want to make sure it has an accurate record on.

A.L.J. LEARY: Did the proxy group D.C.F. R.O.E. change? It -- it looks to me like it might be the same, is that right? Is that number the -- the same -- basically the same number?

MR. FORST: So if you're looking between the Exhibit 649 and what's been -- what's just been provided to the panel, there is a slight change.

MR. MALONEY: A slight change --

A.L.J. LEARY: What --

MR. MALONEY: -- it went from --

A.L.J. LEARY: Hold on.

MR. MALONEY: -- 2 to 8% in the --

A.L.J. LEARY: I'm sorry.

MR. MALONEY: Okay.

A.L.J. LEARY: Slight change meaning what, Mr. Forst?

MR. FORST: So that would be from 8 2 to 8.0.

MR. MALONEY: That -- that's not a slight change, that's -- I believe I could calculate it several million dollars.

A.L.J. COSTELLO: Just so that we have something to refer to. And again, it's not being admitted at this point in time, but we're going to mark it for identification as Exhibit 661.

A.L.J. LEARY: Unfortunately, I -- I hope that it's not another 4 hours of cross because I don't remember you talking for 4 hours about a document like this although you would know better than I what you may have to do. The -- again, I'm going to allow the testimony to be elicited but the question of whether this comes in and potentially any

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testimony that is derived from this document, we're going to defer for another day, but we want to hear on the record what the panel has to say about it.

If you want to ask questions today, you're welcome to do that, but I'm going to give you the opportunity to go discuss with others, your experts and counsel and come back and advise Judge Costello and I if you need to have the Finance Panel come back and for however long you estimate their return being necessary, okay?

MR. DECICCO: Do -- Phil Decicco. Did we get an answer as to when this document was prepared, just so we know?

THE WITNESS: (Duah) If you look on page 3 of the Exhibit on end of December 2019.

MR. DECICCO: So -- so just for the record, this was prepared over a month ago and it's trying to --

MR. FORST: No, no, no. It includes information from December 2019 which is, I believe, the latest available. It was prepared in advance for this hearing.

A.L.J. LEARY: So was it prepared at some point in the last, you know, month or so, a few weeks?

MR. FORST: I would say probably a week or --

A.L.J. LEARY: Week or so?

MR. FORST: Yes.

A.L.J. LEARY: Okay. I do --

MR. MALONEY: Your Honor, I'll be honest with you, you know we -- we sent the Moody's reports that we wanted -- we sent it in discovery before the hearing. This could've been sent to us before the hearing and we would have had the opportunity to cross it.

A.L.J. LEARY: Certainly, that is --

MR. MALONEY: So I -- I'll restate my objection's note, I understand.

A.L.J. LEARY: Well, the -- the reason I rolled over there to talk to Judge Costello was for that very reason. These -- these are the kind of things that, you know -- that got you, not intended, I know that, but these are the kinds of things that we have to look at in terms of the prejudice to the other parties.

Again, I think that prejudice is ameliorated if we provide the company with an adequate opportunity to review it and bring the Finance Panel back to discuss it. And that is what we think will best serve, you know, everyone's interest since this is a belated -- belatedly submitted document.

So let's go forward. And again, we're not -- we're not ruling about anything right now in terms of this -- the admission of this or the testimony associated with it. But I want to hear from the company further, maybe Monday of next week to advise all the parties about whether you believe that further questioning is necessary.

At the end of the hearing, I think we're going to discuss -- before the end of the hearing when we discuss the admission of all of these, we don't a big fight at that time. We want to resolve this hopefully on Monday or Tuesday, if you can take a look at this information between now and then.

And again, you're welcome to continue today if you feel that that's in your best interest, Mr. Maloney.

MR. MALONEY: I'd like to reserve for a later time at this point.

A.L.J. LEARY: Sounds good. Okay, go ahead.

BY MR. FORST: (Cont'g.)

Q. Panel, the document before you includes an additional set of information for October, November and December 2019, is that correct?

A. (Duah) Yes.

Q. And it also includes a number of gas only utilities, is that correct?

A. Yes.

Q. And are these the same 5 gas utilities that were included in the analysis provided in response to N.G.64?

A. Yes.

Q. And what is the impact of the inclusion of those additional months as well as the gas utilities on the overall analysis provided in this document?

A. By including the 5 gas companies brings the recommended R.O.E. to 8-- 8%.

Q. Okay. Thank you. We're going to move on.

MR. MALONEY: Are we going to get a whole new revenue requirement presentation as well with bill impacts et cetera associated with this return? I -- I'd also like to make sure all the discovery is updated, that -- that went to these metric numbers et cetera. I realized I can ask that, but if I have to get back to my office and ask you it would -- it will be Monday.

MR. FORST: So I think the -- the issue here is that the company introduced a response to an -- a res -- a request that was issued in November that included the same specific information here.

A.L.J. LEARY: Be specific about that request --

MR. FORST: Okay.

A.L.J. LEARY: -- what Hearing Exhibit was that?

MR. FORST: That'd be Hearing Exhibit 649. So the company introduced that today during these proceedings which included a table of -- the exact same table which we are now providing with additional months of data for October, November and December.

MR. DECICCO: And -- and it's counsel's position that you conveniently had this with an exhibit number blank ready to be submitted only because we introduced that -- that topic in -- in cross examination.

A.L.J. LEARY: So let me do this, Mr. Decicco, because I'm going to ans -- ask the questions here. I don't want to get into some kind of thing between the company and staff. But I would like to know how come this wasn't provided to the companies and the parties before this moment in time?

MR. GOODRICH: In assessing our risk --

A.L.J. LEARY: Your mic's not on, I don't think.

MR. GOODRICH: Yes, it is.

A.L.J. LEARY: Oh, it is?

MR. GOODRICH: In assessing our risks on cross examination, one of the things that we assess was what if the company asks us, asks our panel about the exclusion of gas companies from the -- the proxy group. And one of the things that we wanted to have available if we -- if that question came up on direct, was an ability to say what the

impacts would be of including the gas companies in the proxy group.

So that is why in assessing what would happen today at the hearing, we did prepare an exhibit to be able to explain what the impact of including the gas companies in our proxy group would be.

A.L.J. LEARY: So this really wasn't as I initially noted, an update to discovery. This was actually in anticipation of the questioning that the companies would present to this panel?

MR. GOODRICH: But it is the same exact information that was provided in response to the discovery request --

MR. MALONEY: No, it's not.

MR. GOODRICH: -- up -- updated for subsequent months.

A.L.J. LEARY: So --

A.L.J. COSTELLO: What -- so would that have been responsive to discovery request by the company? If the --

MR. GOODRICH: If the company had asked for a different time period, yes.

MR. MALONEY: Yes, the company's discovery was actually directed at the testimony that was filed in the case as opposed to anticipating that we might see a whole new set of numbers 4 months down the road. It is very difficult to ask discovery on something that doesn't exist.

A.L.J. LEARY: So this is Hearing Exhibit 649.

A.L.J. COSTELLO: Because this -- this -- I mean, I understand it goes to information that was brought up but it -- it's -- it is like new direct testimony. So I mean we do have to give them an opportunity. They have asked whether information -- discovery-type information can be provided to them. Do you -- do you have an answer to that? Would it be something that you can do relatively quickly if that's -- I just want to hear your --

MR. FORST: Yes, if the company can issue its discovery request, we can be, you know, responsive as we've always been.

MR. MALONEY: What -- what about updating the revenue requirement and the bill impacts et cetera?

A.L.J. LEARY: Yeah, I'm -- I am interested in that question. Is that required here or --

MR. FORST: So as we stated earlier, in preparation for questions we believed we would receive today at the hearing which include whether we wouldn't -- had provided analysis of the inclusion of gas utilities, we updated that analysis for the most time relevant information that we had. That does not mean that we are trying to supplement or replace the testimony that is already currently on the record. We are simply providing an illustrative -- illustrative example of a response to that very question.

A.L.J. LEARY: So I take that as a no in response to your question, Mr. Maloney, that the revenue requirement --

MR. MALONEY: Is it the staff's position that R.O.E. should be 8% or 8.2%?

A.L.J. LEARY: Well, I think that that's an excellent question that you can ask on re-cross after re-direct is completed. And I think that, you know, that --

MR. MALONEY: Well -- well, that goes to whether it's the revenue requirement. I -- I think we should be able to deal with, are they putting in a new position or are they simply providing additional information which is nice but pertains to a period that someone might argue is relevant or not relevant.

MR. FORST: As -- as I stated before, we are not providing an entire new case here on re -- on re-direct. We are simply answering and providing information that is responsive to a question that the panel received during its cross examination.

A.L.J. LEARY: Okay. Let's go ahead with re-direct. I'm -- we'll give you an opportunity to pursue re-cross and again, that's without prejudice to you advising Judge Costello and I that you need additional time and you need the panel to return.

MR. MALONEY: Thank you, Your Honor.

A.L.J. LEARY: But I want a justification if they're coming back, in writing. And you know, some -- some -- yeah, or we can put it on the record Monday or Tuesday whenever you're ready.

MR. MALONEY: Thank you.

A.L.J. LEARY: Okay.

BY MR. FORST: (Cont'g.)

Q. Panel, a doc -- 2 documents were just passed around. They are both 13 pages in length. One is titled S&P Global Ratings RatingsDirect, the Brooklyn Union Gas Company. The second document is S. and P. Global Ratings RatingsDirect, KeySpan Gas East Corporation. Do you have these 2 documents before you?

A. (Duah) Yes.

Q. And are you familiar with these -- no. Okay.

A.L.J. COSTELLO: Yeah. We can mark the first one for the Brooklyn Union Gas Company as Exhibit 662. And the document for KeySpan Gas East Corporation, 663.

MR. FORST: Thank you, Your Honors.

BY MR. FORST: (Cont'g.)

Q. Did you refer to these documents earlier today?

A. Yes.

Q. And are these copies before you true and accurate copies of the reports you referred to earlier?

A. Yes.

MR. FORST: Just bear with me 1 second, Your Honors, I appreciate the time.

BY MR. FORST: (Cont'g.)

Q. Earlier panel, you were asked if -- whether Value Line information including the source of the inputs to that Value Line information -- strike that.

Panel, earlier today you were asked if various inputs to the Value Line information that you rely upon, such as the market risk premium was also sourced from Value Line, could you please clarify your answer to that question?

A. Our market risk premium was not based on Value Line, it was based on Merrill Lynch data and then treasury rates.

MR. FORST: Okay. We have no further questions, Your Honors.

A.L.J. COSTELLO: Mr. Maloney, other than with respect to Exhibit 661, do you have any further re-cross?

MR. MALONEY: If you'd bear with me.

A.L.J. COSTELLO: Sure.

MR. MALONEY: Your Honor, can I ask some limited cross at this time on Exhibit 661?

A.L.J. LEARY: I told you you're -- it's without prejudice to further review outside of this hearing and I'm looking forward to hearing from you next week about what you need further. So you can ask whatever you want today within the bounds of relevancy et cetera --

A.L.J. COSTELLO: Yes. And my re --

A.L.J. LEARY: -- it it's without prejudice to you --

A.L.J. COSTELLO: And my reference --

A.L.J. LEARY: -- coming back.

A.L.J. COSTELLO: I'm sorry, I didn't mean to interrupt. But my reference before was except for 661, that was because my understanding for what you said was you wanted to reserve on that, but you can certainly ask que -- questions at this time.

MR. MALONEY: Thank you, Your Honor.

RE-CROSS EXAMINATION

BY MR. MALONEY:

Q. Good afternoon panel, Ken Maloney for the companies. Earlier today, you submitted sworn testimony that said we believe that the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be electric utilities, is that correct?

A. (Duah) Can you repeat you question?

Q. I -- I think if you can refer to page 54 of your testimony at lines 3 to 6, you'll know what I just asked you.

A. And the question again?

Q. You testified -- and you, I believe, testified and swore today that it was true that you believe the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be electric utilities, is that correct?

A. Yes.

Q. And are you now saying that 33 companies including 5 gas companies is the most suitable proxy group?

MR. GOODRICH: May I? One -- one point your -- one -- one simple point. I think -- I

thought I understood from the testimony we had in the room earlier was that while the universe -- Value Line universe might have been 37 companies that the proxy group -- that staff was using was 28 companies. And I just fear that we're going to confuse the record with the extra number if that's -- if my understanding's correct.

MR. MALONEY: It's correct, there are 28 companies -- electric companies and combination gas and electric companies in the staff proxy group.

THE WITNESS: (Duah) Please, can you repeat the question for us? Sorry.

BY MR. MALONEY: (Cont'g.)

Q. Is it correct that at page 54 of your testimony, you testified that we believe the most suitable proxy group would be derived from a universe of 37 holding companies deemed by Value Line to be electric utilities?

A. We believe it should be electric, based upon our testimony on page 54, of our testimony.

Q. So are you now impeaching your own testimony? Are you testifying that that's not

the case and that there should be 33 companies including 5 gas companies in your proxy group?

A. We are not changing our 8.2, our reco -- recommendation, this is just an illustration of what if you were to include the 5 additional com -- electric companies in our proxy group. But we believe it should be based upon the electric company because as we stated on page 53 of our testimony that the companies are local gas companies even though the investors look at the parent company -- invest in the pen -- parent company and even though it was electric.

Q. So just to be clear, I heard you testify that you're not changing your initial set of recommendations as in your direct testimony?

A. Correct.

MR. MALONEY: I'll stop there for now, Your Honor. Thank you.

A.L.J. COSTELLO: Anything further?

MR. FORST: Nothing further, Your Honor -- Your Honors.

A.L.J. COSTELLO: Okay. What we're going to do is we're not going to excuse the panel at this time. I mean you are free to leave, but you're

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subject to being recalled if there is further cross examination that would be required of you. And we thank you.

A.L.J. LEARY: I do -- I do want to ask if any of the panel members are unavailable next week to return. For any day next week.

THE WITNESS: (Bullock) I won't be available Thursday.

A.L.J. LEARY: Anyone else?

THE WITNESS: (Duah) Me and Abdul will be available.

A.L.J. LEARY: I'm sorry?

THE WITNESS: Abdul and I will be available next -- next week, yeah. We're available.

A.L.J. LEARY: Okay. Monday and Tuesday would be the likely time that you would be recalled. We have hearings scheduled for those dates. So if you could just keep those dates open in your work schedules until you hear from Mr. Goodrich or Mr. Forst that you will not be recalled. We expect to hear from the company hopefully on Monday about whether that will be necessary.

MR. MALONEY: We'll be prepared to respond at that time, Your Honor.

A.L.J. LEARY: Perfect, thank you. Thank you to the panel.

A.L.J. COSTELLO: Yeah. We'll go off the record.

(Off the record, 2:27 p.m.)

(on the record 2:31 p.m.)

THE REPORTER: We're on the record.

A.L.J. COSTELLO: Okay. Mr. Fung?

MR. FUNG: Justin Fung, City of New York.

A.L.J. COSTELLO: Okay. Thank you. And may I ask the witness to please state your name and business address for the record?

MR. CONNOLLY: Christopher Connolly, 40 Sylvan Road, Waltham, Massachusetts 02451.

A.L.J. COSTELLO: And please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MR. CONNOLLY: Yes.

WITNESS; CHRISTOPHER CONNOLLY; Sworn

A.L.J. COSTELLO: Thank you. You may be seated. You may continue.

MR. GAVILONDO: Thank you, Your Honors.

DIRECT EXAMINATION

BY MR. GAVILONDO:

Q. Carlos Gavilondo on behalf of the companies. Good afternoon, Mr. Connolly. Do you have before you a document entitled the Direct Testimony of Christopher J. Connolly dated April 2019 consisting of 42 pages, a cover sheet, a table of contents?

A. Yes.

Q. Thank you. And was that document that you just identified -- I apologize, I skipped over. Do you also have before you a document entitled Rebuttal Testimony of Christopher J. Connolly dated September 18, 2019 consisting of 11 pages, a cover sheet and a table of contents?

A. Yes.

Q. Okay. And now, were those documents that you just identified prepared by you or under your direction and supervision?

A. Yes.

Q. Okay. If I were to ask you the questions contained in those documents today, would your answers be the same?

A. Yes, they would.

Q. Okay. And do you adopt those documents as your sworn testimony in these proceedings?

A. Yes, I do.

MR. GAVILONDO: Your Honors, I ask that the direct and rebuttal pre-filed testimony of Mr. Connolly be transcribed into the record as if given orally.

A.L.J. COSTELLO: The request is granted and at this point the court reporter should insert the following files. It's KEDNY-KEDLI Christopher J. Connolly Direct Testimony and KEDNY-KEDLI Christopher J. Connolly Rebuttal Testimony. #

**Before the Public Service Commission**

**THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAS CORPORATION d/b/a NATIONAL GRID**

**Direct Testimony**

**of**

**Christopher J. Connolly**

**Dated: April 2019**

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## I. Introduction and Qualifications

**Q. Mr. Connolly, please state your full name and business address.**

A. My name is Christopher J. Connolly. My business address is 404 Wyman Street, Waltham, Massachusetts 02451.

**Q. By whom are you employed and in what capacity?**

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid"), as the Business Lead for the Gas Business Enablement ("GBE") Program. In this role, I have overall accountability for the delivery of the GBE Program, which is one of the key strategic priorities for National Grid and its U.S. operating companies, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies").

**Q. Please describe your educational background and professional experience.**

A. I received a Bachelor of Science in Mechanical Engineering Technology from Northeastern University in 1999. I have worked in the energy industry for approximately 20 years in various capacities, first as a contract engineer for DistriGas of Massachusetts Corporation beginning in June 1998 until

October 1999 when I joined Boston Gas Company. From October 1999 through October 2001, I held various engineering and operations supervisory roles at Boston Gas Company, including oversight for gas system relocation and infrastructure modernization activities associated with Boston's Central Artery Tunnel "Big Dig" Project. Following KeySpan Corporation's ("KeySpan") acquisition of Boston Gas Company in 2001, I continued my tenure in operations from 2001 to 2007 with responsibility for the design and execution of complex construction projects across KeySpan's New England service territory in Massachusetts and New Hampshire. In 2007, when National Grid acquired KeySpan, through 2016, I held a number of leadership positions of increasing responsibility within gas engineering and operations. I co-led the development of the Process Excellence Organization in 2013 through 2015, during which time I assembled a process-focused stakeholder team responsible for identifying improvements in safe and reliable gas system operations while ensuring compliance across all jurisdictions. Further, I directed enterprise-wide engineering teams advancing complex engineering, capital work plan strategies, public works projects coordination, and gas growth analysis. In addition, the teams I supervised supported the safe and reliable execution of the gas capital work plan and provided engineering support during emergencies. From February 2015 through July 2015, I took on the role of

Acting Vice President of Gas Systems Engineering and subsequently the role of Director, Gas Project Development from August 2015 through April 2016. On May 1, 2016, I was named Vice President of Process and Business Requirements for the GBE Program, responsible for identifying business requirements and developing and implementing the standardized business processes and new solutions capability and functionality that will support enhanced customer satisfaction, improved safety and compliance performance, and enhanced employee engagement across National Grid's U.S. operating companies. In October 2018, I was named to my current position.

## II. Purpose of Testimony

**Q. What is the purpose of your testimony?**

A. The purpose of my testimony is to present an overview of National Grid's GBE Program, as well as the Companies' proposal for associated cost recovery. I will provide KEDNY's and KEDLI's forecast of rent expense for the capital investments associated with the GBE Program for the twelve-month period ending March 31, 2021 ("Rate Year") and the three subsequent twelve-month periods ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (Data Year 1, Data Year 2, and Data Year 3 are collectively referred to as the

"Data Years"). I will also discuss the operating costs that KEDNY and KEDLI will incur in the Rate Year and Data Years for the GBE Program. Finally, I will detail the efficiency savings expected from implementation of GBE Program capabilities and propose an incentive mechanism tied to the delivery of those benefits.

**Q. Do you sponsor any exhibits as part of your testimony?**

A. Yes. I sponsor the following exhibits that were prepared under my direction and supervision:

Exhibit __ (CJC-1): High-Level Roadmap Showing Phased Implementation and Capabilities;

Exhibit __ (CJC-2): Description of the Specific GBE Capabilities and Benefits that will go In-Service in the Rate Year and Data Years for KEDNY and KEDLI;

Exhibit __ (CJC-3): Incremental Operating Expenses for the GBE Program Allocable to KEDNY and KEDLI in the Rate Year and Data Years;

Exhibit __ (CJC-4): Additional Run the Business Costs to KEDNY and KEDLI to Support the GBE Program Post-Implementation:

Exhibit __ (CJC-5): Total U.S. Type I and Type II Savings Estimates (Capital and O&M) and KEDNY and KEDLI Allocated Type I Savings Estimates Identified in Connection with the GBE Program; and

Exhibit __ (CJC-6): Proposed GBE Performance Incentive Mechanism.

## III. Overview of the GBE Program

**Q. What is the GBE Program?**

A. The GBE Program is a comprehensive framework of new technology solutions and business process changes necessary to strengthen and improve the performance of National Grid's U.S. gas business. Currently, the U.S. gas business faces a number of challenges. These challenges include the need to replace aged computer systems, drive continuous improvement in gas safety performance, deliver an expanding and increasingly complex capital investment program, and meet evolving customer expectations.

The GBE Program was developed through a collaboration among National Grid's U.S. gas business and Information Technology ("IT"), Procurement, Customer, Finance, Shared Services, Customer Meter Services (electric and gas), and the Human Resources functions, among others. The program has been designed as a holistic transformation of National Grid's U.S. gas

business to deliver process improvements across people, systems, and technology to strengthen operational and safety performance and build a modern platform that supports customer demands.

**Q. Why is the GBE Program needed?**

A. Over 90 percent of the systems used by National Grid's U.S. gas business are at or approaching end of life prior to the Rate Year. The average age of these systems today is approximately 15 years compared to an industry average of six. Because the age of these systems limits the ability to make modifications and increases the amount of time the systems are down, it is becoming increasingly difficult to support safe, compliant operations and meet evolving and emerging regulatory obligations. In addition, the current systems, many of which still rely on paper records, no longer support the way today's gas companies must manage work and performance, and provide employees with real time information and effective tools. Modern, supported solutions are also needed to help reliably deliver significant levels of capital investments.

**Q. What are the benefits of the GBE Program?**

A. The GBE Program provides numerous benefits such as:

*Gas Safety.* The GBE Program will strengthen in several respects the Companies' ability to operate a safe, reliable gas distribution system. First, GBE will implement a new enterprise asset management ("EAM") system, creating a single repository for all asset information and enabling a standard process for managing preventative maintenance. The EAM system will be integrated with a geographic information system ("GIS") to improve the Companies' ability to capture, store, access, and analyze geographical asset information concerning its gas distribution network. This will provide a single view of all assets, which will facilitate data-driven investment and maintenance decisions. The GBE Program investments will consolidate information on all required operations and maintenance ("O&M") work, rather than across multiple, manual spreadsheets. Finally, implementing modern, more reliable platforms will provide better records to document compliance and decreases the likelihood of system outages impacting the ability to deliver work.

National Grid's Pipeline Safety Management organization has a central role in the GBE Program to ensure that initiatives have a direct linkage to improving pipeline safety and compliance. For instance, National Grid is in the process of implementing a Pipeline Safety Management System ("PSMS"), a process safety model based on employing and strengthening

the ten essential elements of the American Petroleum Institute's recommended pipeline safety management standards (Recommended Practice 1173 or "API 1173"). GBE Program initiatives have been mapped to API 1173's ten core elements for strong alignment to enhance safety and compliance upon implementation. In addition, National Grid enlisted a third-party consultant (P-Pic) to independently validate that the GBE Program initiatives will strengthen the Company's PSMS.

*Improved Operational Performance*. The main objective of the GBE Program is to consolidate and replace many of the Companies' disparate and aging systems, as well as the associated work processes to achieve a step change in operational performance. The GBE Program investments will also drive continuous improvement in regulatory compliance and transparency with more complete data capture and reporting, less reliance on paper, and greater visibility of required work. The GBE Program aims to create a more performance aligned organization through a focus on end-to-end processes and by implementing business practice changes such as improved data quality and cleansing and field technical training.

*Operations Support*. The GBE Program will implement system improvements and enable the standardization of business processes for enhanced tracking, visibility, and scheduling of work.

*Customer*. Another benefit of the GBE Program is enhanced customer service through improved scheduling and dispatch. This includes enhanced appointment booking and an interactive customer framework (described below), as well as the ability for dispatch and field crews to create a consolidated view of past, scheduled, and potential future work for customers so they will be better equipped to answer customer questions.

**Q. What are the key elements of the GBE Program?**

A. The key elements of the GBE Program include: replacement of aged core systems; enabling customer and employee interaction platforms; and implementing standardized processes and training. I describe each element below.

*Replacement of Aged Core Systems.* Initially, the GBE Program will integrate, standardize, and simplify core delivery processes and systems onto a modern platform (comprising approximately 20 solution components, down from the 100 disparate applications used today).

Specifically, the core systems GBE will design, standardize, and implement include:

- a common industry-standard EAM platform to manage inspection and maintenance of existing assets and capture new asset information in a single database;
- a common scheduling platform to support optimized scheduling, work bundling, and routing of work;
- a common GIS platform with more accurate foundation maps and conversion of gas service records and sketches, available with mobile functionality and enabled for full access viewing from mobile device (Apple iPad);
- a field mobility work management solution accessed from Apple iPads with base capabilities that include receipt and views of work assignment, turn-by-turn directions to work location, access to electronic work packages and customer premise data, updated work status and entry of field completion data, and capabilities to initiate new work in the field, attach pictures, and view assets in GIS maps;
- a standardized Asset Investment Planning and Management ("AIPM") platform for managing the enterprise's project portfolio, including project routing and approval, with the ability to forecast costs; and
- an Asset Risk Ranking/Prioritization application to perform asset condition assessment, assign asset risk ratings, and with the ability to prioritize asset replacements.

The integration of these core systems will support a more holistic management of assets and administration of work. In addition, updating and integrating these core systems will enable full utilization of new tools such as the field mobility work management solution for leak investigation and inspection work orders; drive improvement in gas safety performance;

improve capital delivery effectiveness; and lead to better employee utilization, and ultimately improved customer service.

*Customer & Employee Interaction Platforms.* A flexible interface will be built on top of the core systems to allow customers, call center representatives, and field employees to operate on a common platform and more easily access data. The solution capabilities will be developed and integrated with work management and scheduling solutions that allow customers to interact with National Grid such as by self-service appointment scheduling and re-scheduling, receiving updates (*e.g.,* text or email) based on their preferences for appointments, addressing inquiries for new gas connections and conversions, and accessing information about work on their street or neighborhood. Similarly, capabilities will be developed and further integrated with the asset and work management platforms including scheduling and dispatch, and GIS systems to provide office based employees and field employees with prompts and validations for accurate capture of required information for compliance, field mobility access to asset data, maps, standards and policies, training references, and a consolidated view of relevant customer and premise information to support enhanced customer service. This interface also builds the capabilities

necessary to rapidly adapt processes and address developing channels for customer engagement in the evolving energy marketplace.

*Standardized Processes and Training.* The GBE Program will also support the implementation of standardized operations processes and training that to this point has been fragmented due to the significant complexity of multiple supporting systems. This will reduce the level of requirements that would need to be designed, built, tested and trained, and as a result, mitigate the costs of the new technical solution. In addition, standardized processes and training will further support more consistent delivery and performance reporting.

**Q. Has the GBE Program been considered in any other New York rate proceeding?**

A. Yes. In Cases 17-E-0238 and 17-G-0239, the Commission adopted a Joint Proposal providing for recovery of GBE costs for Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk"), KEDNY and KEDLI's affiliated operating company in New York.

## IV. Program Implementation Approach, Roadmap, and Capabilities

**Q. Please explain National Grid's approach to implementing the GBE Program.**

A. National Grid established a project organization to support the development and implementation of the GBE Program. I am the dedicated Program Lead, overseeing business requirements, solution design, schedule, and budget, with an IT Lead accountable for solution development and delivery. A Chief Transformation Officer is responsible for oversight of all of National Grid's large change programs, including GBE, Advanced Metering Infrastructure, Grid Modernization, Project Volt, SAP S/4 HANA, and CIS to ensure these transformative investments are aligned, prioritized, and built with capabilities to maximize value to customers and support the delivery of anticipated benefits. A separate GBE Steering Committee, comprised of senior executives from the Gas Business Unit, Finance, IT, Regulation and Strategy, Human Resources, and Customer Operations, Safety and Business Excellence oversees the program, including the deployment timeline, benefits realization, and costs. The Chief Operating Officer for the Gas Business Unit is a member of the GBE Steering Committee and Program Sponsor with a focus on the GBE Program's delivery of capabilities and benefits to National Grid's gas operating companies, including KEDNY and KEDLI.

In terms of external support, National Grid worked with two of the top system integrators with experience on similar, large-scale implementations, Accenture and PWC, to complete a high-level design and develop a roadmap that leverages modern system implementation approaches to minimize risk and maximize the likelihood that the desired business outcomes are successfully delivered.

**Q. Please explain the procurement process utilized for the selection of vendors to deliver GBE Program solutions.**

A. The fundamental purpose of the competitive procurement process was to develop the components of the GBE Program using capable and experienced third-party vendors that (i) have the competency to assist in delivering the program on time, on budget, and with the stated capabilities, and (ii) are subject to clearly defined contractual parameters and performance requirements. Specifically, National Grid utilized a competitive, agile sourcing strategy consisting of a pre-defined four-step stage gate process to evaluate and down-select vendors through collaboration to enhance understanding of scope, deliverables, and assessing culture of partnership to reduce the risk of delivery of the GBE Program.

With the benefit of several months of program experience, and in the interest of identifying the optimal delivery model for the next phases of the program, a strategic review of the GBE Program was launched in December 2018. The purpose of the strategic review was to recommend program adjustments, with Steering Committee approval, to strengthen program delivery, improve accountability, reduce risks, provide greater cost assurance, and ensure clarity with regards to program roles and responsibilities going forward. As a result of the strategic review, the GBE Program (i) re-planned the roadmap, as reflected in Exhibit __ (CJC-1), to account for experience to date and re-phase program releases to better manage risks and priorities; (ii) restructured the multi-vendor integrator construct to streamline program delivery and create clearer vendor accountability; and (iii) adjusted program resources to clarify accountability for delivery.

In March 2019, the GBE Steering Committee approved the transition to a single system integrator consultant for the completion of design, development, and delivery of the remainder of the scope of the GBE Program. This decision to transition to one system integrator firmed up the

cost to deliver the GBE Program within budget and contingency, shortened the duration of the overall program with substantial completion of enhanced capabilities by December 2020, and re-confirmed delivery against the planned releases in the revised roadmap with consultant fees at risk for missed delivery dates and a cap on future change orders.

National Grid selected PwC as the single system integrator for the GBE Program. This shift to a single solution integrator model will further ensure end-to-end accountability and help to address challenges the program faced with regards to role clarity, testing, solution performance, and IT integration.

**Q. Please describe the planned implementation approach.**

A. The initial focus of the GBE Program will be development of standardized business processes, validation of jurisdictional requirements coordinated through the end-to-end solution design activities, implementation of asset management, work management, and scheduling applications along with an integrated mapping (*i.e.*, GIS) solution. National Grid will focus on replacing aged core applications and implementing updated solutions as quickly as possible to help reduce the risk associated with critical, unsupported applications. This will create the foundation for building

incremental enhanced capabilities to support safety performance, operational efficiency, the customer experience, and a performance-oriented culture. Examples of such enhanced capabilities include advanced analytics on asset demographic, condition, health, and other information to provide a consolidated view of asset risk geospatially; the customer and employee interaction portals; advanced analytics for work forecasting and planning; and supervisor field mobile capabilities on viewing and tracking crew and work order progress spatially; and auto work notifications to plan and schedule preventative maintenance activities.

A stage-gate methodology is being employed to manage delivery and implementation in the service territories and operating companies once pre-defined thresholds of performance are successfully demonstrated in Rhode Island. The GBE Program is practicing agile development methods wherever it is appropriate to do so. Under this model, business and IT teams work collaboratively in short-cycles to prioritize functionality and get to a minimum viable product (*i.e.*, the simplest solution that can be implemented) allowing earlier release of initial functionality and reprioritization of added features and enhancements based on learning and user feedback.

As shown in Exhibit __ (CJC-1), the GBE Program has developed a well-defined program roadmap to reduce risk in implementation and provide clear visibility of critical path dependencies to assure successful implementation as each phase progresses. The roadmap is defined by the major releases grouped under a total of seven portfolio anchors. Each of the portfolio anchors represents a series of releases with defined capabilities delivered across National Grid's operating companies. The portfolio anchors provide definition to the roadmap timeline to highlight major delivery milestones while signaling the transition of solution focus, jurisdiction focus and resource support for design, development, test, deployment, and solution support activities.

*Portfolio Anchor 1 ("PA1")* (delivered April 2018) – The first GBE solution deployment included a minimum viable product release of solutions for corrosion, instrumentation and regulation, and collections in Rhode Island. The solutions included deployment of the enterprise asset management application (IBM Maximo), and the scheduling and dispatch and field mobility applications under Salesforce Field Service Lightning. The GBE Program roadmap designed the first release at National Grid's Rhode Island gas distribution company, The Narragansett Electric

Company (gas segment), given its significant reliance today on paper-based operations and its manageable scale (*i.e.,* fewer operating yards).

*Portfolio Anchor 2 ("PA2")* (releases from October 2018 through July 2019) – Includes enhancements to the capabilities delivered in PA1 in Rhode Island for corrosion, instrumentation and regulation, and collections plus delivery of the minimum viable product solution for resource management with the first release of the Workforce Time and Attendance application, the first implementation of the GIS solution, enhancements to the AIPM application, and first minimum viable product release of the Asset Risk Management application. This anchor also expands the solution capabilities and functionality to support Customer Meter Services work types and the first implementation of the Customer Contact Center Customer Relationship Management application, Salesforce Service Cloud.

*Portfolio Anchor 3 ("PA3")* (releases from September 2019 through July 2020) – Includes additional enhancements and added features delivered in Rhode Island for PA1 and PA2 along with the release of a minimum viable product solution to support the leak response, investigation, and grading of leaks. This portfolio anchor also expands the releases for the first time to National Grid's New York and Massachusetts operating companies. At the

end of this portfolio anchor, all the corrosion, instrumentation and regulation, customer meter services, customer contact center, resource management, asset risk management, and investment planning solutions are fully released along with minimum viable product solution for the leak process, including the deployment of the standard GIS application and mobile GIS in all operating companies. All field-based operations employees supporting the work will have mobile devices deployed and full use of the applications including the assignment and receipt of orders, field data capture, and order completion information.

*Portfolio Anchor 4 ("PA4")* (releases for Rate Year) – Includes the initial roll-out of the minimum viable product solutions that support construction and maintenance work types and expands the leak process end-to-end to include leak repair and leak re-check to the Rhode Island business. The same software applications previously released in portfolio anchors 1 – 3 are enhanced with new capabilities and features to support construction and maintenance.

*Portfolio Anchor 5 ("PA5")* (releases for Rate Year) – Includes additional enhancements and added features delivered in Rhode Island for PA4 and expands the PA4 releases for the first time to National Grid's New York

and Massachusetts operating companies. These releases also deliver the graphic work design and field redlining capabilities to support construction and maintenance. At the end of this portfolio anchor, all construction and maintenance capabilities are fully released in all operating companies. Field based, operations employees assigned to construction and maintenance will use mobile devices to support work.

*Portfolio Anchor 6 ("PA6") and Portfolio Anchor 7 ("PA7")* (releases for the Rate Year) – Include enhancements to the solutions in the areas of graphic work design and estimating. These anchors also include efforts focused on optimization of the solutions and additional analytics use cases. Many of these activities are driven and supported by IT operations under run the business activities.

**Q. Please describe the capabilities the Program has delivered to date?**

A. To date, the GBE Program delivered three major program releases in April, October, and December 2018 implementing solution capabilities for corrosion, instrumentation and regulation, and collections. The core systems deployed in Rhode Island include enterprise asset management, scheduling and dispatch, field mobile work management, resource management, asset risk management, and GIS solutions. Concurrent with

the delivery of these releases, the GBE Program began preparations for two future releases in 2019 including the customer call center front end, Customer Meter Services scheduling, dispatch and field mobility, and leak response, investigation, and grading solutions. Key program activities included design and development of functionality, planning and initiation of end-to-end integration testing, business stakeholder engagement, solution demonstrations, as well as planning for data loads and other deployment-related activities that have involved users in New York. To date, the GBE Program capabilities have been rolled out to over 800 users, including 77 in New York for purposes of training and supporting their roles requiring access to Rhode Island data.

**Q. Please describe the GBE Program capabilities that have been implemented for New York.**

A. The first GBE Program deployment for Niagara Mohawk, KEDNY, and KEDLI occurred in August 2017 with implementation of the Work Order Lifecycle Redesign project. This project restructured and made more robust the work order interfaces between the SAP, PowerPlan, and existing work management applications (*i.e.*, IBM Maximo in KEDNY and KEDLI and Storms in Niagara Mohawk) and was a pre-requisite for deployment of the GBE Program solution.

The first major New York releases are planned under PA3 for September 2019 for Niagara Mohawk, November 2019 for KEDLI, and June 2020 for KEDNY with the deployment of the capabilities released in Rhode Island. New York operating company employees have been included in the solution design, development, and testing phases to ensure the solution capabilities and features are fit for purpose and meet the requirements for the New York operating companies. Solution development for Niagara Mohawk is complete and solution testing is in progress. Existing solution capabilities and features are being validated through solution demonstrations with KEDLI employees, and design activities have commenced to develop solution enhancements and include requirements specific to KEDNY and KEDLI *(e.g.,* addressing integration architecture requirements with legacy systems).

Notably with respect to KEDNY and KEDLI, recognizing the need and benefits of the GBE capabilities for customers, National Grid and its shareholders have made and will be making the required investments before securing cost recovery for the GBE Program.

**Q. Please describe the specific projects/capabilities that will go in-service in the Rate Year and Data Years for KEDNY and KEDLI.**

A. Implementation is planned for KEDLI prior to the Rate Year, beginning November 2019, and for KEDNY within the Rate Year beginning in June 2020 as discussed above and shown in Exhibit __ (CJC-1).

Exhibit __ (CJC-2) further describes the specific projects and capabilities that will go in-service in the Rate Year and Data Years for KEDNY and KEDLI.

## V. GBE Program Costs

**Q. What is the total cost of the GBE Program?**

A. The total cost of the GBE program for all of National Grid's U.S. operating companies is currently estimated at approximately $539.3 million, of which approximately $384.3 million is capital costs, and $155.0 million is operating expense. While delivery of the GBE Program initiatives is expected to be within the total costs stated herein, it is important to note that GBE Program costs may shift between the Rate Year and Data Years as each of the projects completes detailed design.

The GBE Program follows a sanctioning approval process to provide on-going assurance and transparency of the program delivery, schedule, and budget as the program moves through its various stages. An annual partial sanctioning against the total approved cost estimate is required throughout the duration of the program. The latest sanction approval reflects the total costs referenced above.

**Q. Explain how the current total cost estimate for the GBE Program relates to the $458 million forecast reflected in Niagara Mohawk's rate plan approved in Cases 17-E-0238 and 17-G-0239.**

A. Rates established in Niagara Mohawk's rate plan included $458.1 million forecast for the GBE Program. Although Niagara Mohawk identified a program contingency of $61 million, those costs were not reflected in rates because they were not sufficiently certain. The $539.3 million total cost estimate in this case includes: (i) the $458.1 million forecast reflected in Niagara Mohawk's current rates; (ii) $61.0 million of contingency ($57.4 million of which has been subsequently sanctioned); and (iii) $20.1 million incurred in fiscal year ("FY") 2017 for the initial development of the business case, assessment of processes and applications, and high-level design for the GBE Program. The Companies' revenue requirements in this filing do not reflect the $20.1 million for the initial development of the

business case, $3.6 million in contingency that is not planned for sanction, or operating costs prior to the Rate Year.

As such, as discussed later in my testimony, by the Rate Year, KEDNY and KEDLI collectively will have incurred more than $67.4 million in connection with the GBE Program that is not reflected in their revenue requirements, and therefore are costs borne by shareholders for the benefit of customers.

**Q. Why is National Grid forecasting the need to spend budgeted contingency?**

A. While significant progress has been made in the development and delivery of the GBE Program, National Grid is expecting upward cost pressure on the budgeted contingency. Challenges and complexities of replacing the aged systems or developing legacy interfaces has been a major driver of the need to utilize budgeted contingency.

In addition, delays in development and design work required to enable fully capturing detailed business requirements, including requirements rationalized across operating companies to understand differences, have led to testing delays. Addressing and limiting the scope of these delays has

placed an upward pressure on the budgeted contingency in the form of additional cost for National Grid resources and longer phasing for training and software costs.

Furthermore, due to the phased release of capability across the jurisdictions (one of the key lessons learned from previous programs to mitigate delivery risk), the original estimate of Allowance for Funds Used During Construction ("AFUDC") was significantly under forecast in the original $458 million estimate.

Together, the unanticipated legacy integration costs, incremental National Grid resource, training and software costs, and underestimation of AFUDC have resulted in an increase of $57.4 million against the budgeted contingency.

The refreshed roadmap reflected in Exhibit __ (CJC-1) is the result of the GBE Program's recent strategic review and evaluation of program status discussed earlier, resource requirements and availability, and lessons learned through release implementations. This roadmap further addresses delivery risk by limiting releases to one jurisdiction at a time (*i.e.*, no more than one jurisdiction for a CMS release), separating large releases across

multiple companies (*i.e.*, separate M&C releases for KEDNY and KEDLI), increasing the timeframe between releases to improve testing, solution development, and solution readiness, and avoiding releases during winter operations.

**Q. Does the Historic Test Year include costs for the GBE Program?**

A. Yes, the Historic Test Year includes operating costs for the GBE Program related to operating model assessment and design, business engagement, readiness and change management, training and post go-live support for the three major releases, modifications to existing technology infrastructure, and the day-to-day program management activities, KEDNY and KEDLI have made a normalizing adjustment of $10.910 million and $4.954 million, respectively, to remove these costs from the Historic Test Year and isolate costs for the Rate and Data Years.

**Q. What is the cost of the GBE Program to KEDNY and KEDLI in the Rate Year and Data Years?**

A. Because the GBE Program is a shared investment among National Grid's U.S. gas operating companies, only a portion of the total capital investment will be allocated to KEDNY and KEDLI in the form of an annual rent expense as part of the overall IT service rent expense. The portion of the

annual rent expense attributable to the GBE Program investment for KEDNY is $11.73 million, $18.38 million, $17.64 million, and $16.89 million in the Rate Year and Data Years, respectively as shown in Exhibit __ (RRP-11), Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 3, 6, 9, and 12. The annual rent expense attributable to KEDLI is $8.59 million, $8.34 million, $7.99 million, and $7.65 million in the Rate Year and Data Years, respectively as shown in Exhibit __ (RRP-11), Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 3, 6, 9, and 12.

KEDNY's and KEDLI's allocated share of the $14.2 million, excluding base labor and benefits, total operating expense in the Rate Year, as shown in Exhibit __ (CJC-3) are $4.8 million for KEDNY and $2.2 million for KEDLI. Exhibit __ (CJC-3) also shows the forecast of operating expense allocated to KEDNY and KEDLI for the Data Years.

**Q. Please identify the amount and explain the basis of the incremental operating expense for KEDNY and KEDLI in the Rate Year and Data Years.**

A. The incremental project operating expense included in Exhibit __ (CJC-3) relates to (i) end user training, (ii) data conversion from the legacy applications to the new GBE Program applications, (iii) program

management and leadership to manage the schedule, resources, finance, risks, issues and performance, and (iv) business process documentation that is non-system related.

**Q. Are there any post-implementation run the business costs associated with the GBE Program?**

A. Yes. As shown in Exhibit _ (CJC-4), the Company will incur additional run the business costs to support the GBE Program post-implementation. These costs include (i) a team to support business functions in the use of the new systems, design new processes to take full advantage of the new system, and monitor business controls embedded in the system; (ii) hardware, software, and mobile solutions license maintenance fees and subscriptions; and (iii) support costs to maintain certain legacy applications following implementation until legacy applications are replaced or maintained in an upgraded future state, as appropriate.

Support costs for the legacy applications will decrease from the Rate Year to the Data Years. Additional support costs will be required for legacy applications that will continue to remain after full implementation due to regulatory reporting needs and outstanding legal hold obligations.

Nevertheless, as legacy software systems are retired due to functional replacement as part of the GBE Program, the run the business costs for operating the associated servers, software systems, and field devices will be eliminated. As shown in Exhibit _ (CJC-4), the Companies have netted these costs against the forecast run the business costs expected for the Rate Year and Data Years.

**Q. What are the post-implementation run the business costs associated with GBE in the Rate Year and Data Years?**

A. As shown in Exhibit __ (CJC-4), KEDNY's and KEDLI's allocated share of these costs in the Rate Year are $6.3 million and $3.8 million, respectively. The Companies' respective allocated share of these costs in the Data Years is also shown in Exhibit __ (CJC-4).

**Q. Please explain how costs for the GBE Program are allocated to KEDNY and KEDLI.**

A. Most GBE Program costs have been allocated among all of National Grid's gas operating companies based on the number of gas retail customers, with the exception of those relating to work management scheduling, dispatch/mobility and workforce management, and customer contact

support and interaction, which have been allocated based on the number of gas and electric retail customers.

Exhibit_____(RRP-11), Workpapers to RRP-3, Schedule 9, Workpapers 3, 6, 9, and 12 set forth the allocations of these program costs.

**Q. Do KEDNY's and KEDLI's revenue requirements incorporate any adjustments for potential slippage in costs or timeline for the GBE Program?**

A.` No. The Companies reviewed the concerns Staff raised in the Niagara Mohawk case about potential slippage in costs or schedule for the GBE Program. However, it has been approximately two years since the Niagara Mohawk case, and the KEDNY and KEDLI revenue requirements in this case reflect additional program maturity and the updated program roadmap provided in Exhibit __ (CJC-1). As explained earlier, the roadmap is a result of a detailed strategic review undertaken to strengthen program delivery, reduce risks, and provide greater cost and timeline assurance. As such, the Companies believe that no further adjustment to GBE Program costs or timeline are required. Importantly, as mentioned earlier, by the Rate Year, KEDNY and KEDLI will have incurred more than $67.4 million in connection with the GBE Program that is not reflected in their revenue

requirements due to regulatory lag. A further slippage adjustment on KEDNY's and KEDLI's revenue requirement is unnecessary.

## VI. Forecasted Efficiency and Productivity Savings

**Q. Are there any efficiency and productivity savings associated with the GBE Program?**

A As explained earlier, the main objective of the GBE Program is to consolidate the many duplicate and aging applications and systems across the enterprise. Because GBE is an asset replacement program, the primary benefit is a reduction in operational risk.

The implementation of new asset, work, and mobility systems lays the foundation for enhanced capabilities that will drive a broad range of operational benefits and performance improvements, some of which are anticipated to result in cost reductions. To that end, the GBE Program is an initiative that contributes towards National Grid's Accelerate Program targets, as described in the testimony of the Revenue Requirements Panels. Because of the attention and focus on the GBE Program, benefits have been separately reflected from Accelerate Program savings. Exhibit __ (CJC-5), Page 1 provides the total U.S. benefits (Type I and Type II, and capital and

operating expense benefits) anticipated for the GBE Program as a result of the strategic review discussed earlier.

**Q. Please discuss the business case benefits initially identified by National Grid.**

A. As reflected in Exhibit __ (CJC-5) Page 1, the majority of benefits under the business case will be realized after Data Year 1. Once the enhanced capabilities are fully embedded by FY 2024, the GBE Program estimates a steady state run rate of $50.289 million in total benefits annually. Over a five-year period, the $50.289 million annual benefits will amount to $251 million, which is greater than the approximately $185 million investment in the enhanced capabilities. Specifically, implementation of enhanced capabilities could provide the following benefits:

Type I (Spend Reduction) – the benefit has a direct, quantifiable, and sustainable impact in reducing costs. For example, the GBE Program investments are anticipated to deliver increased clerical and back office productivity beginning in the Rate Year as a result of automation of some manual tasks (*e.g.*, time entry), elimination of paper-based processes, as well as streamlining of data updates performed by clerical staff.

Type II (Capacity Savings) – the benefit is a process improvement that consists of resources freed up or future cost or potential penalty avoidance.

For example, the work and asset management will provide improved scheduling, bundling of work, and enhanced, prescriptive routing for field technicians. In turn, these enhancements will allow optimization of drive time and existing resources freeing additional resource capacity (*i.e.*, additional jobs completed per shift).

**Q. What is the estimate of savings?**

A. As shown in Exhibit ___ (CJC-5), Page 2, approximately $4.4 million, $10.3 million, $10.5 million, and $10.5 million in Type I and Type II O&M savings are allocable to KEDNY in the Rate Year and Data Years, respectively. Approximately $0.8 million, $1.8 million, $2.0 million, and $2.0 million in Type I and Type II O&M savings are allocable to KEDLI in the Rate Year and Data Years, respectively.

**Q. Have forecast cost reductions associated with the GBE Program been reflected in this filing?**

A. Yes. KEDNY and KEDLI have made adjustments to the Rate Year and Data Years to reflect their allocated share of the estimated Type I savings from the GBE program initiatives. KEDNY's adjustment reduces the revenue requirement by $2.9 million in the Rate Year and $6.8 million, $7.1 million, and $7.1 million in Data Years 1, 2 and 3, respectively. KEDLI's

adjustment reduces the revenue requirement by $0.8 million in the Rate Year and $1.7 million, $1.9 million, and $1.9 million in Data Years 1, 2, and 3, respectively. No adjustment is being made for Type II savings because they do not result in a direct cost reduction, but increase capacity for work that otherwise would not be completed. No adjustment is being made for penalty avoidance savings because penalties are not recovered from customers.

**Q. How were initiatives that targeted capital related savings treated in the filing?**

A. With respect to initiatives estimated to result in capital savings, those savings are embedded in the capital plan and not reflected as separate adjustments in the revenue requirement.

## VII. GBE Incentive Proposal

**Q. Please explain the Companies' proposal for an incentive on the delivery of GBE Program benefits.**

A. Recognizing the importance of the GBE Program to National Grid's efforts to improve operations, customer service, and gas safety, National Grid incurred significant costs to begin implementing the five-year program even where costs were not recovered in rates. In particular, because GBE

Program costs were not included in KEDNY and KEDLI's current rate plans, National Grid's shareholders have or will be funding approximately $20.4 million of unrecovered costs through FY 2019 and expect to incur another $24.3 million of unrecovered costs for FY 2020.

Recognizing the value of this shareholder funding for significant program development costs between rate cases and the importance of incenting large investments in critical programs to support operations and deliver customer benefits, the Companies are proposing an incentive mechanism that aims to recover a modest portion of otherwise unrecoverable costs while further incentivizing the delivery of customer benefits from the GBE Program.

**Q. How are the Companies proposing to collect an incentive on the delivery of GBE Program?**

A. The Companies will measure performance on six key performance indicators ("KPIs"), similar to those adopted in Cases 17-E-0238 and 17-G-0239 for Niagara Mohawk, that are intended to demonstrate successful delivery of GBE Program capabilities. The GBE Program KPIs are (i) the average number of completed Customer Meter Services ("CMS") jobs per worker per day; (ii) the average feet of main replaced per Maintenance and Construction ("M&C") worker per day; (iii) work orders processed each

year per each full-time equivalent ("FTE") engineering clerical employee; (iv) total call volume related to customer moves and non-moves per year; (v) total number of gas safety non-compliance occurrences per year; and (vi) the Companies' customer experience (effort) rating based on customer surveys.

At the end of FY 2023, the Companies will measure delivery of GBE Program capabilities based on performance across the FY 2023 KPI targets. The FY 2023 GBE KPIs, weightings, and targets are set forth in Exhibit __ (CJC-6), Schedules 1 and 2. Illustrative examples of the GBE KPI performance measurement are provided in Exhibit __ (CJC-6), Schedules 1 and 2. To the extent the Companies measure improvement against the FY 2023 KPI targets , the Companies will record a regulatory asset for all or a portion of their respective GBE Program investment costs for the period January 1, 2019 through March 31, 2020 ("linking period").

## VIII. Program Delivery, Readiness and Training Elements

**Q. Please describe how Software as a Service ("SaaS") is utilized by the GBE Program, and the benefits of its use.**

A. The GBE Program is exploring the use of SaaS cloud solutions wherever options are available and best meet overall requirements. Examples are in the core systems like enterprise asset management, work management, scheduling and dispatch, and field mobile as well as for data analytics and visualization.

Use of SaaS cloud solutions will provide several benefits including faster implementation and enhancement adoption, fewer upgrades to legacy infrastructure, easier upgrades when needed, reduced risk of obsolescence in the future, and the opportunity to enhance security. SaaS also provides strategic advantages by facilitating external interfaces with third party partners. SaaS can also be more easily scaled for additional capacity when required to enable growth

**Q. How does the GBE Program address cyber security?**

A. Protection of confidential customer information, asset data, and proprietary gas network information is essential to the success of the program. The program team is committed to meet or exceed National Grid's stringent

cyber security requirements, which are based on best practices in the utility and other industries. National Grid's Digital Risk and Security department will provide cyber security guidance in testing and development activities. Digital Risk and Security will also implement device and personnel authentication, monitoring for unauthorized access to information, cloud data security services, malware protection, and identity and access management control such as role-based access, single-sign on, provisioning/de-provisioning, and privileged access management.

The program also has a Cyber Security Architect dedicated to the project beginning in April 2017. In addition, the system integrator, existing partner suppliers, and security analysts will serve as supplemental cyber security experts.

It is important to note that the existing application estate is complex and includes a large number of applications that are at or near end of life. This requires a significant effort to maintain a safe and secure environment. The cloud-based environment the program will implement is far less complex and, therefore, presents less cyber security risk.

**Q. What training will be delivered as part of the GBE Program?**

A. Comprehensive training will be provided to all users of the systems, both field and office workers as well as first line and upper levels of management. Training will be delivered using various media such as computer-based instruction, video, classroom, mobile and written help guides. The GBE Program continues to focus on the upfront change management effort to impacted users through engagement roadshows, demos, and team hub meetings, as well as explore enhanced approaches such as "guided walk me tools" for user support and knowledge retention. Multiple sessions of training classes are scheduled to be delivered for users supporting KEDNY and KEDLI around November 2019.

**Q. How will the program team assess the readiness of the business to begin using the various functional parts of a project?**

A. The GBE Program team has been working with gas business leadership to identify business readiness requirements and develop business readiness checklists and go/no go checkpoints to ensure business readiness by geography. The approach focuses on the business not only being ready to receive the solution but also feeling accountable for making the decision. The business readiness approach utilized across each of the releases to date measures and tracks against an agreed set of go/no go criteria, including readiness of functionality, data, controls, new business processes

**Before the Public Service Commission**

**THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY AND KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID**

**Rebuttal Testimony**

**of**

**Christopher J. Connolly**

**September 18, 2019**

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## I. Introduction

**Q. Please state your name and business address.**

A. My name is Christopher J. Connolly. My business address is 404 Wyman Street, Waltham, Massachusetts 02451.

**Q. Are you the same Christopher J. Connolly who previously submitted testimony in these proceedings?**

A. Yes. Capitalized terms defined in my direct testimony have the same meanings here.

**Q. What is the purpose of your rebuttal testimony?**

A. The purpose of my rebuttal testimony is to respond to certain recommendations presented in the testimonies of the Department of Public Service Staff ("Staff") Policy Panel ("SPP") and Staff Information Technology Panel ("SITP"). Specifically, I will address the following topics:

(i) The SITP's recommendation to apply cost adjustments to the GBE Program, including the proposal to impose an overall cap on the amount of investment the Companies can recover for the GBE Program;

(ii) The SITP's recommendation of a downward-only net utility plant and depreciation expense tracker applicable to the GBE Program;

(iii) The SITP's recommendation to implement specific performance benchmarks related to the GBE Program; and

(iv) The SITP's recommendation to implement additional GBE Program reporting.

**Q. Please summarize the Companies' response to Staff's testimony.**

A. The Companies filed rates for the Rate Year commencing April 1, 2020, and submitted additional information to enable the Public Service Commission (the "Commission") to set rates for the subsequent three years (*i.e.*, April 1, 2021 through March 31, 2024, which are collectively referred to hereinafter as the "Data Years" and individually as a "Data Year"). Staff conducted a thorough review of the Companies' GBE Program; however, the SITP limited its testimony and recommendations to the Rate Year. As I explain below, in some instances, the Companies agree with Staff's recommendations, but in other areas we disagree. Where the Companies accept an adjustment or agree with an issue raised by Staff, such acceptance or agreement is affirmatively noted. Where the Companies disagree with Staff's position, or where more information is needed to clarify the record, I indicate that in my testimony.

## II. Response to Staff's Recommendations and Proposed Adjustments

**Q. Please summarize Staff's recommendations regarding the GBE Program.**

A. Staff's recommended adjustments and changes specific to the GBE Program are described primarily at pages 45-49 of the SITP's testimony. The SITP proposes:

(i) an adjustment to remove $57.4 million of investment from the GBE Program budget;

(ii) an overall cap on the amount of investment the Companies can recover for GBE;

(iii) a downward-only net utility plant and depreciation expense tracker for the GBE Program;

(iv) implementation of specific performance benchmarks related to GBE; and

(v) additional GBE-specific reporting.

In addition to the foregoing, Staff also recommends changes in the rate of return that would apply to National Grid USA Service Company, Inc. ("Service Company") Information Technology ("IT") investments, which also would have an impact on cost recovery by the Companies associated with the GBE Program.

**A. Proposed $57.4 Million Downward GBE Program Adjustment**

**Q. Please address the SITP's proposal to remove $57.4 million of investment from the Companies' GBE Program forecast budget.**

A. The SITP (pages 46-47) proposes an adjustment to reduce the investment forecast for the GBE Program by $57.4 million as follows: (i) $12.2 million reduction related to increased testing and training; (ii) $21.17 million reduction related to increased legacy project integration complexity; and (iii) $24.03 million reduction for allowance for funds used during construction ("AFUDC").

The Companies disagree with these proposed adjustments. As discussed in the Companies' response to Information Request ("IR") DPS-970, a copy of which is included in Exhibit ___ (SITP-1), pages 1438-1441, between the 2017 Niagara Mohawk Rate Case filing and the 2019 KEDNY/KEDLI filing in this proceeding, the total planned budget for the GBE Program increased from $458.1 million to $515.4 million. This amount excludes $20.1 million for the Strategic Assessment work in Fiscal Year ("FY") 2017 for the development of the GBE Program business case for which the Companies are not seeking recovery. At the time rates were set in the 2017 Niagara Mohawk Rate Case, the base budget for the GBE Program was forecast at $458.1 million. In addition to the base forecast, the GBE Program budget reflected

$61 million as a budget contingency in the event of unforeseen scope changes, changing market conditions affecting vendor and procurement costs, and unanticipated program complexity. Although Niagara Mohawk identified the $61 million amount as a budget contingency in its filing, no portion of it was included in Niagara Mohawk's case because the costs were not sufficiently certain at that time.

Since the 2017 Niagara Mohawk Rate Case, National Grid has completed the procurement process as well as launched the Detailed Design and Delivery Phase of the GBE Program, which firmed up the program spending details. As I describe in my direct testimony (pages 25-28), and as detailed in the Companies' responses to IRs DPS-684, DPS-725, and DPS-970, all of which are included in Exhibit __ (SITP-1), the GBE Program has experienced upward cost pressure and unanticipated program complexity due to several factors. These factors include greater than anticipated challenges and complexities replacing aged systems and developing legacy system interfaces for the Companies' two existing customer information systems ($21.168 million), increased workforce and subject matter expert testing and training and associated software costs ($12.196 million), and estimated AFUDC, which had been incorrectly omitted from the original program cost estimates ($24.025 million). Based on the most recent actual and projected cost

information, National Grid sanctioned an additional $57.4 million of costs from what was reflected in the 2017 Niagara Mohawk Rate Case for the GBE Program, and these updated costs are reflected in this case to set rates for KEDNY and KEDLI. Through August 2019, National Grid has incurred nearly half ($27.7 million) of this total sanctioned amount to advance and implement the GBE Program ($7.653 million for testing and training; $8.668 million to address legacy interfaces; and $11.415 on AFUDC).

The SITP's basis for the $57.4 million downward adjustment is to shield customers from increased costs of the GBE Program. Although I recognize the costs to implement the GBE Program are significant, they are being incurred to enable the Companies to continue to provide safe and reliable service for the benefit of customers into the future. Further, as I addressed in my direct testimony (pages 33-36), in addition to replacing aged and duplicate assets to support future operations, the GBE Program also provides the foundation for enhanced capabilities that support operational benefits and performance improvements. The savings to be delivered once the GBE Program is fully implemented are estimated to significantly exceed the investments in the enhanced capabilities of the program, as described in my direct testimony and shown in Exhibit __ (CJC-5). Absent evidence that such costs are unreasonable or imprudent, which Staff has not provided, the

Commission should approve the Companies' request to set its rates based on the updated GBE Program costs, which include the $57.4 million increase. The Commission should reject Staff's proposed adjustment.

**B. Proposed Cap on GBE Program Cost Recovery**

**Q. Please address the SITP's proposal to set an overall cap on the amount of investment the Companies can recover in rates for the GBE Program.**

A. Based on its recommendation to reduce the GBE Program expenditure allowance by $57.4 million, the SITP proposes to impose a cap on the portion of the GBE Program costs the Companies could recover. Specifically, the SITP would limit cost recovery by KEDNY based on maximum capital expenditures of $101.9 million and total operating expenses of $45.2 million. The SITP also would limit cost recovery by KEDLI based on maximum capital expenditures of $47.2 million and total operating expenses of $20.9 million. The Companies disagree with the SITP's recommendation to impose such caps on cost recovery.

As discussed above, the Companies disagree with the SITP's recommended adjustment to reduce expenditures by $57.4 million. These are reasonable and prudent costs to implement the GBE Program for the benefit of customers and are expected to provide efficiency and productivity savings as discussed in my

April 30 direct testimony. Furthermore, as I noted in my direct testimony (page 26), by the Rate Year, KEDNY and KEDLI collectively will have incurred more than $67.4 million in costs in connection with the GBE Program that are not reflected in their revenue requirements and therefore are costs borne by shareholders for the benefit of customers. If the Commission were to adopt the SITP's adjustment, the result would be that the Companies would have incurred in excess of $120 million in unremunerated costs to deliver a project that will have longstanding and significant benefits to customers. Absent a determination that the costs incurred to deliver the GBE Program are unreasonable or imprudent, which Staff has not shown, limiting recovery of costs based on an arbitrary cap that excludes the sanctioned $57.4 million in necessary program costs would be confiscatory and unreasonable.

### C. <u>Proposed Downward-Only Net Utility Plant and Depreciation Expense Tracker</u>

**Q. Please describe the SITP's recommendation that the Companies implement a downward-only net utility plant and depreciation expense tracker for the GBE Program.**

A. The SITP proposes that the Companies implement a downward-only net utility plant and depreciation expense tracker for the GBE Program similar to

what was adopted by the Commission in the 2017 Niagara Mohawk Rate Case.

**Q. Do the Companies agree with the implementation of a downward-only tracker in this case?**

A. No. Although a tracker may be appropriate for a multi-year rate plan, such as the one adopted by the Commission for Niagara Mohawk, the rationale for such a mechanism is less relevant for a litigated one-year case. Additionally, given that the program operates under a cap for Niagara Mohawk, the Companies are already incented to manage GBE Program costs. For these reasons, the Companies do not support a downward-only tracker for the GBE Program in these proceedings.

## D. Proposal to Implement Performance Benchmarks

**Q. What is the position of the Companies regarding the SITP's recommendation to implement specific performance benchmarks related to the GBE Program?**

A. Although performance benchmarks may be appropriate as part of a negotiated resolution of a multi-year rate plan, there is less of a basis for them in a one-year litigated case. Therefore, unless this matter were to proceed to negotiation that resulted in a multi-year resolution, the Companies do not

support establishing specific performance benchmarks for the GBE Program relative to KEDNY and KEDLI. In the event a negotiated resolution of the case was to be pursued, I recommend that any performance benchmarks that may be established reflect updates and learnings since the time of the 2017 Niagara Mohawk Rate Case; and also that the positive incentive proposal described in my direct testimony be incorporated as part of such a performance benchmark program.

**E. <u>Proposal to Modify Service Company Cost of Capital for Rate Setting Purposes</u>**

**Q. What is the position of the Companies on Staff's proposal to use its proposed cost of capital and corporate structure to set the rate of return applicable to GBE Program charges from the Service Company to the Companies?**

A. The Companies disagree with the SITP's proposal to apply the Staff's proposed cost of capital and capital structure to the GBE Program investments of the Service Company. The matter is addressed in the Rebuttal Testimony of Company Witness Jonathan Cohen.

### F. Proposed Reporting Requirements

**Q. Do you agree with the SITP's recommendation that the Companies implement the reporting and process requirements adopted in the Niagara Mohawk Joint Proposal for the GBE Program?**

A. Yes. The Companies agree to implement the same quarterly and annual reporting requirements for the GBE Program as they agreed to provide for Niagara Mohawk. Also, the Companies agree to hold semi-annual meetings with Staff similar as is done on behalf of Niagara Mohawk to (i) discuss budget and actual GBE Program spending to date; (ii) provide an update on the status of GBE Program initiatives that have been undertaken; and (iii) review quarterly filings.

## III. Conclusion

**Q. Does this conclude your testimony?**

A. Yes, it does.

procedures, engagement of all key stakeholders, labor relations, site and governance readiness, and the supporting operational model and the ability to realize expected benefits. The approach also includes incremental readiness checks and regular business forums commencing six months in advance of any go live to ensure sufficient time to manage and address any concerns. The process continues post go live with measurement and tracking of solution sustainment with metrics across people, process, and technology.

**Q. Does this conclude your testimony?**

A. Yes.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Now, with respect to your direct testimony Mr. Connolly, did you sponsor 6 Exhibits identified as C.J.C.-1 through C.J.C.-6?

A. Yes.

Q. Okay. Your Honors, those exhibits have been pre-marked for identification as exhibits -- as Hearing Exhibit numbers 52 through 57. And Mr. Connolly, were those exhibits prepared by you or under your direction and supervision?

A. Yes, they were.

Q. Okay. Thank you.

MR. GAVILONDO: Your Honors, the witness is available for cross examination.

A.L.J. COSTELLO: Thank you. Mr. Forst?

CROSS EXAMINATION

BY MR. FORST:

Q. Good afternoon. Being passed around right now is a response document to an information request titled D.P.S. 1045. It's a 3-page response document submitted or responded to on November 25th, 2019 and includes a -- an additional

31 pages of exhibits for a total of 34 pages. Just take a second to review that. Was this response prepared by you or under your direct supervision?

A. Yes.

Q. And is this a true and accurate copy of the company's response to information request D.P.S. 1045?

A. Yes, it is.

MR. FORST: Your Honors, I'd ask that this be given a pre -- a marked exhibit number?

A.L.J. COSTELLO: Sure, that will be marked as Exhibit 664.

MR. FORST: Thank you.

BY MR. FORST: (Cont'g.)

Q. Can you explain what a portfolio anchor is?

A. Sure. A portfolio anchor as referenced in our roadmap is a place in our timeline where we have a major capability release of the solution into our business.

Q. So another way of stating that is it's similar to a project milestone?

A. Yeah. That would be fair.

Q. And for the Gas Business Enablement Project or G.B.E., these milestones or project anchors are numbered to correspond to specific phases, is that correct?

A. That is correct.

Q. And would phases 1 and 2 refer to project implementation for National Grid's companies in Rhode Island?

A. Yes, that is correct, in addition to Massachusetts electric business.

Q. Can you specify where phases 1 and 2 implement assets for the Massachusetts companies?

A. So the only phase where we implemented in the Massachusetts electric business as released 2.3 July of 2019.

Q. Now you state in your response to D.P.S. 1045 that the portfolio anchor that are -- for the releases related to G.B.E. in Rhode Island were delayed. Can you explain to what date these releases were delayed?

A. The original release date for release 2.3 was April of 2019.

Q. And so I think you just stated earlier that 2.3 was completed in July of 2019?

A. That is correct.

Q. And the delays were necessary to address what you referred to in the response as defect resolution, is that correct?

A. Yes, that is correct.

Q. And were these defect resolutions meant to address specific issues related to National Grid's Rhode Island companies?

A. Correct, yes. National Grid's Rhode Island company as well as Massachusetts Electric.

Q. Is it correct that phase 3 refers to implementation in New York?

A. Correct.

Q. So in your response to D.P.S. 1045 specifically your response to questions 1 and 4 which I believe is on page 2 of the response document, you state that portfolio anchor 3.1 which refers to implementation in upstate New York was delayed from September to October 2019. And then you state it was further delayed to October 2020, is that correct?

A. I'm just reviewing here the response, give me a moment please.

MR. DECICCO: Counsel, did you mean April 2020, Phil Decicco for the company?

MR. FORST: If I mis -- if I misspoke and said April 2019, it's April 2020.

THE WITNESS: Thank you. That -- that's -- that's where I was confused in re -- reviewing the response again.

BY MR. FORST: (Cont'g.)

Q. Sorry. Just to clarify the record, that should read September 2019 to October 2019, to then April 2020.

A. And -- and that would be correct.

Q. Yeah. And what is the current anticipated date for the implementation of portfolio anchor 3.1?

A. The current anticipated release date for portfolio anchor 3.1 for our Niagara Mohawk Power Corporation Operating Company is April of 2020.

Q. Does anchor 3.1 include any of -- any other National Grid New York operating companies?

A. It does not.

Q. Referring to page 2 -- sorry, the first page of the attachments to the response document. This is the G.B. roadmap current. If you refer down to portfolio anchor 3, there's a purple box with then 2 purple arrows. I believe the bold language on the right-side of the roadmap states release 3.1 and 3.2 U.N.Y.N.L.I., April 2020. Could you clarify what that refers to?

A. Sure. So release 3.1 refers to our Niagara Mohawk Power Corporation, upstate New York U.N.Y. release. Release 3.2 refers to our Long island or KEDLI release.

Q. So would you say that the further anchor phases for example, you know, 3.2, 3.3, 4, 5, 6 are referred to activities which include some or all of the other National Grid New York operating companies?

A. Yes, that is correct.

Q. And in your response to D.P.S. 1045, you stated that these further phases are undergoing or have undergone various replanning activities, is that correct?

A. Yes, it is.

Q. And you stated that this was caused by cascading impact, is that correct?

A. Correct.

Q. Would you consider these cascading impacts to be a factor which contributes to the upward pressure on the budget contingency?

A. Yeah, it is, yes.

Q. And has there been any pro -- has there been any progress in the replanning activity?

A. There has been progress, yes. The team has progressed what we referred to as our Gas Business Enhancement reset or G.B.E. reset. The team is actively working through that reset and it is not completed at this time.

Q. When is the anticipated completion date of the G.B.E. reset?

A. The anticipated completion is approximately mid-March, during which time the recommendations will be presented to the senior leadership team.

Q. And that's March 2020?

A. March 2020, correct.

Q. And when you stated that the recommendations will be presented, is that

recommendations to be implemented or a battery of recommendations that need to be considered and then implemented?

A. It'll be -- well, it'll be a recommendation presented to the senior leadership for implementation.

Q. And included in those recommendations are there updated release dates for the further portfolio anchors listed in the roadmap?

A. Yes, there will be updated release dates.

Q. And are those updated release dates further or later in time than those indicated on the roadmap included in the response to D.P.S. 1045?

A. They will change. I can't speak to today what those changes are as I'm not involved in those efforts through the reset.

Q. Referring to your response to D.P.S. 1045 specifically question 3. You stated that the -- there's an update to the last projected G.B.E. budget of 569 million dollars, is that correct?

A. Yes, that is correct.

Q. Is there an update to that figure at this time?

A. As part of the reset, the team has continued to work on what the final cost to deliver the program will be. Again, I'm not part of those conversations or activity, so I don't know what the update is at this time.

Q. Has there been any reconsideration of the percentage of costs associated with the G.B.E. allocated to both KEDNY and KEDLI?

A. No, not to my knowledge.

Q. And haven't -- and to your knowledge, has any other National Grid company had their percentage share re -- redistributed or changed based on these updates?

A. No.

Q. Okay.

MR. FORST: No further questions, Your Honor.

A.L.J. COSTELLO: Any re-direct?

A.L.J. LEARY: Can I just clarify something before you do redirect because I heard a couple of dates KEDLI, Long Island, April 2020 for

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas release. I'm looking at the G.B.E. roadmap, is that correct?

THE WITNESS: The KEDLI release is now currently scheduled for May of 2020. So where it refers to here in the roadmap as April of 2020 is now May of 2020.

A.L.J. LEARY: And because of the reset, will that be further extended? That May 2020 date?

THE WITNESS: No. It -- it will not.

A.L.J. LEARY: Okay.

THE WITNESS: So what's happening right now is the team is currently in user acceptance testing preparing for both the April and May releases. And the decision to move forward, will be subject to a go no-go decision by both the program and the business. But they're currently on target for those release dates.

A.L.J. LEARY: Okay. And then I see what I think is a reference to KEDNY in New York City for October 2020 as a release. And that looks like it's 3.4?

THE WITNESS: That's correct.

A.L.J. LEARY: So there are actually 2 dates here relevant in the proceeding which would be May 2020 and October 2020?

THE WITNESS: Correct.

A.L.J. LEARY: Okay. Thank you.

MR. GAVILONDO: Your Honor, if you --

A.L.J. LEARY: Sorry. As a result of the reset, will the October 2020 date be further extended?

THE WITNESS: It -- it likely will. Again, I'm not involved in the most recent reset discussions and activities so I -- I can't speak -- I can't speculate on when that date will be.

A.L.J. LEARY: Well, are we talking about a month or 6 months?

THE WITNESS: I -- I don't know, I can't --

A.L.J. LEARY: Okay.

THE WITNESS: -- answer to that question.

A.L.J. LEARY: Okay. Thank you. Sorry about that.

MR. GAVILONDO: No, it's quite all right, Your Honor. Can I just have 2 minutes to --

A.L.J. LEARY: Yeah.

MR. GAVILONDO: Thank you.

A.L.J. COSTELLO: Off the record.

(Off the record 2:45 p.m.)

(On the record 2:46 p.m.)

MR. GAVILONDO: Okay. The company has no re-direct for Mr. Connolly.

A.L.J. COSTELLO: Okay. Thank you. Mr. Connolly, thank you for your testimony, and you're excused.

THE WITNESS: Thank you.

A.L.J. LEARY: Let's call D.P.S. Staff Policy Panel.

MR. GOODRICH: Your -- Your Honors, I just -- be -- I was wondering if given time constraints I know that the companies have brought up Mr. Williard who's an S.I.R. witness. And I know that if you want to -- if -- if it is desired to reshuffle a little bit, I know that the Staff Policy Panel would be available on, for example, Monday if -- if -- you know, they don't get on by the end of today. So if we wanted to reshuffle a little bit, that would be --

A.L.J. COSTELLO: Yeah, that's fine. If everybody --

MR. GOODRICH: -- possible?

A.L.J. COSTELLO: Is that agreeable --

A.L.J. LEARY: Is that okay?

A.L.J. COSTELLO: -- to the company to --

A.L.J. LEARY: To go ahead with Mr. Williard? Is that okay with everyone else because I see --

A.L.J. COSTELLO: PULP.

A.L.J. LEARY: -- PULP. You have questions A.R.P. has waived their questions, is that okay Ms. Wheelock and Mr. Berkley?

MS. WHEELOCK: Yes, Your Honors.

A.L.J. LEARY: Thanks. That was a good idea, I think.

(Discussions)

A.L.J. COSTELLO: All right. We'll go back on the record. Could the witness please state your name and business address for the record?

MR. WILLIARD: My name is Charles Williard.

A.L.J. COSTELLO: Okay. Please stand and raise your right hand. Do you swear or affirm that the testimony you'll provide is the truth?

MR. WILLIARD: Yes, Your Honor.

WITNESS; CHARLES F. WILLIARD; Sworn

A.L.J. COSTELLO: Thank you. You may be seated. And you may proceed.

DIRECT EXAMINATION

BY MR. EUTO:

Q. Good afternoon Mr. Williard. Can you hear me okay?

A. Yes.

Q. Do you have before you a document entitled Direct Testimony of Charles F. Williard dated April 2019, consisting of 28 pages, a table of contents and a cover sheet?

A. Yes, I do.

Q. Do you also have before you a document entitled Rebuttal Testimony of Charles F. Williard dated September 18th, 2019, consisting of 5 pages and a cover sheet?

A. Yes.

Q. Were all these documents prepared by you or under your supervision?

A.L.J. COSTELLO: Could -- can I just -- for clarification purposes, I -- the testimony was filed under the 2 separate documents I believe the direct testimony?

THE WITNESS: Yeah -- yeah. To be clear, Your Honors and apologies for that. I am speaking --

THE REPORTER: You need to speak in to your microphone as well.

THE WITNESS: I am speaking for case 19-G-0309 for the KEDNY docket.

A.L.J. COSTELLO: Okay. Thank you.

MR. EUTO: Thank you for that.

BY MS. EUTO:

Q. All right Mr. Williard, if I ask you the questions contained in those documents today, would your answers be the same?

A. (Inaudible)

Q. Do you adopt these documents as your sworn testimony as if given orally in these proceedings?

A. Go ahead, sorry. Can you repeat that?

Q. Do you adopt these documents as your sworn testimony as if given orally in these proceedings?

A. Yes.

Q. Your Honor, I ask that the direct and rebuttal testimony of Charles F. Williard be transcribed into the record as if given orally in the KEDNI docket.

A.L.J. COSTELLO: Okay. That request is granted. If you can also lay the foundation for the other documents I will just put them all in at one time.

MR. EUTO: Okay.

BY MR. EUTO: (Cont'g.)

Q. All right Mr. Williard, I'm going to turn to case 19-G-0310, the KEDLI docket. Do you have before you a document entitled Direct Testimony of Charles F. Williard dated April 2019 consisting of 17 pages, a table of contents and a cover sheet?

A. Yes.

Q. Do you also have before you a document entitled Rebuttal Testimony of Charles F. Williard dated September 18th, 2019, consisting of 5 pages and a cover sheet?

A. Yes.

Q. Were all these documents prepared by you or under your direction and supervision?

A. Yes.

Q. And if I ask you the -- the questions contained in these documents today, would your answers be the same?

A. Yes.

Q. Do you adopt these documents as your sworn testimony as if given orally in these proceedings?

A. Yes.

Q. Your Honor, I ask that the direct and rebuttal testimony of Charles F. Williard be transcribed into the record as if orally given in the KEDLI docket.

A.L.J. COSTELLO: That request was granted. At this point in the record the court reporter should insert the following files.

The first one is KEDNY Charles F. Williard direct testimony. The second is KEDLI Charles F. Williard direct testimony and the last is KEDNY-KEDLI Charles F. Williard rebuttal testimony. #

**Before the Public Service Commission**

**THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY**

**Direct Testimony**

**Of**

**Charles F. Willard**

**Dated: April 2019**

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## I. Introduction and Qualifications

**Q. Please state your name and business address.**

A. My name is Charles F. Willard. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

**Q. By whom are you employed and in what capacity?**

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid"), and currently hold the position of Director, Environmental Management. My responsibilities include overseeing the environmental and site investigation and remediation ("SIR") programs for National Grid's operating companies, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or "Company") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI").

**Q. Please describe your educational background and business experience.**

A. I am a graduate of the State University of New York at Geneseo with a Bachelor of Arts degree in Geology. In addition, I hold a Masters in Engineering Geology with a concentration in Environmental Engineering from Drexel University and a Masters in Business Administration from LeMoyne University.

I have been with National Grid since 1996. Prior to my appointment as Director, Environmental Management in 2014, I held the positions of Lead Environmental Engineer, Manager of New York SIR, and Director of SIR. Before joining National Grid, I held various management level positions in the field of environmental engineering and worked on projects such as environmental investigations, feasibility studies, remedial designs, and construction at large Environmental Protection Agency ("EPA") Superfund, Resource Conservation and Recovery Act, and New York State Superfund sites.

**Q. Have you previously testified before the New York State Public Service Commission (the "Commission")?**

A. Yes. I testified on behalf of KEDLI and KEDNY in Cases 16-G-0058 and 16-G-0059 (collectively, the "2016 KEDNY and KEDLI Rate Cases"), and on behalf of Niagara Mohawk Power Corporation d/b/a National Grid in Cases 17-E-0238 and 17-G-0239.

**Q. What is the purpose of your testimony?**

A. The purpose of my testimony is to support the SIR costs included in KEDNY's revenue requirement. Specifically, I will:

(i) provide an overview of KEDNY's SIR program;

(ii) discuss the historic level of SIR costs incurred by KEDNY, including those incurred during the twelve months ended December 31, 2018 ("Historic Test Year"), as well as the forecast costs for the twelve months ending March 31, 2021 ("Rate Year") and the twelve months ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (collectively, the "Data Years");

(iii) present KEDNY's proposal for recovery of SIR costs in the Rate Year and Data Years, including its proposal to address the significant level of projected SIR costs at the Newtown Creek and Gowanus Canal Superfund sites; and

(iv) discuss KEDNY's cost control procedures.

I will also address KEDNY's compliance with the rate case filing requirements adopted by the Commission in its November 28, 2012 Order in Case 11-M-0034 ("SIR Generic Order").

**Q. Do you sponsor any exhibits as part of your testimony?**

A. Yes. Attached to my testimony are the following exhibits and appendix that were prepared under my direction and supervision:

(i) Exhibit __ (CFW-1) provides details on work progress at KEDNY's

manufactured gas plant ("MGP") and Superfund sites;

(ii) Exhibit __ (CFW-2) is an example of a New York State Department of Environmental Conservation ("DEC") Order on Consent;

(iii) Exhibit __ (CFW-3) is the Unilateral Administrative Order for the remedial design of the Gowanus Canal Superfund Site;

(iv) Exhibit __ (CFW-4) is the Administrative Settlement Agreement and Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund Site;

(v) Exhibit __ (CFW-5) provides examples of changes to DEC remedy decisions following discussions with KEDNY;

(vi) Exhibit __ (CFW-6) is a copy of the DEC work schedules for fiscal years 2018, 2019, and 2020;

(vii) Exhibit __ (CFW-7) sets forth KEDNY's compliance with existing timetables and DEC requirements;

(viii) Exhibit __ (CFW-8) provides KEDNY's past SIR program spend on an annual basis for 2016 and 2017;

(ix) Exhibit __ (CFW-9) sets forth KEDNY's SIR program spend in the Historic Test Year;

(x) Exhibit __ (CFW-10) sets forth KEDNY's forecast SIR program spend in the Rate Year and Data Years; and

(xi) Appendix 1 sets forth the Company's cost control efforts.

## II. SIR Program Overview

**Q. Please provide a brief overview of KEDNY's SIR program.**

A. KEDNY's SIR program includes activities in connection with the management, investigation, and remediation of MGP and Superfund sites that have been contaminated by the past release of substances from property owned or formerly owned by KEDNY or its predecessors. KEDNY has responsibility for 27 MGP and two federal Superfund sites associated with waterways (Gowanus Canal and Newtown Creek) adjacent to former MGP sites. Both Superfund sites are in the remedial planning process. Of the 27 MGP sites, seven sites are in site characterization or remedial investigation, five sites are in remedial planning, one site is in remedial action and nine sites are have been either remediated but are subject to future monitoring and/or are subject to site management obligations. Five sites have received a "No Further Action" determination following site characterization and/or remediation, with four of the five no longer reported to DEC. As I will discuss later in my testimony, remediation and investigation activities at the Gowanus Canal and Newtown Creek Superfund sites are projected to cause KEDNY to incur significant SIR costs in the Rate Year, Data Years, and beyond.

Exhibit __ (CFW-1) describes the sites currently being managed by

KEDNY. The exhibit includes the background of each site, and details the investigation and remediation activities that have taken place over the last three years from January 2016 through the end of the Historic Test Year.

**Q. What types of costs does KEDNY incur under its SIR program?**

A. Costs under KEDNY's SIR program include consultant and contractor costs, remediation activities aimed at reducing the volume, toxicity, or mobility of pre-existing contamination, and incremental external costs (including insurance and legal costs) incurred to seek recovery from third parties or to otherwise mitigate the Company's SIR costs or liabilities.

**Q. What roles do the DEC and other regulatory agencies play with regard to the scope and timing of investigation and remediation work conducted at MGP and Superfund sites?**

A. The DEC and EPA control the scope and timing of work at MGP and Superfund sites. The scope of KEDNY's site investigations, work plans, clean-up, and field work decisions are reviewed, approved, and/or expanded by the DEC and/or the EPA pursuant to various orders. An example Order on Consent for 23 of KEDNY's MGP sites is provided in Exhibit __ (CFW-2). The Unilateral Administrative Order for the remedial design of the Gowanus Canal Superfund site and the Administrative Settlement

Agreement and Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund site are provided in Exhibit __ (CFW-3) and Exhibit __ (CFW-4), respectively. Although the Company has limited control over the scope and timing of SIR activities, KEDNY challenges the DEC and EPA when a more cost effective and equally protective remedy is available and, as described later in my testimony, manages costs in the areas it can control, such as contracting procedures. Exhibit __ (CFW-5) provides examples of changes to DEC remedy decisions and value engineering related modifications following discussions with KEDNY that resulted in cost savings to customers.

**Q. How is the schedule for work at MGP sites determined?**

A. Prior to the beginning of each New York State fiscal year (April 1 through March 31), KEDNY and the DEC meet to discuss the upcoming work schedule. The DEC approves the work schedule for the upcoming fiscal year only. As a result, the amount of spending in a given year is highly dependent upon the DEC and other third parties, including private property owners, permitting authorities, *et cetera*. While the DEC only approves a one-year schedule, in the course of meetings, the Company and the DEC discuss the anticipated logical progression of work at each site for an

additional year beyond the upcoming fiscal year, which gives the Company a sense of the projected work that will be required.

The DEC-approved schedule for the twelve months ended March 31, 2018, March 31, 2019 and March 31, 2020 is provided in Exhibit __ (CFW-6). The schedule includes KEDNY's 23 actively tracked MGP sites subject to DEC Orders on Consent. One site Jamaica Gas Light received a "No Further Action" determination during the past year. Three sites (Jamaica Holder Station, Keap Street Holder Station, and Rutledge Street Holder Station) have received "No Further Action" determinations and are no longer tracked with the DEC. Remediation at a fourth site, the Newtown Holder Station, was previously completed and the site closed in approximately 2002.

The schedule for work at the Gowanus Canal and Newtown Creek Superfund sites as required by the EPA is discussed later in my testimony.

**Q. Has the Company prepared an exhibit demonstrating that its remediation process is in compliance with existing timetables and DEC and EPA requirements?**

A. Yes. Exhibit __ (CFW-7) shows the status as of the end of calendar year

2018 for each site, indicates whether there was a difference between the scheduled activities and the activities actually completed and, if there was a difference, explains the reason. Variance from the timetable has occurred at six of KEDNY's MGP sites and were due to delays resulting from site access constraints, negotiated work scope changes due to changes in regulatory or site conditions, and regulatory reviews or approvals, all of which were outside of KEDNY's control.

**Q. Has KEDNY discussed the differences identified in Exhibit __ (CFW-7) with the DEC and EPA?**

A. Yes. KEDNY updates the DEC bi-monthly and the EPA monthly on the status of each site. Further, the Company's project managers communicate with the DEC's and EPA's project managers on project schedule and progress of work as necessary. Based on these updates, it is my understanding that the DEC and EPA are satisfied with KEDNY's progress.

## III. <u>SIR Program Costs</u>

**Q. What level of SIR costs does KEDNY currently recover annually through base rates?**

A. KEDNY currently recovers (i) historic, unrecovered SIR expenditures in an amount equal to one-tenth of the forecast SIR deferral balance as of

December 31, 2016 ($18.521 million per year) and (ii) forecast SIR costs of $53.872 million in calendar year 2017, $45.653 million in calendar year 2018, and $46.767 million in calendar 2019) (the "Forecast Rate Allowance"). These amounts were approved in the Commission's December 16, 2016 Order in the 2016 KEDNY and KEDLI Rate Cases. Because remediation and investigation activities at the Gowanus Canal and Newtown Creek sites were in their early stages at the time of the 2016 KEDNY and KEDLI Rate Cases, it was difficult to accurately forecast SIR expenses attributable to clean-up those sites. Consequently, forecast SIR costs associated with the Gowanus Canal and Newtown Creek sites are not currently included in base rates. However, given the uncertain timing and potential magnitude of these costs, the Joint Proposal adopted by the Commission authorized KEDNY to defer any costs related to those sites and implemented an SIR Recovery Surcharge that would be triggered if KEDNY incurred total SIR costs above a set threshold.

**Q. Does KEDNY fully reconcile SIR costs?**

A. Yes. Each year, KEDNY fully reconciles its actual SIR expense (inclusive of Gowanus Canal and Newtown Creek costs) to the Forecast Rate Allowance. Any under or over expenditures are deferred for future refund to or recovery from customers (with the exception of the Citizens site).

Under the Joint Proposal, KEDNY agreed to absorb 10 percent of the remaining investigation costs for the Citizens site pursuant to the Stipulation and Agreement Resolving Corporate Structure Issues and Establishing Multi-Year Rate Plan, dated June 25, 1996 in Case 95-G-0671.

**Q. Please explain how the SIR Recovery Surcharge operates.**

A. Beginning in calendar year 2018, if the difference between actual SIR expense (inclusive of Gowanus Canal and Newtown Creek costs) and the Forecast Rate Allowance exceeds $25 million on a cumulative basis, KEDNY is authorized to recover through the SIR Recovery Surcharge (i) the difference between actual SIR expense and the Forecast Rate Allowance and (ii) any amount that was not recovered in the prior rate year's SIR Recovery Surcharge (if triggered) because the cumulative difference between actual SIR costs and the Forecast Rate Allowance did not exceed the $25 million threshold and/or the amount would have increased KEDNY's aggregate revenues by more than two percent.

To date, KEDNY has not triggered the SIR Recovery Surcharge. However, as explained below, given the size of the costs associated with Gowanus Canal and Newtown Creek, the same circumstances that

supported Commission approval of KEDNY's SIR Recovery Surcharge still exist and warrant continuation of the mechanism.

**Q. What have the Company's historic SIR costs been?**

A. Prior to the Historic Test Year, during calendar years 2016 through 2017, KEDNY conducted investigation, remediation, and operations, maintenance and monitoring ("OM&M") activities pursuant to Orders on Consent or EPA Administrative Agreements under the DEC's and EPA's remedial programs at 23 of its MGP and Superfund sites. KEDNY incurred approximately $47 million and $52 million of SIR costs in 2016 and 2017, respectively, for these activities. While investigation and OM&M activities contributed to a steady and significant level of costs during this time period, increased spending was primarily due to the remedial construction activities at the Williamsburg MGP site and remedial program spending related to the Gowanus Canal and Newtown Creek Superfund sites.

**Q. What SIR costs did KEDNY incur in the Historic Test Year?**

A. KEDNY incurred $32.682 million in SIR costs in the Historic Test Year, as shown in Exhibit __ (CFW-9). Of this amount, $12.486 million was incurred for design work at Gowanus Canal and $7.489 million for remedial

investigation activities at Newtown Creek (totaling $19.975 million). The remaining $12.707 million was for investigation and remediation activities at 24 of KEDNY's MGP sites. These activities included: remedial planning, remedial action, or OM&M activities at 17 sites; site characterization or remedial investigation activities at seven sites; and SIR Program activities.

**Q. What is KEDNY's forecast of SIR costs for work at the Company's MGP sites?**

A. As shown in Exhibit __ (CFW-10), KEDNY forecasts SIR costs for work at MGP sites to be $66.088 million in the Rate Year and $62.635 million, $44.916 million, and $30.040 million in Data Years 1, 2 and 3, respectively. These forecasts do not include projected costs for Gowanus Canal or Newtown Creek, which are addressed separately below.

**Q. Please explain how KEDNY developed its forecast of SIR costs for work at its MGP sites.**

A. The DEC schedule is not yet available for the Rate Year or Data Years. Therefore, to develop the forecast, KEDNY utilized the current approved DEC schedule as well as the anticipated work schedule for fiscal year 2020. KEDNY took into account the status of each site and knowledge derived

from project managers regarding site progress and potential delays to determine the projected work on a site-by-site basis. KEDNY projected costs for each site based on the nature of the work at the site and the expected remedy for those sites where a remedy has been selected (*e.g.,* excavation of site soils, containment recovery, installation of barrier walls, treatment of groundwater, *et cetera*), site specific conditions, and past spending and estimates provided by consultants for the anticipated work. In instances where the project manager anticipated delays (*e.g.,* potential site access issues, permitting delays, or onsite condition constraints), the timing of the work was adjusted in the forecast.

**Q. What are the major drivers of SIR costs in the Rate Year for KEDNY's MGP sites?**

A. While KEDNY has made progress at most of its MGP sites, SIR spending above the Historic Test Year level is expected to continue through the Rate and Data Years. The increase in SIR costs at KEDNY's MGP sites (*i.e.,* excluding Gowanus Canal and Newtown Creek) from $12.707 million in the Historic Test Year to $66.088 million in the Rate Year is due to remedial action work that KEDNY anticipates will take place, and work that was delayed in the Historic Test Year that will now take place in the Rate Year. While the Company will continue its investigation and remediation

activities at all 24 sites worked on in the Historic Test Year, approximately 90 percent of the spending forecast for the Rate Year is due to activity at four sites and MGP Program activities. Specifically, KEDNY will be conducting significant remedial activities at the Citizens Gas Works, Fulton Municipal Gas Works, Metropolitan Gas Works, and Greenpoint sites, as well as MGP Program activities. This work is described below. Portions of the remedial plans at three of these sites (*i.e.,* Citizens, Fulton Municipal Gas Works, and Metropolitan Gas Works) must be completed on a schedule coordinated with the EPA's schedule for the Gowanus Canal remediation.

Citizens and Fulton Municipal Gas Works - Both sites were in the remedial planning (design) phase during the Historic Test Year, have commenced procurement of remedial contractor services, and are expected to progress to the remedial action phase in the Rate Year and Data Years resulting in a significant ramp up in spending. The work will include construction of deep barrier walls along the Gowanus Canal and soil excavation activities as part of the DEC-selected site remedies.

Metropolitan Gas Works - The Metropolitan site was in the remedial planning (remedy selection) phase during the Historic Test Year and is expected to progress to the remedial action phase in the Rate Year and Data

Years. The work is expected to include remedial measures that are protective of the Gowanus Canal. The DEC has not yet selected a remedy for the Metropolitan site but is expected to support the EPA's remedy schedule for the Gowanus Canal.

Greenpoint - Greenpoint was in the remedial investigation phase during the Historic Test Year. Given the size of this 100-acre site, the remedial investigation is being performed in multiple phases that may continue into the Rate Year. It is expected that, in consultation with the DEC, specific remedial objectives will be established, and interim remedial measure will be conducted, to begin addressing environmental conditions within specific areas at the site. In addition, Site Management Plan support activities are expected to address site related contamination on an as necessary basis through this period. The site is expected to transition into the remedial planning phase in the Data Years.

SIR Program Activities – Activities including pursuit of cost recovery from other parties and legacy insurance policies is expected to continue along with other SIR Program activities.

**Q. What are the major drivers of SIR costs in the Data Years for KEDNY's MGP sites?**

A. For Data Year 1, KEDNY forecasts spending to remain significant and at levels consistent with the Rate Year, at approximately $62.635 million. In Data Year 1, the four sites listed above and the MGP Program activities are expected to continue, as explained above. In addition, the Nassau site is expected to have higher spend for remedial construction and investigation and site management support activities.

In Data Year 2, KEDNY anticipates spending approximately $44.916 million. Significant spending is expected to continue at the Fulton site, and increased spends at the Greenpoint, Williamsburg, Jamaica Gas Light, and Nassau sites in support of remedial activities.

For Data Year 3, KEDNY forecasts spending to remain significant but lower than the Rate Year, Data Years 1, and Data Year 2 levels, at approximately $30.040 million. Significant spending is expected to continue at the Fulton and Greenpoint sites, but decrease at the Williamsburg and Nassau sites. Moderate spending at the Fulton site will continue in support of remedial activities.

**Q. Could actual SIR costs differ from the Company's forecast for the MGP sites?**

A. The forecast reflects the Company's best estimate of SIR costs. While KEDNY believes that its forecast is reasonable, SIR costs are subject to a high degree of variability, through no fault of the Company, as projects are subject to schedule and scope modifications by the DEC and the EPA, as well as site access issues with property owners or delays resulting from onsite condition constraints. In addition, estimates of future costs for SIR activities can be significantly influenced by the pace of redevelopment in areas where former MGP facilities are located. Areas of Brooklyn that historically were used for industrial and commercial purposes, particularly along waterfronts, are being redeveloped rapidly for residential and mixed use. Redevelopment at these properties or adjacent properties could significantly influence the schedule and scope of remedial activities at a number of sites. KEDNY's forecast reflects costs that it expects will be incurred in the Rate Year and Data Years; however, there is a risk that actual costs could be more or less. Accordingly, as discussed below, KEDNY is proposing to continue fully reconciling these costs.

**Q. Turning to the Gowanus Canal and Newtown Creek Superfund sites, please provide some background on these sites.**

A. Gowanus Canal - KEDNY had three former MGP facilities located along the Gowanus Canal – Fulton, Citizens, and Metropolitan. These former MGP facilities are individual remediation projects (as discussed above) and, along with other industries and processes located on or near the Canal, have contributed to the need for environmental remediation of the Canal itself, which is listed on the EPA's Superfund National Priorities List.

In 2013, the EPA identified a remediation plan for the Canal that involves removing contaminated sediment via dredging, installing a cap, and other activities. In 2014, the EPA issued a Unilateral Administrative Order, provided in Exhibit __ (CFW-3), to KEDNY and 26 other parties to perform the remedial design of the EPA's selected remedy in accordance with the schedule set by the EPA. Because of pressures to redevelop the area around the Gowanus Canal, the EPA led and expedited the completion of the remedial investigation and feasibility study in a little more than two years. The cleanup is being designed in multiple phases. Once the final design for each phase is approved, it is expected that active construction to remediate the Canal will begin and take place over a five to seven-year period. In addition to KEDNY, at least 26 other parties are responsible for

the remedial design of the Gowanus Canal under the Unilateral Administrative Order and could be potentially responsible for construction and cleanup under future EPA orders. KEDNY has been proactive in its efforts to research and work with the EPA to identify and engage these and other potentially responsible parties to equitably share the cost of investigation and remediation. KEDNY was initially approached to complete the work on its own, but refused, citing the multiple other parties that share responsibility.

Newtown Creek - At the Newtown Creek Superfund site, the remedial investigation and feasibility study commenced in 2011 and is in progress. This project is proceeding more slowly than Gowanus Canal because the EPA is pursuing a traditional remedial investigation conducted by a group of responsible parties. KEDNY and five other responsible parties have entered an Administrative Settlement Agreement and Order on Consent, provided in Exhibit __ (CFW-4), with the EPA to conduct the Newtown Creek remedial investigation and feasibility study. The remedial investigation of this site is in its final stages and the feasibility study to develop and evaluate remedial alternatives has commenced. However, it is likely to be some time before a remedy is evaluated, selected, and designed, and construction commences. During the remedy evaluation

process, it is expected that some supplemental investigation will be required. Given the size and complexity of this site, the remainder of the investigation, design, and construction is likely to be lengthy, extensive, and costly. The responsible parties have proposed an Early Action to the EPA and are working with to develop the scope and objectives for this expedited remedial work that will target a portion of Newtown Creek.

KEDNY continues to be very active in discussions with the other responsible parties and with the EPA to manage the process and ensure a fair and equitable characterization of impacts and allocation of responsibility. KEDNY also is working with the other responsible parties who executed the settlement agreement to develop information regarding the operations and resulting environmental impacts of still other responsible parties to enable KEDNY to seek cost recovery proportionate to its equitable share of liability.

**Q. Does the Company expect to incur SIR costs in the Rate Year and Data Years for the Gowanus Canal and Newtown Creek sites?**

A. Yes. KEDNY expects to incur significant SIR costs for work performed in the Rate Year and Data Years for both Gowanus Canal and Newtown Creek. The precise amount of these costs, however, remains difficult to

estimate with reasonable certainty given the dynamic scope of these projects and the number of parties involved.

With respect to the Gowanus Canal, the process of dredging and capping sediments in the first of three remediation areas along with the remedial design for areas two and three are expected to occur in the Rate Year. The completion of the remedial design, bulkhead construction along the canal, and the start of sediment dredging and capping in remediation areas two and three is anticipated during Data Years 1, 2, and 3. The work will be conducted in accordance with the EPA Record of Decision, which estimated the total project cost at $506 million. The pace of progress and the associated costs are highly dependent on regulatory drivers and third-party access and cooperation along the entire length of the canal.

As to Newtown Creek, work is expected to continue through the Rate Year and Data Years and involve the completion of both the Remedial Investigation and Feasibility Study process, with the projected issuance of a Record of Decision in 2024. The longer-term costs associated with the feasibility study and remedy selection process, the early action in a portion of Newtown Creek, and other project costs are highly uncertain given the changing scope and schedule of the project, which is being driven by the

EPA. While near term spending at the Gowanus Canal is expected to be higher, the total costs for Newtown Creek are likely to exceed those of Gowanus Canal, as Newtown Creek is larger, more polluted, and will require extensive design and remediation work.

**Q. What is KEDNY's proposal for recovery of its SIR costs?**

A. KEDNY proposes to include the following three cost components in base rates:

(i) forecast MGP-related SIR costs of $66.088 million in the Rate Year, $62.635 million in Data Year 1, $44.915 million in Data Year 2, and $30.040 million in Data Year 3;

(ii) $20 million in each of the Rate Year and Data Years for costs associated with remediation of Gowanus Canal and Newtown Creek, based on the combined spending on these sites during the Historic Test Year; and

(iii) the current amortization of $18.521 million annually, representing one-tenth of the forecast deferral balance at December 31, 2016.

While it is difficult to estimate the costs for the Gowanus Canal and Newtown Creek Superfund sites, there are certain baseline project costs that are expected to continue in the Rate Year and Data Years and can be

estimated with reasonable confidence. The combined spending for the Gowanus Canal and Newtown Creek during the Historic Test Year totaled approximately $20 million and is projected to be representative of the baseline spending for these two sites over the next several years. Consequently, the Company's forecast includes $20 million as the reasonable estimate of baseline costs for the Superfund sites.

KEDNY also proposes to continue the current SIR reconciliation, whereby any difference between actual SIR expense (including actual Gowanus Canal and Newtown Creek costs) would be reconciled to the base rate allowance (exclusive of the amortization), with any difference deferred for future recovery from, or credit to, customers. The provision in the current reconciliation related to the Citizen's site would continue. In addition, KEDNY proposes to continue the current SIR Recovery Surcharge.

**Q. Why does KEDNY believe that its proposal is consistent with the Commission's guidance and in the best interests of customers?**

A. The rate recovery proposal is consistent with the SIR Generic Order as it is tailored to address the specific conditions at KEDNY – namely, the rate allowance reflects significant forecast SIR costs combined with a means to recover a portion of KEDNY's historic deferral balance. In the SIR

Generic Order, the Commission noted its intent to retain the flexibility to "tailor the rate treatment of SIR costs to the concrete conditions of each utility" and moderate annual bill impacts by spreading the amortization of costs over a longer period "as needed and as dictated by the costs in specific rate cases." Increasing the rate allowance for forecast MGP-related SIR costs and including a baseline projection of costs for Gowanus Canal and Newtown Creek allows the Company to recover costs going forward and avoids adding to an already substantial deferral balance. Maintaining the SIR Recovery Surcharge enables the Company to mitigate bill impacts by not reflecting an overly aggressive forecast in base rates for Gowanus Canal and Newtown Creek, while addressing the uncertainty in forecasting expenditures should the costs for these two Superfund sites accelerate in the Rate Year and Data Year. These same circumstances justified the Commission's adoption of the SIR Recovery Surcharge in the 2016 KEDNY and KEDLI Rate Cases. The recovery proposal reflects the specific SIR cost issues affecting the Company and is intended to avoid future rate shock to customers.

## IV. SIR Cost Control Efforts

**Q. What steps has the Company undertaken to control its SIR costs and liabilities?**

A. KEDNY follows the cost management practices set forth in the Inventory of Best Practices for Utility SIR Programs established in the SIR Generic Order and as clarified by the DEC. Appendix 1 describes the Company's cost control efforts.

**Q. Has the Commission previously reviewed KEDNY's cost control measures?**

A. Yes. In its November 28, 2012 Order in Case 06-G-1185, which authorized KEDNY to recover a portion of its SIR deferral balance, the Commission found that KEDNY had employed cost effective measures and that the costs through December 31, 2009 had been prudently incurred. In its October 19, 2015 Order in Case 15-G-0323, the Commission reviewed the Company's oversight and cost containment procedures and found that they appeared reasonable. More recently, in its December 16, 2016 Order in the 2016 KEDNY and KEDLI Rate Cases, the Commission noted that Staff had reviewed and concurred that the Company was pursuing all appropriate cost control efforts.

## V. Compliance with Rate Case Filing Requirements

**Q. Are you familiar with the rate case filing requirements adopted by the Commission in the SIR Generic Order?**

A. Yes, I am. At page 30 of the SIR Generic Order, the Commission required that "[in] any future rate filing in which a utility seeks to recover SIR expenses, the utility must provide sworn testimony (1) establishing that the remediation process is in compliance with existing timetables and DEC requirements, or providing explanations for any divergence; (2) discussing the utility's SIR cost control efforts, including an attestation to utility compliance with the best practices inventory; and (3) indicating the results of any internal process the utility may have conducted with respect to review of SIR procedures, and in particular explaining how internal controls are brought to bear on site investigation and remediation projects."

**Q. Has the Company complied with each of these requirements in this filing?**

A. Yes, it has.

**Q. Please discuss the Companies' compliance with each of these requirements.**

A. Earlier in my testimony, I discussed the Company's compliance with existing timetables and DEC requirements, with reference to Exhibit __ (CFW-7). The Company's cost control efforts are also discussed earlier in my testimony and in Appendix 1, which details SIR cost control efforts.

With respect to the Company's internal controls and how they apply to SIR projects, National Grid has conducted internal reviews of the administration and management of the SIR group, as well as the adequacy of controls to meet significant environmental obligations. Further, as discussed above and in Appendix 1, the Company utilizes established procedures for selecting environmental consultants and contractors as well as a rigorous process for reviewing and approving consultant and contractor invoices as additional internal controls on SIR projects.

**Q. Does this conclude your testimony?**

A. Yes.

**Before the Public Service Commission**

**KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID**

**Direct Testimony**

**Of**

**Charles F. Willard**

**Dated: April 2019**

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## Introduction and Qualifications

**Q. Please state your name and business address.**

A. My name is Charles F. Willard. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

**Q. By whom are you employed and in what capacity?**

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid"), and currently hold the position of Director, Environmental Management. My responsibilities include overseeing the environmental and site investigation and remediation ("SIR") programs for National Grid's operating companies, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI" or "Company").

**Q. Please describe your educational background and business experience.**

A. I am a graduate of the State University of New York at Geneseo with a Bachelor of Arts degree in Geology. In addition, I hold a Masters in Engineering Geology with a concentration in Environmental Engineering from Drexel University and a Masters in Business Administration from LeMoyne University.

I have been with National Grid since 1996. Prior to my appointment as Director, Environmental Management in 2014, I held the positions of Lead Environmental Engineer, Manager of New York SIR, and Director of SIR. Before joining National Grid, I held various management level positions in the field of environmental engineering and worked on projects such as environmental investigations, feasibility studies, remedial designs, and construction at large Environmental Protection Agency ("EPA") Superfund, Resource Conservation and Recovery Act, and New York State Superfund sites.

**Q. Have you previously testified before the before the New York State Public Service Commission (the "Commission")?**

A. Yes. I testified on behalf of KEDLI and KEDNY in Cases 16-G-0058 and 16-G-0059 (collectively, the "2016 KEDNY and KEDLI Rate Cases"), and on behalf of Niagara Mohawk Power Corporation d/b/a National Grid in Cases 17-E-0238 and 17-G-0239.

**Q. What is the purpose of your testimony?**

A. The purpose of my testimony is to support the SIR costs included in KEDLI's revenue requirement. Specifically, I will:

(i) provide an overview of KEDLI's SIR program;

(ii) discuss the historic level of SIR costs incurred by KEDLI, including those incurred during the twelve months ended December 31, 2018 ("Historic Test Year"), as well as the forecast costs for the twelve months ending March 31, 2021 ("Rate Year") and the twelve months ending March 31, 2022 ("Data Year 1"), March 31, 2023 ("Data Year 2"), and March 31, 2024 ("Data Year 3") (collectively, the "Data Years");

(iii) present KEDLI's proposal for recovery of SIR costs in the Rate Year and Data Years; and

(iv) discuss KEDLI's cost control procedures.

I will also address KEDLI's compliance with the rate case filing requirements adopted by the Commission in its November 28, 2012 Order in Case 11-M-0034 ("SIR Generic Order").

**Q. Do you sponsor any exhibits as part of your testimony?**

A. Yes. Attached to my testimony are the following exhibits and appendix that were prepared under my direction and supervision:

(i) Exhibit __ (CFW-1) provides details on the progress of work at KEDLI's manufactured gas plant ("MGP") sites;

(ii) Exhibit __ (CFW-2) is an example of a New York State Department of Environmental Conservation ("DEC") Order on Consent;

(iii) Exhibit __ (CFW-3) provides examples of changes to DEC remedy decisions following discussions with KEDLI;

(iv) Exhibit __ (CFW-4) is a copy of the DEC work schedules for fiscal years 2018, 2019, and 2020;

(v) Exhibit __ (CFW-5) sets forth KEDLI's compliance with existing timetables and DEC requirements;

(vi) Exhibit __ (CFW-6) provides KEDLI's past SIR program spend on an annual basis for 2016 and 2017;

(vii) Exhibit __ (CFW-7) sets forth KEDLI's SIR program spend in the Historic Test Year;

(viii) Exhibit __ (CFW-8) sets forth KEDLI's forecast SIR program spend in the Rate Year and Data Years; and

(ix) Appendix 1 describes the Company's cost control efforts.

## I. SIR Program Overview

**Q. Please provide a brief overview of KEDLI's SIR program.**

A. KEDLI's SIR program includes activities in connection with the management, investigation, and remediation of MGP sites that have been contaminated by the past release of substances from property owned or formerly owned by KEDLI or its predecessors. KEDLI is responsible for 29 MGP sites, one of which, Belle Harbor, involves other potentially responsible parties ("PRPs") that share responsibility for the site with KEDLI. As of the end of the Historic Test Year, 16 sites were actively tracked with the DEC, including one site in the site characterization/remedial investigation phase and 15 sites in remedial planning, remedial action, or operations, maintenance and monitoring ("OM&M") phases of work. Seven of the 15 sites have received varying degrees of DEC regulatory closure. It is important to note that despite regulatory closure or a "No Further Action" finding, these sites may continue to have ongoing regulatory obligations for operations and maintenance and/or site management requirements. As of the end of the Historic Test Year, a total of 13 of the 29 sites have received a No Further Action finding, have no ongoing obligation and are no longer tracked with DEC.

Exhibit __ (CFW-1) describes the sites currently being managed by KEDLI. The exhibit includes the background of each site, and details the investigation and remediation activities that have taken place over the last three years from January 2016 through the end of the Historic Test Year.

**Q. What types of costs does KEDLI incur under its SIR program?**

A. Costs under KEDLI's SIR program include consultant and contractor costs, remediation activities aimed at reducing the volume, toxicity, or mobility of pre-existing contamination, and incremental external costs (including insurance and legal costs) incurred to seek recovery from third parties or to otherwise mitigate the Company's SIR costs or liabilities.

**Q. What role does the DEC play with regard to the scope and timing of investigation and remediation work conducted at MGP sites?**

A. The DEC controls the scope and timing of work at MGP sites. The scope of KEDLI's site investigations, work plans, clean-up, and field work decisions are reviewed, approved, and/or expanded by the DEC pursuant to Orders on Consent. An example Order on Consent for eight of KEDLI's MGP sites is provided in Exhibit __ (CFW-2). Although the Company has limited control over the scope and timing of SIR activities, KEDLI challenges the DEC when a more cost effective, and equally protective

remedy is available and, as described later in my testimony, manages costs in the areas it can control, such as contracting procedures. Exhibit __ (CFW-3) provides examples of changes to DEC remedy decisions following discussions with KEDLI that resulted in significant cost savings to customers.

**Q. How is the schedule for work at MGP sites determined?**

A. Prior to the beginning of each New York State fiscal year (April 1 through March 31), KEDLI and the DEC meet to discuss the upcoming work schedule. The DEC approves the work schedule for the upcoming fiscal year only. As a result, the amount of spending in a given year is highly dependent upon the DEC and other third-parties, including private property owners, permitting authorities, *et cetera*. While the DEC only approves a one-year schedule, in the course of meetings, the Company and the DEC discuss the anticipated logical progression of work at each site for an additional year beyond the upcoming fiscal year, which gives the Company a sense of the projected work that will be required.

The DEC-approved schedule for the twelve months ended March 31, 2018, March 31, 2019, and March 31, 2020 is provided in Exhibit __ (CFW-4). The schedule includes all 16 of KEDLI's active MGP sites subject to DEC

Orders on Consent. Thirteen sites (Bellmore, Brentwood, East Hampton GVF, East Hampton Hortonsphere, Garden City, Long Beach, Lynbrook, Manhasset, Pinelawn, Port Jefferson, Riverhead, Saltaire and Southold) have received "No Further Action" determinations and are no longer tracked with the DEC.

**Q. Has the Company prepared an exhibit demonstrating that its remediation process is in compliance with existing timetables and DEC requirements?**

A. Yes. Exhibit __ (CFW-5) shows the status as of the end of calendar year 2018 for each MGP site and includes a column to show whether there was a difference between the scheduled activities and the activities actually completed. There has been no variance from the timetable for the 16 sites that were actively tracked by DEC during 2018. Of the 16 sites, twelve are in the OM&M phase where either active maintenance and/or monitoring are taking place, or site management or inspections and reporting are required. Only four sites remain in the Remedial Investigation or Remedial Planning phases.

**Q. Please describe communications with the DEC for the work activities described in Exhibit __ (CFW-5)?**

A. KEDLI updates the DEC on a bi-monthly basis on the status of each site. Further, the Company's project managers communicate with the DEC's project managers on project schedule and progress of work, as necessary. Based on these updates, it is my understanding that the DEC is satisfied with KEDLI's progress.

## II. SIR Program Costs

**Q. How much SIR costs does KEDLI currently recover annually through base rates?**

A. KEDLI currently recovers (i) its historic, unrecovered SIR expenditures in an amount equal to one-tenth of the forecast SIR deferral balance as of December 31, 2016 ($14.168 million per year) and (ii) forecast SIR costs of $13.402 million in calendar year 2017, $7.442 million in calendar year 2018, and $4.648 million in calendar year 2019. These amounts were approved in the Commission's December 16, 2016 Order in the 2016 KEDNY and KEDLI Rate Cases.

Each year, KEDLI fully reconciles its actual SIR expense to the rate allowance for forecast SIR costs set forth above. Any under or over

expenditures of the rate allowance are deferred for future refund to or recovery from customers.

**Q. What is the Company's recent experience with SIR costs?**

A. During calendar years 2016 and 2017, KEDLI conducted investigation, remediation and OM&M activities under the DEC's remedial program at 20 of its MGP sites. KEDLI incurred approximately $19 million of SIR costs from 2016 through 2017 for these activities. As illustrated in Exhibit __ (CFW-6), KEDLI observed a declining level of spending from 2016 to 2017. The lower spending was attributable to less remediation construction activities during this period due to the completion of field construction programs at the larger sites, receipt of "No Further Action" determinations, and low levels of activity at the four sites with remaining future remedial obligations.

**Q. What SIR cost did KEDLI incur in the Historic Test Year?**

A. Actual spending during the Historic Test Year was approximately $8.3 million (compared to a rate allowance of $7.4 million) as shown in Exhibit __ (CFW-7). Pursuant to Orders on Consent with the DEC, KEDLI incurred costs for site investigation and remediation activities at 16 of 29 MGP sites during the Historic Test Year. The SIR activities during the

Historic Test Year included: remedial investigation at one site (Inwood Holder); remedial planning activities at three sites (Babylon, Belle Harbor, and Patchogue); and OM&M (including site management activities) at twelve sites.

**Q. What is KEDLI's forecast of SIR costs for work at the Company's MGP sites?**

A. As shown in Exhibit __ (CFW-8), KEDLI forecasts SIR costs of $6.6 million in the Rate Year and $6.6 million, $3.8 million, and $2.6 million in Data Years 1, 2 and 3, respectively.

**Q. Please explain how KEDLI developed its forecast of SIR costs for work at its MGP sites.**

A. The DEC schedule is not yet available for the Rate Year or Data Years. Therefore, KEDLI developed the forecast utilizing the current approved DEC schedule as well as the anticipated work schedule for fiscal year 2020. KEDLI took into account the status of each site and knowledge derived from project managers regarding site progress and potential delays to determine the projected work on a site-by-site basis. KEDLI projected costs for each site based on the nature of the work at the site and the expected remedy for those sites still requiring remedial activities (*e.g.,* excavation of site soils, containment recovery, installation of barrier walls, treatment of

groundwater), site specific conditions, and past spending and estimates provided by consultants for the anticipated work. In instances where the project manager anticipated delays (*e.g.,* potential site access issues, permitting delays, or on-site condition constraints), or negotiated with DEC to delay work to best manage project or program resources, the timing of the work was adjusted in the forecast.

**Q. What are the major drivers of SIR costs in the Rate Year and Data Years?**

A. Spending in the Rate Year is expected to decrease slightly from Historic Test Year levels, primarily due to work at three sites (Patchogue, Belle Harbor, and Babylon) progressing through remediation. The most costly of the remediation projects is the Patchogue site, where a portion of construction activities are expected to be performed in the Rate Year.

The SIR activities anticipated in the Data Years are comprised principally of OM&M related activities with some remaining remedial construction activities associated with the Inwood Holder site.

**Q. Could actual SIR costs differ from the Company's forecast?**

A. The forecast reflects the Company's best estimate of SIR costs. While KEDLI believes that its forecast is reasonable, SIR costs are subject to a high degree of variability, through no fault of the Company, as projects are subject to schedule and scope modifications by the DEC and environmental regulators, as well as site access issues with property owners or delays resulting from onsite condition constraints. KEDLI's forecast reflects costs that it expects will be incurred in the Rate Year and Data Years; however, there is a risk that actual costs could be more or less. Accordingly, as discussed below, KEDLI is proposing to continue to fully reconcile these costs.

**Q. What is KEDLI's proposal for recovery of its SIR costs?**

A. KEDLI proposes to recover SIR costs in the same manner as it does presently under its current rate plan. Specifically, KEDLI proposes a base rate allowance of $6.6 million, $6.6 million, $3.8, and $2.6 million in the Rate Year and Data Years 1, 2 and 3, respectively, which represents forecast SIR costs. In addition, to enable the continued recovery of historic SIR expenditures, base rates will continue to include the current amortization of $14.168 million annually, representing one-tenth of the forecast SIR deferral balance at December 31, 2016. KEDLI also proposes to maintain

the current SIR reconciliation, whereby any difference between actual SIR expense and the level of forecast SIR costs in rates would be reconciled, with any difference deferred for future recovery from, or credit to, customers. Finally, KEDLI proposes to continue the provision in its current rate plan that allows the Company to petition the Commission to recover unanticipated, incremental SIR costs through the SIR Recovery Surcharge.

**Q. Why does KEDLI believe that its proposal is consistent with the Commission's guidance and in the best interests of customers?**

A. The proposal is consistent with the SIR Generic Order because it is tailored to address the specific conditions at KEDLI. Specifically, in the SIR Generic Order, the Commission noted its intent to retain the flexibility to "tailor the rate treatment of SIR costs to the concrete conditions of each utility" and moderate annual bill impacts by spreading the amortization of costs over a longer period "as needed and as dictated by the costs in specific rate cases." KEDLI's proposal reduces its base rate allowance to align with a forecast decrease in SIR costs, while providing for the continued recovery of historic SIR expenditures in a manner that should fully resolve the Company's prior deferral balance.

## III. SIR Cost Control Efforts

**Q. What steps has the Company undertaken to control its SIR costs and liabilities?**

A. KEDLI follows the cost management practices set forth in the Inventory of Best Practices for Utility SIR Programs established in accordance with the SIR Generic Order and as clarified by the DEC. Appendix 1 describes the Company's cost control efforts.

**Q. Has the Commission reviewed KEDLI's cost control measures?**

A. Yes. In its November 28, 2012 Order in Case 06-G-1186, the Commission found that KEDLI had employed cost effective measures and that the costs through December 31, 2009 had been prudently incurred. In its October 19, 2015 Order in Case 15-G-0323, the Commission reviewed KEDNY's oversight and cost containment procedures, which are the same procedures as KEDLI's, and found they appeared reasonable. More recently, in its December 16, 2016 Order in the 2016 KEDNY and KEDLI Rate Cases, the Commission noted that Staff had reviewed and concurred that the Company was pursuing all appropriate cost control efforts.

## IV. Compliance with Rate Case Filing Requirements

**Q. Are you familiar with the rate case filing requirements adopted by the Commission in the SIR Generic Order?**

A. Yes, I am. At page 30 of the SIR Generic Order, the Commission required that "[in] any future rate filing in which a utility seeks to recover SIR expenses, the utility must provide sworn testimony (i) establishing that the remediation process is in compliance with existing timetables and DEC requirements, or providing explanations for any divergence; (ii) discussing the utility's SIR cost control efforts, including an attestation to utility compliance with the best practices inventory; and (iii) indicating the results of any internal process the utility may have conducted with respect to review of SIR procedures, and in particular explaining how internal controls are brought to bear on site investigation and remediation projects."

**Q. Has the Company complied with each of these requirements in this filing?**

A. Yes, it has.

**Q. Please discuss the Company's compliance with each of these requirements.**

**A.** Earlier in my testimony, I discussed the Company's compliance with existing timetables and DEC requirements, with reference to Exhibit __ (CFW-5). The Company's cost control efforts are also discussed earlier in my testimony and in Appendix 1, which details SIR cost control efforts. With respect to the Company's internal controls and how they apply to SIR projects, National Grid has conducted internal reviews of the administration and management of the SIR group, as well as the adequacy of controls to meet significant environmental obligations. Further, as discussed above and in Appendix 1, the Company utilizes established procedures for selecting environmental consultants and contractors as well as a rigorous process for reviewing and approving consultant and contractor invoices as additional internal controls on SIR projects.

**Q. Does this conclude your testimony?**

A. Yes.

**Before the Public Service Commission**

**THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY**
**AND**

**KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID**

# Rebuttal Testimony

**of**

**Charles F. Willard**

September 18, 2019

## I. Introduction

**Q. Please state your name and business address.**

A. My name is Charles F. Willard. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

**Q. Are you the same Charles F. Willard who previously filed direct testimony in these proceedings?**

A. Yes. The terms defined in my direct testimony have the same definitions here.

**Q. Mr. Willard, what is the purpose of your rebuttal testimony?**

A. I am responding to testimony from the Department of Public Service Staff ("Staff") Site Investigation and Remediation ("SIR") Panel recommending downward adjustments to the Companies' proposed rate allowances for SIR programs, and testimony from the Public Utility Law Project ("PULP) witness William Yates regarding impacts to customers in the vicinity of certain clean-up sites.

**Q. Do you agree with the Staff SIR Panel's recommendation (at 29) to reduce KEDNY and KEDLI's Rate Year forecast of SIR expense by**

**$15.168 million and $1.734 million, respectively?**

A. Not entirely. The Companies acknowledge that SIR work schedules are subject to a high degree of uncertainty and reflected this variability in the SIR forecasts. As described in my direct testimony, the Companies took into account the status of each site and potential delays to determine projected work on a site-by-site basis. As part of this evaluation, the Companies reviewed the last six years of historic spend and adjusted the forecast for KEDNY based on that historical underspend. In addition, the latest available information indicates that key elements driving the Companies' respective forecasts, including increased investigation and remediation activities at a number of active SIR sites, are still anticipated to occur. Therefore, the Companies believe that their forecasts, which are based on an extensive review of SIR activities at each individual MGP and Superfund site, are the best indicator of SIR Program spend in the Rate Year.

**Q. Mr. Willard, please summarize key elements driving the Rate Year forecast for KEDNY.**

A. For KEDNY, the main drivers for Rate Year expenditures are investigation and remediation activities at the Fulton, Citizens, Metropolitan and Greenpoint sites along with related SIR program

spending, including insurance recovery efforts. Work at the Citizens, Fulton, and Metropolitan sites is performed on a schedule that is being coordinated by the EPA. Since my direct testimony was filed, construction contracts have been issued for construction of barrier walls at the Citizens and Fulton sites, and for soil remediation at the Citizens site. Field work associated with both contracts has begun and is expected to extend into the Rate Year as originally anticipated. Discussions with property owners regarding the remediation at the Metropolitan site has commenced, which could facilitate an increase in work at that site. Further, KEDNY's insurance recovery litigation trial has been scheduled for the Fall of 2020, during the Rate Year, which will drive legal and expert expenses associated with trial preparation during the Rate Year.

**Q. Mr. Willard, please summarize key elements driving the Rate Year forecast for KEDLI.**

A. For KEDLI, the main drivers for Rate Year expenditures are activities at the Bay Shore, Babylon, Belle Harbor, and Patchogue sites. The Bay Shore site is in the operation, maintenance, and monitoring phase with predictable spends, and KEDLI has recently obtained access to the Babylon site, with pre-design investigations planned for Fall 2019. Further, the Patchogue site is currently in the remediation construction

phase at this time, so the work projected for the Rate Year for this site is unlikely to be delayed or impacted.

**Q. Do you believe that the Staff SIR Panel's proposed adjustments are necessary to more closely align rate recovery with project spending?**

A. No. Based on latest available information, the Companies believe that key work activities driving the Companies' forecast of SIR expenditures during the Rate Year will occur, and thus, the Companies proposed rate allowances should be used in favor of the Staff SIR Panel's proposed alternatives.

**Q. Do you agree with Mr. Yates assertions (at 22-23) regarding recovery of clean-up costs from customers who live in the vicinity of superfund sites based on assertions that the impacts on such customers are only negative?**

A. No. Mr. Yates' asserts (at 22) that it is against the public interest to expect customers who live in the vicinity of superfund sites to pay for the costs to clean-up these sites. Mr. Yates suggests (at 22-23) that these efforts bear only negative impacts for such customers. In the Order in the 2016 KEDNY and KEDLI Rate Cases, the Commission noted that utilities are required by law to incur SIR expenses and they should "therefore be

treated as normal costs of doing business in today's society" (at 83-84). Mr. Yates' assertions ignore the fact that the Companies' efforts to clean up these sites benefit both the sites and the surrounding communities. The Companies' efforts not only remedy historical impacts, but also enable various projects with substantial benefits for customers in the vicinity of such sites, such as the creation and/or improvement of public parks and support for brownfield redevelopment. For these reasons, Mr. Yates' assertions should be rejected.

**Q. Does this conclude your testimony?**

A. Yes.

BY MR. EUTO: (Cont'g.)

Q. Okay. Moving on to your exhibits. For your direct testimony, did you also sponsor in the KEDNY docket 11 exhibits pre-marked for identification as Exhibits 172 through 181 consisting of C.F.-1 through C.F.W.-10 -- let me -- let me back up, C.F.W.-1 through C.F.W.-10 and Exhibit 613, which is appendix 1-KEDNY cost control efforts?

A. Yes.

Q. Were these exhibits prepared by you or under your supervision and direction?

A. Yes.

Q. All right. In addition for your direct testimony in the KEDLI docket, did you also sponsor 9 exhibits pre-marked for identification as Exhibits 182 through 189 consisting of C.F.W.-1 through C.F.W.-8. And Exhibit 614 consisting of Appendix 1-KEDLI cost control efforts?

A. Yes.

Q. Were these exhibits prepared by you or under your supervision and direction?

A. Yes.

Q. Thank you. Your Honors, the witness is available for cross examination.

A.L.J. COSTELLO: Okay. Thank you. PULP?

MS. WHEELOCK: Thank you, Your Honors.

CROSS EXAMINATION

BY MS. WHEELOCK:

Q. Good afternoon, Mr. Williard. My name is Laurie Wheelock. I'm an attorney with the Public Utility Law Project. My questions are going to focus on your -- primarily your KEDNY initial testimony. And today I'm going to be asking questions about Site Investigation Remediation cost. For the record, those are SIR cost and I do apologize I'll probably revert to the acronym a lot.

So with the first question, if you could turn to page 5 in the KEDNY direct testimony and focus on lines 6 through 9 please. And you can let me know when you found that.

A. I'm sorry page?

Q. 5, lines 6 through 9 please.

A. Okay.

Q. Can you clarify what the current status is for both superfund sites?

A. Sure. The 2 super -- superfund sites I believe you're referring to are the Gowanus Canal and Newtown Creek. Currently, Gowanus is in design, our -- we are nearing submission of our final design to the E.P.A. that should be submitted next week. And we are expecting -- right now, we are scheduled to begin the workout in the canal in August. For Newtown Creek, it is still under investigation. We don't expect a -- an evaluation of alternatives until another year or 2.

Q. Thank you. Now if you could turn to page 10 and review lines 20 through 21 that then turns into page 11, lines 1 through 4 please.

A. Okay.

Q. To your knowledge, are there any other sites in either KEDNY or KEDLI'S territory like the citizen site where the company has ever agreed to absorb any percentage of the remaining investigation costs?

A. No, these sites were entered into prior to the generic proceeding.

Q. Has KEDNY ever refunded customers when there are any under-expenditures for SIR expenses?

A. Not to my knowledge.

Q. And if not, what has KEDNI done with the unexpended funds when there are under-expenditures?

A. Historically, it was a deferral balance. So we are currently recovering 10 years' worth of expenses prior to the last rate agreement, have been amortized over a 10-year period. So those were not collected initially, they were collected after the work was completed.

Q. Thank you. You can turn now to page 19 in your testimony reviewing lines 21 that run over to page 20, lines 1 through 3 please.

And this part of your testimony is about the Gowanus Canal.

A. Okay.

Q. Has the number of potentially responsible parties for the Gowanus Canal Superfund site changed at all since April 2019 to your knowledge?

A. No.

Q. Okay. Moving on to Page 20, lines 3 through 8. Do you expect any of the potentially responsible parties that have been

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identified to help pay for the investigation or remediation costs in this rate case year?

A. Yes. This year, that has yet to be determined.

Q. Okay. What about in future years has the company?

A. Yes.

Q. Thank you. Now moving on to Newtown Creek which is also on Page 20 but looking at lines 14 through 17.

A. Okay.

Q. Are all 5 of the potentially responsible parties that have been identified in the administrative settlement agreement in order on consent still active or have any dropped out?

A. Since the beginning of the work, one party has dropped out.

Q. Do you know if there is any other additional parties that have been identified potentially in the rate year to come?

A. I'm not aware of any, no.

Q. Okay. Turning now to your rebuttal testimony, Page 3. And this will -- I guess it's both KEDNY and KEDLI.

A. Page 3?

Q. Yeah, lines 10 through 12, please. So here it says further KEDNY's insurance recovery litigation trial has been scheduled for the fall of 2020 during the rate year, which will drive legal and expert expenses associated with trial preparation during the rate year. Can you just clarify which insurance recovery litigation trial this is?

A. Can -- can you restate where -- where it's --

Q. Of course.

A. -- located on that page? Sorry.

Q. So -- no problem. In your rebuttal testimony.

A. Okay.

Q. It's page 3, lines 10 through 12.

A. I'm having problems locating it, but if you can just read it to me I'll be able.

Q. Of course and I can --

A. Yeah.

Q. -- give you my copy.

A. Could you -- thank you. Okay.

Q. So if you could just clarify which trial that is for the record.

A. We are pursuing in court, both Munich Insurance and Century.

Q. And are those tied to any specific sites to your knowledge?

A. They are for the Gowanda's Canal sites.

Q. Thank you.

A. Gowanda canal and -- and the sites associated.

Q. So you can turn back now to your additional testimony page 27. And once you're at page 27, if you could just review lines 18 through 21, please.

A. Okay.

Q. To your knowledge has KEDNY conducted any analysis on the bill impacts associated with its SIR costs, including the recovery surcharge in this rate year?

A. Our rates requirement group has done that. Yes.

Q. And has KEDNY ever considered sharing any of the costs associated with the SIR expenses in this rate year?

A. We've been operating under the generic order from 2012. So we have been following the guidelines that are included in that order from the commission to ensure that we are following the guidelines in that -- that order. So no.

Q. And my last question has to do with the Environmental Protection Agency. Living in uncertain times with the federal level and changes that occur to federal rules, laws and regulations, what, if any, planning has the company done in case there are dramatic changes to the federal superfund law?

A. We don't anticipate significant changes to the superfund law. There are changes in agency fundings, but the work that -- work undergoing are funded separately through orders on consent or unilateral orders. So it's separate from the federal government’s funding.

MS. WHEELOCK: No further questions.

A.L.J. COSTELLO: Do you --

A.L.J. LEARY: I have a couple of questions.

A.L.J. COSTELLO: Sure.

A.L.J. LEARY: On the -- I believe it's Gowanda so you -- you're down to now P.R.P. group of 4 plus KEDNY or are you at 5 plus -- plus KEDNY?

THE WITNESS: So --

A.L.J. LEARY: 3, I'm sorry.

THE WITNESS: So all the parties still exist. The latest order that came out from the E.P.A., there was -- there has been a series of orders that have come out from the E.P.A. unilateral orders.

A.L.J. LEARY: Uh-huh.

THE WITNESS: The earlier orders had named all of the parties, but now that we're getting towards the implementation, they're naming the larger parties, which are 5 other parties in addition to KEDNY.

A.L.J. LEARY: And so it's not 4 plus 1 --

THE WITNESS: It's --

A.L.J. LEARY: -- it's 5 plus 1.

THE WITNESS: -- 5 plus 1, but all the other parties are still in -- in it, but not necessarily named in the order.

A.L.J. LEARY: Not necessarily named in the order.

THE WITNESS: Yeah, they are not named in the last order. They were named in the prior orders.

A.L.J. LEARY: And that's a remedial design order?

THE WITNESS: No, the latest or -- latest 2 orders so they're -- the first order for design was a preliminary order to get us out into the field to get preparations done in order to keep the schedule. The order that just came out last month was to complete the upper end of the canal, so the first of the third sections of the canal.

A.L.J. LEARY: And who are those other parties and let me ask you this. Do you have a joint defense agreement with those other parties, the 5 plus KEDNY?

THE WITNESS: For implementation, we don't have anything in place at this point.

A.L.J. LEARY: Do you have some kind of agreement that has percentages designated for each of those parties in terms of liability?

THE WITNESS: For -- for the remediation, no we have -- we have just received the order as of last month. So those discussions are ongoing.

A.L.J. LEARY: Was there any percentage that you had on the remedial investigation side, those costs? You're smiling Mr. Williard.

THE WITNESS: You know, just -- just from -- from the confidentiality perspective, we -- we are ongoing in discussions with the other parties for the -- for the remediation, the prior work. There is -- it is subject to confidentiality. But the more information we provide in a public forum, the more it puts us at risk in our --

A.L.J. LEARY: Very helpful.

THE WITNESS: -- negotiations.

A.L.J. LEARY: I don't want to press you on this, but I'm going to request that this question be answered in and shared with the parties who are subject to the confidentiality agreement. It's -- it gives me context, it helps me understand

what's up with KEDNY as well as the other parties prospectively even in just the rate year.

So I guess I have to assume that that percentage will have some correlation with what the percentage may be under the -- the new E.P.A. unilateral 113 order, right?

THE WITNESS: It --

A.L.J. LEARY: It may -- it may bear some.

MR. EUTO: Your Honor, there's another issue that I think we need to raise at this point and that is that one of the parties to this proceeding is also a potential P.R.P., being the city.

A.L.J. LEARY: Oh, oh, the city.

MR. EUTO: And also a signatory to the protective order and so if we --

MR. O'BRIEN: Patrick O'Brien. So that sort of complicates the company responding to that request.

A.L.J. LEARY: Okay. I'm just going to stay my request for now. Let me talk to Judge Costello about that, and so I was going to ask who the other parties are. The city is another party and -- and who are the other parties?

THE WITNESS: If I could recall off the top of my head, Honeywell was one of them. And I'd have to take a look at the list of other parties. A lot of the parties that you would have, you know, that are on the order, I think, I'm not Sure if Kraft is one of the larger ones. But a lot of the names aren't necessarily traditionally what you would think of as a polluting party, but they've morphed over the years.

A.L.J. LEARY: Right.

THE WITNESS: So I'd have to take a look at the list on the order.

A.L.J. LEARY: Okay. Thanks. That's very helpful. And again, I'm not asking the company to do anything at this point, but I may need further information in this regard, which I'm wondering how I get without somehow prejudicing the company's negotiations on the P.R.P. agreement. And I don't want to do that. So that's all I have. Ms. Wheelock, you have something else resulting from what I just asked the witness.

MS. WHEELOCK: Just briefly, Your Honor, as far as the name of potentially responsible parties, there was an exhibit. If you just want me

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to direct you to that, within your -- your testimony, it's Exhibit C.F.W.-3 page 1 of 25 it's pre-marked as Exhibit 184.

A.L.J. LEARY: I see. But this is the old unilateral 113 order and it's 1 of 6 order and it's not the 5 -- it's not directed, right, at the most recent.

THE WITNESS: Correct.

A.L.J. LEARY: Was this the most recent?

THE WITNESS: It -- it was the latest information that came out last month.

A.L.J. LEARY: So I think that, you know, this is -- but in the record, we don't have that more limited P.R.P. group. Is that right, we don't have that order --

THE WITNESS: Correct.

A.L.J. LEARY: -- publicly available.

A.L.J. COSTELLO: That's iden --

THE WITNESS: It's publicly available.

A.L.J. COSTELLO: -- identified.

THE WITNESS: It is publicly available.

A.L.J. COSTELLO: It is publicly available.

A.L.J. LEARY: Okay.

THE WITNESS: Yeah.

A.L.J. LEARY: Thanks.

A.L.J. COSTELLO: D.P.S. staff.

MR. FORST: Yes, we just have one document we'd like to get on the record. So we'll pass that out now.

CROSS EXAMINATION

BY MR. FORST:

Q. So being passed around right now is a document request response, titled D.P.S. 1031. It consists of a 2-page response document issued on November 15, and 4 pages of attachments for a total of 6 pages. Can you just review that really quickly?

A. Yes.

A.L.J. COSTELLO: Okay. And we're going to mark that as Exhibit 665.

BY MR. FORST: (Cont'g)

Q. And was this exhibit prepared by you or under your direct supervision?

A. Yes.

Q. And is the information contained in this document true and correct to the best of your knowledge?

A. Yes.

Q. Excellent.

MR. FORST: No further questions, Your Honor.

A.L.J. COSTELLO: Any redirect, you want to take a moment -- moment or?

MR. EUTO: No, Your Honor, no redirect. Thank you.

A.L.J. COSTELLO: Okay. Thank you. Mr. Willard, thank you for your testimony and you're excused.

A.L.J. LEARY: Yeah, I think we need a break right. Is that okay, off the record.

(Off the record 3:09 p.m.

(On the record 3:25 p.m.)

A.L.J. COSTELLO: I'm going to ask the witnesses to please one at a time, state your name and business address for the record.

MR. CASTANO: John Castano, 3 Empire State Plaza, Albany, New York.

MR. FLAUM: Jeremy Flaum, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MR. CASTANO: Yes.

MR. FLAUM: Yes.

WITNESS; JOHN CASTANO; Sworn

WITNESS; JEREMY FLAUM; Sworn

A.L.J. COSTELLO: Thank you. You may be seated. Counsel.

DIRECT EXAMINATION

BY MR. FORST:

Q. Panel members, before you is a document entitled prepared testimony of Staff Site Investigation and Remediation Panel consisting of a cover page and 30 pages of question and answers dated August 2019, which has both a redacted and confidential version as well as 4 exhibits submitted with your testimony labeled S.I.R.-1 redacted, S.I.R.-1 confidential, S.I.R.-2 and S.I.R.-3, is that correct?

A. (Flaum) That's correct. Three exhibits.

Q. MR. FORST: Yeah. These exhibits have been pre-marked as exhibits 506 through 509. Was this set of testimony and exhibits prepared by you or under your direct supervision?

A. Yes.

Q. Do you have any changes or corrections to make to that tes -- to make to that testimony?

A. No.

Q. If you were asked the same questions today under oath, would you answer them the same way?

A. Yes.

Q. And do you affirm the information contained in your testimony and exhibits true to the best of your knowledge, information and belief?

A. Yes.

MR. FORST: Your Honors, I would like to move that the pre-filed testimony of the Staff Site Investigation Remediation Panel be entered into the record as if given orally today during the hearing.

A.L.J. COSTELLO: That request is granted. And at this point, in the public version of

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the transcript, the court reporter should insert D.P.S. staff, S.I.R. Panel redacted testimony and in the separate confidential trans -- transcript it should be inserted the following file D.P.S. staff S.I.R. Panel confidential testimony. #

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

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In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

August 2019

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Prepared REDACTED Testimony of:

Staff Site Investigation and
Remediation Panel

John P Castano
Auditor 2 (Public Utilities)
Office of Accounting, Auditing
and Finance

Jeremy Flaum
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State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Please state the names, employer, and business address of the Staff Site Investigation and Remediation (SIR) Panel.

A. Our names are John Castano, and Jeremy Flaum. We are employed by the New York State Department of Public Service (Department). Our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Mr. Castano, are you a member of the Staff Revenue Requirement Panel and are your credentials contained in that testimony?

A. Yes.

Q. Mr. Flaum, what is your position at the Department?

A. I am employed as a Utility Analyst 3 in the Environmental Certification and Compliance Section of the Office of Electric, Gas and Water.

Q. Please briefly describe your educational background and professional experience.

A. I graduated from the State University of New York College at Cortland in 2003 with a Bachelor of Science degree in Geology. I also earned a Master of Science degree in Environmental

Management from the University of Maryland, University College, in 2008. I joined the Department in 2009. Prior to joining the Department, I held Geologist positions at two environmental consulting firms where I performed field investigations, oversight, and data analysis for multiple environmental remediation sites.

Q. Please describe your responsibilities with the Department.

A. My primary responsibilities include evaluating environmental and land use impacts for the siting of electric and gas transmission facilities and electric generating facilities under Articles VII and 10 of the New York State Public Service Law. Additionally, I have been assigned to review utility SIR cost control practices and have provided recommendations for previous rate cases before the Public Service Commission of the State of New York (Commission).

Q. Have you provided testimony in previous Commission proceedings?

A. Yes. I previously testified as part of

Department Staff's SIR Panels for several rate cases. Most recently, I testified in Cases 18-E-0067 and 18-G-0068, Orange and Rockland Utilities, Inc.; 17-E-0459 and 17-G-0460, Central Hudson Gas and Electric Corporation; and Cases 17-E-0238 and 17-G-0239, Niagara Mohawk Power Corporation d/b/a National Grid. I have also testified before the Commission regarding geologic and water resource impacts of proposed electric transmission projects in Cases 08-T-0034 and 10-T-0139, and before the New York State Board on Electric Generation Siting and the Environment regarding geologic and water resource impacts, and as part of the Staff Policy Panel, for proposed wind energy facilities in Cases 14-F-0490, 15-F-0122, 16-F-0062, 16-F-0328, 16-F-0559, and 16-F-0205.

Q. Please summarize the scope of the SIR Panel's testimony.

A. Our testimony consists of our review of the SIR programs and costs of the KeySpan Gas East Corporation d/b/a National Grid (KEDLI) and the Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY), including the Companies' SIR

practices, historic and forecasted expenditures, and rate allowance and SIR surcharge requests. We then present our findings regarding the effectiveness of the Companies' SIR cost mitigation strategies and procurement practices for SIR work; the Companies' compliance with regulatory requirements applicable to their SIR sites; and the reasonableness of the Companies' projected SIR expenditures and proposed rate allowances related to the recovery of their SIR program costs. Our findings concerning these issues form the basis for our recommendations to the Commission.

Q. Is the Panel sponsoring any exhibits?

A. Yes, we are sponsoring three exhibits; Exhibit___(SIR-1), Exhibit___(SIR-2), and Exhibit___(SIR-3).

Q. Please describe Exhibit___(SIR-1).

A. Exhibit___(SIR-1) includes the information request (IR) responses we relied upon and have referenced throughout our testimony. We will refer to these IRs by the number assigned by the Department, for example DPS-100.

Q. Please describe Exhibit___(SIR-2).

A. Exhibit___(SIR-2) consists of the workpapers that were created in calculating the adjustments referenced in our testimony.

Q. Please describe Exhibit___(SIR-3).

A. Exhibit___(SIR-3) consists of the Companies' budgeted versus actual amounts for calendar years 2016, 2017, and 2018.

**Regulation and Oversight**

Q. Briefly describe the regulation and oversight of SIR programs in New York State.

A. SIR programs are primarily overseen by the New York State Department of Environmental Conservation (DEC), in accordance with 6 NYCRR Part 375 regulations. The United States Environmental Protection Agency (EPA) regulates and oversees certain sites regulated pursuant to the Federal Comprehensive Environmental Response, Compensation and Liability Act of 1980 ("Superfund"). Under orders or agreements with utilities, DEC regulates the cleanup of former manufactured gas plan ("MGP") sites within the State.

Q. How are SIR activities for the utilities' MGP sites within New York State prioritized?

A. In consultation with the New York State Department of Health, DEC's prioritization of remedial activities at MGP sites is ongoing. DEC prioritizes sites based on several considerations, including existing land use of the site and nearby properties, proximity of the site to sensitive environmental receptors, cultural and recreational resources in close proximity to the site, reliance on private and public water supply wells in close proximity to the site, and potential reuse of the site.

Q. Please summarize your understanding of the Commission's Order Concerning Costs for Site Investigation and Remediation, issued November 28, 2012, in Case 11-M-0034 (SIR Order).

A. During that proceeding, the Commission considered the possibility of adopting a generic policy with respect to the sharing of SIR costs between ratepayers and shareholders. The Commission concluded that a generic cost-sharing requirement should not be applied; however, the Commission stated that cost-sharing should be considered in future rate cases if it is determined that a utility is not exerting

appropriate efforts to control SIR costs. The SIR Order required the joint major electric and gas companies to collaborate and file with the Commission an inventory of best practices for SIR cost containment. In addition, the SIR Order required that, in any future rate filing in which a utility seeks to recover SIR expenses, the utility must provide testimony discussing its SIR cost control efforts and include an attestation to the utility's compliance with the inventory of best practices for SIR cost containment. Furthermore, the SIR Order directed that testimony be provided establishing that the remediation process is in compliance with existing timetables and DEC requirements, explaining the results of any internal process the utility may have conducted to review its SIR procedures, and describing how internal controls are utilized for SIR projects.

**SIR Program Overview**

Q. What type of expenses are included in the Companies' SIR program?

A. SIR expenses primarily relate to the costs of assessment, monitoring, cleanup and restoration

of sites containing environmental contamination for which the Companies have been found to be wholly or partially responsible, pursuant to Superfund and the Companies' Consent Orders with DEC, to perform SIR activities associated with the cleanup of environmental contamination resulting from the former operation of MGP sites owned or operated by the Companies and their predecessor companies.

Q. How many sites are currently included in KEDLI's and KEDNY's SIR Programs?

A. According to the testimony of Company Witness Willard, KEDLI is wholly or partially responsible for 29 MGP sites, of which 13 have received a "No Further Action" determination with no ongoing obligation. KEDNY is wholly or partially responsible for 27 MGP sites, five of which have received a "No Further Action" determination, and two federal Superfund sites associated with the Gowanus Canal and Newtown Creek waterways.

Q. Have the Companies identified, or been notified, of any other potential SIR sites for which they may be wholly or partially responsible?

A. According to the Company's response to IR DPS-411 included in Exhibit___(SIR-1), KEDNY is currently aware of four additional SIR sites identified by DEC for which it may be wholly or partially responsible, however, no formal determination of liability for these sites have been made by DEC. Additionally, according to the responses to IRs DPS-411 and DPS-412, included in Exhibit___(SIR-1), both Companies received notifications from the EPA on September 21, 2018 regarding alleged responsibility for potential contamination at the Pure Earth Superfund Site, a former waste disposal facility located in Vineland, New Jersey. Both KEDNY and KEDLI are evaluating EPA requests to join a potentially responsible parties (PRP) group, yet, no formal determination of liability have been made for either Company with respect to the site.

Q. Are there any other environmental contamination sites for which the Companies are responsible?

A. Yes. According to the Companies' response to IR DPS-411, and DPS-412, both KEDLI and KEDNY have responsibility for the cleanup of petroleum and

other hazardous materials released as part of their operations, however, these remediation projects are treated by the Companies as operational expenses and the costs are not included in their SIR deferral balances for future recovery.

Q. Has the Panel reviewed the Companies' forecasted schedules and costs for SIR activities for each of their SIR sites?

A. Yes, we have reviewed all relevant materials associated with the Companies' SIR schedule and cost forecasts. Based on our experience in examining SIR programs, we conclude that these costs are appropriate for the scopes of work anticipated by the Companies, however, we also note that schedule slippages have historically occurred, resulting in spending delays, and should be considered when evaluating SIR cost projections on an annual basis.

Q. Have KEDLI and KEDNY demonstrated that their SIR programs are in compliance with all applicable regulatory requirements and timetables?

A. Yes. Company witness Willard provides an attestation of the Companies' overall compliance

with existing timetables and DEC requirements in his testimonies for both KEDNY and KEDLI. Additionally, Exhibit___(CFW-7) of Charles F. Willard's KEDNY testimony and Exhibit___(CFW-5) of Charles F. Willard's KEDLI testimony provide explanations for any variances from the scheduled activities and the activities actually performed in calendar year 2018.

Q. Do the Companies describe their practices and strategies for reducing and/or minimizing SIR costs?

A. Company witness Willard provides attestations that both of the Companies' SIR programs comport with the Inventory of Best Practices for Utility SIR Programs, filed March 28, 2013, in Case 11-M-0034, pursuant to the Commission's direction in the SIR Order. Company witness Willard summarizes the Companies' cost control efforts for their SIR programs in Appendix 1 of his testimony for both Companies.

Q. Describe the Companies' competitive bidding and procurement processes for consultants and remediation contractors for their SIR programs.

A. KEDLI and KEDNY use the same competitive bidding

and procurement processes. Their processes for the competitive bidding and procurement of consultant and remediation contractor services for the SIR programs are described in Appendix 1 of the testimony of Company witness Willard for both Companies. Based on the information provided, we understand that the Companies use a competitive bidding process to procure consultant and remediation contractor services for all of their SIR sites. The procurement processes for both environmental consultants and remediation contractors involve the same general process, which includes a project-specific request for proposals. For certain smaller projects or routine work, the Companies sometimes use Blanket Purchase Orders to solicit competitive bids. The Companies require that competitive bids include specific details breaking down the cost, including labor hours, subcontractor costs, and other direct costs for each task. Consultant and contractor bids are then reviewed, and selections are made based on cost, technical merit and personnel qualifications. According to Appendix 1 of the

testimonies of Mr. Willard for both KEDLI and KEDNY, the Companies established new Master Services Agreements (MSAs) for routine SIR-related environmental consulting work and for construction management work in 2018. The new MSAs issued in 2018 resulted in an expansion of the Companies' consulting pool from five to seven firms. Proposals for routine SIR work are evaluated based on the negotiated rates under these agreements. The use of MSAs allows the Companies to maintain consistency with SIR consultants during multi-phased long-term remediation projects and minimize overhead costs by reducing costs of bidding and achieving savings through negotiated rates and volume discounts. According to Appendix 1 of Mr. Willard's testimonies, the Companies separately perform interviews and establish rates with a single vendor for circumstances where specialized consulting services are required, particularly in support of legal matters.

Q. Do the Companies' procurement and competitive bidding practices enable them to effectively implement their SIR programs while mitigating

consultant and remediation contractor costs?

A. Based on the information provided by KEDLI and KEDNY in this proceeding, we believe that the consultant and remediation contractor procurement and competitive bidding practices enable the Companies to effectively carry out the SIR programs, while utilizing competitive bidding processes that minimize costs for individual sites.

**Cost Control Efforts**

Q. Do KEDLI and KEDNY describe the Companies' mechanisms for reducing and/or minimizing SIR costs?

A. Yes. Appendix 1 of the testimonies of witness Willard and the Companies' responses to IRs in this proceeding identify numerous cost control efforts that are utilized in the SIR programs. The Companies' cost control efforts include implementation of flexible investigation work plans that reduce costs associated with recurrent work plans and mobilization of contractors and equipment, development of remediation solutions that contemplate current and planned uses of a site, and reuse of

excavated materials where feasible and acceptable.

Q. Do the Companies perform audits or other internal reviews of their SIR Programs or individual SIR sites?

A. According to the Companies' response to IR DPS-410, included in Exhibit___(SIR-1), over the past five years, National Grid conducted five audits for work conducted under the MSA contracts. Further, National Grid's SIR Project Managers performed 44 compliance assessments on active SIR sites between June 2014 and June 2019. According to the response to DPS-410, National Grid also performs multiple audits per year of facilities the Companies use for disposal of materials from SIR sites in order to minimize risk of future environmental liabilities. National Grid's SIR group also retained an external consultant to assess active SIR project sites.

Q. How do the Companies address deficiencies and recommendations identified in the reports of audits of their SIR programs?

A. According to the Companies' response to IR DPS-

410, the SIR group reviews audit recommendations or deficiencies with the audit team and identifies appropriate corrective actions. Corrective action items are tracked on a monthly basis and documentation is provided to the audit team upon completion.

Q. Do the Companies pursue cost-sharing or cost recovery opportunities?

A. Yes. According to the testimonies of witness Willard, both KEDLI and KEDNY routinely perform reviews of historic land uses of specific sites and neighboring properties and perform investigations to identify other PRPs.

Q. Describe the results of the efforts by the Companies to identify other PRPs at their SIR sites.

A. As described in the Companies' redacted response to IR DPS-413, included in Exhibit___(SIR-1), both KEDLI and KEDNY have successfully identified PRPs at some of their SIR sites, including Gowanus Canal, Newtown Creek, Williamsburg, Sag Harbor, and College Point. The Companies' efforts have included successful demonstrations to DEC that existing

contamination is not attributable to the Companies' legacy operations and were alternatively caused by historic operations from other former property occupants or historic operations at adjacent properties. These efforts to identify other PRPs have resulted in decreases in the liabilities of KEDLI and KEDNY at these sites. For example, as noted in the redacted response to DPS-413, KEDNY was initially tasked to assume full responsibility for remediation costs at the Gowanus Canal. KEDNY's efforts to identify other PRPs for this site have resulted in the formation of a group of 21 parties that have agreed to fund the remedial design on an interim basis.

Q. Do KEDLI and KEDNY pursue insurance cost recovery opportunities for investigation and remediation of their MGP sites?

A. Yes. As noted in the Companies' response to DPS-227 and the confidential response to DPS-905, both of which are included in Exhibit___(SIR-1), KEDLI has successfully settled MGP claims with a number of insurance companies for approximately

**BEGIN CONFIDENTIAL INFORMATION <** [redacted]

[redacted]

[redacted]

[redacted] **> END CONFIDENTIAL INFORMATION**

KEDNY has settled MGP claims with a number of insurance companies for approximately **BEGIN CONFIDENTIAL INFORMATION <** [redacted] **> END CONFIDENTIAL INFORMATION**

Q. Are the Companies currently pursuing insurance cost recoveries with any other excess liability insurance carriers?

A. In its response to IR DPS-227, included in Exhibit___(SIR-1), KEDNY indicates that it is involved in litigation with Century Indemnity Company and Munich Reinsurance America, involving questions of insurance coverage for 17 of the Company's MGP sites. According to KEDNY, the amount of the claim is unknown at this point and will be based on a number of factors. KEDLI has not identified any additional insurance claims that remain outstanding.

Q. Are there any other cases involving claims made by or against the Companies relating to their SIR sites?

A. Yes. According to the response to DPS-227, KEDLI is defending multiple lawsuits brought on behalf of property owners and residents of Bay Shore, primarily associated with alleged diminution of the values of residential and commercial properties proximal to the Bay Shore MGP site. The response to DPS-227 further indicates that KEDNY filed a federal Superfund lawsuit in January 2017 against 15 parties from which it seeks cost recovery and contribution for SIR costs in the area of KEDNY's Williamsburg MGP site. Litigation in this proceeding is ongoing.

**SIR Program Costs & Cost Recovery**

Q. What level of SIR costs does KEDNY currently recover through base rates?

A. KEDNY's annual base rate recovery for SIR costs is $65.288 million. This includes annual recovery of one-tenth of the forecast SIR deferral balance as of December 31, 2016, or $18.521 million, and $46.767 million for forecasted expenses at its MGP sites.

Q. Is KEDNY currently recovering costs associated with the Gowanus Canal and Newtown Creek?

A. No. According to the direct testimony of Charles F. Willard, at the time of the 2016 Rate Case, it was difficult to accurately forecast SIR expenses for those sites. Therefore, the costs associated with the Gowanus Canal and Newtown Creek are not currently included in base rates. However, KEDNY is authorized to defer any costs related to the Gowanus Canal and Newtown Creek, and an SIR Recovery Surcharge would be triggered if KEDNY incurred costs above a set threshold. The mechanics regarding the SIR Recovery Surcharge are discussed below.

Q. Has KEDNY incurred costs above the set threshold triggering the SIR Recovery Surcharge Mechanism?

A. No. KEDNY has underspent its base rate allowance. As of December 31, 2018, KEDNY’s actual post-2016 SIR deferral balance is deferred credit in the amount of ($14.761) million.

Q. What level of SIR costs does KEDLI currently recovery through base rates?

A. KEDLI’s annual base rate recovery for SIR costs is $18.816 million. This includes annual recovery of one-tenth of the forecast SIR

deferral balance as of December 31, 2016, or $14.168 million, and $4.648 million for forecasted expenses at its MGP sites.

Q. Please describe KEDLI's recent SIR spending?

A. According to the direct testimony of Charles F. Willard, KEDLI experienced a declined level of spending from 2016 to 2017, largely attributable to less remediation construction activity. Based on KEDLI's recent historic spending, and spending forecast through Fiscal Year (FY) 2024, we do not anticipate any significant increase in SIR spending.

Q. Has KEDLI also been underspending its base rate allowance?

A. Yes. As of December 31, 2018, KEDLI's actual post-2016 SIR deferral balance is deferred credit in the amount of ($5.771) million.

**Rate Year Recovery**

Q. Explain how the Companies developed their Rate Year forecasts.

A. According to the direct testimonies of Charles F. Willard, the Companies' Rate Year forecasts utilized the current approved DEC schedule, and the anticipated work schedule for fiscal year

2020. The work schedules take into account the status of each site, and the knowledge derived from project managers regarding site progress and potential delays to determine the projected work on a site-by-site basis. The projected costs for each site are based on the anticipated scope of work at the site and the projected remedy for those sites still requiring remedial activities. KEDNY's forecast of MGP costs for the Rate Year is $66.088 million. KEDNY also forecasted $20.0 million for its Superfund sites (Gowanus Canal, and Newtown Creek) based on Historic Test Year spending. The resulting total of Rate Year recovery is $86.088 million. KEDLI's forecast of SIR costs for the Rate Year is $6.6 million.

Q. Please explain the Companies' proposal for recovery of SIR expenditures during the Rate Year.

A. The Companies propose to include the forecast of SIR costs for the Rate Year in base rates. In addition, the Companies propose to continue to annually recover one-tenth of their respective SIR deferral balance, as of December 31, 2016.

Further, KEDNY proposes the continuation of the SIR Recovery Surcharge.

Q. Explain how the SIR Recovery Surcharge is supposed to function.

A. In the event the difference between actual SIR costs (inclusive of the Gowanus Canal and Newtown Creek), and the SIR base rate allowance exceeds $25 million on a cumulative basis, KEDNY is authorized to recover through the SIR Recovery surcharge (1) the difference between actual SIR expense and the SIR rate allowance (2) any amount that was not recovered in the prior year’s SIR Recovery Surcharge because the cumulative difference between actual SIR costs, and the base rate allowance did not exceed the $25 million threshold and/or the amount would have increased KEDNY’s aggregate revenue by more than two percent.

Q. Do you agree with the continuation of the SIR Recovery Surcharge Mechanism for KEDNY?

A. Yes. KEDNY has demonstrated that the substantial anticipated costs associated with the Gowanus Canal and Newtown Creek Superfund sites warrant continuation of the SIR Recovery

Surcharge Mechanism. The SIR Recovery Surcharge Mechanism would enable the Company to appropriately recover significant incremental costs associated with these sites in a manner that would not unreasonably impact ratepayers.

Q. Do you agree with KEDLI's proposal to continue the provision, allowing KEDLI to petition the Commission to recover unanticipated, incremental SIR costs through the SIR Recovery Surcharge?

A. Yes. While we recognize KEDLI is currently in a maintenance phase for many of its remediation sites, facilitating more reliable forecasting of SIR costs, it remains a reasonable possibility that KEDLI may have to recover considerable incremental SIR expenses.

Q. Do you agree with the Companies' proposal to recover forecasted SIR costs for the Rate Year?

A. Not entirely. We do not agree with the Companies' proposed Rate Year recovery for their MGP sites.

Q. Please explain.

A. Based on the information provided in the Companies' responses to IRs DPS-234 and DPS-235, included in Exhibit___(SIR-1), and the

Companies' SIR Annual Reports filed in Case 11-M-0034, the Companies actual expenditures for its SIR sites have historically been below its cost budget. A comparison of forecasted versus actual spending for calendar years 2016 through 2018 is provided in Exhibit___(SIR-3).

Q. Has the Panel compared KEDNY's actual MGP costs to KEDNY's rate allowance for its MGP sites, per its latest rate plan, Case 16-G-0059?

A. Yes. For Rate Year 1, or calendar year 2017, KEDNY had a rate allowance of $53.871 million, and incurred actual MGP costs of $23.383 million. For Rate Year 2, or calendar year 2018, KEDNY had a rate allowance of $45.643 million, and incurred actual MGP costs of $12.706 million. This results in a combined rate recovery for KEDNYs MGP sites of $99.514 million, and combined actual spending of only $36.089 million, or a difference of $63.425 million.

Q. Has the Panel compared KEDLI's actual MGP costs to KEDLI's rate allowance for its MGP sites, per its latest rate plan, Case 16-G-0058?

A. Yes. For Rate Year 1, or calendar year 2017,

KEDLI had a rate allowance of $13.402 million, and incurred actual MGP costs of $7.301 million, resulting in difference of $6.101 million. For Rate Year 2, or calendar year 2018, KEDLI had a rate allowance of $7.442 million, and incurred actual MGP costs of $8.259 million. This results in a combined rate recovery for KEDLI's MGP sites of $20.844 million, and combined actual spending of only $15.56 million, or a difference of $5.284 million.

Q. What could potentially cause SIR spending to be lower than what is forecasted under the DEC schedule?

A. Several factors could contribute, including: (1) schedule and scope modifications that occur on a year to year basis, which are outside of the Companies' control; (2) construction delays due to site access issues with property owners; (3) permitting requirements and regulatory approvals; and (4) investigation, design, and implementation delays associated with unanticipated site conditions. Further, as stated in the testimonies of Charles F. Willard, SIR costs are subject to a high degree of

variability, which leads to a high degree of risk that actual costs could be more or less than anticipated.

Q. What is the Panel's recommendation?

A. We are concerned about the accuracy of the Rate Year MGP forecasts. Therefore, we recommend the use of a projected average for the Companies' MGP sites based on their projections for Fiscal Year(s) 2021 through 2024.

Q. Explain why the Panel disagrees with the Companies' proposed rate year forecast.

A. Historically, the Companies' projections have been significantly inaccurate. While we acknowledge that the Companies have attempted to consider potential delays and other inhibiting factors in their forecasts, we do not have confidence that the forecasts can be relied upon given the large margin between the Companies' MGP rate allowance, and what costs were actually incurred during the previous rate period. The use of a projected average levelizes the costs the Companies expect to incur.

Q. Please explain why the use of a projected average to levelized costs is appropriate.

A. Based on the historical differences between the Companies' MGP forecasts and actual MGP spending, our approach will more closely align rate recovery with project spending and reduce the year to year variability. We expect this approach to stabilize rate impacts as SIR costs increase in coming years.

Q. Why does the Panel anticipate SIR costs to increase in coming years?

A. KEDNY is anticipating significant liabilities at two Superfund sites: Gowanus Canal and Newtown Creek. KEDNY's liabilities for these sites are unknown at this time. Therefore, we are unable to reasonably forecast the expenses for these sites. In the event KEDNY incurs significant incremental costs, the SIR Recovery Surcharge is readily available.

Q. Is the Panel proposing any adjustment to KEDNY's Rate Year forecast relating to the Gowanus Canal and Newtown Creek?

A. No. We agree with KEDNY's proposal to utilize the amount of $20.0 million, incurred for the Gowanus Canal, and Newtown Creek in the Historic Test Year be used as the basis to forecast the

Rate Year. As previously stated, KEDNY is unable to reasonably forecast its future liabilities at these sites at this time. However, we recognize that the Company is currently incurring costs for both sites that are generally consistent with the proposed amount of $20.0 million.

Q. Please summarize the SIR Panel's recommendations.

A. Based on the information provided in this proceeding, we believe that the Companies have sufficiently satisfied the informational requirements of the SIR Order and adequately demonstrated compliance with the Inventory of Best Practices for Utility SIR Programs. We recommend the use of a projected average to forecast the Companies' MGP spending for the Rate Year. Our proposal utilizes the Companies' MGP forecasts for FY 2021 through FY 2024. Therefore, we have reflected a downward adjustment to SIR expense in the amount of $15.168 million for KEDNY, and $1.734 million for KEDLI. This results in a Rate Year forecast for KEDNY in the amount $70.920, and a Rate Year

forecast for KEDLI in the amount of $4.896 million.

Q. Does the SIR panel propose any other adjustments or recommendations?

A. No.

Q. Does this conclude your testimony at this time?

A. Yes it does.

MR. FORST: Excellent. I now proffer the S.I.R. panel or Site Investigation Remediation panel for cross examination.

A.L.J. COSTELLO: Okay. Okay. We're going to start with the companies.

CROSS EXAMINATION

BY MR. EUTO:

Q. Good afternoon panel, my name is Jeremy Euto for National Grid. Could you please describe the process that you use to conduct discovery of the company's S.I.R. programs?

A. Could you please clarify what you mean by the process?

Q. Sure. For example, when you were undertaking to conduct discovery on certain confidential information that was related to the S.I.R. program, could you just describe the process that you followed, how did you actually do that?

A. With respect to discovery that was requested, if any discovery was referencing information that was confidential per this proceeding that was so noted in the discovery question to the company.

Q. In addition to marking or noting that discovery -- certain discovery responses were confidential, did you also conduct any in-camera reviews?

A. Yes, we did.

Q. Thank you.

MR. EUTO: No further questions, Your Honors.

MR. O'BRIEN: Okay. Ms. Wheelock.

MS. WHEELOCK: Thank you, Your Honors.

CROSS EXAMINATION

BY MS. WHEELOCK:

Q. Good afternoon, panel. My name is Laurie Wheelock. I'm an attorney with the Public Utility Law Project and I'm going to ask you just a few questions about your testimony. To begin, is the panel familiar with the ordering Case 11-M-0034 the proceeding on motion of the commission to commence review and evaluation of the treatment of the states regulated utilities, site investigation and remediation or SIR costs, which was issued and effective on November 28th, 2012?

A. (Flaum) Generally, yes.

MS. WHEELOCK: I believe Your Honor for the record I just wanted to mention that this is Exhibit SIR-1, attachment 1 D.P.S. response to I.R. question 234 and it's pre-marked as Exhibit 506. We do have copies if anyone else would like one in the room.

THE WITNESS: (Castano) Yes, please.

BY MS. WHEELOCK: (Cont'g)

Q. And once you have the order, if you could open it up to Page 12?

A. (Flaum) Thank you.

Q. Please review the last full paragraph which starts with, in so deciding. And again, that's page 12 of the order, the last full paragraph starting with, in so deciding and you can let me know once you review that.

A. Okay. We've reviewed that paragraph.

Q. Thank you. While reviewing Mr. Williard's testimony for the company, did you consider whether KEDNY should share any portion of the SIR related costs?

A. We did consider that as we do for every rate case, requesting S.I.R. cost recovery in -- consistent with the order that you provided.

Q. And ultimately did the panel decide that the company should share some sort of proportion of the SIR costs.

A. We recommended that no -- no sharing of SIR costs be required for this proceeding.

Q. And briefly plan -- panel, can you explain why that determination was made?

A. (Castano) If you please refer to our direct testimony on page 10, line -- excuse me -- line 10, page 10, we state, we have reviewed all relevant materials associated with the company's S.I.R. schedule and cost forecasts. Based on our experience and examining S.I.R. programs we conclude that these costs are appropriate for the scopes of work anticipated by the companies.

However, we also know that the schedule surprises have historically occurred resulting in spending delays and should be considered when evaluating SIR cost projections on an annual basis.

Q. Thank you. Now turning to page 27, if you could review.

A. Excuse me, our testimony or the element of your --

Q. Of you testimony.

A. Thank you.

Q. So reviewing page 27 of your testimony, lines 10 through 22, please.

A. (Flaum) For the record, may I just request that you clarify, did you mean to refer to lines 9 through 21?

Q. I have lines 10 through 22, but I'd be happy to read it to make sure that we have it accurate for the record.

A. Yes, please.

Q. Question, explain why the panel disagrees with the company's proposed rate year forecast. Answer, historically, the company's projections have been significantly inaccurate. While we acknowledge that the companies have attempted to consider potential delays and other inhibiting factors in their forecasts, we do not have confidence that the forecast can be relied upon given the large margin between the company's M.G.P. rate

allowance and what costs were actually incurred during the previous rate period. The use of projected average levelizes the cost the company expect to incur.

A. Thank you.

Q. Of course. The question I have following that is since the panel's testimony was filed and panel being you in August 2019, has the company provided staff with any additional information that changes the panel's conclusion?

A. No.

Q. And so therefore, in the panel's opinion, the rate year forecasts remain inaccurate?

A. I think inaccurate would not be the term that we use. Forecast are by nature, not 100% definitive and our review of previous forecasts, combined with our general experience in these types of reviews for S.I.R. programs is what -- is the basis of our recommendation as indicated in our testimony that you just read aloud.

Q. Thank you. Remaining on that page 27 please review lines 23 through 24 that carry over to page 28 lines 1 through 7. And specifically lines 5 through 7 on page 28 read, we expect this

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approach to stabilize rate impacts as SIR costs increase in coming years. Can the panel please explain how this approach stabilizes rate impacts as SIR costs are expected to increase?

A. So I'm just going to refer to Exhibit S.I.R.-2 page 2 of 3. As you can see for fiscal year '21, the forecast for KEDNY's total site spending is 80 -- approximately 86 million dollars, for fiscal year '22, approximately 82 million dollars, for fiscal year '23, it drops down to approximately 65 million dollars and for fiscal year '24, it drops down to approximately 50 million dollars.

And based on past experience and what the rate allowance was in the 2016 case, and what the company actually incurred, we felt that this approach as reiterated in our testimony will stabilize the right impacts as the costs associated with a Superfund sites increase in the later years.

Q. Thank you.

MR. FORST: Your Honors, just for clarity of the record. The 2016 cases I believe referred to 16-G-0059 and -0058 as we've discussed previously.

A.L.J. LEARY: Okay.

BY MS. WHEELOCK: (Cont'g)

Q. And my last question to the panel is, did the panel consider or conduct any analysis to determine what the bill impacts will be from the SIR costs including the recovery surcharge in conjunction with the proposed rate and increase in this case?

A. We did not provide testimony on -- in addressing that question.

Q. Thank you.

MS. WHEELOCK: No further questions, Your Honors.

A.L.J. COSTELLO: Thank you. Mr. Forst, do you need a moment or?

A.L.J. LEARY: I have a question because I'm actually not completely satisfied that you did what I was hoping that you would do. And I think you know that, right? No. Okay. So off the record, we had a colloquy among counsel as well as the witness panel about how discovery was conducted and how information was exchanged between the companies and D.P.S. staff.

And I'm going to ask the panel if you could, to identify what was meant by, quote in camera

review of information that was provided to you by the company. What did that -- what does that mean?

THE WITNESS: (Castano) We had a meeting with the company during which the company provided certain information that was shared during the meeting, no documentation was provided for us to leave the meeting with. So there was no exchange of documentation, other than for the purposes of and during the meeting and no I.R. responses or any other documentation was provided regarding the information shared during that in-camera meeting any time after that meeting.

A.L.J. LEARY: And what was your understanding about why that was the protocol that was necessary to be followed as -- as -- as an explanation provided by the company.

THE WITNESS: The company reiterated to the panel that this information was extremely, extremely sensitive. And once again, given that there is other parties involved in this case that may have some sort of allocation associated with these sites.

A.L.J. LEARY: Liability allocation.

THE WITNESS: Liability -- liability allocation yes, that it would be best to do this in-camera review.

A.L.J. LEARY: And so it's your understanding that this protocol was necessary to -- to -- to protect an interest of the company, is that correct?

THE WITNESS: That is correct, Your Honor.

A.L.J. LEARY: Okay, that's all I have. I just want to make sure that we're clear. And I -- if anyone has any other questions about this, please if it -- if you don't think it's clear, please feel free to inquire of the panel. Anybody? Okay. Anything?

A.L.J. COSTELLO: Mr. Forst.

A.L.J. LEARY: Mr. Forst.

MR. FORST: I'm going to go --

A.L.J. LEARY: You want to take a minute?

MR. FORST: We'll take one minute.

A.L.J. LEARY: Okay. We're off the record.

(Off the record 3:45 p.m.

(On the record 3:48 p.m.)

MR. FORST: Yes, Your Honors. We have one question for redirect.

A.L.J. COSTELLO: Okay.

RE-CROSS EXAMINATION

BY MR. FORST:

Q. Panel, in the course of your in-camera review of certain information, can you clarify what types of information you reviewed during that in-camera review?

A. (Flaum) Yes, we reviewed cost and design information with respect to potential allocations and liabilities for remedial design work for Gowanda and mostly Gowanda but also Newtown Creek sites and our understanding is that both of those are still subject to litigation outside of this proceeding.

Q. And when you speak of allocations, you're referring to allocations amongst P.R.P.s or potentially responsible parties.

A. That is correct.

MR. FORST: No further questions, Your Honors.

A.L.J. LEARY: Thanks.

A.L.J. COSTELLO: Ms. Wheelock.

MS. WHEELOCK: No further questions.

A.L.J. COSTELLO: Okay. We thank the panel for its testimony. You're excused. We'll just go off the record for a moment.

(Off the record 3:50 p.m.)

A.L.J. COSTELLO: And I'm just going to ask the panel members beginning with the gentleman, the furthest seated from -- from me just to state your name and business address for the record.

MR. CALKINS: My name is Ron Calkins. My address is 3 Empire State Plaza, Albany, New York.

MR. RIDER: Aric Rider, my business address is 3 Empire State Plaza, Albany, New York.

MR. DUAH: Kwaku Duah, 3 Empire State Plaza, Albany, New York.

MS. MAMMENS: Kathryn Mammens, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Please stand and raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MR. CALKINS: Yes.

WITNESS; RONALD CALKINS; Sworn

MR. RIDER: Yes.

WITNESS; ARIC RIDER; Sworn

MR. DUAH: Yes.

WITNESS; KWAKU DUAH; Sworn

MS. MAMMENS: Yes.

WITNESS; KATHRYN MAMMENS; Sworn

A.L.J. COSTELLO: Thank you. You may be seated. Mr. Forst -- Mr. Goodrich, I'm sorry.

MR. GOODRICH: It's actually back to me. I just want to note one thing. On our testimony we have listed as -- as one of the panel members is Chelsea Krueger and while she is a -- a panel member she is -- is out for -- for medical reasons and -- but we, the panel is -- is fully prepared to proceed.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Thank you.

DIRECT EXAMINATION

BY MR. GOODRICH:

Q. So staff policy panel before you is a document entitled prepared testimony of Staff Policy Panel consisting of a cover page and 57 pages of questions and answers dated August 2019. Was that testimony prepared by you or under your direct supervision?

A. (Rider) Yes.

Q. Also before you is a document entitled prepared supplemental testimony of Staff Policy Panel consisting of a cover page and 17 pages of questions and answers dated January 2020. Was that testimony prepared by you or under your direct supervision?

A. Yes.

Q. Do you have any changes or corrections to make to your prepared testimony?

A. No.

Q. Do you affirm that the information contained in the panel's prepared testimony is true and correct to the best of your knowledge?

A. Yes.

Q. If you are asked the same questions today under oath, would you answer them in the same way?

A. Yes.

MR. GOODRICH: Your Honor, I ask that the prepared testimonies of the staff policy panel be entered into the record as if given orally.

A.L.J. COSTELLO: That's granted. And at this point the court reporter should insert the following files. DPS Staff Policy Panel direct testimony and D.P.S. Staff Policy Panel supplemental testimony. #

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

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In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

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Prepared Testimony of:

Staff Policy Panel

Aric Rider
Chief, Consumer Advocacy

Chelsea Kruger
Utility Analyst 2
Office of Consumer Services

Ronald Calkins
Supervisor, Utility Accounting and Finance

Kwaku Duah
Principal Utility Financial Analyst

Office of Accounting Audits and Finance

Kathryn Mammen
Utility Analyst 3
Office of Markets & Innovation

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Department of Public Service (Department) staff (Staff) Policy Panel (Panel), please state your names, employer and business address.

A. Our names are Aric Rider, Chelsea Kruger, Ronald Calkins, Kwaku Duah, and Kathryn Mammen. We are employed by the Department of Public Service. Our business address is Three Empire State Plaza, Albany, New York, 12223-1350.

Q. Mr. Rider, in what capacity are you employed by the Department?

A. I have been recently promoted to Chief, Consumer Advocacy for the Office of Consumer Services.

Q. Mr. Rider, please provide a summary of your educational background and professional experience.

A. I hold a Bachelor of Science Degree in Civil Engineering Technology, which I received in 2001 from the State University of New York Institute of Technology at Utica/Rome. Within the Office of Electric, Gas and Water, I was previously assigned to the Gas and Water Rates and Supply, Gas and Water Rates, Major Utility Rates, Gas Rates, Gas Safety, Gas Policy and Supply, and

Electric Rates Sections. My duties involve the engineering analysis of utility operations as they relate to the ratemaking process, as well as participating in various reviews of local distribution companies' activities.

Q. Mr. Rider, have you previously provided testimony in proceedings before the Commission?

A. Yes, I have provided testimony in several proceedings before the Commission regarding sales forecasts, revenue imputations, operation and maintenance expenses, depreciation, capital planning, development of net plant, cost of service, revenue allocation, rate design, merchant function charges, revenue decoupling mechanisms, gas safety performance mechanisms, and tariff issues.

Q. Ms. Kruger, are you the same Chelsea Kruger testifying as part of the Staff Consumer Services Panel?

A. Yes. I discuss my credentials in that Panel's testimony.

Q. Mr. Calkins, are you the same Ronald F. Calkins testifying as part of the Staff Accounting Panel?

A. Yes. I discuss my credentials in that Panel's testimony.

Q. Mr. Duah, are you the same Kwaku Duah testifying as part of the Staff Finance Panel?

A. Yes. I discuss my credentials in that Panel's testimony.

Q. Ms. Mammen, are you the same Kathryn Mammen testifying as part of the Staff Efficiency and Sustainability Panel?

A. Yes. I discuss my credentials in that Panel's testimony.

**SCOPE OF TESTIMONY**

Q. Panel, what is the purpose of your testimony?

A. We will: (1) Explain the impact of the Northeast Supply Enhancement project on these rate proceedings; (2) present an overview of Staff's recommended revenue requirements for The Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid LI's (KEDLI) (collectively, the Companies); (3) present an overview of Staff's recommended return on equity, common equity ratio and earnings sharing mechanism; (4)

provide a summary of the Companies' proposed capital investment plans; (5) provide an overview of the proposed Information Technology (IT) capital and Operation and Maintenance (O&M) expense plans; (6) summarize Staff's recommendations regarding earning adjustment mechanisms, performance metrics and incentives; and, (7) discuss KEDNY's Newtown Creek renewable natural gas project and make recommendations for rate treatment of the project.

Q. In your testimony, will you refer to, or otherwise rely on, any information obtained during the discovery phase of this proceeding?

A. Yes. We rely on several IR responses provided by the Companies. These responses are included in Exhibit___(SPP-1), and will be referred to using the Department's assigned request number (e.g., DPS-121). This Exhibit includes a Table of Contents that identifies each IR included within our testimony, and the page on which it may be found in the Exhibit.

Q. Are you sponsoring any other exhibits?

A. Yes. We are also sponsoring the following

exhibits: Exhibit___(SPP-2), which contains a summary of Staff's recommended incentive mechanisms; and, Exhibit___(SPP-3), which contains our calculation of the Newtown Creek Project Revenue Imputation.

Q. What is the Historic Test Year in these proceedings?

A. The Historic Test Year is the twelve months ending December 31, 2018.

Q. What is the Rate Year in these proceedings?

A. The Rate Year is the twelve months ending March 31, 2021. This period coincides with KEDNY and KEDLI's fiscal year (FY) 2021.

Q. Did the Companies include additional data for periods beyond the Rate Year?

A. Yes. The Companies provided data for the three subsequent twelve-month periods ending March 31st after the Rate Year, or what they called Data Years 1, 2 and 3.

**NORTHEAST SUPPLY ENHANCEMENT PROJECT**

Q. What is Transcontinental Gas Pipe Line LLC's (Transco) Northeast Supply Enhancement (NESE) project?

A. As described by Ms. Arangio, KEDNY and KEDLI signed precedent agreements for 100 percent of the incremental pipeline capacity with Transco to deliver an additional 400,000 dekatherms (dth) of natural gas per day to the downstate New York area. Transco plans to expand its existing pipeline system along Pennsylvania, New Jersey, and New York to connect to the Companies' system in the Rockaway Peninsula.

Q. Did the Companies assume the NESE project would be built during the Rate Year when they filed the instant cases?

A. Yes. As described by Ms. Arangio, KEDNY and KEDLI's rate filing assumes that NESE will be completed and available by the 2020/2021 winter heating season. This would be in the second half of the Rate Year.

Q. Does the Panel agree that rates should be set in these proceedings assuming that the NESE project will be built during the Rate Year?

A. No. This Staff team has no indication of whether or not the NESE project will be constructed, and if so, when. At present the NESE Project is under review at the New York

State Department of Environmental Conservation and the New Jersey Department of Environmental Protection. Given this situation, we believe that the most reasonable course of action for these rate proceedings is to assume that the NESE Project will not be available during the Rate Year. Accordingly, we have instructed the other Staff witnesses and panels to present recommendations based on the assumption that the NESE project will not be available during the Rate Year.

Q. What if the circumstances around the NESE project change prior to the Commission issuing a final rate order in these proceedings?

A. Each of the Staff witnesses and panels has endeavored to provide information sufficient for the Commission to set rates for the Rate Year under either circumstance, whether or not the NESE project will be available during the Rate Year. For example, the Staff Gas Infrastructure and Operations Panel (SGIOP) provides adjustments for both scenarios.

**SUMMARY OF STAFF'S REVENUE REQUIREMENTS**

Q. Please summarize Staff's recommendation regarding KEDNY's requested gas revenue requirement?

A. Staff recommends a gas base rate revenue requirement increase of $2.205 million, or approximately $173.5 million less than the $175.703 million requested by KEDNY.

Q. How does Staff's recommendation compare to KEDNY's on an overall total base delivery revenue increase?

A. Compared to KEDNY's $195.6 million base delivery revenue increase, Staff's recommendation results in a total base delivery revenue increase of $28.294 million, or a difference of approximately $167.3 million. The $28.294 million reflects a base delivery revenue requirement increase of $2.205 million, less incremental gross revenue taxes of $0.078 million, plus the impact of removing both the currently in place 2017 Tax Cuts and Jobs Act sur-credit of $44.646 million and the energy efficiency surcharge of $18.636 million as Staff also recommends including both of these

components in the base rate revenue requirement. The Staff Revenue Requirements Panel describes the major reasons for the differences between KEDLI's requested and Staff's recommended Rate Year revenue requirement.

Q. Please summarize Staff's recommendation regarding KEDLI's requested gas revenue requirement?

A. Staff recommends a base delivery revenue requirement decrease of $51.852 million, or approximately $91.1 million less than the $39.234 million requested by KEDLI.

Q. How does Staff's recommendation compare to KEDLI's on a total base delivery revenue increase basis?

A. Compared to KEDLI's $61.2 million total base delivery revenue increase, Staff's recommendation results in a total base delivery revenue decrease of $28.740 million, or a difference of approximately $89.9 million. The $28.740 million reflects a base delivery revenue requirement decrease of $51.852 million, less decremental gross revenue taxes of $0.772 million, plus the impact of removing both the

currently in place 2017 Tax Cuts and Jobs Act sur-credit of $31.834 million and the energy efficiency surcharge of $9.478 million as Staff also recommends including both of these components in the base rate revenue requirement. The Staff Revenue Requirements Panel describes the major reasons for the differences between KEDLI's requested and Staff's recommended Rate Year revenue requirement.

**Low Income Program Deferral Credits**

Q. What are KEDNY and KEDLI's deferral credit balances related to the low income programs?

A. As of December 31, 2018, KEDNY has a deferral credit of $10.9 million and KEDLI has a credit of $9.8 million.

Q. What do you recommend regarding these deferral credits?

A. Based on Staff's proposed revenue requirement, we believe use of the low income deferral credits is not needed during the Rate Year. However, if the revenue requirement for both Companies does increase, we recommend the Commission apply a quarter of the deferral

credit balances to the Energy Affordability Program budgets in the Rate Year to minimize bill impacts on ratepayers. The remainder of the deferral credits can be reserved to offset the Energy Affordability Programs in the future, as the program's bill discounts and budgets fluctuate.

## COST OF CAPITAL

Q. What return on equity (ROE) and common equity ratio did the Companies request?

A. The Companies requested a 48.0 percent common equity ratio, with a 9.65 percent return on equity and overall pre-tax rate of return of 8.53 percent for KEDNY and 8.51 percent for KEDLI.

Q. What does Staff recommend for a return on equity and a common equity ratio?

A. The Staff Finance Panel recommends a common equity ratio of 48.0 percent, with a return on equity of 8.20 percent for both Companies. These result in an overall pre-tax rate of return of 7.58 percent for KEDNY and 7.54 percent for KEDLI.

Q. What is the impact of Staff's recommended rate of return on the Companies' revenue requirements?

A. The impact of our recommended rate of return is to decrease the Companies' revenue requirements by approximately $49.00 million and $33.0 million for KEDNY and KEDLI, respectively.

Q. What is an earnings sharing mechanism (ESM)?

A. An ESM provides for the sharing of any earnings in excess of a threshold above a company's allowed return on equity. Such a mechanism is useful in two ways. First, by requiring that a portion of such excess earnings be retained for customers, the mechanism helps to protect customers from paying for an excessive return. Second, as shareholders will receive the benefit of a portion of these excess earnings, the structure of the ESM maintains an incentive for a company to increase efficiencies.

Q. Do you recommend that the Commission institute an ESM in the event that the Commission sets rates for a one year period, i.e., the Rate Year?

A. An ESM should not be instated during the Rate

Year, however, if the Commission sets rates for a single Rate Year, we recommend that the Commission institute an ESM for the period after the end of the Rate Year. Additionally, we note that the Commission instituted an ESM for the period following the rate year when it last set rates for National Fuel Gas Distribution Corporation in Case 16-G-0257. Specifically, we recommend that the Commission require the implementation of an ESM for each 12-month period following the Rate Year, i.e., the 12-month period ending March 31, 2022 and so on. The ESM would only go into effect in the event the Companies do not file for new rates to go into effect on or before October 1, 2021. By doing so, the Commission can ensure that the Companies' rates remain just and reasonable in the event there are unanticipated consequences that may unreasonably increase the Companies' actual earnings.

Q. What structure do you recommend for the ESM?

A. We recommend structuring the ESM in four bands. For the first band, we recommend that the Companies be allowed to retain 100 percent of

earnings up to and including 50 basis points above their allowed ROE. For the second band, defined as earnings greater than 50 basis points above their allowed ROE and up to 100 basis points above their allowed ROE, we recommend that 50 percent of any earnings be deferred for the benefit of customers and the remaining 50 percent be retained by the Companies. For the third band, defined as earnings greater than 100 basis points above their allowed ROE and up to 150 basis points above their allowed ROE, we recommend that 75 percent of any earnings be deferred for the benefit of customers and the remaining 25 percent should be retained by the Companies. For the last band, defined as any earnings in excess of 150 basis points above their allowed ROE, we recommend that 90 percent of any earnings should be deferred for the benefit of customers and the remaining 10 percent be retained by the Companies.

## CAPITAL INVESTMENT PLANS

Q. Please briefly describe how many gas customers the Companies serve and the areas in which they

operate.

A. KEDNY serves approximately 1.2 million customers in Brooklyn, Staten Island and Parts of Queens. KEDLI serves approximately 590,000 customers in Nassau and Suffolk Counties and the Rockaway Peninsula.

Q. Have the Companies proposed investment plans to address the conditions of their gas distribution systems?

A. Yes. KEDNY and KEDLI plan to invest $856 million and $568 million, respectively, in the Rate Year, as shown in the Companies' Exhibits___(GIOP-1CU), in order to support maintaining safe, reliable and resilient delivery service. The budgets exclude the cost of removal, which the Companies show in their presentations for cash flow purposes. The Companies' investment plans include the retirement of leak prone pipe (LPP), programs to respond to conditions on their transmission system, as well as preparing for the major rehabilitation of their liquified natural gas (LNG) tanks.

Q. What investment levels did the Commission

authorize in the 2016 Rate Order, issued on December 16, 2016 in Cases 16-G-0058 and 16-G-0059?

A. As shown in Appendix 1, Schedule 5 of the Joint Proposal adopted through the 2016 Rate Order, the Commission approved KEDNY's budgets of $603 million, $654 million, $650 million and $629 million for calendar years 2017 through 2020, respectively. As shown in Appendix 2, Schedule 5 of the Joint Proposal, the Commission approved KEDLI's budgets of $322 million, $377 million, $396 million and $406 million in 2017 through 2020, respectively.

Q. Please explain the major drivers of the increases in the capital budgets in the instant cases.

A. The major drivers are the need for the following investments: (1) to meet customers' requests for service; (2) the City/State construction activity; (3) the removal of LPP; (4) integrity management programs to identify and address the gas system; and (4) to start major reliability projects – the LNG tank rehabilitations.

Q. What are the current drivers of the incremental

increases in the O&M budgets associated with the capital investment plans?

A. The Companies identified the drivers as: (1) increasing capital investments and increased costs for executing the Companies' capital plans; (2) changes to Companies' O&M workload; and, (3) initiatives the Companies are undertaking in the Rate Year to address new or expanding safety requirements and to implement lessons learned from recent incidents throughout the gas distribution industry.

Q. What are the proposed costs of the incremental O&M programs?

A. As shown on Exhibits__(GIOP-8CU), KEDNY proposes $12.3 million and KEDLI proposes $8.3 million in incremental O&M expenses for the Rate Year.

Q. Did the Companies propose increases to the number of FTE positions to deliver the capital and O&M programs?

A. Yes. As shown on Exhibits__(GIOP-9CU), the Companies included an additional 145.7 and 85.3 FTE positions for KEDNY and KEDLI, respectively, in the Rate Year in various areas including: service line inspection, contractor safety

inspection, enhanced inactive account, instrumentation and regulation, pipeline safety, materials lab testing, meter inspection, gas control standard operating procedure training, first responder training, high emitter methane detection, operator qualification program, support for capital investments, integrity management programs, storm hardening, and research and development.

Q. What did the Companies forecast for incremental O&M expenses related to the new FTEs?

A. As shown on Exhibits___(RRP-3CU), Schedule 27, Page 6 of 8, the Companies forecast total incremental gas expense of $8.3 million and $6.5 million for KEDNY and KEDLI, respectively.

Q. Did the SGIOP conduct a comprehensive review of the Companies' Capital and O&M budgets?

A. Yes. The SGIOP performed a review of the Companies' process and procedures, including project planning, estimating, project management, budgeting and the approval process. The SGIOP also reviewed the historic capital expenditures and expenses, forecast capital expenditures and expenses and changes to O&M

expenses related to changes to the forecast capital expenditures.

Q. Does the SGIOP have recommendations with regard to the gas capital investment plans?

A. Yes. The SGIOP recommends that the Rate Year capital expenditure budgets, inclusive of cost of removal, be reduced by $124.3 million and $157.5 million for KEDNY and KEDLI, respectively.

Q. Does the SGIOP have O&M expense recommendations with regard to the new initiatives and FTEs?

A. Yes. The SGIOP reduced the Companies' Rate Year O&M expense budgets by $4.806 million and $3.010 million for KEDNY and KEDLI, respectively, and reduced the Companies' FTE requests by 11.6 and 7.9 for KEDNY and KEDLI, respectively.

**INFORMATION TECHNOLOGY INVESTMENT PLANS**

Q. Please briefly describe the proposed Information Technology capital investment plan.

A. The Companies' IT Panel explained that IT serves a wide range of services, from critical gas transmission / distribution support systems to standard office desktop applications, and is

vital for the “safe, reliable, and secure physical and commercial operation of KEDNY and KEDLI’s gas distribution business”. IT also provides the necessary software applications to serve the customer needs and to efficiently and effectively manage and operate KEDNY and KEDLI.

Q. Where in National Grid’s corporate structure do IT investments take place.

A. Generally, National Grid USA Service Company, Inc (Service Company) makes investments in IT and then allocates the cost of those investments to the operating utilities, including KEDNY and KEDLI through Service Company Rents.

Q. What investment levels do KEDNY and KEDLI propose?

A. As shown in the Exhibit___(ITP-4CU), the proposed IT spending levels are $243.8 million, $382.8 million, $180.8 million and $299.2 for the Rate Year, FY 22, FY 23, and FY 24, respectively, for the Service Company. This is a significant increase as compared to the five-year average of annual capital spending of $97.5 million. The Companies also forecast Service Company-level “run the business” expenses and

operating expenses of approximately $318.7 million for all projects in the Rate Year. This is also a significant increase, compared to $244.2 million of run the business and operating expenses incurred in the Historic Test Year.

Q. Please explain the major drivers of the investment in IT.

A. The Companies state that in addition to the baseline IT needed to run the business, there are several significant IT programs and initiatives: (1) Technology Modernization; (2) Cyber and Physical Security; (3) Customer Information System Replacement; (4) SAP S/4 HANA; (5) Gas Business Enablement; and (6) and Customer Transformation.

Q. Did the Staff IT Panel conduct a comprehensive review of the Companies' proposal?

A. Yes. For the proposed new IT projects the Staff IT Panel performed a thorough review of the IT budgeting process, proposed IT investment plan, and the cost estimation and implementation planning process.

Q. Describe the Staff IT Panel's recommendations regarding the IT investments.

A. The Staff IT Panel recommends revenue requirement adjustments to reflect: (1) the removal of certain IT projects, (2) an adjustment to KEDNY and KEDLI's forecast of the Service Company-level IT capital budget, and (3) an adjustment to Service Company return on IT capital investments. The Staff IT Panel also recommends a downward-only reconciliation of capital expenditures associated with KEDNY and KEDLI's Service Company Rent Expense and adding reporting requirements to allow for enhanced oversight of the Companies' IT programs.

Q. Did the Staff IT Panel have concerns with the Companies' IT proposals?

A. Yes. Due to the Companies' budgeting process and the timing of the rate filings, many projects do not have sanction papers for Staff to review. It is, therefore, extremely difficult for Staff to determine if the proposed Service Company budgets are reasonable and appropriate.

Q. How can the Companies improve on their processes?

A. We recommend the following process improvements:

(1) proposed budgets included in a rate filing should be supported by sanction papers; (2) once the sanction papers are developed, National Grid's CIO should attest to the costs and benefits of the programs that make up the approved budget; (3) after the projects are closed, National Grid's CIO should attest to and report on the actual costs and benefits achieved through the IT investment plan, and, (4) these improvements should be incorporated into the IT reporting requirements.

Q. Why are these improvements reasonable?

A. If the Companies filed sanction papers in the rate case, Staff could assess the reasonableness of the IT budgets. Moreover, having the CIO attest to the costs and benefits would provide a level of accountability for the Companies to deliver on their promises.

## EFFICIENCY AND SUSTAINABILITY

### Energy Efficiency

Q. Describe Staff's recommendations regarding energy efficiency.

A. The Staff Efficiency and Sustainability Panel

recommends using the Companies' proposed energy efficiency budgets as placeholders until the Commission authorizes budgets in Case 18-M-0084, at which time the authorized budgets should replace the Companies' proposed budgets. In addition, the Staff Efficiency and Sustainability Panel recommends amortizing the Companies' unspent energy efficiency related funds over five years to offset the energy efficiency collections.

**Gas Supply**

Q. Did the Staff Gas Supply Panel address the Companies proposed Green Gas Tariff?

A. Yes. The Staff Gas Supply Panel supports the proposed program with two modifications: the proposed FTEs be reduced from two to one FTE for the Rate Year and the Companies' proposed reconciliation mechanism be modified given the innovative nature of the Green Gas Tariff.

Q. What does Staff recommend regarding the Clean Conversion Program and the Low and Moderate Income Conversion Program.

A. The Staff Efficiency and Sustainability Panel

addresses the Companies' conversion programs. That Panel recommends that as we are assuming that the NESE project will not be built in the Rate Year, these programs should be denied.

Q. If the NESE Project does get built, do you recommend approving the Clean Conversion Program or the Low and Moderate Income Conversion Program?

A. No.

Q. Please explain.

A. Despite its name, the Clean Conversion Program is no different than the former Neighborhood Expansion Program, except for the provision of information regarding renewable natural gas and geothermal heat pumps. Even if supply is available, we recognize that these programs ask existing firm customers to subsidize providing gas service to new customers. Accordingly, we recommend discontinuing them.

Q. What does Staff recommend regarding the Fuel Switching Calculator, Demand Response program expansion, and expanded utility-owned Geothermal Demonstration Project?

A. The Staff Efficiency and Sustainability Panel

makes recommendations related to each of these topics. Specifically, the Staff Efficiency and Sustainability Panel recommends: (1) approval of the Fuel Switching Calculator, provided that the costs of the platform are spread amongst the other National Grid operating utilities; (2) approval of the expanded Demand Response Program as requested by the Companies, with the exception that recovery of such costs occur as a line item through the Delivery Service Adjustment surcharge mechanism; and (3) that expansion of the utility-owned Geothermal Demonstration Project be rejected.

## PERFORMANCE METRICS

### Earnings Adjustment Mechanisms

Q. Describe Exhibit___(SPP-2).

A. Exhibit___(SPP-2) is a summary of all gas incentive mechanisms recommended in these proceedings.

Q. What types of incentive mechanisms does Staff recommend?

A. Various Staff panels recommend earnings adjustment mechanisms (EAMs), and performance

metrics with associated negative and/or positive revenue adjustments.

Q. Please briefly summarize the EAMs that the Staff EAMs Panel (SEAMP) recommends for KEDNY and KEDLI in these proceedings.

A. The SEAMP recommends that the Commission institute a peak reduction metric for each Company. The SEAMP also recommends that the Commission institute a “Share the Savings” energy efficiency metric for each Company.

Q. Please describe the peak reduction metric.

A. The SEAMP recommends high level concepts for a peak reduction metric that would measure peak-day usage reductions from firm customers on a per-Heating Degree Day basis. However, the SEAMP believes that input from the Companies and other parties will be necessary to fully develop this metric. Further, the SEAMP recommends that each Company be allowed to earn up to 10 basis points under this metric.

Q. Please describe the Share the Savings energy efficiency metric.

A. At a high level, each Company could earn a 30 percent share of the savings the Company creates

by improving the financial efficiency of its energy efficiency programs.

Q. How do the SEAMP recommendations compare to the Companies' EAM proposals?

A. KEDNY and KEDLI proposed multiple, sometimes overlapping, EAMs. Associated with their proposed metrics, KEDNY and KEDLI proposed to earn up to 78 basis points and 77 basis points, respectively.

Q. Does the SEAMP address non-pipe alternatives (NPAs)?

A. Yes.

Q. Please briefly summarize the SEAMP Panel's NPA recommendations.

A. The SEAMP recommends instituting an NPA mechanism through which the Companies would consider NPA alternatives to traditional capital projects. The SEAMP Panel recommends implementing a structure for the Companies to develop and propose NPAs in these proceedings, but would leave consideration of an actual incentive to a separate filing, when one of the Companies has a concrete NPA proposal.

Q. Please briefly summarize the platform service

revenues (PSRs) that the SEAMP recommends for KEDNY and KEDLI.

A. The SEAMP recommended that the Companies be allowed to retain 20 percent of the net revenues generated from their Utility Energy Services Contracts program.

Q. How do the SEAMP recommendations compare to the Company's proposed PSRs?

A. The Companies proposed to retain 20 percent of the revenues generated from their E-Commerce Platform, as well as one-third of the gross revenues generated from their Utility Energy Services Contracts program.

**Customer Service Quality**

Q. Please briefly summarize the Customer Service Quality Program (CSQP) as recommended by the Staff Consumer Services Panel in these proceedings.

A. The Staff Consumer Services Panel recommends continuing the Customer Service Quality Program presently in place for KEDNY and KEDLI, with modifications to the targets. The CSQP includes targets for: (1) the annual PSC Complaint Rate

per 100,000 customers; (2) Residential Customer Transaction Satisfaction, measured through a customer satisfaction survey; (3) Percentage of Calls Answered within 30 Seconds; and (4) Percentage of Adjusted Customer Bills. The CSQP includes potential negative revenue adjustments (NRAs), which are incurred should the company fail to meet minimum performance targets in each measure. A total of $11.7 million and $9.9 million, in potential gas revenue adjustments are currently at risk annually for KEDNY and KEDLI, respectively, divided among the four performance measures. The Staff Consumer Services Panel recommends maintaining the NRAs at their current levels.

Q. Please describe the Staff Consumer Services Panel’s recommended incentive for terminations and uncollectibles.

A. The Companies proposed to continue the positive-only revenue adjustment to reduce residential service terminations and uncollectibles, with adjusted targets. The Staff Consumer Services Panel recommends adjusting targets and adding a third metric for residential arrears, whereby

the Companies could earn a Positive Rate Adjustment (PRA) if it reduces residential terminations, uncollectible expenses, and arrears to specified targets.

**Pipeline Safety**

Q. Describe the Staff Pipeline Safety Panel's testimony in these proceedings.

A. The Staff Pipeline Safety Panel's testimony addresses the Companies' proposals for performance metrics in the areas of leak prone pipe removal, leak management, damage prevention, emergency response, and compliance with the pipeline safety regulations. The Staff Pipeline Safety Panel also addresses first responder training, inactive accounts, residential methane detectors, the service line proceeding, New York City Local Law 152, outside meters, plastic fusion, the methane leak pilot program, enhanced contractor safety inspections, buried vent lines, historic revenue adjustments and associated program enhancements, and annual reporting requirements.

Q. Do KEDNY and KEDLI propose to continue the

incentives to remove LPP mileage in addition to the minimum amount they are required to remove?

A. Yes. The Companies propose to earn two basis points for each full mile of LPP it removes, beginning with the second full mile of LPP removed above the specified threshold levels. As we will discuss, this incentive should be eliminated.

Q. Please describe the Companies' proposal for positive incentives to repair additional Type 3 leaks.

A. The Companies proposed to receive one basis point for eliminating 50 additional of the highest volume emitting Type 3 leaks. This positive revenue adjustment would be capped at five basis points and a corresponding annual reduction of 250 leaks in the Companies' respective total leak backlogs. The Staff Pipeline Safety Panel recommends that for every 100 additional leaks repaired beyond the total leak management targets, the Companies could earn two basis points, capped at six basis points, or 300 additional leak repairs.

Q. Please describe the Staff Pipeline Safety

Panel's recommendation for a positive revenue adjustment related to damage prevention?

A. The Staff Pipeline Safety Panel recommends restructuring the damage prevention metric to focus on the total damage prevention targets with associated positive revenue adjustments. KEDNY and KEDLI reported that 31 and 34 percent, respectively, of their damages in 2018 were a direct result of either mismarks, Company error, or Company contractor error. This means that the majority of damages were a direct result of either excavator error, or for excavators failing to provide notice of their intent to excavate to the one-call notification system. With the goal of the damage prevention measures being the reduction of total damages, the Staff Pipeline Safety Panel believes that the Companies should focus their efforts on, and be encouraged to, improve in the areas which will drive down total damages.

Q. Please describe the Staff Pipeline Safety Panel's recommendation for a positive revenue adjustment related to emergency response?

A. The Staff Pipeline Safety Panel concurs with the

Companies' proposals for positive revenue adjustments in the area of emergency response. Specifically, the Companies could each earn up to six basis points should they respond to a higher percentage of gas leak, odor, and emergency reports within 30 minutes.

Q. Are there currently metrics with associated negative revenue adjustments for KEDNY and KEDLI's performance in the areas of LPP removal, leak management, damage prevention, and emergency response?

A. Yes.

Q. Does the Staff Pipeline Safety Panel recommend continuing these metrics?

A. Yes. The Staff Pipeline Safety Panel recommends continuing these metrics with updated targets and associated adjustments, as described in its testimony.

Q. Are there currently metrics with associated negative revenue adjustments regarding the Companies' compliance with the Commission's pipeline safety regulations?

A. Yes.

Q. Does the Staff Pipeline Safety Panel recommend

continuing these metrics?

A. Yes. However, the Staff Pipeline Safety Panel recommends continuing these metrics with updated targets and associated adjustments, as described in its testimony.

Q. Does this Panel recommend any changes to incentives available to the Companies?

A. Yes. Staff seeks to ensure that the incentives that are provided to natural gas utilities to maintain safe and reliable service are appropriate and produce desired outcomes. Since the Companies filed their rate cases, Governor Cuomo signed into law the Climate Leadership and Community Protection Act (CLCPA). While the CLCPA does not contain any provisions directly relating to the provision of natural gas service in the Companies service territories, it enacts goals for carbon emissions reductions which will be difficult to accomplish without taking some steps to address the carbon emissions from space and water heating. Additionally, the Companies are currently facing supply constraints.

Q. Does the Panel expect that the Public Service Commission will address these issues in the

future?

A. Yes, although it is difficult to assess the timing of any Commission actions.

Q. Given that New York State has ambitious climate leadership goals and immediate supply constraints, what issues should be addressed for KEDNY and KEDLI in these proceedings?

A. The Companies incur significant costs addressing the safety and reliability of their systems. The Companies also face the problem of aging infrastructure which manifests itself in the highest leak backlog in New York State at KEDLI. The methane which is emitted from these leaks is a greenhouse gas, in addition to being a safety threat to the residents of these service territories. The Companies' safety programs include items like leak prone pipe replacement and regulator station replacements in their capital budget, as well as leak repairs and other O&M activities which are expensed. In total, these programs are significant in the Rate Year.

Q. What does the Panel recommend to address these issues?

A. We propose the following changes to recognize current issues: (1) Eliminate the proposed positive revenue adjustment for leak prone pipe replacement over and above the annual target as shown in Exhibit___(GIOP-3); (2) Eliminate the proposed positive revenue adjustment for leak prone pipe unit costs as shown in Exhibit___(GIOP-3); (3) Eliminate the proposed positive revenue adjustment for customer connections as shown in Exhibit___(GIOP-3); and (4) implement a positive revenue adjustment for the Companies if they can deliver their investment plans - a minimum amount of leak prone pipe replacement and a reduction in their respective year end leak backlogs - for less than the budgeted program amounts.

Q. How should the new positive incentive be developed?

A. The Commission should require the Companies to file: (1) their algorithm used to risk rank leak prone pipe segments for Staff review within 30 days of the date the Commission issues a rate order in these proceedings; (2) at the beginning of every year, the complete list of all leak

prone pipe segments in order of highest to lowest risk per the risk ranking; (3) at the end of every year, the list of all pipeline segments that were replaced in the year that just ended, indicating what position each segment had on the risk ranking at the beginning of the year. The Companies should develop a Safety and Reliability Program, which would contain all capital and expense budget items that contribute to the safety and reliability of the system. If the Companies can deliver their investment plans for less than the budgeted program amounts, they should be awarded an incentive.

Q. How could the positive incentive be developed?

A. Assuming that the Companies meet the metrics of minimum mileage of leak prone pipe replacement and year end leak backlog, and the Companies were able to do so for less than the budgeted program amount, an incentive could be developed by determining the revenue requirement savings of the program. The Commission could allow the Companies to retain a percentage of the revenue requirement savings.

Q. Would the Capital Investment Reconciliation

Mechanism have to be adjusted?

A. Yes. There would be an interplay between the two mechanisms.

Q. Why is this new positive incentive reasonable?

A. This incentive is designed to encourage the Companies to make decisions to ensure safety and reliability and to decrease carbon emissions, while increasing customer choice and decreasing costs.

Q. What other desired outcomes should the Commission encourage?

A. As described in the Staff Earned Adjustments Mechanisms Panel’s testimony, the Companies are encouraged to begin responding to system needs with non-traditional solutions as quickly as possible, and any LPP removed from service in this manner would be included in meeting the LPP mileage metric.

**NEWTOWN CREEK PROJECT**

Q. What is the Newtown Creek Project?

A. On page 77 of the direct testimony of the Future of Heat Panel it explains that KEDNY has developed a project with the City of New York to

capture biogas generated from the City's Newtown Creek Wastewater Treatment Plant. The biogas consists of approximately 60 percent methane and 40 percent $CO_2$. KEDNY will filter it through a purification facility and upgrade it to pipeline quality natural gas, which will then be odorized and injected into the gas system for direct use by customers.

Q. Was the Newtown Creek Project in the last rate case regarding KEDNY, Case 16-G-0059?

A. Yes. The SGIOP in that case explained that KEDNY would own specific assets: gas purification systems that consist of vessels, a compressor, vacuum skids, a transformer and other electrical equipment, gas analyzer and a chromatograph, odorization system, tail gas tank, cooler, thermal oxidizer, metering and regulation system, air system, air sensors, fire suppression equipment, and electronic communication equipment. The Newtown Creek Project also was projected to have ongoing incremental O&M expenses.

Q. In Case 16-G-0059, what rate treatment did KEDNY propose for the assets and incremental O&M

expenses associated with the Newtown Project.

A. KEDNY proposed that the assets be placed in to plant in service and the incremental O&M expenses be reflected in the cost of service.

Q. In Cases 16-G-0058 and 16-G-0059, how did the Companies propose customers be charged for the gas injected into KEDNY's distribution system?

A. KEDNY and KEDLI proposed that their sales customers pay for the gas through the Gas Adjustment Clause (GAC) at the monthly weighted average cost of gas (WACOG). KEDNY's sales and transportation customers would receive a credit equal to the gas charges paid by KEDNY's and KEDLI's sales customers. KEDNY's customers would also receive credit for any environmental attributes or third-party sales.

Q. Please explain the credits associated with environmental attributes or third-party sales.

A. In Case 16-G-0059, KEDNY's Gas Infrastructure and Operations Panel explained that the renewable gas from the project may qualify for various federal and state emissions credits depending on how the gas is utilized by the end user, under New York's Renewable Portfolio

Standard program or the Environmental Protection Agency's Renewable Fuel Standard Program. KEDNY claimed that it would work to identify and leverage available emissions credits to offset the cost of the Newtown Creek Project.

Q. In the last rate case, did Staff raise concerns with the Newtown Creek Project?

A. Yes. The Staff Infrastructure and Operations Panel's testimony stated that, from an environmental perspective, it did not have any concerns, however from a ratemaking perspective, it did not believe that the in-service date of the project would occur in the Rate Year. The Staff Infrastructure and Operations Panel's testimony also recommended that the Commission consider the following ratemaking options: (1) limit the level of investments reflected in the cost of service to be more in line with the projected revenues; (2) require the use of any excess revenues to write down the assets in lieu of the proposed sharing mechanism; and, (3) encourage the City of New York to provide full property tax abatement for 20 years, which would improve the economics of the project.

Q. Did the Commission adopt a Joint Proposal in Case 16-G-0059?

A. Yes. The Commission adopted a joint proposal and set rates for KEDNY and KEDLI in the Rate Order it issued on December 16, 2016 in Cases 16-G-0058 and 16-G-0059.

Q. How was the Newtown Creek Project addressed in the Joint Proposal that the Commission adopted?

A. Beginning on page 22 of the Joint Proposal, provision 5.6, the language states: “KEDNY will continue to develop the Newtown Creek biogas purification project. Because the ultimate in service date and final capital costs are not known at this time, the costs are not included in the revenue requirement. Once the project is in service, KEDNY is permitted to defer the return of and return on the Newtown Creek project and have those costs reviewed for potential recovery in its next base rate filing. The revenue requirement for the Newtown Creek project will be subject to a $1.6 million annual exclusion (prorated from the in service date) for 20 years. The $1.6 million amount and the 20 year time period will be subject to review

for possible modification in KEDNY's next rate case. The revenue requirement associated with the project will reflect a 40 year depreciable life. [The City of New York] agrees to support KEDNY's reasonable efforts to mitigate the effects of the revenue requirement exclusion, including, but not limited to, supporting KEDNY's application(s) for property tax abatements, to the extent such efforts are permitted by applicable law. Further, any revenues from KEDNY's share of the sale of gas at a premium (i.e., in excess of the weighted average cost of gas) or environmental credits generated from the project will be used to offset the revenue requirement exclusion."

Q. Did KEDNY explain the current status of the Newtown Creek Project in its filing in the instant cases?

A. Yes. The Future of Heat Panel testimony states that by June 2018, KEDNY secured all necessary permits and subsequently began project construction in July of 2018. The Future of Heat Panel states that efforts are currently on schedule, and KEDNY expects to bring the project

online in November 2019. In addition to the construction activities, KEDNY issued a request for proposals in 2018, seeking a firm to monetize the environmental attributes KEDNY expects to generate from the injection of RNG into the local distribution system. KEDNY selected Element Markets to complete this task. KEDNY and Element Markets are currently collaborating to register the project under the federal Renewable Fuel Standard Program as a generator of Renewable Identification Numbers (RINs).

Q. What ratemaking treatment did KEDNY propose for the Newtown Creek Project in the instant cases?

A. As explained on the Future of Heat Panel's testimony, beginning on page 79, KEDNY proposed including a total capital cost of approximately $32 million in plant in service; recovery of associated property tax expense, net of abatement; recovery of depreciation expense reflective of a 20-year depreciation life; and recovery of O&M costs of $0.668 million. KEDNY also included a revenue estimate of $1.9 million per year, comprised of $1.0 million from the gas

sales and $0.9 million from the sale of environmental attributes. KEDNY proposed to true-up the actual revenues for recovery from or return to customers. Further, KEDNY proposed that any revenues realized above the level necessary to fully reimburse customers for the project costs would be shared evenly between customers and NYC beginning in the fifth year after the project becomes operational. KEDNY proposed that revenue sharing would then be assessed every year thereafter for the remainder of the project's life.

Q. Why did KEDNY propose to true up the actual revenues to the level imputed in this case?

A. Beginning on page 81 of the Future of Heat Panel's testimony, KEDNY explained that the value of the environmental attributes will vary due to market changes, and KEDNY expects significant variability over time. Moreover, KEDNY explained that the actual revenue is highly dependent on the volume of biogas produced and the proportion of the volume assigned to the environmental attributes.

Q. Does KEDNY have a contract with the City of New

York associated with the Newtown Creek Project?

A. Yes, the contract was provided in response to IR DPS-736.

Q. Does the contract explain how KEDNY's customers will receive the benefit of emission credits?

A. Yes. According to the contract, every year KEDNY will compare the implied cost (revenue requirement) to the market value (annual project output multiplied by an annual WACOG), including environmental credits, to offset the project's revenue requirement. If, at the end of year five, there are cumulative excess benefits, the City of New York would receive 50 percent of the excess benefits and the other 50 percent of the excess benefits is to be credited to KEDNY's firm customers. Section 6.5(c) of the contract also states that the City of New York shall maintain all right, title and interest to any and all Air Emission Credits for reduced emissions of constituents regulated by Title V Air Permit and/or any related state air permit earned by the City of New York in connection with this Project.

Q. When was the Newtown Creek Project first

sanctioned?

A. The response to IR DPS-734 states that the project was first sanctioned in August 2009.

Q. What was the estimated cost of the project and in-service date in August 2009?

A. KEDNY projected the cost at $8.3 million, with an in-service date of March 2011.

Q. How many times has the Newtown Project been re-sanctioned and what were the estimated costs and associated in-service dates?

A. The project has been re-sanctioned a total of six times, as shown in the response to IR DPS-734. The re-sanction dates, estimated costs and in-service dates are as follows: April 2012, with an estimated cost of $14.4 million and in-service date of March 2014; August 2014, with an estimated cost of $19.9 million and an in-service date of September 2016; October 2015, with an estimated cost of $32.8 million and an in-service date of February 2017; February 2016, with an estimated cost of $34.1 million and an in-service date of July 2017; and November 2017, with an estimated cost of $37.9 million and an in-service date of December 2018.

Q. Did KEDNY report on a potential tax credit for the project?

A. Yes. On page 18 of Attachment 8 to the response to IR DPS-734, KEDNY reported that due to the extensive delays to start construction a $4 million tax credit was jeopardized. In the response to IR DPS-971, KEDNY stated $3.1 million in real property tax savings will be awarded because the completion date deadline was extended to February 29, 2020. Thus, the delays appear to have caused KEDNY to lose $0.9 million in real property tax savings.

Q. What is the current status of the Newtown Creek Project?

A. As explained in the response to IR DPS-737, the Project is 65 percent complete and KEDNY now anticipates an in-service date of December 31, 2019.

Q. What is the most recent cost benefit analysis for the Newtown Creek Project?

A. In response to IR DPS-734, KEDNY stated that the benefits of the project include: expanding the concept of renewable technology; reducing an estimated 90,000 metric tons of $CO_2$ emissions;

providing a solution to food waste management; and, improving diversity of supply. KEDNY also provided an economic analysis presentation dated January 22, 2016 that compares the cost of the Project to the potential revenues. On slide 10, KEDNY estimated a base case, a worst case, and a base case with full property tax abatement.

Q. Please explain the base case.

A. KEDNY's base case assumes an annual production of 277,500 dth, $33 million capital costs and five years of property tax abatement on $14.4 million of the capital investment. Under the base case the levelized revenue requirement is $5.7 million per year, which equates to $20 per dth of gas injected into the distribution system.

Q. Why is the property tax abatement calculated on only $14.4 million of the capital investment?

A. Our understanding is that the property tax abatement agreement was based off of the 2012 cost estimate, which has not been updated.

Q. For reference, what is KEDNY's current WACOG?

A. Per KEDNY's GAC Statement 252, effective August 1, 2019, KEDNY's current WACOG is 40.453 cents

per therm, or $4.0453 per dth.

Q. Please explain the worst case.

A. KEDNY’s worst case assumes annual production of 208,000 dth, a 20% reduction in renewable credits, $36 million capital costs and again assumes five years of property tax abatement on $14.4 million of the capital investment. Under the worst-case scenario the levelized revenue requirement is $8.2 million per year, which equates to $29 per dth of the gas injected into the distribution system.

Q. What is the last scenario KEDNY provided?

A. KEDNY’s last scenario is the base case, but assuming full property tax abatement. This means receiving property tax abatement for 20 years. Under this scenario, the levelized revenue requirement is $4.2M per year, which equates to $15 per dth of gas injected into the distribution system.

Q. Does the Panel have any concerns with the Newtown Creek Project?

A. Yes. We are concerned with the overall cost of this project and the resulting ongoing annual revenue requirement impact to customers.

Q. What is the estimated capital cost of the project?

A. According to the response to IR DPS-972, the estimated capital cost of the project is $37.898 million plus or minus ten percent.

Q. Did the Panel ask KEDNY how it attempted to control the costs of the project?

A. Yes. According to the response to IR DPS-735, KEDNY pursued several measures to reduce the total cost of the project, including using in-house engineering, collaborating with the City to pay lower rates for electricity and securing a five-year property tax abatement.

Q. Does KEDNY currently own or operate any gas production facilities?

A. According to the response to IR DPS-736, no.

Q. Will the equipment owned by KEDNY be located on KEDNY’s property?

A. According to the response to IR DPS-736, no. The equipment will be located on property owned by the City of New York.

Q. Why does KEDNY believe it is reasonable for the City of New York to charge property taxes for the assets used for this project?

A. According to the response to IR DPS-736, KEDNY claims that these assets are subject to be taxed by the City of New York in the same manner as any other utility-owned assets. KEDNY further stated that it has secured a tax exemption for the Project from City of New York property taxes for the first five years of the Project operation, based on a discount rate of 6.25 percent per annum and a total project cost of $14.4 million, which is based on the 2012 cost estimate.

Q. Did the Panel ask KEDNY to identify the annual revenue requirement impact of the Newtown Creek Project?

A Yes. In response to IR DPS-738, KEDNY provided Attachment 1 that shows the net Rate Year revenue requirement impact of $2.9 million using the following assumptions: a total capital cost of $32.2 million; depreciation over a 20 year service life; an 8.53 percent pre-tax weighted cost of capital; $0.668 million for O&M expense; no property taxes; and $1.9 million in revenues from gas sales and environmental attributes.

Q. Are the total capital costs accurate?

A. No. As previously mentioned, the estimated capital cost of the project is $37.898 million plus or minus ten percent.

Q. Why does KEDNY believe a 20-year depreciable life is reasonable?

A. According to the response to IR DPS-736, KEDNY believes a 20-year depreciable life is reasonable because it matches the term of the lease agreement with the City of New York.

Q. Does the 8.53 percent pre-tax weighted cost of capital reflect KEDNY’s proposed return on equity and capital structure?

A. Yes.

Q. Why are there no property taxes included in the Rate Year?

A. There is a lag between when assets are placed in service and when property tax assessments begin. KEDNY estimates that property taxes will be assessed beginning in FY 22.

Q. How did KEDNY develop the $1.9 million in revenues from gas sales and environmental attributes?

A. KEDNY provided its workpapers in the response to IR DPS-739. These show that KEDNY estimated

annual gas revenues of $0.939 million and annual RIN revenues of $0.973 million.

Q. Does the Panel agree with the Companies' proposed ratemaking treatment of the Project?

A. No.

Q. Why not?

A. There is a significant amount of uncertainty with this project and under KEDNY's proposal and all of the risk is borne by KEDNY's customers. Indeed, it seems the only certainty is that annual revenues from the Newtown Creek Project will not meet or exceed the annual revenue requirement of the project.

Q. How should the Commission protect customers regarding the Newtown Creek Project?

A. We recommend the Commission impute a level of revenue in KEDNY's revenue requirement to equal the actual annual costs of the project. KEDNY should test annually to determine if the actual revenues exceed the actual revenue requirement of the project. If the actual revenues do not exceed the actual revenue requirement of the project, no reconciliation should be permitted. If the actual revenues exceed the actual revenue

requirement, shareholders can retain any revenues from previous cumulative shortfalls.

Q. Why should shareholders bear the risk of generating sufficient revenues to cover the cost of the Newtown Creek Project?

A. KEDNY has known throughout its sanctioning process that the projected levelized cost of any natural gas produced from the project was much higher than the market value of the natural gas and environmental attributes, yet KEDNY chose to continue the project.

Q. Does the Panel recommend a level of Newtown Creek Project revenues to impute in the Rate Year?

A. Yes. We recommend imputing $3.3 million in revenues from the Newtown Creek Project.

Q. How did you develop this imputation?

A. As shown on Exhibit___(SPP-3), we used KEDNY's response to IR DPS-738, Attachment 1 and made the following modifications: (1) updated the rate of return to 7.59% to reflect the Staff Finance Panel's cost of capital recommendations; and (2) updated the capital costs of the Newtown Creek Project to $37.9 million based on the

response to IR DPS-973.

Q. Did you provide this adjustment to the Staff Revenue Requirements Panel?

A. Yes.

Q. Does this conclude your testimony at this time?

A. Yes, at this time.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

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In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

January 2020

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Prepared Supplemental Testimony of:

Staff Policy Panel

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Q. Members of the Staff Policy Panel (Panel), please state your names.

A. Aric Rider, Chelsea Kruger, Ronald Calkins, Kwaku Duah, and Kathryn Mammen.

Q. Has the Panel previously submitted testimony in these proceedings?

A. Yes, we submitted pre-filed direct testimony on August 30, 2019.

Q. In this testimony, will you refer to or rely on any responses to information requests (IRs)?

A. Yes. We have included the responses to IRs that we have relied on in Exhibit___(SPP-4). We will refer to those responses in our testimony by the designation given to them when asked, for example, DPS-123.

**Second Supplemental Filing**

Q. What was the purpose of the Companies' second supplemental testimonies?

A. The purpose of KEDNY and KEDLI's second supplemental testimony was to update the Companies' capital and O&M expense forecasts to incorporate changes necessary to implement the recent settlement agreement adopted by the

Commission in Case 19-G-0678. The settlement concerned gas supply constraints in their service territories and lifting the moratorium on new customers connections that the Companies imposed.

Q. Did the Companies also propose revisions to certain contractor cost projections?

A. Yes.

Q. Please explain why the Companies propose to revise their contractor costs.

A. As described on page 15 of the Second Supplemental Testimony of the Gas Instructure and Operations Panel (GIOP), the Companies have completed contract renegotiations with certain contractors that will provide necessary services in the Rate Year and Data Years. According to the Companies, these negotiations have resulted in a known and measurable decrease to the unit costs of KEDLI's Proactive Main Replacement (LPP) capital expenditure program, and an increase to the damage prevention mark-out operation and maintenance (O&M) costs for both Companies. The Companies propose to use these new costs to revise their capital and O&M

expense forecasts.

Q. Please describe the revision to the forecast for KEDLI's LPP Program.

A. As described on page 15 of the Second Supplemental Testimony of the GIOP, KEDLI renegotiated its agreement with a contractor that provides services for the Proactive Main LPP capital program, and this new agreement resulted in a reduction of the Companies' projection of the program's unit cost from $306 per foot, to $304 per foot. This revision in the cost per food reduces KEDLI's capital expenditure forecast by $1.9 million in the Rate Year.

Q. Do you have any adjustments relating to KEDLI's change to the forecast for its Proactive Main LPP capital program.

A. We are not addressing that request. The Staff Gas Infrastructure and Operations Panel addresses this change in its supplemental testimony.

Q. Please explain damage prevention mark-outs and how the O&M costs of these mark-outs are reflected in the Companies' rate filings.

A. As stated on pages 16 and 17 of the Second Supplemental Testimony of GIOP, the Companies' damage prevention mark-out work is required to support KEDLI's projects, and when customers, third parties, and/or municipalities contact New York State's Dig Safe One Call Center. KEDNY uses internal employees to locate facilities on Company-related jobs and a contractor for all third-party Dig Safe mark-out requests. KEDLI sends all Dig Safe mark-out requests, whether internal or external, to a contractor to mark-out facilities. The costs associated with damage prevention mark-outs are included in KEDNY's O&M expense and KEDLI's O&M expense and capital expenditure forecasts. To prepare the rate case forecasts in their initial filing, the Companies increased the O&M expense components from the Historic Test Year by standard annual inflation to arrive at their Rate Year projections.

Q. Please explain the recent developments that the Companies' claim will affect the contractor cost component of damage prevention mark-out costs.

A. As explained on page 17 of the Second Supplemental Testimony of the GIOP, on August 19, 2019, the Companies solicited contractor bids for their damage prevention mark-out work. Bids were received from contractors on September 16, 2019. Thereafter, bidders were evaluated both commercially and, on their ability, to execute technical requirements. Extensive negotiations were also conducted with bidders between September 16, 2019 and November 8, 2019.

Q. Have the Companies selected contractors based on the bids received?

A. Yes. As described on page 17 of the Second Supplemental Testimony of the GIOP, the Companies' are in the process of executing agreements with two contractors to provide damage prevention mark-out services from February 2020 to December 2022. As of the date of the GIOP's testimony, the agreements were expected to be executed on or about January 1, 2020.

Q. What effect will the new agreements have on the Companies' contractor costs?

A. The Companies allege the contracts will raise the labor rates for these contractors in excess of the standard annual inflation rate used to derive the Companies' Rate Year forecasts.

Q. What are the drivers of the higher contractor bids?

A. As explained on page 18 of the Second Supplemental Testimony of the GIOP, the Companies indicate the increase is attributable to three factors: (1) the prior contractor labor rates were based on long-term, dated agreements; (2) recent unionization of the contractor labor force; and (3) an increased scope of work to incorporate locator assurance technology. The Companies indicate they have not bid this work in over a decade. Furthermore, the Companies explain that while the prior agreements provided for minimal price increases, the rates paid to those contractors were generally lower than market prices at the time. Furthermore, the Companies claim that contractors were no longer willing to accept below-market rates during the recent negotiations. Additionally, the Companies explain that the contractors providing

services under the selected agreements unionized earlier in 2019, and while the Companies indicate they were generally aware that these labor forces were unionizing, the increase in labor rates experienced was significantly higher than anticipated or reflected in the Companies' initial filing. The Companies also indicate the new agreements require that the contractors use updated technology (locator assurance technology that enables real-time tracking of technician performance against industry standards to improve overall effectiveness of locating field work) when marking out facilities, and with this requirement the scope of work required in prior agreements is expanded and will require additional training and inventory for these contractors.

Q. What are the Companies' projected impacts to KEDNY's Rate Year O&M expense forecast due to the changes to the contracts for damage prevention mark-out services?

A. This change increased KEDNY's O&M expense forecast, specifically the cost element Other Initiatives expense by $6.8 million.

Q. What are the Companies' projected impacts to KEDLI's rate year capital and O&M expense forecasts due to the changes to the contracts for damage prevention mark-out services?

A. This change increased KEDLI's O&M expense forecast, specifically the cost element Other Initiatives expense, by $1.9 million, increased Depreciation expense by $0.011 million, and increased Net Utility plant by $1.3 million.

Q. Are there any revisions required to the Company's incremental damage prevention mark-out costs include in the Companies' Second Supplemental Filing.

A. Yes, based on the Companies' response to DPS-1067, there are revisions that need to be made for both KEDNY and KEDLI.

Q. Explain why the revisions need to be made.

A. As explained in response to IR DPS-1067, the Companies explain that the incremental cost calculation applies current vender contract rates, and future vender contract rates under the new contract, to Historic Test Year units (or number of tickets). For KEDNY, certain Historic Test Year tickets were worked by in-

house crews rather than an outside contractor. Inadvertently, the in-house crew ticket counts were not removed from the Historic Test Year in the original calculation of the incremental cost provided in the Companies' second supplemental filing. Therefore, when KEDNY applied the current and new contract rates, the incremental costs were overstated. In addition, a small portion of Historic Test Year tickets had a separate call-out fee rate that was not included in the original calculation, resulting in an understated incremental cost. For KEDLI, the Companies' indicate the original incremental cost calculation did not include the effect of both the separate call-out fee rate as well as a separate time and materials rate, resulting in an understated incremental cost.

Q. Do the Companies provide the impact of the identified calculation revisions that should be made to the Rate Year forecasts?

A. Yes, in the response to IR DPS-1067, the Companies indicate the calculation revisions result in a reduction to KEDNY's incremental O&M expense of $1.888 million, and an increase to

KEDLI's incremental O&M expense of $0.168 million and an increase in incremental capital expenditures of $0.392 million.

Q. Do you agree with the Companies' request to revise their contractor costs relating to damage prevention mark-outs?

A. No. The Companies' request to revise their contractor costs relating to damage prevention mark-outs is too late in the process and puts Staff and the intervening parties in this case at a significant disadvantage. Therefore, we recommend that the Commission deny the Companies' request to revise its contractor costs associated with damage prevention mark-outs.

Q. Explain why you disagree with the Companies' request to revise its contractor costs associated with damage prevention mark-outs.

A. The Companies have known that their contracts associated with their damage prevention program were expiring long before their April 2019 filing. Indeed, well in advance of their April 2019 filing, they began the process of actively procuring contractors.

Q. As of what date were the Companies aware of the need to update the contracts for the contractors performing their damage prevention mark-outs.

A. As shown in Attachment 1, in response to IR DPS-1068, the Companies have been aware for over two years now, that they planned to update their contracts associated with their damage prevention program. In fact, the initial Stakeholder meeting regarding this revision dates to January 1, 2018. Furthermore, not only were the Companies' aware of the potential revision but were also fully aware that the existing contracts were set to expire in May of 2019, which is approximately 22 months prior to the end of the Rate Year, March 31, 2021. However, regardless of the Companies knowledge, it never identified this issue in these proceedings prior to its second supplemental testimony.

Q. Did the Companies provide any testimony in their original, first supplemental, corrections, or rebuttal filings regarding a potential revision to contractor costs related to the damage prevention mark-out contracts.

A. No, the Companies failed to mention, or discuss anything related to a potential revision to their contractor costs related to damage prevention mark-out contracts in any of their prior submissions in these proceedings.

Q. If the costs were not known at that time, could they have addressed this issue in any of their four prior filings in these proceedings?

A. Yes, the Companies could have addressed these issues in their initial, first supplemental, corrections and updates, or rebuttal filings in these proceedings. Whether or not they included forecasts of the resulting increases in contractor costs, the Companies could have provided much background information, including but not limited to, the requirements that the contractors use new equipment and technology. Had the Companies included discussion of this issue in any of their previous filings in these proceedings, the parties to these cases would have had a greater opportunity to more thoroughly examine the issues and to assess the reasonableness of any resulting increases.

Q. Are there any other reasons why the Commission

should deny the Companies' request to revise the rate year forecast for damage prevention mark-out costs?

A. Yes. In the Companies' calculation of incremental damage prevention costs that they expect to result from the new contractor contracts, there is no acknowledgement or reflection of any potential savings that will be realized. In the Companies' response to DPS-1072, they indicate the sole benefits to customers would be enhanced public safety and the safety of the facilities.

Q. What type of savings could result from the new contractor agreements?

A. In their second supplemental testimony, the Companies indicate that one of the reasons for the increased mark-out costs is attributable to the new contractor agreements requiring contractors to use updated technology when marking out facilities. The Companies state the updated technology will "enable real-time tracking of technician performance against industry standards to improve overall effectiveness of locating field work." Thus,

one would expect that, with the use of new technology, there would be some benefits realized such as operational efficiencies of the damage prevention program and increased productivity by the workforce. The Companies have not factored savings and benefits such as, but not limited to, these into their incremental cost calculation. Instead, the Companies' incremental cost calculations assume no benefits or savings would result, and this assumption is untested and unrealistic. If the Companies are not going to consider, in a comprehensive manner, the true incremental costs of the new contractor agreements, which would include applying offsetting benefits/savings that would be realized, then the Companies' current calculation of incremental costs should be rejected. It is not equitable or fair to expect customers to fund the expected incremental costs of the new contractor agreements and allow the Companies' to benefit from savings that will result.

Q. In their Second Supplemental Filing, did the Companies provide an analysis, or perform a

review of all its existing contractors to determine where there was a potential for other cost savings opportunities?

A. No. The Companies failed to consider the possibility of other contract agreements, where savings opportunities could have been achieved.

Q. Didn't KEDLI also revise its LPP program forecast?

A. Yes, however, this was a correction of a cost element forecast that was identified early in the rate making process. For both KEDLI and KEDNY, the change in LPP contractor costs had already been estimated and reflected in KEDNY's initial filing and KEDLI's corrections and updates filing. Referring to the Corrections and Updates Testimony of the GIOP, on page 6, KEDLI states "A correction to the forecast is required because the LPP unit costs were understated due to a formula error in the workpaper used to develop the unit costs." The change in KEDLI's contractor cost forecast for LPP replacement work could be considered a correction, although the Staff Gas Infrastructure and Operations Panel is

addressing the merits of KEDLI's LPP contractor costs.

Q. Did you ask the Companies to provide any additional analysis of their review of other contracts during the pendency of these proceedings?

A. Yes. In IR DPS-1073, we asked the Companies to provide a list of contractors "for which the Company recently reviewed the current contract agreement to determine if there was a necessity and/or cost savings opportunity to renegotiate the agreement." This is similar to the justification the Companies provided for having renegotiated the damage prevention mark-out contracts. In response to IR DPS-1073, the Companies' explained that the request was for "an analysis that has not been performed" and objected to the request. They did state that, notwithstanding the objection, the GIOP "is not aware of any agreements that have been recently negotiated, or that the Companies are preparing to renegotiate that would have a material impact on the Companies' costs in the Rate Year."

Q. What conclusions do you draw from this IR

response?

A. We conclude that the Companies have chosen to try to revise their rate filings for the mark-out costs without assessing whether there are cost savings that may offset the increased costs in other areas of their original filings.

Q. What do you recommend?

A. We recommend disallowing the incremental O&M expense and capital cost associated with the contractor cost revisions for the damage prevention program. For KEDNY, this adjustment reduces Other Initiatives expense by $4.950 million. For KEDLI this adjustment reduces Other Initiatives expense by $1.937 $million, Depreciation expense by $0.001, and Rate Base by $1.290 million.

Q. Does this conclude your testimony at this time?

A. Yes.

BY MR. GOODRICH: (Cont'g.)

Q. Panel, also before you are exhibits identified at the time of filing as S.P.P. 1, S.P.P. 2, S.P.P. 3 and S.P.P. 4. These have now been marked for identification purposes as Exhibits 489, 490, 491 and 510 respectively. Were those exhibits prepared by you or under your direct supervision?

A. Yes.

Q. Do you affirm that the information contained in those exhibits just identified is true to the best of your knowledge?

A. Yes.

MR. GOODRICH: Your Honors, the Staff Policy Panel is now available for cross-examination.

A.L.J. COSTELLO: Okay. Thank you. Mr. Maloney.

MR. MALONEY: Thank you, Your Honor.

CROSS EXAMINATION

BY MR. MALONEY:

Q. Panel, it's -- my name is Kenneth Maloney by the way, and I'll be asking you some questions this afternoon hopefully relatively briefly. It's correct, isn't it that the companies

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas filed testimony and exhibits and supported rate changes for the year -- rate year ending March 31, 2021. And also included cost projections for an additional 3 data years ending March 31, 2022, '23 and '24?

A. Yes.

Q. But in proceeding with the litigation of the case, the staff is only addressing the rate year ending March 31, 2021. Is that correct?

A. Yes.

Q. From the panel's perspective recognizing that it's been the prior practice of the commission to litigate cases on a 1-year basis. Is there any reason why the panel believes a 3 -- a 2, 3 or 4-year case couldn’t be litigated?

MR. GOODRICH: Your Honors.

A.L.J. LEARY: Yes.

MR. GOODRICH: I'm -- I'm not objecting to the panel providing whatever answer they can provide to this. I would just note that there are potentially legal ramifications that I know that counsel's office has discussed from time to time over the years, not recently.

But I mean just that, that there are potentially legal aspects to this question that the panel wouldn't be able to answer.

A.L.J. LEARY: Can we just go off the record for 1 minute?

(Off the record 3:51 p.m.)

(On the record 3:51 p.m.)

A.L.J. COSTELLO: Indicate on the record that you withdrew that question.

MR. MALONEY: I -- I will withdraw the last question.

BY MR. MALONEY: (Cont'g.)

Q. But I would like to refer you to pages 12 and 13 of your proposed testimony where you talk about a proposed earning sharing mechanism in this case.

A. That's pre-filed right? That's the original on order.

Q. I'm sorry. Yes, your direct testimony.

A. Okay, we're there.

Q. Okay. Now, it's my understanding that earning sharing mechanisms have normally been

adopted in the context of negotiated settlements. Is that the panel's understanding as well?

A. It is our understanding that generally earning sharing mechanisms are adopted in a multiyear settlement. However, as we note on page 13, the commission has adopted an earning sharing mechanism in case 16-G-0257 and a 1-year case.

Q. And that would involve National Fuel, is that correct?

A. That's correct.

Q. And were there specific facts about the company such as a recent history of extensive over earnings or anything else the panel can identify that caused the panel to propose an earning sharing mechanism in this case?

A. Can you rephrase? I'm not sure when you say the company. We were just discussing National Fuel and I'm not sure whether you mean, you know --

Q. I am talking --.

A. National Fuel or -- or what?

Q. Let -- let me try again. Other than the National Fuel case, it's -- is there any other circumstance that you're aware of in which the

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas commission adopted an earning sharing mechanism in a litigated case?

A. I think we answered that in the last question. It's generally done in a multiyear settlement, however, on page 13, we -- we noticed that -- or identified that it has been done for National Fuel.

Q. But National Fuel is the only litigated case you're aware of where it's been done?

A. Yes.

Q. So then my next question was with respect to the companies KEDNY and KEDLI, were there particular circumstances that the panel identified that caused them to propose in this case which would recom -- which would represent the second time that an earning sharing mechanism has been imposed in a litigated case that caused the panel to propose it here?

A. So our recommendation is because the company, to the extent that it believes earnings are too low, can file a rate case and we believe that customers need protection to the extent that earnings are too high, and the companies stay out.

Q. Is it the policy panel's understanding that to the extent the commission believes that a utility's rates are no longer just reasonable, the commission has the power to issue a show cause order directing a utility to demonstrate why the commission should not reduce the utilities rates?

A. The commission does have that power. However, I would note that sometimes there's a lag between information when the company files its earnings, staff has to do an analysis, that could take some time. And having a mechanism upfront would protect customers immediately.

Q. Is it also though, the panel's understanding that if the commission believes that the utilities rates are excessive, it has the authority to make a portion of the rates temporary pending the outcome of its show cause order?

MR. GOODRICH: So I'm going to object. I mean, this is just talking about essentially what are provisions in the public service law.

MR. MALONEY: Where are we going with this, Your Honors, is -- it seems to me, the public service law was enacted sometime around 1907 and we

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas

went about a 100 years without earning sharing mechanisms being imposed. And I'm just trying to explore with the commission -- with the panel why that may have -- what's changed. Why are we now requiring this in cases?

A.L.J. LEARY: I think you can ask the question that you just stated but hold on a second. This is what I'll say. These are legal -- legally available options for the commission. I understand where you're going, but this is not a legal panel in terms of being able to speak for the commission, for the general counsel, for staff counsel in terms of when they're going to do what with the show cause order or set temporary rates.

So I mean I -- I think you can explore this area with the panel, but I -- I'm just cautioning that this is not their -- as far as I know, there's no lawyer on this panel that really would know what the extent of the commission's authority is and so forth. So the best --

MR. MALONEY: I'll move on. Thank you.

A.L.J. LEARY: Okay.

BY MR. MALONEY: (Cont'g.)

Q. Is it correct that the only way that the companies could avoid the -- the earnings mechanism is to file a new -- new rates that go into effect on or before October 1, 2021? And I -- I'd refer you to page 13 of your direct testimony, lines 13 to 15, I believe.

A. Can you repeat the question again?

Q. Is it correct that the only way the companies could avoid the earnings mechanism is to file new rates that go into effect on or before October 1, 2021?

A. Yes.

Q. Okay. Now, to avoid asking a legal question, but at the same time, were you present for the cross examination of the Staff Finance Panel this morning?

A. Took a few years off my life, but yes.

Q. Okay. It's correct, isn't it that the staff is recommending a return on equity of 8.2%. Is that correct?

A. (Duah) That's correct.

Q. And the commission has to make a determination in these cases as to the appropriate return on equity, is that correct?

A. That's correct.

Q. Okay. And in setting the appropriate R.O.E., I believe I asked this morning. The commission should establish an R.O.E. that provides the company with an opportunity to earn a return that is comparable to the returns earned on by entities that have corresponding risk to the companies?

MR. GOODRICH: Objection. I'm -- I'm failing to understand the relevance of this to -- to discussion of an earnings sharing mechanism that doesn't go into a -- that wouldn't go into effect but for the company not filing for new rates when it legally has the ability to do so.

A.L.J. LEARY: So the -- the panel -- the panel does talk in their testimony about the R.O.E. So is that -- help me out here. What -- what -- how would you --

MR. MALONEY: Well --

A.L.J. LEARY: -- address Mr. Goodrich's objection.

MR. MALONEY: The -- the -- the point is the panel -- the case is setting it R.O.E. for a rate year, which is the rate year beginning April 1. The panel is now proposing that the R.O.E. extend beyond the rate year. And what I'm going -- where I'm going with this is I'm going to explore with them why they believe it's appropriate to extend the R.O.E. beyond and -- and see whether that's reasonable or not.

MR. GOODRICH: It's -- but it's not proposing to extend the R.O.E. Rates are set based on a rate year, but they continue in effect until change.

A.L.J. LEARY: You stay out, right?

MR. GOODRICH: That's your choice.

A.L.J. LEARY: So if you stay out, you -- you make the decision that the R.O.E. is going to continue I -- I think, right? Am I missing something?

MR. GOODRICH: Yes.

MR. MALONEY: Well -- well the companies -- and what I want to establish is that I want to compare what the results would be under the earning sharing mechanism over a couple of years

19-G-0309/19-G-0310 - 2-20-20 - Brooklyn Union Gas compared to what would be the returns earned by companies with corresponding or commensurate risks. That's where I'm ultimately going with this.

MR. GOODRICH: I guess though, my -- my thing is he's talking about the returns outside of the rate year and the -- the staff testified and not even this panel that the finance panel. I think the reference to R.O.E. in this panel is simply summarizing what -- what's in the rest of staff's testimony.

But the Staff Finance Panel testified to an R.O.E. for the rate year. No one is testifying to an R.O.E. for that -- that would be appropriate --

A.L.J. LEARY: Indefinite --

MR. GOODRICH: -- in the future.

MR. MALONEY: Nonetheless, the panel is testifying to a mechanism that would require the -- that would permit the commission to confiscate a portion of the company's earnings.

A.L.J. LEARY: No, confiscate?

MR. MALONEY: Yes.

A.L.J. LEARY: Whoa, those are fighting words.

MR. MALONEY: Well, it's a government -- it's a government imposed mechanism that takes part of the company's earnings. Isn't that what it does?

A.L.J. LEARY: It -- and it also incentivizes a number of other matters, but let's not get into that discussion. This last -- you just said something that I -- I have a better understanding of where you're going. But I'm going to just let the panel answer whatever question was then pending and you can renew your objection, Mr. Goodrich, because I understand -- I just want the record to be clear as I think you actually, Mr. Goodrich have clarified it and my understanding of it.

But I -- I want to give the company some latitude here because I think there's a piece here that they're entitled to ask about. Don't ask me what that is.

BY MR. MALONEY: (Cont'g.)

Q. Let me try this.

A. (Rider) Okay.

Q. It's correct isn't it, that the proposed earnings mechanism is tied to the company's

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earned returns on equity for periods beyond the rate year?

A. Yes.

Q. And in this case, the Staff Finance Panel is only proposing an R.O.E. that would be just and reasonable for the rate year. Is that correct?

A. Yes.

Q. To the -- to the know -- to the knowledge of the panel the Staff Finance Panel has not attempted to estimate what the appropriate R.O.E. would be for the companies in for example, F.Y. 2022, F.Y. 2023 or F.Y. 2024?

A. The goal of this -- this case is to set rates in the rate year.

Q. Right.

A. The company has the option to file at the end of the rate year to the extent it believes that it needs additional revenues to run its business in a safe and efficient manner.

Q. But if the companies were to find themselves in a situation where their cost of equity has increased, but nonetheless they found efficiencies to offset those cost increases and still

continue to earn a compensatory R.O.E., wouldn't the mechanism that you're imposing require them to come in and file and incur the costs of doing that in order to reset rates for no reason?

MR. GOODRICH: Objection, speculative. There's so many moving parts in there. It's like impossible to -- to --

MR. MALONEY: And -- and it goes to the whole fairness of the mechanism which is it establishes a -- okay.

A.L.J. LEARY: We don't entertain argument. We just deal with objections. Mr. Goodrich has again asserted an objection which is that the question is speculative.

MR. GOODRICH: It calls for speculation in that there are so many moving parts that it's impossible I would -- I would argue it's impossible for the panel to provide an answer that is going to provide any sort of meaningful insight.

A.L.J. LEARY: Well, that -- let the panel not provide meaningful insight. But I do recommend a breakdown of a number of the moving parts, even if you just break it into two. So the hypothetical setup --

MR. MALONEY: Let me pass out an exhibit and -- and -- let me pass this out. And this is an exhibit that was prepared by me. I'm just wondering as to how the earning sharing mechanism would work.

A.L.J. LEARY: Well, I think it's on page 14 of their testimony. So it's 8.2 plus 50 basis points is the first band, right? Is that right? Am I reading that correctly?

MR. MALONEY: Yes.

A.L.J. LEARY: Thank you.

BY MR. MALONEY: (Cont'g.)

Q. So panel, do you have the exhibit in front of you?

A. Yes.

Q. And is it correct and --

MR. MALONEY: Can I ask that this be marked for identification, I'm sorry.

A.L.J. COSTELLO: We'll mark for identification as Exhibit 666.

A.L.J. LEARY: And just --

MR. MALONEY: Somehow that's appropriate I suppose.

A.L.J. LEARY: You're going to ask the panel to identify this.

MR. MALONEY: I'm going to ask them whether the information in it is correct. I'm trying simply to simplify, for purposes of cross-examination whether this -- the data presented on this piece of paper is correct as to how they understand their proposed mechanism works.

A.L.J. LEARY: Okay.

BY MR. MALONEY: (Cont'g.)

Q. So panel, is it correct that from 8.2 to 8.7 the company share of -- of earnings would be 100%?

A. I'll point you to page 13 of our testimony where it says, we recommend that the companies be allowed to retain a 100% of earnings up to and including 50 basis points above their allowed R.O.E.

Q. And so this is correct?

A. That's right.

Q. Okay. And -- and from 8.7 to 9.2, the company would keep 8.7% up to 25 basis points or a cap of 8.95%, is that correct? From 8.7% to 9.2% the company would keep 50% and that would

mean that they would keep 8.7% up to 25 basis points or a cap of 8.95%?

A. Yes.

Q. Okay. Now, from -- from 9.2% to 9.7%, it would be 25%, so that effectively the companies would be able to keep out of 9.7%, 8.95% up to 12 and a half basis points or a cap of 9.075%, is that correct?

A. Yes.

Q. Okay. And finally from 9.7% to infinity, the companies would keep 10% above 9.7% and would earn above 9.075%, is that correct?

A. I don't know about infinity, but --

Q. As high as it could go.

A. I would say greater than --

Q. Yes.

A. -- 9.7, yes.

Q. So just to do the math, to earn a 10.5% R.O.E. for shareholders, the companies would earn -- need to earn a total R.O.E. of approximately 24%. Is that correct?

A. What's the calculation behind that? I don't -- I -- I'd have to see the math.

Q. And I would go down there below shareholders --

A.L.J. LEARY: One at the time.

MR. GOODRICH: And I would also object as to the relevance of a 10.5% R.O.E.

MR. MALONEY: Well, we're going to talk about that in a second.

A.L.J. LEARY: Here -- here is the problem I'm having. This is not a document that was prepared by this panel or your panel -- well, one of your panels. This is a document you prepared, correct Mr. Maloney?

MR. MALONEY: And I'm just asking as to whether the information in it is correct. I don't need to move this into evidence. I'm just trying to simplify the cross examination.

A.L.J. LEARY: Okay. The -- I think Mr. Rider has indicated he would have to see sort of the background on the math for where -- what your next question asked.

BY MR. MALONEY: (Cont'g.)

Q. Well, if -- if you -- as -- as we said, shareholders would keep 9.075 of a 9.7% R.O.E. and they would keep 10% above, which is 1.4% of an

additional 14% R.O.E., would you agree with that math?

A. If you provide me the calculation, I will verify the numbers.

Q. Well, I just -- we agreed that from 9.7% to infinity, right, the number was 9. -- at -- at 9.7% the company would keep 9.075%, is that correct?

A. Again, if you provide me the math, I can verify the numbers. I'd -- I'm not following how you're -- you're getting the -- the percentages that you have on this -- this Exhibit 666.

Q. Okay. Would the panel agree that in order to earn, for example, an equity return of 10.5% the companies would have to for the shareholders, under your earnings sharing mechanism, the company would have to earn an R.O.E. far in excess of 10.5%?

A. Yes, and that's why we would have an earning sharing mechanism to protect customers in case that situation occurred.

Q. Now this morning, we talked about the fact that the Staff Finance Panel testified and

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is it correct that in the Staff Finance Panel testimony the staff created a return on equity as part of its sustainable growth calculation for the year 2022 to 2000 -- I'm sorry, bear with me.

From 2023 that according to -- for its proxy group on average was 11.14% to on an average at a median of 10.64% and I'm referring you to Exhibit 440.

MR. GOODRICH: Could counsel please clarify the period? I was -- I was confused by the -- the year.

MR. MALONEY: Yeah, I'm sorry. It was for 2023.

MR. GOODRICH: For calendar year 2023?

MR. MALONEY: Yes, 2023. Yeah, I can pass out copies of the exhibit.

MR. GOODRICH: I don't have that exhibit.

MR. MALONEY: Go ahead and pass those up.

THE WITNESS: (Rider) So we have the exhibit, can you point us to --

BY MR. MALONEY: (Cont'g.)

Q. Sure. If you could look at page 2 of 2.

A. Okay.

Q. There is a column, the second column and it states that the return on equity, it calculates a return on equity. And I believe that was a number prepared by the Staff Finance Panel for each member of the proxy group. And at the bottom it says that the average is 11.14% and the median is 10.64%.

A. That's what the document says.

Q. Okay. And this proxy group represents a calculation by staff of what the proxy group is forecast to earn in 2023, is that correct on average, with data that was obtained from Value Line?

A. I believe the Staff Finance Panel testified to this. So are you asking our opinion of their testimony, I'm not --

Q. No, I'm asking you -- I'm asking -- I know Mr. Duah was on the finance panel and on this panel. So I'm asking is that in fact what this shows?

A. (Duah) Yeah. That's the one I was going to develop this one.

A. (Rider) Right.

Q. And so the question is, okay, under your earning sharing mechanism coupled with the 8.2% return on equity, would the company -- is it the company's testimony that the returns being offered to the company, an 8.2% return and the earning sharing mechanism that we've described would allow the company in 2023 to earn a commensurate return with the average or median of the proxy group?

A. I'm confused. You said company.

Q. The companies.

A. The companies?

Q. Is it the panel's testimony?

A. Which panel? I'm confused. Can you -- can you re --

Q. Sure. I'm sorry. It's been a long day. Is it the panel's testimony that at 10 point -- an 8.2% proposed return on equity, coupled with the proposed earnings sharing mechanism would afford the companies an opportunity to earn a return commensurate with the average return that the staff proxy group -- that the Staff Finance Panel identified for the staff proxy group in 2023?

A. So you're miss -- mixing two periods of time, right? So the Staff Finance Panel is -- is recommending a return in the rate year. And the earning sharing mechanism does not kick in until after the rate year.

Q. In 2023, it would apply?

A. That's correct.

Q. And so you're setting a return for 2023?

A. No, the company has the --

Q. You are?

A. No. The company has an opportunity and it makes a decision after the rate year to file for new rates. And if the company --

Q. So the company -- I'm sorry.

A. -- does not file for new rates, then it's accepting the 8.2 and it continues to operate its business.

Q. But I'm asking you whether those returns are commensurate, do you believe that they are? The staff proxy group was thought to be a comparable group to the companies, is that correct?

MR. GOODRICH: So again, objection. Look, as -- as the panel has explained, the E.S.M. is

for periods after the rate year. If this case leads to a 1-year litigated case, staff would not be doing an analysis of what the appropriate R.O.E. would be in that future period unless and until the company seeks to increase its rates. So I mean, this is -- you're talking about 2023 is the last three months of the year after the rate year, if I -- if I understand correctly.

MR. MALONEY: While the staff may not be proposing a – an earned return for that period, they are proposing a rate making mechanism which would deprive the company of the ability under any set of reasonable circumstances to achieve the rate returns that are even close to what's forecast for the proxy group. That's my point.

MR. GOODRICH: It's -- it doesn't deprive the company of anything. The company has the option to file as the panel has testified to repeatedly.

A.L.J. LEARY: And is it -- am I correct in understanding, if in fact the company files, they don't stay out, they come back in, the R.O.E. would be at that point evaluated by staff?

MR. GOODRICH: Correct. Just like in this case.

A.L.J. LEARY: Mr. Maloney, a couple concerns. I think you're asking questions about Exhibit 440 that are perfect for the finance pine -- panel and not particularly perfect for this panel. This is a public policy can -- panel and they don't set the R.O.E. What they do is something different so that -- that may be part of the frustration here that you're talking to the wrong panel maybe.

So it seems pretty stark to me that the company doesn't like this earning sharing mechanism. I'm -- I understand why, everybody in the room understands why. Is there something that this panel -- that you can ask this panel that goes to their testimony or the bands that are outlined for the E.S.M. on page 13 that may help you make your point?

BY MR. MALONEY: (Cont'g.)

Q. I'll just ask one more question and move on which is simply, for the company to file a case, requires it to incur millions of dollars of expenses, as well as hundreds of thousands of man hours, I mean, hundreds or thousands of man hours of

-- of time, not only from the company's perspective, but from parties like staff and other parties?

A. There are rate cases expenses incurred by the company and paid by ratepayers when utilities file for rates.

Q. Okay. Let's move on to page 29 of your testimony. At page 29, you discuss certain changes to the customer service quality N.R.A., Negative Revenue Adjustments, is that correct?

A. The testimony filed in August?

Q. Yes.

A. Yeah. That is a summary of the Staff Consumer Service Panel.

Q. And on page 34 of your -- of the same testimony, you discuss certain changes to the pipeline safety metrics, is that correct?

A. That's correct.

Q. Would you agree that all other things being equal, the changes that staff is proposing to the customer service metrics and the service quality metrics increase the company's risks and the risk specifically of not earning the return on equity?

A. So this testimony is a summary of the Consumer Services and the Gas Safety Panel metrics. So those panels really are testifying to the targets. However, I would note that the company is provided in rates a level of expenses to achieve its goals. And the panel believes that if it operates as it has, then it could achieve those goals and it doesn't necessarily increase the risk of the company.

Q. Can the panel identify any specific funding that the company -- that the staff is proposing to provide the company in this case, to enable it to meet the enhanced metrics that we are discussing?

A. What metrics?

Q. The service quality and safety metrics that we were just discussing.

A. I believe those panels, the Consumer Service Panels and the Gas Safety Panels reviewed the cost the company had proposed in the rate year and they would be best offered that question.

Q. Well, to the extent that it were not possible to identify any specific incremental

dollars that the company received to meet these revised metrics, I'll come back to my original question, wouldn't you be imposing risks without offering the company additional funds to achieve those risks.

MR. GOODRICH: So objection. The panel has stated, I think twice now that this was a summary of testimonies provided by other panels. The Customer Service Panel was already up, and the company had an opportunity to cross examine that panel regarding the targets that the -- the panel set and the basis for those targets. The company is seeking a second bite at a panel that is -- has merely summarized staff's position on -- with regard to these items.

MR. MALONEY: They're testifying to the positions that they are the policy panel, which connotes at least in my mind, that they're the broad -- this is why we do these things in the aggregate sort of panel. And my question led to the overall change in risk associated with these proposals which they summarized through their testimony.

A.L.J. LEARY: They summarize the risk in their testimony?

MR. MALONEY: No, they summarize the proposals that I'm asking them about what the impact of those proposals are.

THE WITNESS (Rider) No, you're asking me whether I -- what specific cost elements and not -- that was addressed in each of those panels.

MR. MALONEY: Right. But -- but --

A.L.J. LEARY: So risk is what I'm sort of struck with and that to me, it seems, is another panel's testimony but --

BY MR. MALONEY: (Cont'g.)

Q. In this case, it's correct isn't it that the companies may not get an order -- a rate order until sometime in the middle of the year?

A. It's possible that the commission could decide sometime in July.

Q. Okay. And given that, when would the staff propose that the revised metrics, safety metrics and -- and customer service quality metrics take effect?

A. I believe that was -- that testimony is in the Gas Safety Panel or the consumer service metrics or Consumer Service Panel when they become effective.

Q. Well, to the extent that they proposed that they be effective on a calendar year basis, would it be the Staff's Policy Panel's recommendation that the metrics would not take effect until January 1st, 2021?

A. We're not usurping the -- those panel's testimony. I -- I believe their positions speak for themselves.

Q. Okay. I want to move on and ask you some questions about Newtown Creek and your proposal there. Is it the panel's understanding that the joint proposal in case 16-G-0058, I believe, which I think was KEDNY's last case, was signed on September 7th, 2016, is that correct?

A. I don't know the exact date it was signed but it's -- it seems -- okay, yes.

Q. Okay. If you could look at your testimony, your direct testimony at page 48.

A.L.J. COSTELLO: And I just want to clar -- are you talking about KEDNY?

MR. MALONEY: KEDNY, yes.

A.L.J. COSTELLO: Okay. I think that's 16-G-0059.

MR. MALONEY: Thank you, Your Honor. They were backwards in that case.

A.L.J. LEARY: Are you still on Newtown Creek because that starts at 39.

MR. MALONEY: And I'm referring him to his testimony at page 48.

A.L.J. LEARY: 48, okay.

THE WITNESS: (Rider) I'm there.

BY MR. MALONEY: (Cont'g.)

Q. And looking at lines 20, 21, 22, at the time that the joint proposal in the company's last case, it's correct isn't it, that the Newtown Creek project was sanctioned with an estimated cost of $34.1 million, is that correct? And I'm looking at lines 20 and 21 for this.

A. Those dates seem to line up. I would note though that when a joint proposal was signed and what information staff has throughout that process may be somewhat different.

Q. Will the staff -- does the panel recall what it was aware of at the time it signed the J.P. with respect to the projected cost of the Newtown Creek project?

A.L.J. LEARY: You want to ask if they were involved in that J.P. first or have some awareness?

BY MR. MALONEY: (Cont'g.)

Q. Mr. Rider, were you involved in the negotiation of that J.P.?

A. Of course, I was, yes.

A.L.J. LEARY: Sorry. Dumb question.

MR. MALONEY: I'm sorry.

A.L.J. LEARY: Just need to protect the record here, okay.

BY MR. MALONEY: (Cont'g.)

Q. So what is --

A. I don't -- I don't --

Q. -- your understanding of the amount that the Newtown Creek project was likely to cost at the time the J.P. was executed?

A. A very large amount of money.

Q. So north of $30 million?

A. Yes.

Q. Would that be fair? Thank you. And at the time that the joint proposal was executed, was it staff's understanding that the annual revenue record -- from sales of gas from the Newtown Creek

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project would exceed the annual revenue requirement for the Newtown Creek project?

A. It was our general understanding however, the project was not complete at that point in time.

Q. It was your understanding that the revenue from the sales would exceed?

A. Would not exceed.

Q. Would not exceed. Thank you. In proposing the revenue amputation in this case, has the staff panel identified any potential market for the R.N.G. produced by the Newtown facility that would permit KEDNY to recoup the costs of the Newtown Creek project?

A. Staff asked the company, I.R. D.P.S. 739 to provide work papers to support the rate year revenue estimate of $1.9 million in the rate year and they provided a response, an attachment. That's the extent of the information we have on the -- the amount of revenue that would be generated from the project.

MR. MALONEY: Thank you. I believe we have some I.R.s we'd like to mark with respect to Newtown Creek.

A.L.J. LEARY: This is the Newtown Creek R.N.G. project as opposed to the --

MR. MALONEY: Yes, that's correct, Your Honor.

A.L.J. LEARY: -- hazardous waste site?

MR. MALONEY: Yes. And I will say I'm very close to finishing so.

A.L.J. LEARY: That's okay. Take your time. Thank you.

MR. GAVILONDO: We'll mark this as Exhibit 667 for identification.

A.L.J. LEARY: Do you want to mark them as one or you want to keep them separate?

MR. GAVILONDO: Whatever is easiest for you, thanks.

A.L.J. LEARY: We would love to get to a 1000 exhibits.

MR. GOODRICH: From staff's perspective, I prefer to keep them separate since they're separate documents and might float apart at some point.

A.L.J. LEARY: As I said, we're happy to get to a 1000 exhibits. Do you want to do all of them, or you want to start with one?

MR. MALONEY: We could do it however, if I can go ahead if you want to start and we can do this.

BY MR. MALONEY: (Cont'g.)

Q. Panel, you've been handed Exhibit 667, is this a true and correct version of your response to I.R. request N.G. 13?

A. Yes.

Q. And would your answer to this be the same today if I were to ask you?

A. Yes.

Q. Does the panel have before it the response to N.G. 52?

A.L.J. COSTELLO: We don't have it before us so.

MR. MALONEY: So -- they're telling me to go with them.

A.L.J. LEARY: We have N.G. 13 marked as 667. Thank you.

A.L.J. COSTELLO: And we have N.G. 53, was that the one that you're going to next?

MR. MALONEY: 52, I believe.

A.L.J. COSTELLO: 52?

MR. MALONEY: I have 52.

A.L.J. COSTELLO: We have 53 and 54.

MR. GAVILONDO: Those were the last two, Your Honor.

A.L.J. COSTELLO: Okay.

MR. GAVILONDO: But there were four total of N.G. including 13 --

A.L.J. COSTELLO: Thank you.

MR. GAVILONDO: -- 52, 53 and 54.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: We don't have 52. Yeah, we do.

A.L.J. COSTELLO: All right. So we'll mark N.G. 52 as Exhibit Number 668.

A.L.J. LEARY: 53 is --

A.L.J. COSTELLO: And we'll do 50 -- N.G. 53 will be Exhibit Number 669 and N.G. 54 will be 670.

MR. MALONEY: May I proceed?

A.L.J. LEARY: Yes, please.

MR. MALONEY: Thank you.

BY MR. MALONEY: (Cont'g.)

Q. Panel, do you have before you the documents that have been -- the responses to N.G. 52, 53 and 54 which have been marked as Exhibits 668, 669 and 670?

A. (Rider) Yes.

Q. And are these true and correct copies of your responses to these information requests?

A. Yes.

Q. And do you affirm that the responses would be the same if I ask you the questions today?

A. Yes.

Q. Okay. Is the staff panel aware that in Con Edison's most recent rate case, case 19-G-0066, the commission approved a provision that authorized Con Edison to purchase renewable natural gas within the company service territory and acknowledged that such purchase may be more costly than conventional gas supplies?

A. No.

Q. Okay. And I just -- final couple questions. Turning to the staff supplemental testimony, your policy panel. And I'm referencing

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pages 7 and 8 where you -- it's correct, isn't it that the policy panel --

A. Hold on a second. Hold on.

Q. I'm sorry.

A. Okay.

Q. Is it -- is it correct that the panel proposes to disallow certain damage prevention costs that were included in the company's supplemental testimony?

A. Yes.

Q. Okay. Is it -- does the staff contend that allowing the proposed cost changes to be included in rates in these proceedings would be inconsistent with the commission's policy on test periods and major rate proceedings?

A. Can you rephrase? I'm not sure I understand.

Q. Does the staff panel believe that or contend that allowing the proposed cost changes for damage prevention to be included in rates would be inconsistent with the commission's policy on test periods and major rates proceedings? And what I mean by that is do you believe the update came too late under the policy statement?

A. I don't believe our testimony says that.

Q. It's correct, isn't it that the staff has had nearly two months since the company filed testimony concerning the increased damage prevention costs to review those costs?

A. Not necessarily. The company filed in December -- mid-December and staff had to turn around testimony at the end of January. So there was a limited amount of time and if you compare that to when the company actually knew when these expenses were going to be incurred, they started this process in early 2018. They went through a process where they determined that these costs or the -- these certain contractors and costs needed to be updated.

And if you go to their exhibit, I think it lays out the timeline there. So the company had plenty of time to explore these costs, then inform staff. What staff is really concerned about is that the company knew that these changes were forthcoming, but didn't inform staff, didn't educate staff or engaged in -- in staff in the development of these -- these changes.

Q. When the company files an update to its cost to service, would you agree that the cost should be known and measurable?

A. If the company proposes to recover costs from customers, yes, it should be known and measurable. However, I -- I'd give you an example of a cost that was on the horizon, N.E.N.Y., right. Staff had proposed in its testimony that there could be changes. It informed all of the parties that there could be changes.

And when the commission actually issued its order, then folks knew that these changes were, you know, what they were, and they wanted -- the commission wanted the -- these changes to be incorporated into the record.

A.L.J. LEARY: I'm sorry, Mr. Rider, you referred to a couple things that I just need to have clarified on the record. One of which was a reference to something that the company has provided. It was -- if you look at their schedules, what was that Mr. Goodrich, do you know?

MR. GOODRICH: So I don't know, but I know that I was just trying to -- Aric, you had said something like if you look at the company's exhibit.

A.L.J. LEARY: Exhibit, yes.

THE WITNESS: I can point you to an exhibit that lays out the timeline of when the company knew about damage prevention changes and give me a second and I'll identify that for you.

MR. GOODRICH: Is that the I.R. responses? I think it's our exhibit that's I.R. responses.

MR. RIDER: So I was referring to Exhibit S.P.P. 4, page 21 of 48. I believe the second -- Katie help me here on the order for N.E.N.Y.

A.L.J. LEARY: Yeah, The acronym, can you just speak into the microphone?

MS. MAMMENS: Yes. N.E.N.Y. is New Efficiency New York and the order was issued December 13th, I believe, 2019. Yes, we reflected that and the fact that it was forthcoming in our initial testimony.

MR. GOODRICH: And I would just add, Mr. Rider referenced S.P.P. 4 and that I believe is marked as Exhibit 510.

A.L.J. LEARY: Thank you. Mr. Maloney.

BY MR. MALONEY: (Cont'g.)

Q. Going to what you just testified to, knowing that the N.E.N.Y., because I can't remember what the acronym means, order was coming, how did that help the company, or anyone develop costs for it until the order itself was finally issued?

A. It -- it put all of the parties on notice that this cost was forthcoming to the extent that there was any questions or folks needed to coordinate. We knew -- folks know that, that this cost was forthcoming. The difference is that the company had known about damage prevention changes since 2018 and could have engaged staff early on in the process.

Staff could have understood what those changes were and the – what -- what the potential savings could be from -- from changes in these contracts. And to the extent that when we did get the contracts, we could have been more informed and able to review the I.R. responses that were provided in January by the company.

A.L.J. LEARY: And I just want to clarify. There was a previous energy efficiency order in 2018 or 2019. Is that right, Mr. Rider?

MS. MAMMENS: That's correct.

A.L.J. LEARY: So did that help the company to understand costs, targets, budgets, et cetera?

MS. MAMMENS: That order set out presumptive targets and I was incorrect. The order you're referring to is December 13th, 2018 and the order I was referring to was January of 2020.

A.L.J. LEARY: That's what I thought.

MS. MAMMENS: So that December 13th, 2018 order set out presumptive targets and budgets which -- which the company was able to use to predict the direction that commission was going ultimately in the January order.

A.L.J. LEARY: Yeah. I just got thrown by the date -- two dates because I heard you say -- I wasn't sure what you said but differentiating -- that's helpful to differentiate those two orders, thanks.

BY MR. MALONEY: (Cont'g.)

Q. At this point, does the panel have any reason to doubt that the company will incur the increased damage prevention costs identified by the companies in their testimony?

A. (RIDER)I don't know. One concern I have is that the number of tickets that the company used to forecast the -- the expense hasn't been vetted yet. There -- there may be -- should be normalized, I don't know. But really our concern is all around savings.

The utility said -- the company said that the damage prevention contractors would use updated and newer technology and it really hasn't been vetted from staff's perspective on whether there should be or could be savings on the company's side from implementing this new contract.

Q. If there were some savings, they might offset part of the increased costs, is that your understanding?

A. I haven't done that analysis. I don't know whether it'd be part of or in full.

Q. And when I asked you about the -- the increased cost, for -- for the damage prevention

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contractors themselves though, you don't dispute that those rates are going up, do you?

A. That's what the contracts state.

MR. MALONEY: Okay. Thank you, panel. I have nothing further.

A.L.J. COSTELLO: Mr. Goodrich, do you need some time?

MR. GOODRICH: I think we would like to take a few minutes, yes.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Very few.

A.L.J. COSTELLO: Say 10 minutes. Is that -- or five minutes, what do you?

A.L.J. LEARY: Can you do it in 10 minutes and I don't have to send Mr. Rider out because he is not going to be sent -- able to be sent out. We're off the record.

A.L.J. COSTELLO: So 10 minutes off the record.

(Off the record 5:03 p.m.

(On the record 5:08 p.m.)

A.L.J. COSTELLO: Mr. Goodrich.

MR. GOODRICH: We have no redirect.

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A.L.J. COSTELLO: Okay. Panel, thank you very much for your testimony and you're excused. And we're off the record.

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STATE OF NEW YORK

I, KAYLA ALLEN, do hereby certify that the foregoing was reported by me, in the cause, at the time and place, as stated in the caption hereto, at Page 1 hereof; that the foregoing typewritten transcription consisting of pages 1 through 3884, is a true record of all proceedings had at the hearing.

IN WITNESS WHEREOF, I have hereunto subscribed my name, this the 26th day of February, 2020.

Kayla Allen, Reporter